Workshop Manual

Crafter 2017 ➢
TGE 2017 ➢
e-Crafter 2019 ➢
e-TGE 2019 ➢

Running gear, axles, steering

Edition 10.2018

List of Workshop Manual Repair Groups

Repair Group

Technical information should always be available to the foremen and mechanics, because their careful and constant adherence to the instructions is essential to ensure vehicle road-worthiness and safety. In addition, the normal basic safety precautions for working on motor vehicles must, as a matter of course, be observed.

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00 – Technical data

1 Safety information

(VRL012137; Edition 10.2018)

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1.1 Safety measures when working on vehicles with a start/stop system

Risk of injury due to unexpected motor start

If the vehicle’s start/stop system is activated, the engine can start unexpectedly. A message in the dash panel insert indicates whether the start/stop system is activated.

– Deactivate start/stop system by switching off the ignition.

1.2 Safety precautions when working on subframe

♦ It is not permitted to weld or straighten load-bearing or wheel-guiding components of the suspension.

♦ Always renew corroded nuts and bolts.

♦ Bonded rubber bushes can be twisted only to a limited extent. Therefore, tighten the bolted connections of components with bonded rubber bushes only when the wheel bearing housing has been raised (unladen state) ⇒ page 8 .

1.3 Safety precautions when working on high-voltage system

Danger to life from high voltage

The high-voltage system is under high voltage. Severe or fatal injury from electric shock.

– Persons with life-preserving or other electronic medical devices in or on their body must not perform any work on the high-voltage system. Such medical devices include internal analgesic pumps, implanted defibrillators, pacemakers, insulin pumps and hearing aids.

– The high-voltage system must be de-energised by a suitably qualified technician.

Risk of injury due to unexpected engine start

On electric and hybrid vehicles, the operational readiness of the vehicle is difficult to detect. There is a risk of parts of the body becoming trapped or drawn in.

– Switch off ignition.

– Always store the ignition key outside the vehicle.

Risk of damage to high-voltage cables

Improper handling of high-voltage cables or high-voltage connectors may result in damage to their insulation.

– Never support body weight on high-voltage cables or high-voltage connectors.

– Never support any tools on high-voltage cables or high-voltage connectors.

– Never kink or severely bend high-voltage cables.

– Always observe the coding when connecting high-voltage connectors.

1.4 Safety precautions when working in the vicinity of high-voltage components

Danger to life from high voltage

The high-voltage system is under high voltage. Damage to high-voltage components can result in severe or fatal injury from electric shock.

– Perform visual check of high-voltage components and high-voltage cables.

– Never use cutting or forming tools, or any other sharp-edged tools.

– Never use heat sources such as welding, brazing, soldering, hot air or thermal bonding equipment.

1.5 Safety precautions when tow-starting and towing

Risk of irreparable damage to electric drive motor

Risk of destroying electric drive motor when towing incorrectly.

– Position selector lever in “N” position when vehicle is being towed.

– 50 km/h must not be exceeded when towing.

2 General information

⇒ “2.1 General notes on tyre pressure indicator”, page 3

2.1 General notes on tyre pressure indicator

- It is extremely important to adhere to the instructions and warnings in the following descriptions.
- Check whether the tyre pressure sensor should also be replaced ⇒ Vehicle diagnostic tester.

Note

- *Make sure that the tyre and the tyre pressure sensor do not touch each other during removal or assembly work.*
- *The tyre pressure sensor must not come into contact with water or be blown upon with compressed air when the wheel is cleaned.*

3 Repair instructions

3.1 General information

- ♦ The engine control unit has a self-diagnosis capability. Before carrying out repairs and for fault finding, first read event memory.
- ♦ For trouble-free operation of electrical components, a voltage of at least 11.5 volts is necessary.
- ♦ Do not use sealants containing silicone. Particles of silicone drawn into the engine will not be burnt in the engine and will damage the lambda probe.
- ♦ Vehicles are fitted with a crash fuel shut-off circuit. It reduces the danger of a fire in a crash as the fuel pump is switched off via the fuel pump relay.
- ♦ The system also improves the starting characteristics of the engine. When the driver door is opened, the fuel pump is activated for 2 seconds to build up pressure in the fuel system.

3.2 General repair instructions

To achieve the desired results when performing repairs, it is important to work with the greatest possible care and cleanliness, and to use proper tools which are in perfect condition. Of course, the basic rules for safety also apply during repair work.

A number of general notes on the individual repair procedures, which can otherwise be found in the relevant sections of the workshop manual, are summarised here. They apply for this particular workshop manual.

Special tools

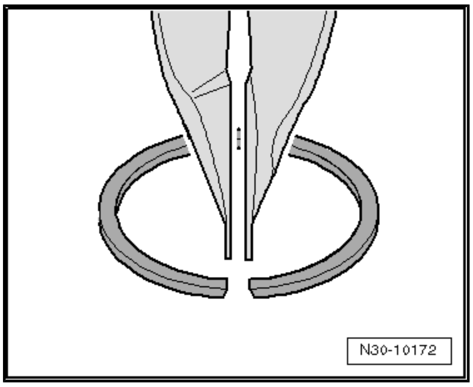
For a complete list of special tools used in this workshop manual, see ⇒ Workshop equipment and special tools .

Gearbox

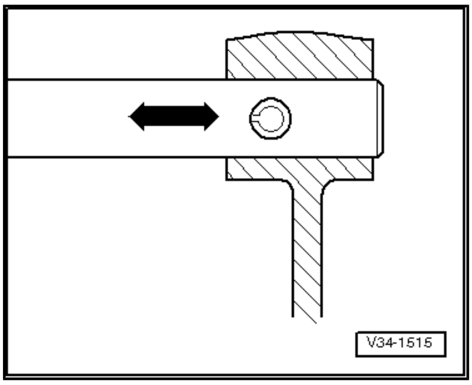
- ♦ When installing mounting brackets or waxed components, clean the contact surfaces. Contact surfaces must be free of wax and grease.
- ♦ Bolts and other components are replacement parts: ⇒ Electronic parts catalogue (ETKA) or ⇒ webMantis for MAN TGE.

- ♦ Check gearbox oil level after installing engine/gearbox. ⇒ Rep. gr. 34 ; Controls, housing; Gear oil; Checking gear oil level

Locking devices



- ♦ Renew retaining rings.
- ♦ Do not overstretch retaining rings.
- ♦ Retaining rings must locate properly in grooves.
- ♦ Renew spring pins. Installation position: slot must be in line with direction of force -direction of arrows-.



3.3 Rules for cleanliness

- ♦ Before disconnecting, thoroughly clean connections and the surrounding areas.
- ♦ Use commercially available, lint-free cloths for cleaning.
- ♦ Seal open lines and connections immediately with clean plugs from engine bung set - VAS 6122- .
- ♦ Place removed parts on a clean surface and cover them over. Use plastic foils or lint-free cloths.
- ♦ If repair work cannot be performed immediately, cover new parts which have been removed from their packing.
- ♦ Install only clean parts; do not remove new parts from packaging until immediately before installing.
- ♦ Protect disconnected electrical connectors from dirt and water, and reconnect them only when dry.
- ♦ Rinse off any escaped brake fluid from components using water.
- ♦ Dusty components should only be cleaned using compressed air and an extraction unit.

3.4 Rules for cleanliness when working on high-voltage system

When working on the high-voltage system, pay careful attention to the following rules for cleanliness:

- ♦ Thoroughly clean all connections/inspection holes and corresponding surrounding areas before disconnecting/opening them.

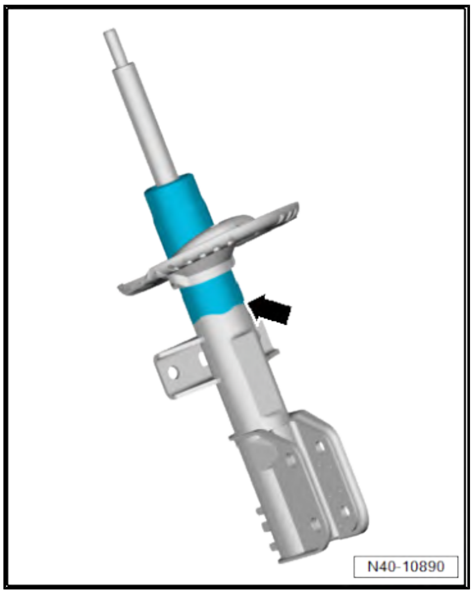
- ♦ Place removed parts on a clean surface and cover them over. Use lint-free cloths only.
- ♦ If repair work cannot be performed immediately, cover new parts which have been removed from their packing.
- ♦ Install only clean components.
- ♦ Remove packing from replacement parts immediately prior to installation and not before.
- ♦ Do not use parts which have been stored in unpackaged condition (e.g. in tool boxes, etc.).
- ♦ Transportation and protective packaging and sealing caps must be removed only immediately before fitting.
- ♦ Do not work with compressed air if the system is open. Do not move the vehicle.

3.5 Leaks at shock absorbers

Shock absorbers are often replaced because of externally visible leakage. Inspections on the test rig and in the vehicle have shown that in the majority of cases this replacement is not justified.

Slight loss of fluid (“sweating”) at the piston rod seal is not a reason for replacing a shock absorber. An oil sweating shock absorber is deemed OK under the following conditions:

- ♦ Fluid seepage (as shown in the blue part of the illustration) is visible, but the fluid is dull and possibly dried by dust
- ♦ The fluid seepage extends only from the top shock absorber seal (piston rod oil seal) down to the bottom spring plate -arrow-.



3.6 Noises from shock absorbers

Shock absorbers are often replaced because of noises (rumbling, etc.) experienced by the customer. Inspections on the test rig and in the vehicle have shown that in approx. 70% of these cases there is no case for complaint. The replacement was not justified.

In the case of complaints regarding rumbling or knocking noises please proceed as follows.

– Carry out a road test with the customer - if possible on a dry, uneven road - to establish when, where and how the noises occur.

Note

These noises are only very rarely caused by the shock absorbers.

3.7 Checking shock absorbers when removed

Defective shock absorbers can be identified by loud rumbling noises when driving, caused by wheel hopping, especially on bad roads. Heavy fluid leakage is an additional visual indication.

Note

Shock absorbers are maintenance-free; shock absorber fluid cannot be topped up.

After removal, a shock absorber can be checked by hand as follows:

– Compress shock absorber by hand.

• The piston rod should move smoothly over the entire stroke with uniform resistance and without jolts.

– Release piston rod.

• If the shock absorber has sufficient gas pressure the piston rod will return by itself to its original position.

Note

♦ *If this is not the case, the shock absorber does not necessarily need to be renewed. Provided there has been no major loss of fluid, it will still be as effective as a conventional shock absorber.*

♦ *Even without gas pressure, the shock absorber will provide full damping effect as long as there has been no major loss of fluid. However, it may produce more noise.*

3.8 Steering rack

To achieve the desired results when performing repairs on the steering rack it is important to work with the greatest possible care and cleanliness, and to use proper tools in good condition. Also note the basic rules on safety when performing repair procedures.

A number of general notes on the individual repair procedures, which can otherwise be found in the relevant sections of the workshop manual, are summarised here. They apply for this particular workshop manual.

♦ Thoroughly clean all unions and the adjacent areas before disconnecting.

♦ When installing the steering rack, make sure that dowel sleeves between bracket and steering rack are seated correctly.

♦ Place removed parts on a clean surface and cover them to prevent them from getting dirty. Use sheeting and paper for this purpose. Use lint-free cloths only.

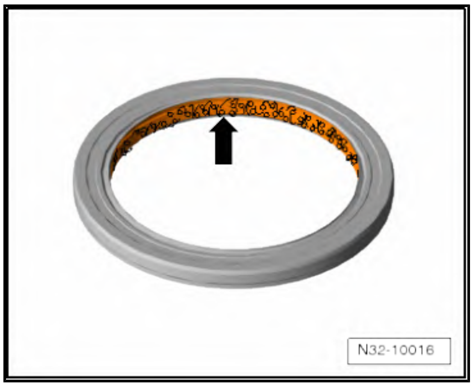
♦ Install only clean parts; do not remove new parts from packaging until immediately before installing.

♦ Use only the grease and sealants with specified part numbers.

♦ If repair work cannot be performed immediately, cover new parts which have been removed from their packing.

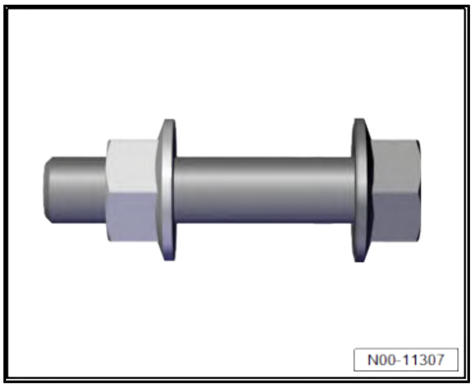
3.9 Gaskets and seals

- Always renew seals and gaskets.
- When seals have been removed, check contact surface on housings or shafts for burrs and damage and rectify as necessary.
- Completely remove all residue from liquid sealants from the sealing surfaces, making sure that no residual sealant gets into the steering rack housing.
- Open side of seals faces oil.



3.10 Nuts and bolts

- Avoid canting sensitive parts, such as servo motor with control unit. Loosen and tighten them diagonally in stages.
- Specified torques given are for unoiled nuts, bolts and screws.
- Always renew self-locking nuts and bolts.
- Always replace nuts and bolts with turning further angle.



3.11 Routing and attachment of lines

Risk of damage to lines

Lines may become damaged by moving or hot components.

– Route lines in their original positions.

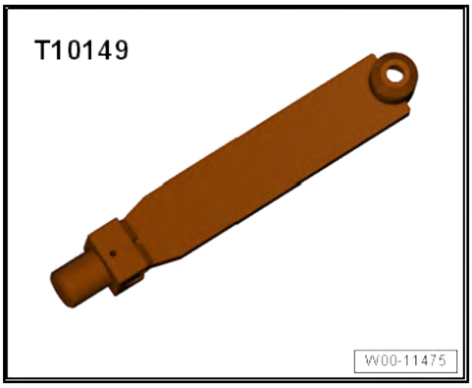
– Ensure there is sufficient clearance to moving or hot components.

3.12 Raising wheel suspension to unladen position

Special tools and workshop equipment required

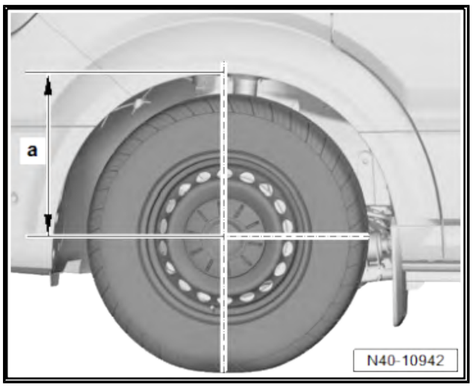
- Engine and gearbox jack - VAS 6931-

- Support - T10149-



The measurement must be performed in the unladen position (normal position).

– Before starting work, measure distance -a- from centre of wheel to edge of wheel housing using e.g. a tape measure.



– Note measured value.

Raising wheel suspension

Note

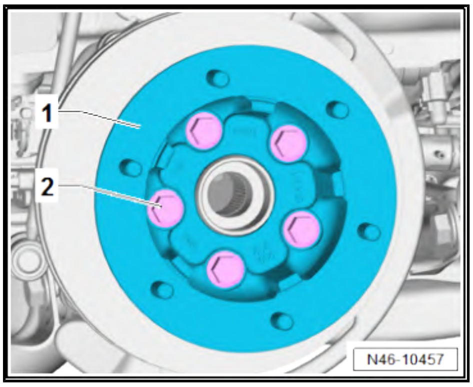
- *All bolts on running gear components with bonded rubber bushes may be tightened only when the component is in the unladen position (normal position).*
- *Axle components with bonded rubber bushes must therefore be brought to a position equivalent to the operating (unladen) position before tightening.*
- *To simulate this position on the lifting platform, raise the respective wheel suspension with the engine and gearbox jack - VAS 6931- and support - T10149- .*

– Raise vehicle ⇒ Maintenance ; Booklet 10.3 ; Description of work; Raising vehicle with lifting platform and trolley jack .

Vehicles with twin wheels

– Unscrew bolts -2-.

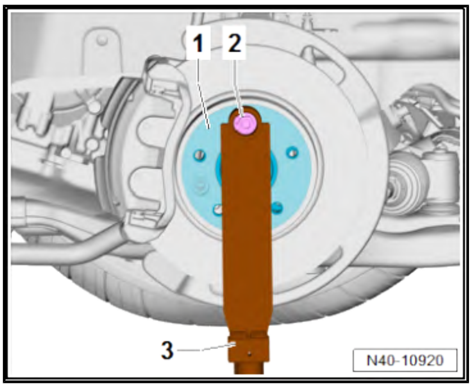
– Remove spacer -1-.



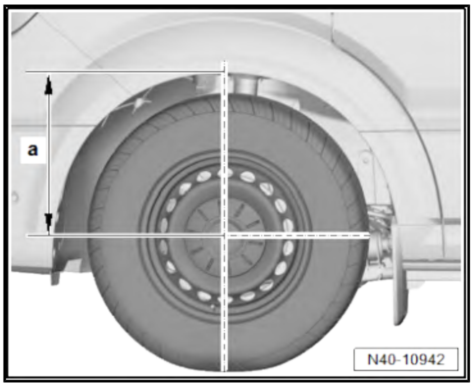
Continued for all vehicles

– Turn wheel hub -1- until one of the wheel bolt holes is at the top.

– Attach support - T10149- -3- with a wheel bolt -2-.



The respective nuts and bolts may be tightened only when dimension -a- between the centre of wheel hub and edge of wheel housing, measured before starting work, has been attained.



– Raise wheel bearing housing using engine and gearbox jack - VAS 6931- and support - T10149- until dimension -a- is attained.

– Tighten respective nuts and bolts.

– Lower wheel bearing housing.

– Pull engine and gearbox jack - VAS 6931- out from under vehicle.

– Detach support - T10149- .

Specified torques

♦ Front brake; Assembly overview – front brake ⇒ Brake system; Rep. gr. 46 ; Front brake; Assembly overview – front brake

4 Evaluating accident vehicles

⇒ “4.1 Check list for evaluating running gear on accident vehicles”, page 11

4.1 Check list for evaluating running gear on accident vehicles

Damage to running gear may go unnoticed during repairs to load-bearing and suspension parts of accident vehicles. Under certain circumstances, this undiscovered damage could lead to serious consequential damage during later vehicle operation. Therefore, the following parts of accident vehicles must be examined in the manner and order described independent of wheel alignment which may have to be performed. If no deviations from specifications are measured during wheel alignment, there are no deformations of the running gear.

Visual and functional examination of steering system

- ♦ Visual examination for deformation and cracks
- ♦ Examination for play in track rod joints and steering rack
- ♦ Visual examination for tears in boots
- ♦ Examine electrical and hydraulic lines and hoses for chafing, cuts and kinks.
- ♦ Check hydraulic lines, threaded connections and steering rack for leaks.
- ♦ Check steering rack and lines for secure seating
- ♦ Check for flawless function from lock to lock by moving the steering from stop to stop. In the process, the steering wheel must turn with a constant force without resistance.

Visual inspection and functional check of running gear

- Adhere to the sequence of the following inspection steps!
- ♦ Check all components shown in the assembly overviews for deformation, cracks and other damage.
- ♦ Renew damaged parts.
- ♦ Check the vehicle geometry on a wheel alignment stand approved by the manufacturer.

Visual and functional check of wheels and tyres

- ♦ Check for true running and imbalance ⇒ page 193 .
- ♦ Check tyres for cuts and impact damage in the tread and on the flanks.
- ♦ Check tyre inflation pressure; see tyre inflation pressure sticker or ⇒ Maintenance ; Booklet 10.3 ; Checking tyres: condition, tyre wear pattern, inflation pressure, tread depth .

If rim of wheel and/or tyre is damaged, renew tyre. This also applies if the circumstances of the accident and the damage to the vehicle indicate possible damage which is not visible.

A further factor in the decision is the age of the tyre. Tyres should not be older than 6 years.

If in doubt, the following always applies:

- Whenever a safety risk cannot be excluded, the tyre(s) must be renewed.

Complete vehicle

Also check other vehicle systems such as:

- ♦ Brake system including ABS
- ♦ Visual and functional examination of exhaust system and passenger protection

Specifications for testing and adjusting can be found in the respective workshop manual in ELSA.

The accident vehicle check described here applies to the running gear. This makes no claim to completeness for the overall vehicle.

Electronic vehicle systems

Safety-relevant systems like, for example, ABS/EDL; airbags; electronically regulated suspension systems; electromechanical or electrohydraulic steering and other driver assist systems, must be read with the vehicle diagnostic, testing and information system - VAS 5051B- for possible stored fault messages. If faults are saved in the event memories of the systems mentioned above, repair them according to instructions in workshop manuals in ELSA. Following repairs, read event memories of the affected systems again to ensure that complete function has been restored.

5 Waste disposal

⇒ “5.1 Releasing gas and draining front gas-filled shock absorbers”, page 13

⇒ “5.2 Releasing gas and draining rear gas-filled shock absorbers”, page 14

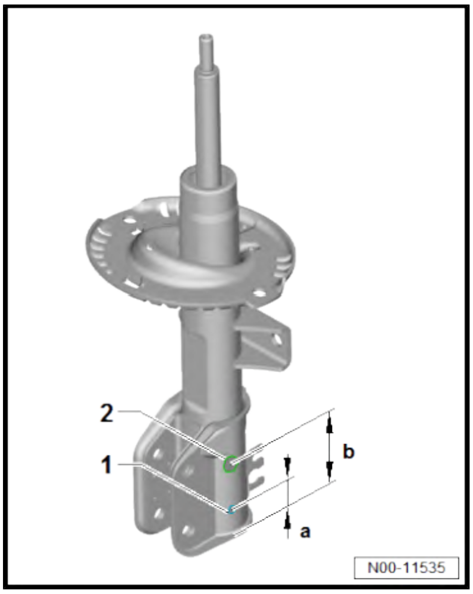
5.1 Releasing gas and draining front gas-filled shock absorbers

– Clamp gas-filled shock absorber vertically in vice, with piston rod pointing downwards.

Note

Safety goggles must be worn when drilling.

– Drill a hole of 3 mm in diameter -1- through the outer tube of shock absorber with a spacing of 20 mm -a-.



Note

Gas escapes upon penetration.

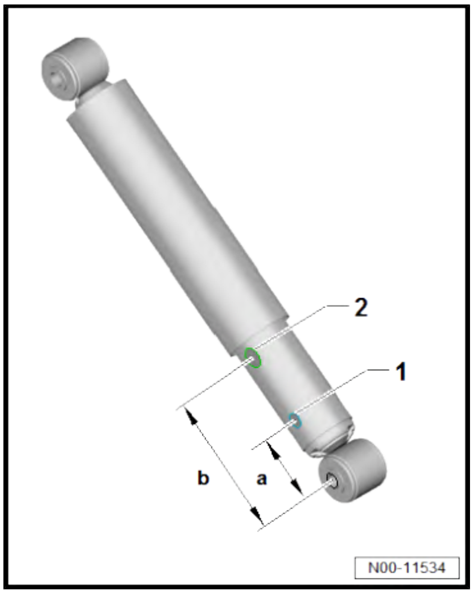
– Continue drilling until the inner tube has been penetrated (approx. 25 mm deep).

– Drill a second hole of 6 mm in diameter -2- through the outer tube of shock absorber and inner tube of shock absorber with a spacing of 60 mm -b-.

– Hold gas-filled shock absorber over a drip tray. Move the piston rod back and forth over its full extension until no further oil escapes.

5.2 Releasing gas and draining rear gas-filled shock absorbers

– Clamp gas-filled shock absorber vertically in vice.



Note

Safety goggles must be worn when drilling.

– Drill a hole of 3 mm in diameter -1- through the outer tube of shock absorber with a spacing of 50 mm -a-.

Note

Gas escapes upon penetration.

– Continue drilling until the inner tube has been penetrated (approx. 25 mm deep).

– Drill a second hole of 6 mm in diameter -2- through the outer tube of shock absorber and inner tube of shock absorber with a spacing of 150 mm -b-.

– Hold gas-filled shock absorber over a drip tray. Move the piston rod back and forth over its full extension until no further oil escapes.

6 Classification of dangers of the high-voltage system

DANGER

The vehicle's high-voltage system and the high-voltage battery are dangerous and can cause burns or other injuries and even lead to a fatal electric shock.

- **Any work on the high-voltage system, or on systems which could be indirectly affected by it, must only be carried out by properly trained and qualified expert personnel.**
- **In the event of queries or uncertainties regarding the terms "high-voltage technician" or "high-voltage expert", or those concerning the high-voltage system, the responsible importer must be contacted prior to the start of any work.**
- **Any repair work must be performed in accordance with applicable laws and regulations, the state-of-the-art technology, any relevant accident prevention regulations (in Germany, including but not limited to the BGI/GUV-I 8686 – Training for work on vehicles with high voltage systems), as well as this workshop manual.**

Before work on the high-voltage system is started, a high-voltage technician must de-energise the high-voltage system ⇒ Electric drive motor (270, LS2); Rep. gr. 93 ; De-energising high-voltage system .

The types of work for which the high-voltage system has to be de-energised are indicated in the list entitled "Work on the high-voltage system" ⇒ page 15 .

Work during which the high-voltage system has to be de-energised:

- ♦ Only the HVT ⇒ page 16 is authorised to de-energise the high-voltage system in a certified manner.
- ♦ All work on an electric drive vehicle must be carried out only by those who are at least qualified as electrically instructed persons (EIP) ⇒ page 16 .
- ♦ Regardless of the work to be performed, visually inspect high-voltage components in the work area.
- ♦ The high-voltage cables must not be extensively bent or kinked.
- ♦ In the event of conspicuous findings or uncertainties, the high-voltage technician (HVT) or the high-voltage expert (HVE) ⇒ page 16 must be consulted.
- ♦ Any work involving metal-removing, deforming and sharp-edged tools or heat sources such as welding, soldering, hot air, thermal bonding and infrared drying in the vicinity of high-voltage components and cables is prohibited. In this case the high-voltage system must be de-energised and the respective component removed or sufficiently protected.
- ♦ All listed work refers to the removal and installation or the renewal of the individual components.
- ♦ For safety reasons, no further work should be performed in or on the vehicle during the charging process.
- ♦ The high-voltage system does not have to be de-energised during regular maintenance work.

Explanation of qualifications

Working on high-voltage system

Qualification VW	Qualification MAN	Qualification DGUV-I 200-005	Scenario
EIP (electrically instructed person)	Person made aware of high-voltage issues in connection with MAN Truck & Bus AG	Chapter 2	Induction training for non-electrical work (< 60 VDC)
HVT (high-voltage technician)	Qualified vehicle electrician of the MAN Truck & Bus AG (for vehicles not being intrinsically safe in terms of high voltage)	Chapter 3.1 b	Work on inherently safe HV production vehicles ♦ Vehicles are de-energised exclusively with certification. ♦ Complete contact protection in place.
HVE level 1 (high-voltage expert)	Qualified vehicle electrician with authorisation to work in energised vehicles of MAN Truck & Bus AG (for vehicles not being intrinsically safe in terms of high voltage)	Chapter 3.2 b	Work on not inherently safe HV production vehicles ♦ Vehicles are also de-energised without certification or there is no complete contact protection in place, e.g. in the event of an accident.
HVE level 2 (high-voltage expert)		Chapter 3.3	Work on live accumulators ♦ Work on live parts, inevitably with no contact protection, for fault finding, component replacement, etc.

Observe the minimum qualification for the following work:	Minimum qualification:
De-energise high-voltage system	HVT ⇒ page 16

Work on power unit

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Three-phase current drive - VX54-	X			EIP
Electric drive motor - V141-	X			EIP
Temperature sender/rotor position of electric drive motor -G712- / -G713-	X			EIP

Work on power and control electronics for electric drive

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Power and control electronics for electric drive - JX1- with ♦ Electric drive control unit - J841- ♦ Intermediate circuit capacitor 1 - C25- ♦ Voltage converter - A19- ♦ DC/AC converter for drive motor - A37-	X			EIP
Air conditioner compressor fuse - S355-	X			EIP

Working on high-voltage system

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Potential equalisation lines (earth) with connection to HV components	X			HVT
Measuring insulation resistance		X		HVT
TW – Service plug for high-voltage system (service plug, service disconnector)			X	EIP
PX – High-voltage lines (orange) in vehicle	X			HVT

Work on heating and air conditioning system

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Electrical air conditioner compressor - V470-		X		EIP
High-voltage heater (PTC) - Z115-		X		EIP
Work on heating and air conditioning system in passenger compartment			X	EIP
Refrigerant lines in the vehicle periphery (work which is not directly on the AC compressor and can be carried out without opening the refrigerant circuit, e.g. loosening and securing refrigerant lines)			X	EIP

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Air conditioning performance test (for checking the pressure levels in the refrigerant circuit by means of air conditioning service equipment).			X	EIP
Refrigerant lines directly on electrical air conditioner compressor - V470-		X		EIP
Extracting, evacuating or replenishing refrigerant			X	EIP
Heat exchanger for high-voltage battery - VX63-			X	EIP

Work on charging connection

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
High-voltage battery charging socket 1 - UX4- with ♦ Temperature sender for charging socket 1 - G853- ♦ LED module for charging socket 1 - L263- ♦ Actuator for locking		X		EIP
Immediate charge button - E766-			X	EIP
Emergency release for charging connector		X		EIP
Charging high-voltage battery in workshop area			X	EIP

Work on charging unit

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Charging unit 1 for high-voltage battery - AX4- with control unit for high-voltage battery charging unit - J1050-		X		EIP

Work on high-voltage battery

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Removing and installing underbody cladding			X	EIP
Removing and installing high-voltage battery 1 - AX2-		X		HVT
Disconnecting high-voltage network connection		X		HVT
Disconnecting low-voltage connection of high-voltage battery			X	EIP
Battery regulation control unit - J840- (service flap)			X	HVT
Module monitor control unit for batteries - J497- (battery removed)				HVE
Switching unit for high-voltage battery - SX6- (battery removed)				HVE
Opening high-voltage battery 1 - AX2- (battery removed)				HVE
Bonding high-voltage battery 1 - AX2- (battery removed)				HVE
Battery module 1 - J991- and other battery modules (battery removed)				HVE

Working on accident vehicles

When working on the following components:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Preliminary evaluation of degree of danger				HVT
Body work (with straightening jig)	X			EIP
Painting vehicle – observe instructions in paint manual.			X	EIP

Work in vicinity of high-voltage components

When working on the following components or for following work:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Body work (assembly work as well as glass and dent repairs)			X	EIP
Gearbox with electric drive motor	X			EIP
Steering rack		X		EIP
Front subframe		X		EIP

When working on the following components or for following work:	Must the HVT de-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Front brakes			X	EIP
Rear brakes			X	EIP
Brake servo	X			EIP
Brake pressure accumulator for energy recovery of brake system	X			EIP
Rear axle and running gear			X	EIP
Underbody cladding			X	EIP
When welding, cover high-voltage components and visually inspect following work	X			EIP
Work involving metal-removing, deforming and sharp-edged tools or heat sources such as welding, soldering, hot air, thermal bonding and infrared drying in the vicinity of high-voltage components and cables	X			EIP
Work for which the motor is lifted, right side (e.g. motor mounting)			X	EIP
Work for which the motor is lifted, left side (e.g. gearbox mounting)	X			EIP
Control units and electric components of 12-V system (except airbag)			X	EIP
Battery - A-			X	EIP
Front left headlight - MX1-			X	EIP
Front right headlight - MX2-			X	EIP
Changing bulb in headlight			X	EIP

General work

When working on the following components:	De-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Draining and replenishing fluids (coolant and oils)			X	EIP
Coolant circuit and coolant expansion tank			X	EIP
Changing tyres			X	EIP
Miscellaneous work on 12-V system (except airbag)			X	EIP
Work on earth points of 12-V system (without potential equalisation lines)			X	EIP
Tail light clusters			X	EIP
Renewal or repair of glazing			X	EIP
Repairs in vehicle interior			X	EIP
Repairs in roof area			X	EIP
Repairs on rear lid			X	EIP

When working on the following components:	De-energise high-voltage system before work is started?			Minimum qualification: ⇒ page 16
	Yes (diagnostic and manual power disconnection)	Yes (only diagnostic power disconnection)	no	
Repairs on bumpers			X	EIP
Moving lock carrier to service position		X		EIP
Removing and installing lock carrier		X		EIP
Repair work on lock carrier		X		EIP

40 – Front suspension

1 Front axle

⇒ “1.1 Overview of fitting locations - front axle”, page 22

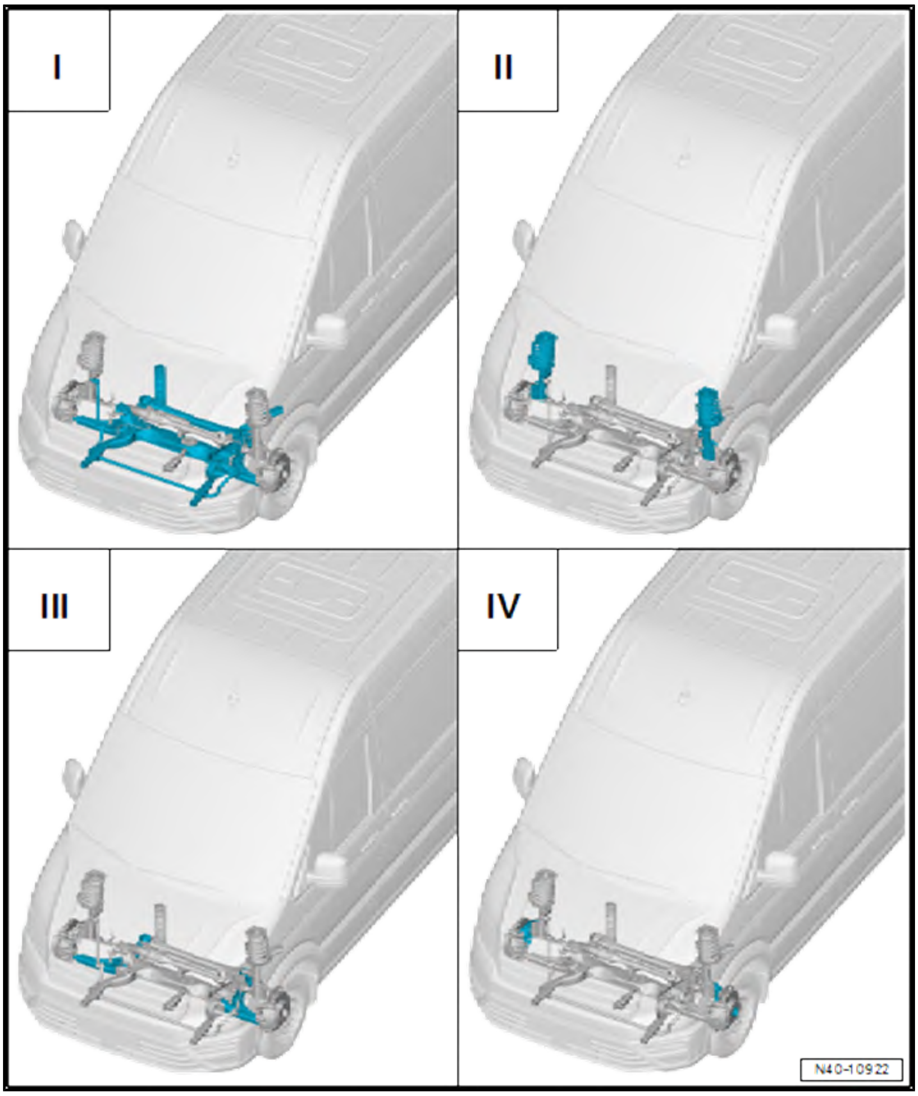
1.1 Overview of fitting locations - front axle

I - Subframe ⇒ page 23

II - Suspension strut, upper suspension link ⇒ page 59

III - Lower suspension link, swivel joint ⇒ page 65

IV - Wheel bearing ⇒ page 75



2 Subframe

⇒ “2.1 Assembly overview - subframe”, page 23

⇒ “2.2 Removing and installing subframe with steering rack”, page 29

⇒ “2.3 Repairing subframe”, page 37

⇒ “2.4 Removing and installing anti-roll bar”, page 49

⇒ “2.5 Renewing anti-roll bar bushes”, page 50

⇒ “2.6 Removing and installing coupling rod”, page 51

⇒ “2.7 Fixing position of subframe”, page 51

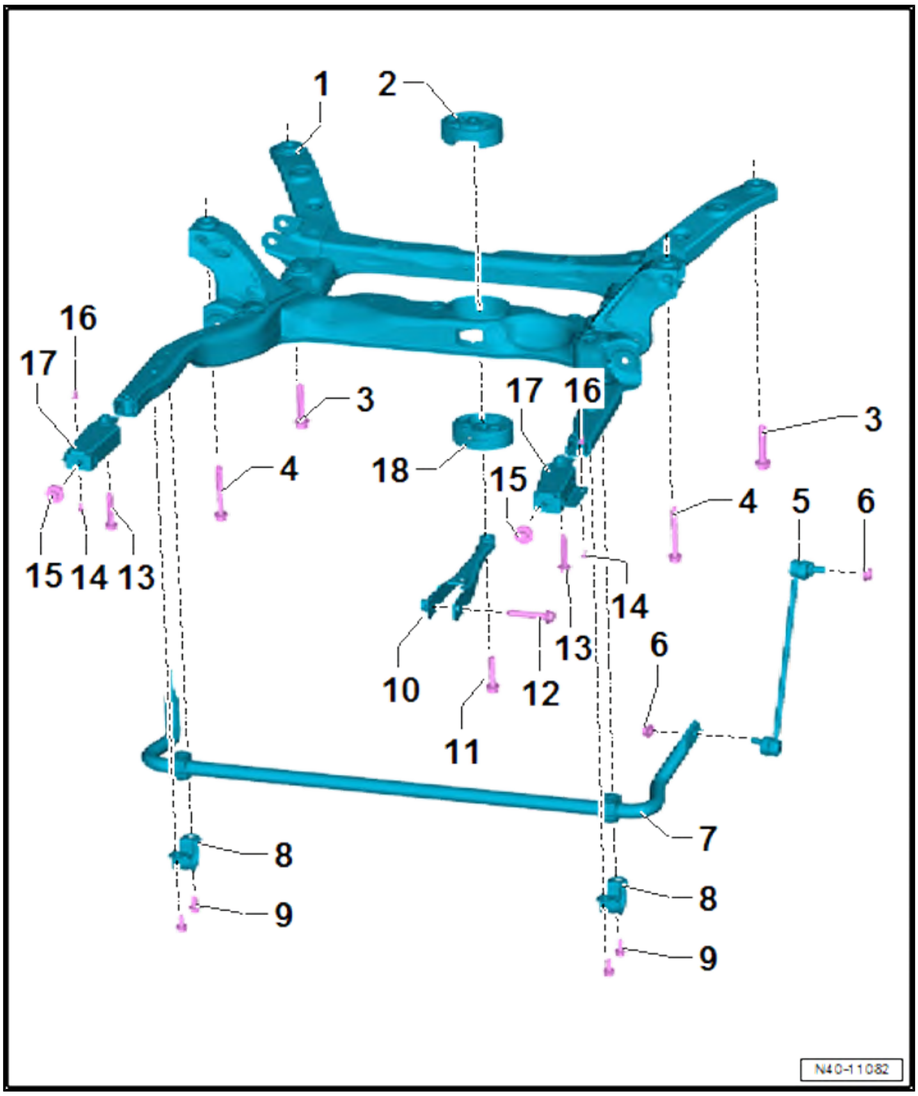
2.1 Assembly overview - subframe

⇒ “2.1.1 Assembly overview - subframe, transverse engine”, page 23

⇒ “2.1.2 Assembly overview - subframe, longitudinal engine”, page 25

⇒ “2.1.3 Assembly overview - subframe, e-Crafter or e-TGE”, page 27

2.1.1 Assembly overview - subframe, transverse engine



1 - Subframe
- ❑ Removing and installing ⇒ page 29

2 - Bonded rubber bush
- ❑ Top
- ❑ Removing and installing ⇒ page 37

3 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ M16x1.5x85
- ❑ Initially tighten to 180 Nm +90°, loosen by 120°, final torque 180 Nm +180°

4 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ M14x1.5x110
- ❑ Initially tighten to 130 Nm +90°, loosen by 120°, final torque 130 Nm +180°

5 - Coupling rod
- ❑ Removing and installing ⇒ page 51

6 - Nuts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 100 Nm +45°

7 - Anti-roll bar

❑ Removing and installing ⇒ page 49

8 - Clip

9 - Bolts

❑ Qty. 4

❑ 110 Nm

10 - Pendulum support

❑ Removing and installing ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing pendulum support

11 - Bolt

❑ Renew after removal

❑ Tightening torque ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings

12 - Bolt

❑ Renew after removal

❑ Tightening torque ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings

13 - Bolts

❑ Qty. 2

❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 50 ; Lock carrier; Assembly overview - lock carrier

14 - Bolts

❑ Qty. 2

❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 66 ; Wheel housing liner; Assembly overview - front wheel housing liner

15 - Bolts

❑ Qty. 2

❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 50 ; Lock carrier; Assembly overview - lock carrier

16 - Plastic nut

❑ Qty. 2

❑ 20 Nm

17 - Connecting piece for frame

❑ For frame

❑ Qty. 2

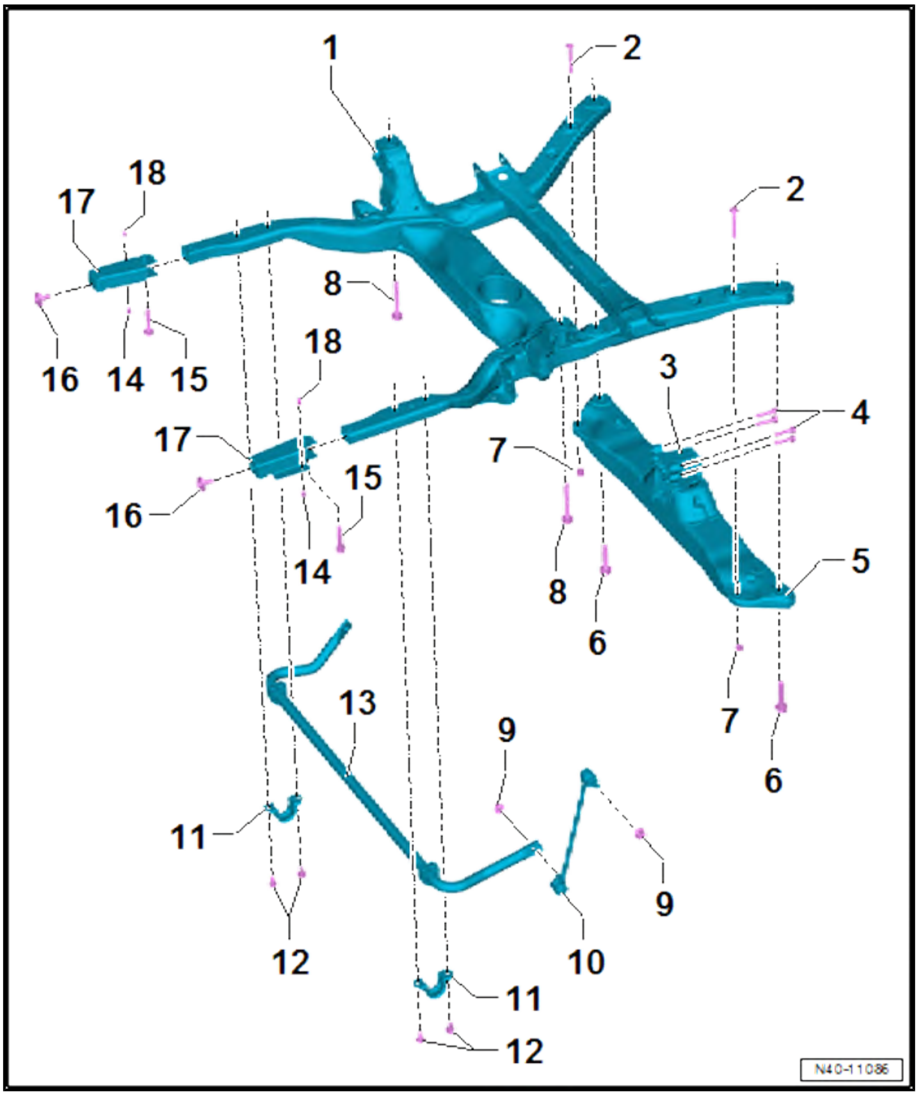
❑ Removing and installing ⇒ Rep. gr. 50 ; Body - front; Lock carrier; Removing and installing connecting pieces for frame

18 - Bonded rubber bush

❑ Bottom

❑ Removing and installing ⇒ page 37

2.1.2 Assembly overview - subframe, longitudinal engine



1 - Subframe

- ❑ Removing and installing ⇒ page 29

2 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ Rep. gr. 34 ; Assembly mountings; Assembly overview - assembly mountings

3 - Gearbox mounting

- ❑ Removing and installing ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing gearbox mounting

4 - Bolts

- ❑ Qty. 4
- ❑ Specified torque ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings

5 - Gearbox carrier

- ❑ Removing and installing in case of manual gearbox ⇒ Rep. gr. 34 ; Assembly mountings; Removing and installing gearbox carrier
- ❑ Removing and installing in case of automatic gearbox ⇒ Rep. gr. 37 ; Assembly mountings; Removing and installing gearbox carrier
- ❑ Specified torques in case of manual gearbox ⇒ Rep. gr. 34 ; Assembly mountings; Assembly overview - assembly mountings
- ❑ Specified torques in case of automatic gearbox ⇒ Rep. gr. 37 ; Assembly mountings; Assembly overview - assembly mountings

6 - Bolts

- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ page 26

7 - Nuts

- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ Rep. gr. 34 ; Assembly mountings; Assembly overview - assembly mountings

8 - Bolts

- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ page 26

9 - Nuts

- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 100 Nm +45°

10 - Coupling rod

- ❑ Removing and installing ⇒ page 51

11 - Clamps

12 - Bolts

- ❑ Qty. 4
- ❑ 110 Nm

13 - Anti-roll bar

- ❑ Removing and installing ⇒ page 49

14 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 66 ; Wheel housing liner; Assembly overview - front wheel housing liner

15 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 50 ; Lock carrier; Assembly overview - lock carrier

16 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 50 ; Lock carrier; Assembly overview - lock carrier

17 - Connecting piece for frame

- ❑ For frame
- ❑ Qty. 2
- ❑ Removing and installing ⇒ Rep. gr. 50 ; Body - front; Lock carrier; Removing and installing connecting pieces for frame

18 - Plastic nuts

- ❑ Qty. 2
- ❑ 20 Nm

Specified torques for bolts securing subframe to longitudinal member

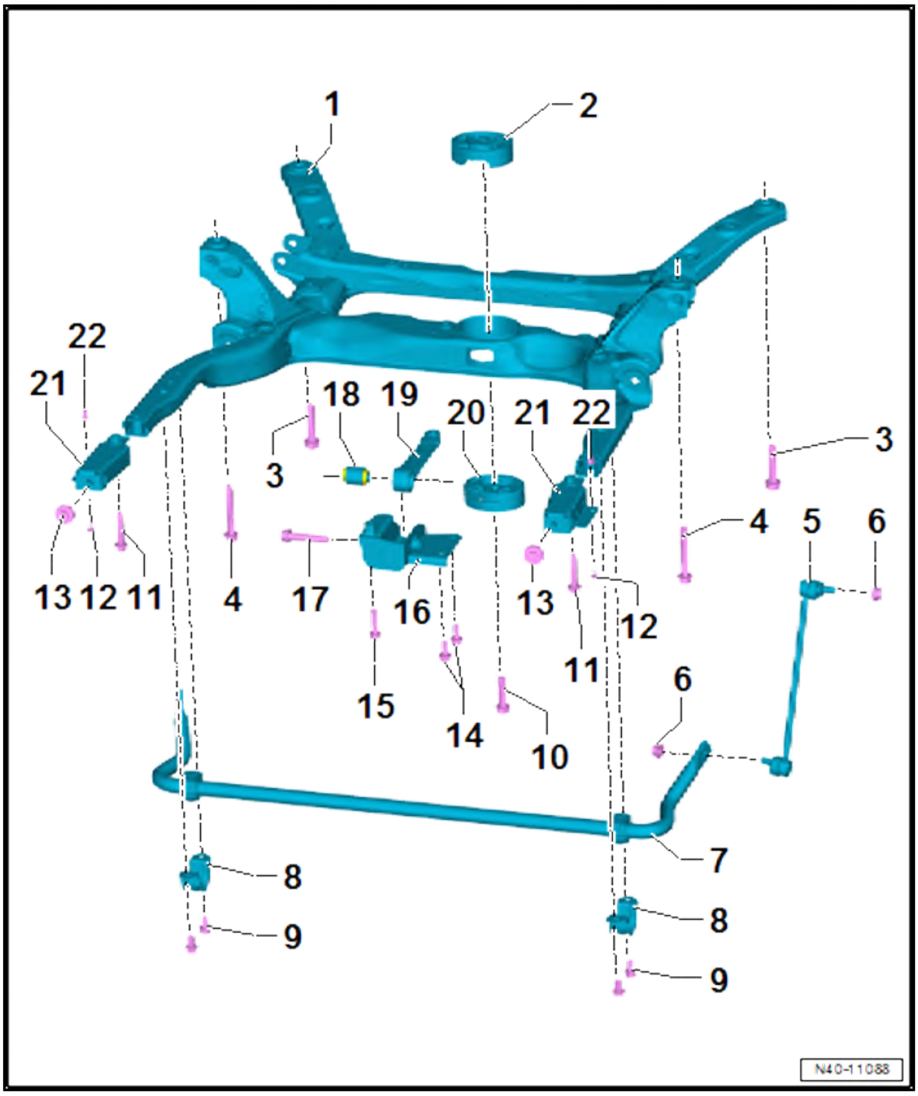
Stage	Bolts	Specified torque/turning further angle
1.	-6-	180 Nm +90°
2.	-6-	Loosen 120°
3.	-6-	180 Nm + turn 180° further

Specified torques for bolts securing subframe to longitudinal member

Stage	Bolts	Specified torque/turning further angle
1.	-8-	130 Nm +90°
2.	-8-	Loosen 120°

Stage	Bolts	Specified torque/turning further angle
3.	-8-	130 Nm + turn 180° further

2.1.3 Assembly overview - subframe, e-Crafter or e-TGE



1 - Subframe
- ❑ Removing and installing ⇒ page 33

2 - Bonded rubber bush
- ❑ Top
- ❑ Removing and installing ⇒ page 37

3 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ page 28

4 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ page 28

5 - Coupling rods
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 51

6 - Nuts
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 100 Nm + 45°

7 - Anti-roll bar
- ❑ Removing and installing ⇒ page 49

8 - Clamps
- ❑ Qty. 2

9 - Bolts
- ❑ Qty. 4
- ❑ 110 Nm

10 - Bolt
- ❑ Specified torque ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Assembly overview - pendulum support

11 - Bolts
- ❑ Qty. 2
- ❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 50 ; Lock carrier; Assembly overview - lock carrier

12 - Bolts
- ❑ Qty. 2

- ❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 66 ; Wheel housing liner; Assembly overview - front wheel housing liner

13 - Bolts

- ❑ Qty. 2
- ❑ Specified torque ⇒ General body repairs, exterior; Rep. gr. 50 ; Lock carrier; Assembly overview - lock carrier

14 - Bolts

- ❑ Specified torque ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Assembly overview - pendulum support

15 - Bolt

- ❑ Specified torque ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Assembly overview - pendulum support

16 - Bracket

- ❑ Specified torque ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Assembly overview - pendulum support

17 - Bolt

- ❑ Specified torque ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Assembly overview - pendulum support

18 - Bonded rubber bush

- ❑ Not an individual part

19 - Pendulum support

- ❑ Removing and installing ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Removing and installing pendulum support

20 - Bonded rubber bush

- ❑ Bottom
- ❑ Removing and installing ⇒ page 37

21 - Connecting piece for frame

- ❑ For frame
- ❑ Qty. 2
- ❑ Removing and installing ⇒ Rep. gr. 50 ; Body - front; Lock carrier; Removing and installing connecting pieces for frame

22 - Plastic nuts

- ❑ Qty. 2
- ❑ 20 Nm

Specified torques for bolts securing subframe to longitudinal member

Stage	Bolts	Specified torque/turning further angle
1.	-3-	180 Nm +90°
2.	-3-	Loosen 120°
3.	-3-	180 Nm + turn 180° further

Specified torques for bolts securing subframe to longitudinal member

Stage	Bolts	Specified torque/turning further angle
1.	-4-	130 Nm +90°
2.	-4-	Loosen 120°

Stage	Bolts	Specified torque/turning further angle
3.	-4-	130 Nm + turn 180° further

2.2 Removing and installing subframe with steering rack

⇒ “2.2.1 Removing and installing subframe with steering rack”, page 29

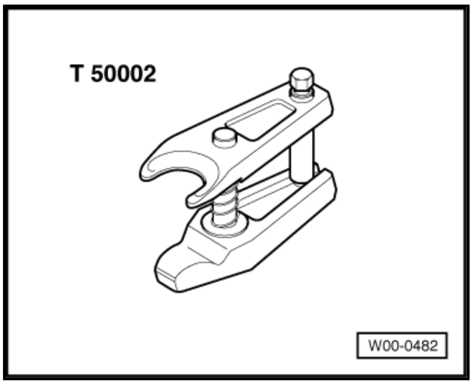
⇒ “2.2.2 Removing and installing subframe with steering rack, e-Crafter or e-TGE”, page 33

2.2.1 Removing and installing subframe with steering rack

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1332-

- ♦ Ball joint puller - T50002-



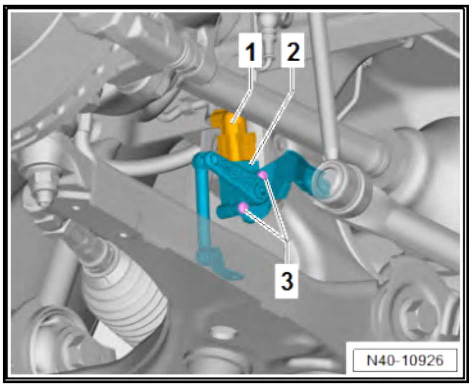
Removing

– Turn wheels to straight-ahead position.

– Remove front wheels ⇒ page 150 .

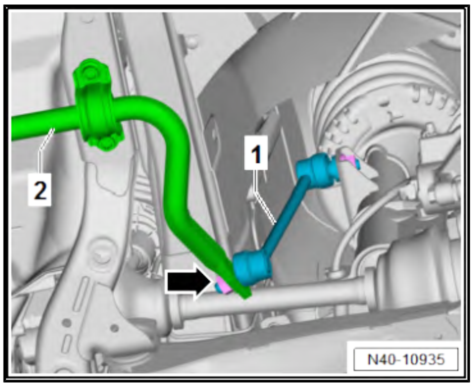
Vehicles with right front vehicle level sender - G289-

– Separate electrical connector -1-.
– Unscrew bolts -3-.
– Lay right front vehicle level sender - G289- -2- aside.

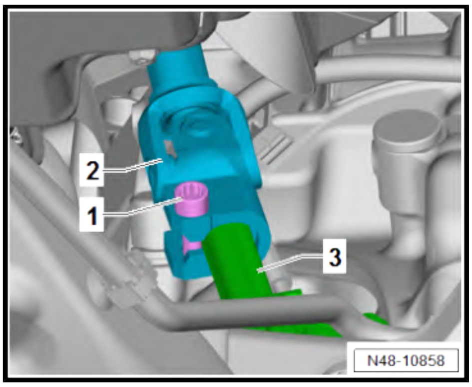


Continued for all vehicles

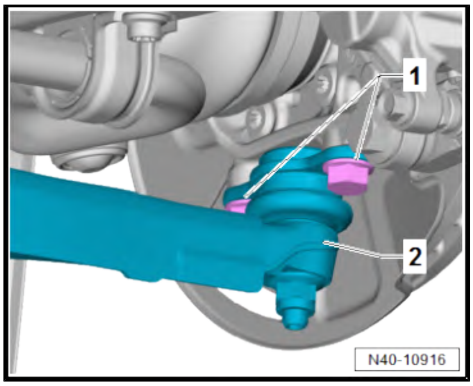
– Unscrew nut -arrow-.
– Pull coupling rod -1- out of anti-roll bar -2-.



– Unscrew bolt -1-.
– Pull off intermediate steering shaft -2- from steering rack -3-.



– Unscrew bolts -1-.
– Pull off suspension link -2-.

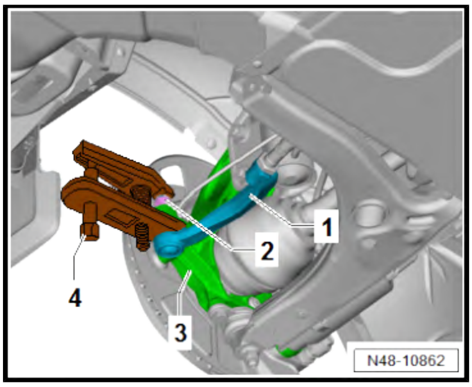


– Loosen nut -2-.

– Press track rod ball joint -1- off swivel joint -3- using ball joint puller - T50002- -4-.

– Unscrew nut -2-.

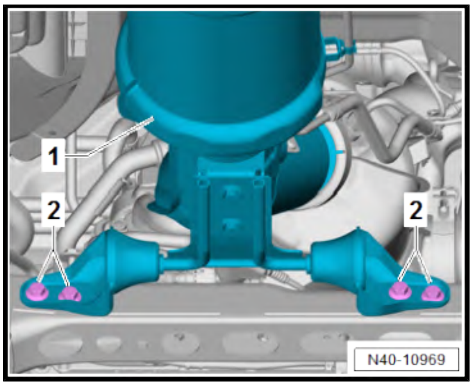
– Pull off track rod ball joint.



Vehicles with transverse engine

– Unscrew bolts -2- from exhaust system -1-.

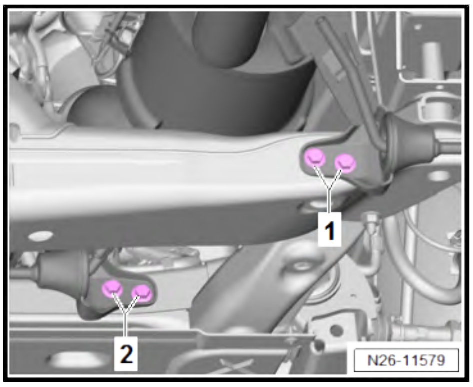
– Remove pendulum support ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing pendulum support .



Vehicles with longitudinal engine

– Unscrew bolts -1- and -2-.

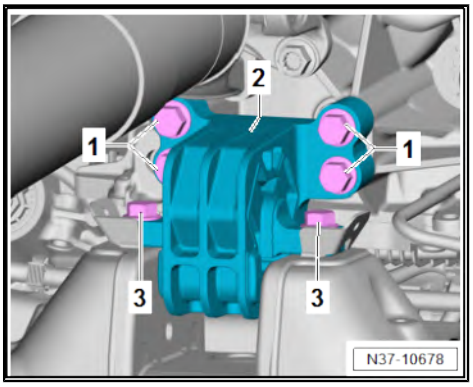
– Support gearbox using commercially available support.



Note

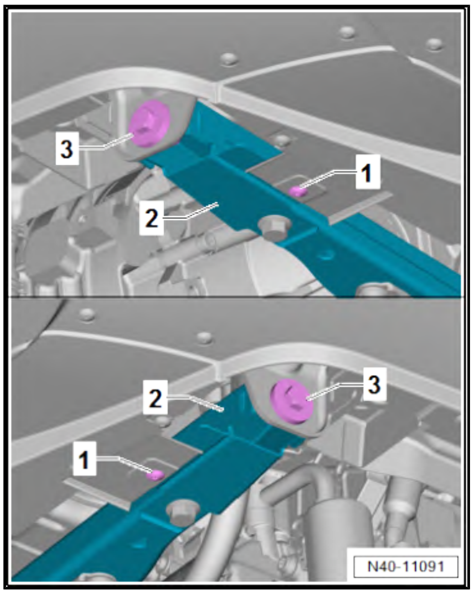
Disregard items 1 and 2.

– Unscrew bolts -3-.

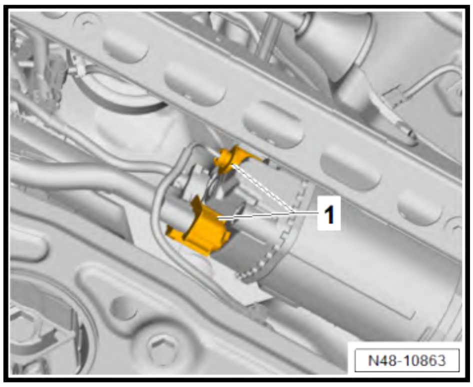


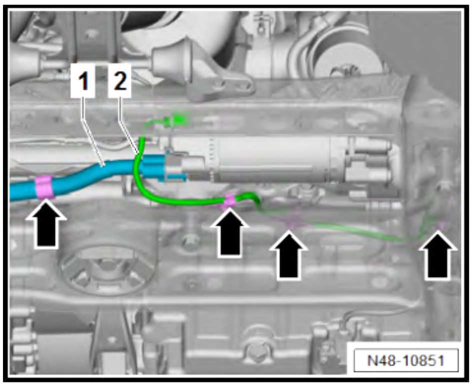
Continued for all vehicles

– Unscrew bolts -1- and -3- from connecting piece for frame -2-.



– Separate electrical connectors -1-.





– Unclip lines -1- and -2- from retainers -arrows-.

– Fix position of subframe ⇒ page 51 .

Installing

Install in the reverse order of removal, observing the following:

Vehicles with transverse engine

If subframe is to be renewed:

– Use cleaning solvent or cleaning agent to dewax bearing seat of bonded rubber bushes for pendulum support ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Note

After cleaning, make sure not to touch the interior of the bearing seat with your hands or gloves if there is still any wax residue on them.

Vehicles with longitudinal engine

– Align gearbox mounting free of stress.

Continued for all vehicles

Specified torques

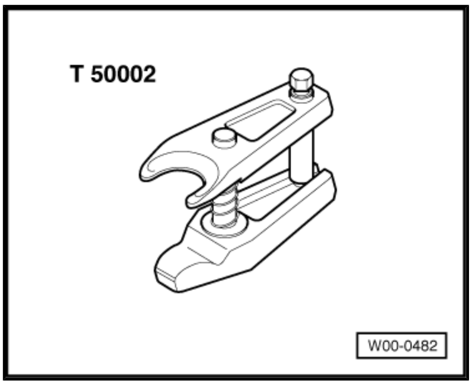
- Emission control; Assembly overview - emission control ⇒ Rep. gr. 26 ; Emission control; Assembly overview - emission control
- Assembly mountings; Assembly overview - assembly mountings ⇒ Rep. gr. 10 ; Assembly mountings; Assembly overview - assembly mountings .
- Assembly mountings; Assembly overview - assembly mountings ⇒ Rep. gr. 34 ; Assembly mountings; Assembly overview - assembly mountings
- Assembly mountings; Assembly overview - assembly mountings ⇒ Rep. gr. 37 ; Assembly mountings; Assembly overview - assembly mountings
- ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145
- ⇒ “2.1 Assembly overview - subframe”, page 23
- ⇒ “3.1 Assembly overview - steering rack”, page 229
- ⇒ “2.1 Assembly overview - steering column”, page 223
- ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65

2.2.2 Removing and installing subframe with steering rack, e-Crafter or e-TGE

Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Ball joint puller - T50002-



Removing

DANGER

Danger to life from high voltage.

Severe or fatal injury from electric shock.

– **The high-voltage system must be de-energised by a suitably qualified technician.**

– De-energise high-voltage system ⇒ Rep. gr. 93 ; De-energising high-voltage system .

– Turn wheels to straight-ahead position.

– Remove front wheels ⇒ page 150 .

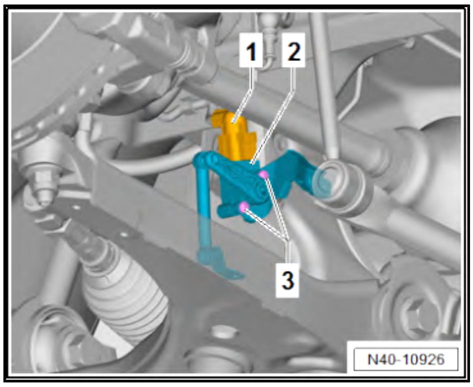
– Remove underbody cladding ⇒ Rep. gr. 66 ; Exterior equipment; Underbody cladding; Removing and installing underbody cladding .

Vehicles with right front vehicle level sender - G289-

– Separate electrical connector -1-.

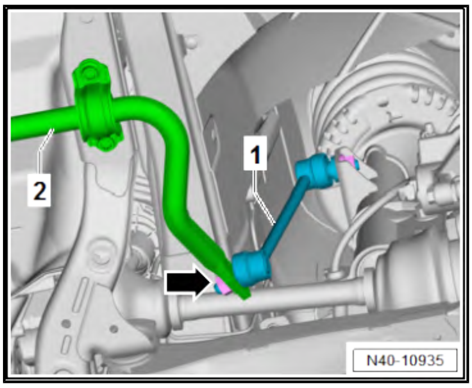
– Unscrew bolts -3-.

– Lay right front vehicle level sender - G289- -2- aside.

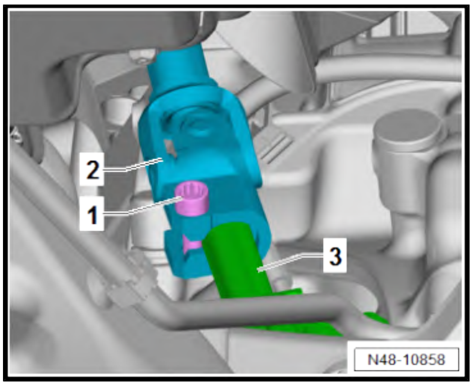


Continued for all vehicles

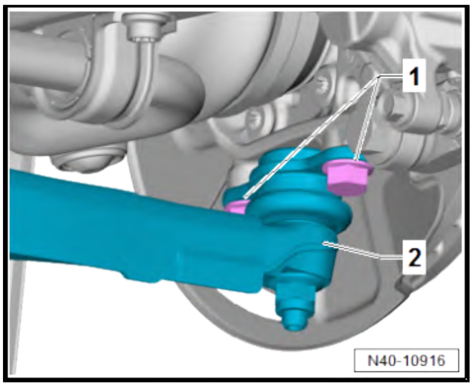
– Unscrew nut -arrow-.
– Pull coupling rod -1- out of anti-roll bar -2-.



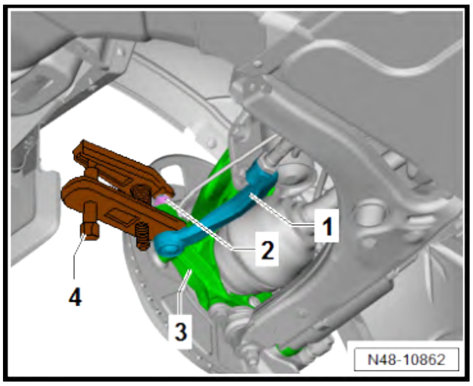
– Unscrew bolt -1-.
– Pull off intermediate steering shaft -2- from steering rack -3-.



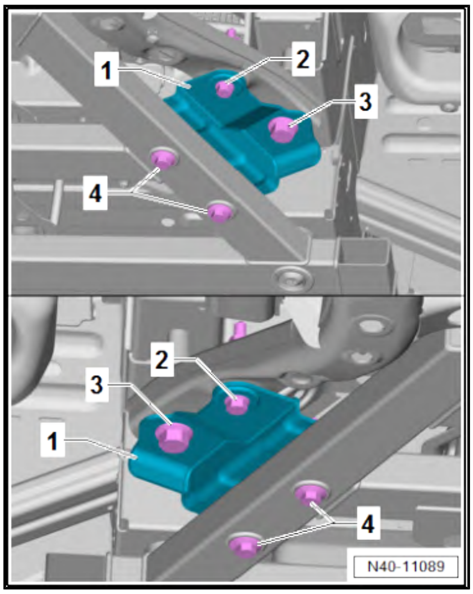
– Unscrew bolts -1-.
– Pull off suspension link -2-.



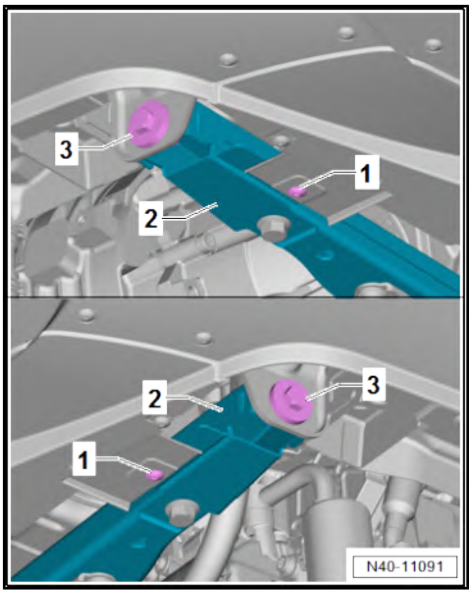
– Loosen nut -2-.
– Press track rod ball joint -1- off swivel joint -3- using ball joint puller - T50002- -4-.
– Unscrew nut -2-.
– Pull off track rod ball joint.
– Remove pendulum support ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Removing and installing pendulum support .



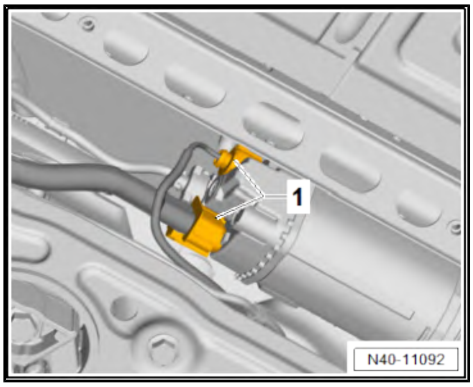
- Unscrew bolts -2-.
- Unscrew bolts -3-.
- Unscrew bolts -4-.
- Remove brackets -1-.



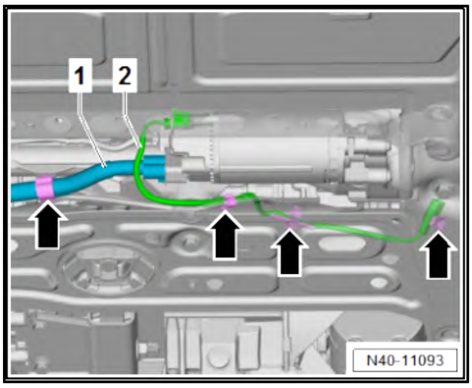
- Unscrew bolts -1- and -3- from connecting piece for frame -2-.



- Separate electrical connectors -1-.



– Unclip wiring harness -1- and -2- from retainers -arrows-.



– Fix position of subframe ⇒ page 51 .

Installing

Install in the reverse order of removal, observing the following:

WARNING

Danger to life from high voltage.

Risk of severe or fatal injury due to electric shock.

– Have a qualified technician re-energise the high-voltage system.

– Re-energise high-voltage system ⇒ Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

Specified torques

♦ Installing pendulum support ⇒ Rep. gr. 93 ; Electric drive; Electric drive motor; Removing and installing pendulum support .

♦ Underbody cladding; Overview of fitting locations - underbody cladding ⇒ Rep. gr. 66 ; Exterior equipment; Underbody cladding; Overview of fitting locations - underbody cladding

♦ Assembly overview - ladder frame ⇒ Electric drive motor (270, LS2); Rep. gr. 93 ; High-voltage battery unit; Removing and installing high-voltage battery 1 -AX2-

♦ ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145

♦ ⇒ “2.1.3 Assembly overview - subframe, e-Crafter or e-TGE”, page 27

♦ ⇒ “3.1 Assembly overview - steering rack”, page 229

♦ ⇒ “2.1 Assembly overview - steering column”, page 223

♦ ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65

2.3 Repairing subframe

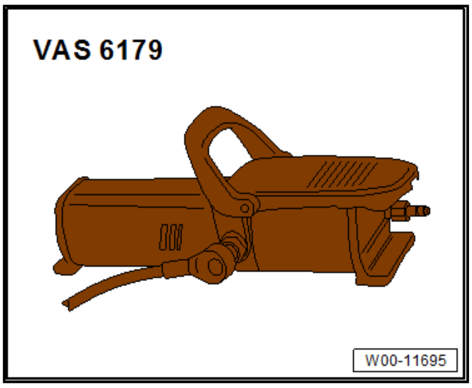
⇒ “2.3.1 Removing and installing bonded rubber bush”, page 37

2.3.1 Removing and installing bonded rubber bush

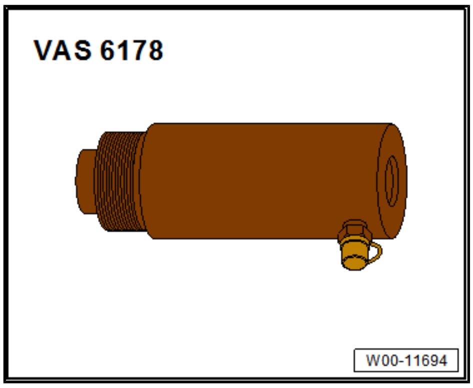
Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1331-

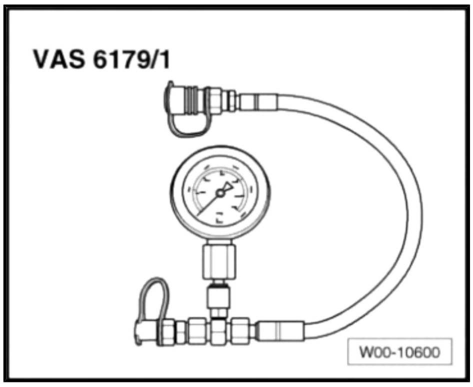
- Foot pump - VAS 6179-



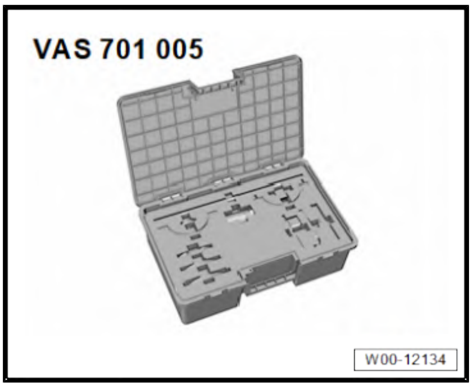
- Hydraulic cylinder - VAS 6178-



- Pressure gauge with connection line - VAS 6179/1-



- Assembly tool - VAS 701005-



- Engine and gearbox jack - VAS 6931-

Removing

Vehicles with electric drive

DANGER

Danger to life from high voltage.

Severe or fatal injury from electric shock.

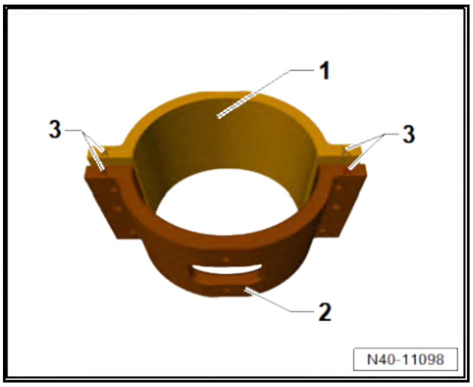
– The high-voltage system must be de-energised by a suitably qualified technician.

– De-energise high-voltage system ⇒ Electric motor (270, LS2); Rep. gr. 93 ; De-energising the high-voltage system .

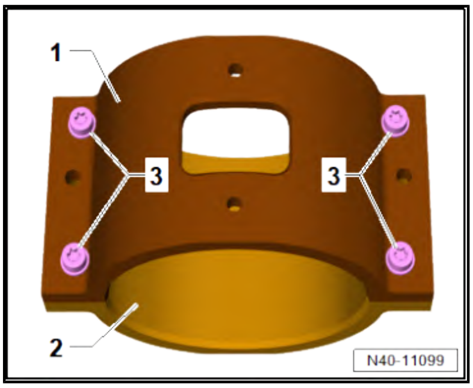
Continued for all vehicles

– Remove pendulum support ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing pendulum support .

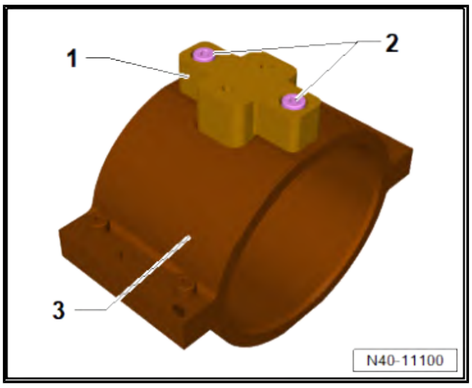
– Bring together jaw 1 - VAS 701005/2- -1- and jaw 2 - VAS 701005/3- -2- so that X markings -3- are opposite each other.



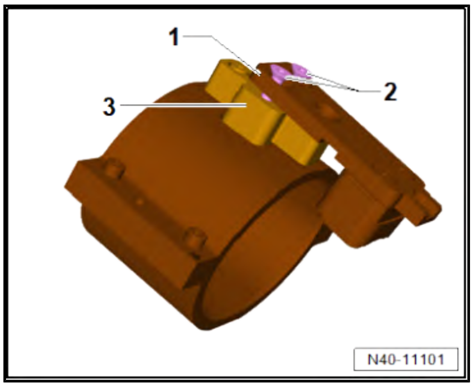
– Join jaw 1 - VAS 701005/2- -1- and jaw 2 - VAS 701005/3- -2- with bolts -3- and washers ⇒ page 49 .



– Screw jaw element - VAS 701005/4- -1- on jaw 2 - VAS 701005/3- -3- with bolts -2- ⇒ page 49 .



– Screw positioning plate with subframe element - VAS 701005/5a- -1- on jaw element - VAS 701005/4- -3- one to two turns with bolts -2-.

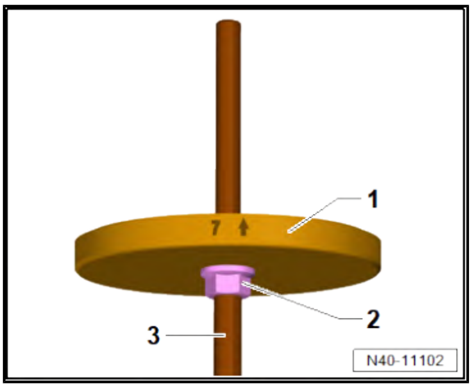


– Screw nut -2- with flange on threaded spindle - VAS 701005/6- -3- on side of threaded spindle without hexagon.

• Flange of nut must point upwards.

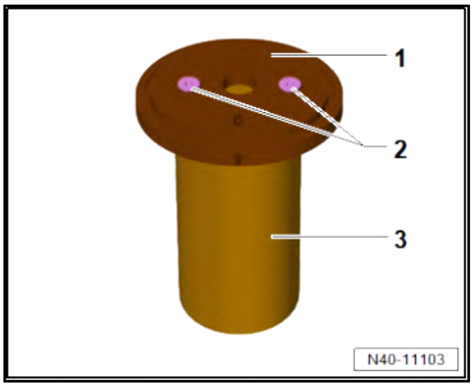
– Fit assembly thrust piece - VAS 701005/7- -1- on threaded spindle.

• -Arrow- on assembly thrust piece - VAS 701005/7- must point upwards.



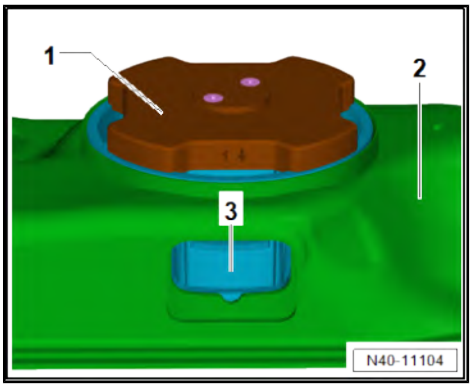
– Screw thrust piece - VAS 701005/8- -1- on hydraulic press - VAS 6178- -3- with bolts -2- ⇒ page 49 .

• -Arrow- on thrust piece - VAS 701005/8- -1- must be on opposite side to that of hydraulic connection.



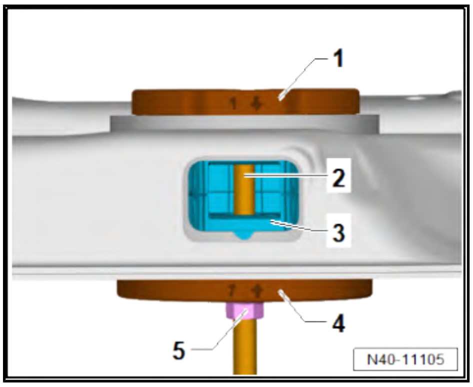
– Place disassembly thrust piece - VAS 701005/1- -1- on bonded rubber bush -3-, ensuring when doing so that disassembly thrust piece - VAS 701005/1- engages and is firmly seated.

• -Arrow- on disassembly thrust piece - VAS 701005/1- -1- must point in direction of opening in subframe -2-.



– Guide threaded spindle - VAS 701005/6- -2- through bonded rubber bushes -3- and screw in disassembly thrust piece - VAS 701005/1- -1- as far as stop.

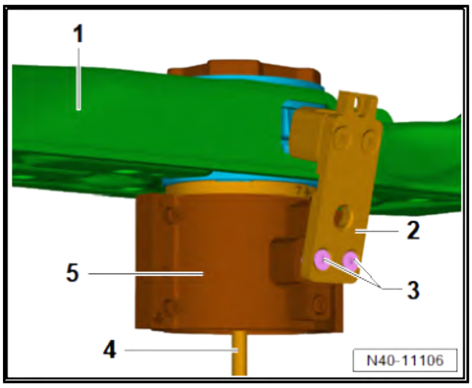
– Engage assembly thrust piece - VAS 701005/7- -4- in lower bonded rubber bush and clamp using nut with flange -5-.



– Guide preassembled jaws -5- over lower end of threaded spindle - VAS 701005/6- -4-.

– Push positioning plate with subframe element - VAS 701005/5a- -2- into opening on subframe -1- as far as stop.

– Tighten bolts -3-.

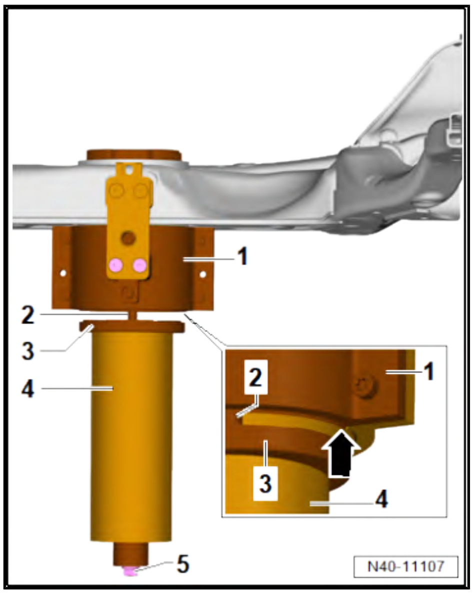


– With thrust piece - VAS 701005/8- -3- leading, fit hydraulic press - VAS 6178- -4- on threaded spindle - VAS 701005/6- -2- and secure with nut - VAS 701005/9- -5- to prevent it from falling down.

– Carefully guide thrust piece - VAS 701005/8- up to centring of jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- -1- until it is flush. Clamp with nut - VAS 701005/9- when doing this.

• Lower edge on jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- -arrow- must not be damaged!

– Check that assembly tool is mounted correctly.



Mounting not correct:

– Loosen nut - VAS 701005/9- and rectify seat of assembly tool.

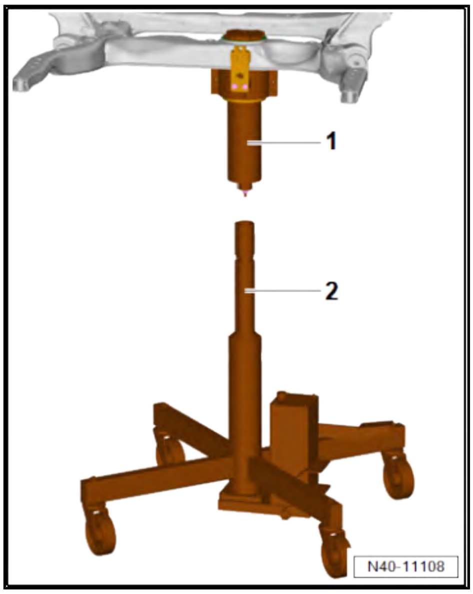
Mounting correct:

– Tighten nut - VAS 701005/9- by hand.

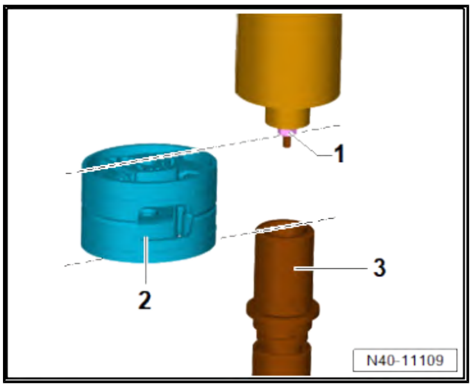
– Fit hose of foot pump - VAS 6179- on pressure gauge with connection line - VAS 6179/1- .

– Connect pressure gauge with connection line - VAS 6179/1- to pressure fitting of hydraulic press - VAS 6178- .

– Bring engine and gearbox jack - VAS 6931- -2- into position under hydraulic press - VAS 6178- -1- without additional adapter. Engage anti-roll devices when doing so.



– Extend engine and gearbox jack - VAS 6931- -3- enough to create clearance for new bonded rubber bushes -2- between engine and gearbox jack - VAS 6931- and nut - VAS 701005/9- -1-.



Engine and gearbox jack - VAS 6931- is used to safeguard complete assembly tool - VAS 701005- .

Note

As the bonded rubber bushes are pressed out, the nut - VAS 701005/9- should sink into the opening of the gearbox jack.

– Start pressing out sequence.

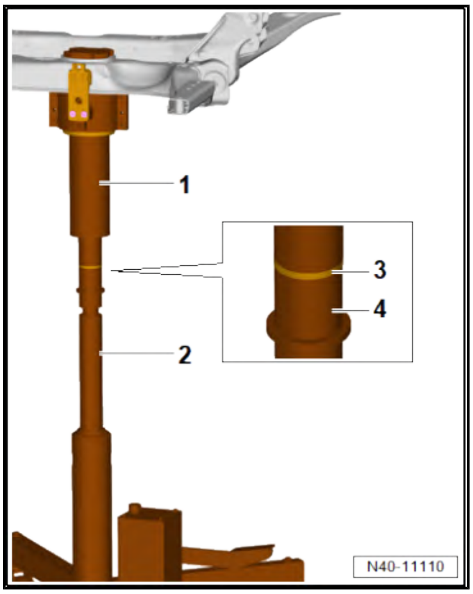
Note

As it is pressed out, the bonded rubber bush is damaged beyond repair, causing the removing tool to move and bursting noises to be emitted.

– Continue the pressing out sequence until a pressure increase of approx. 80-100 bar is reached on the pressure gauge with connection line - VAS 6179/1- .

– Relieve the pressure but do not let the piston rod of the hydraulic press - VAS 6178- retract yet.

– Secure removing tool -1- with engine and gearbox jack - VAS 6931- -2- to prevent it from falling down.



Collar of nut - VAS 701005/9- -3- rests on tube of engine and gearbox jack - VAS 6931- -4-.

– Unscrew bolts -1-. Remove positioning plate with subframe element - VAS 701005/5a- -2- when doing so.

– Hold removing tool against jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- . Lower engine and gearbox jack - VAS 6931- when doing this.

– Remove removing tool along with hydraulic press - VAS 6178- .

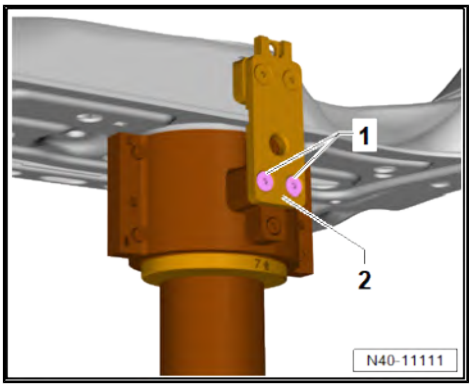
– Retract piston from hydraulic press - VAS 6178- .

– Release hose from pressure fitting.

– Loosen nut - VAS 701005/9- . When doing this, remove hydraulic press - VAS 6178- with thrust piece - VAS 701005/8- .

– Remove threaded spindle - VAS 701005/6- and nut with flange.

– Remove disassembly thrust piece - VAS 701005/1- and assembly thrust piece - VAS 701005/7- from jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- .

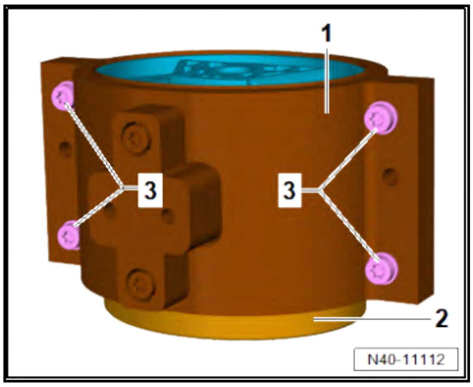


– Carefully place jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- -1- and pressed out bonded rubber bushes with centring on thrust piece - VAS 701005/8- -2-.

• Lower edge on jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- -1- must not be damaged!

– Loosen bolts -3- alternately.

– Separate jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- from each other. Remove pressed out bonded rubber bushes when doing this.

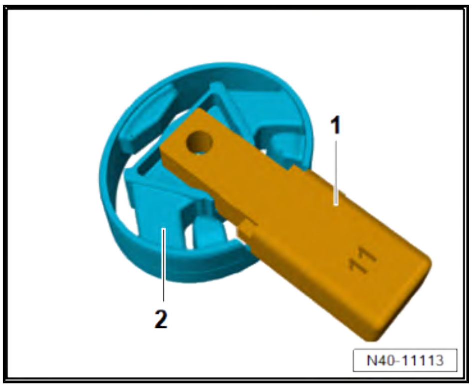


Installing

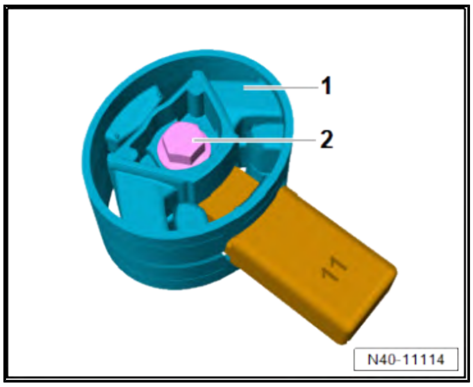
– Insert positioning pin - VAS 701005/11- -1- with respective bearing marking in opening of bonded rubber bush 2N0 199 868 -2-.

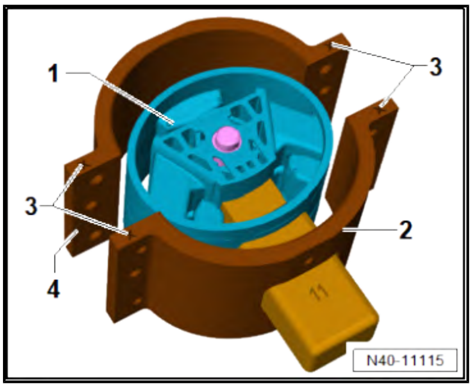
– Bring both bonded rubber bushes together.

– Position upper bonded rubber bush -1-.



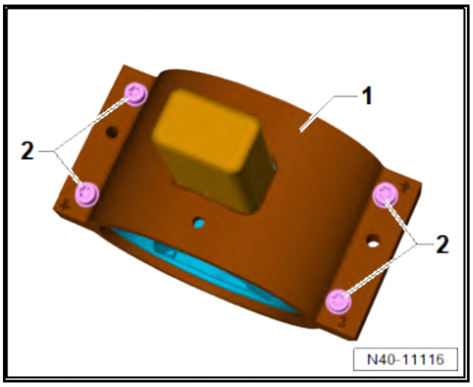
– Screw upper and lower bonded rubber bush together using unscrewed pendulum support bolt -2- and tighten to prescribed torque ⇒ page 49 .





– Insert bonded rubber bush -1- in jaw 2 - VAS 701005/3- -2- so that lower bonded rubber bush 2N0 199 868 is on side with X markings -3-.

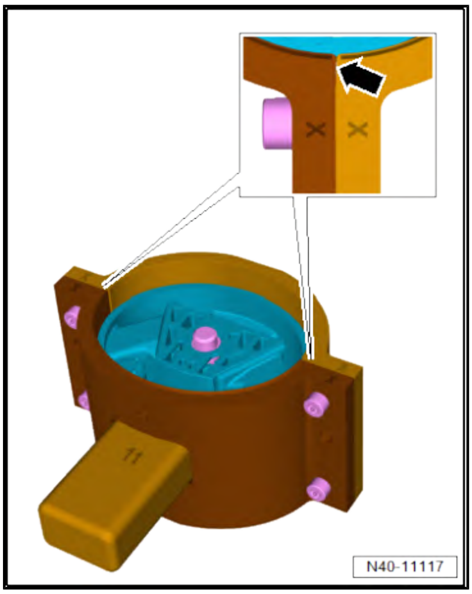
– Add jaw 1 - VAS 701005/2- -4- so that it is aligned to X markings -3-.



– Screw jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- -1- together with bolts -2- and washers.

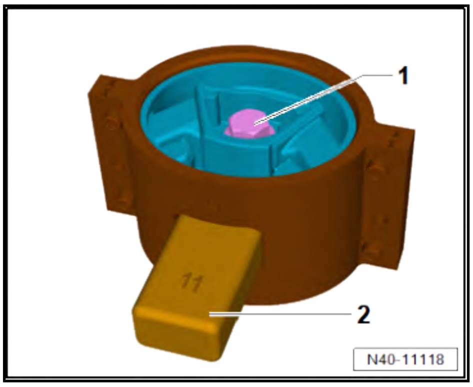
• Lower edge on jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- must not be damaged!

– Tighten bolts alternately to prescribed torque until both jaws are closed flush on the outer sides ⇒ page 49 .



Note

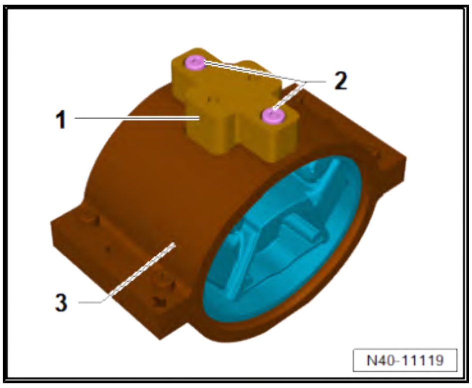
A small gap in the area marked with an -arrow- is permissible on both sides.



– Unscrew bolt -1- from bonded rubber bush.

– Pull out positioning pin - VAS 701005/11- -2-.

– Screw jaw element - VAS 701005/4- -1- on jaw 2 - VAS 701005/3- -3- with bolts -2- ⇒ page 49 .

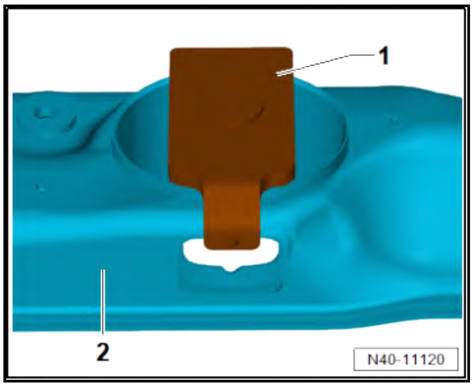


– Clean bearing seat in subframe with petroleum ether or cleaner ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

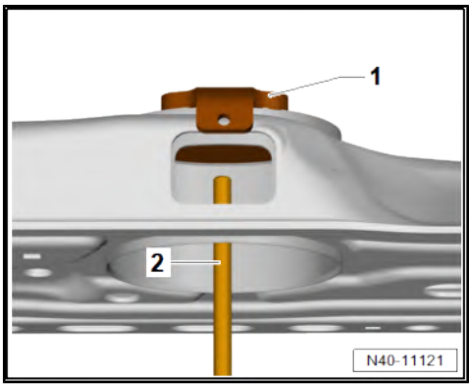
Note

Do not contaminate bearing seat in subframe with wax.

– Place counterhold tool - VAS 701005/10- -1- on subframe -2-.

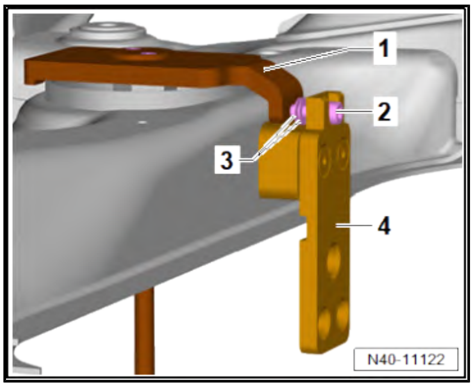


– Screw threaded spindle - VAS 701005/6- -2- as far as stop from underneath into counterhold tool - VAS 701005/10- -1-.



– Screw positioning plate with subframe element - VAS 701005/5a- -4- on counterhold tool - VAS 701005/10- -1- using bolt -2-, a flat M8 washer and an 8×14×0.5 shim -3- ⇒ page 49 .

• Washer and shim must be between positioning plate with subframe element - VAS 701005/5a- and counterhold tool - VAS 701005/10- .

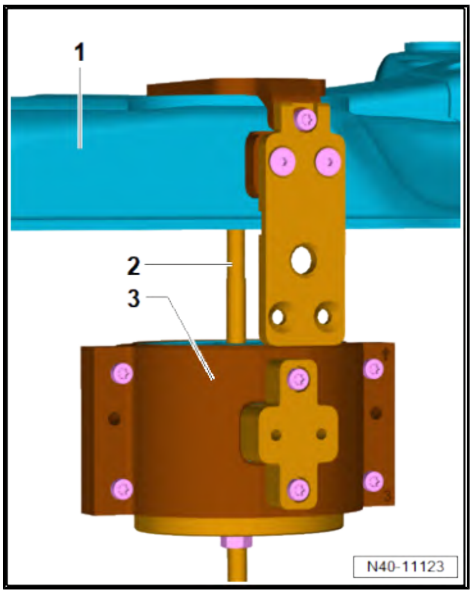


– Push preassembled jaw package -3- on threaded spindle - VAS 701005/6- -2-.

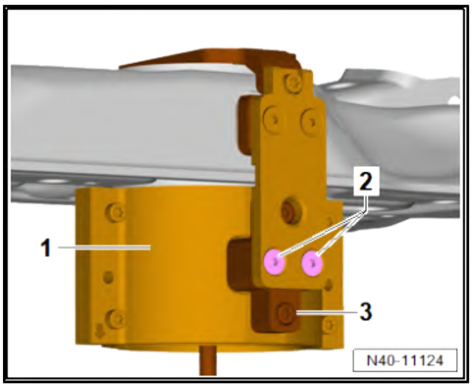
-Arrow- on jaw 2 - VAS 701005/3- must point in pressing in direction.

Edges of jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- must sink into tube from subframe -1-.

• Edge on jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- must not be damaged!

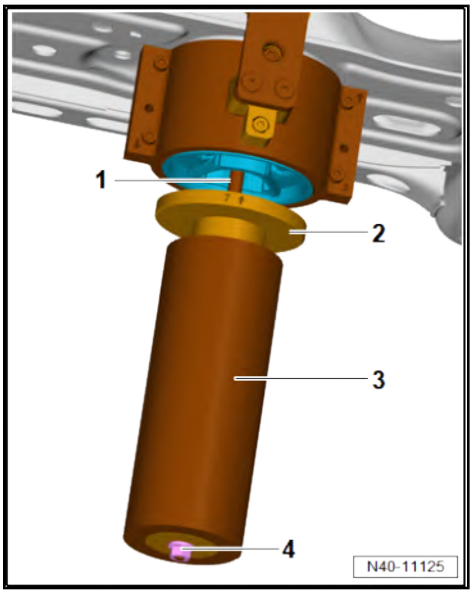


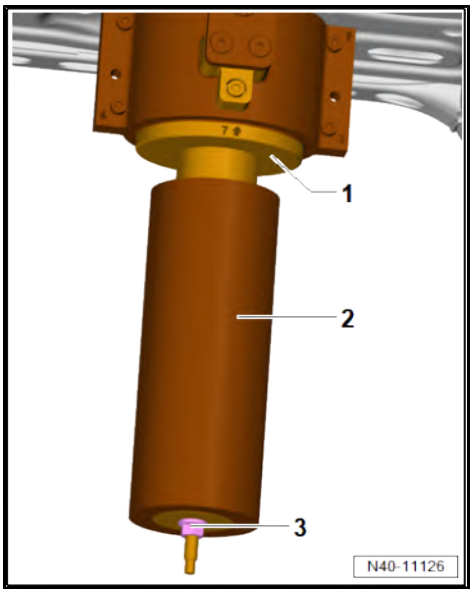
– Secure preassembled jaw package -1- on jaw element - VAS 701005/4- -3- using bolts -2- ⇒ page 49 .



– Push assembly thrust piece - VAS 701005/7- -2- and hydraulic press - VAS 6178- -3- onto threaded spindle - VAS 701005/6- -1- with piston rod side up.

– Secure hydraulic press - VAS 6178- with nut - VAS 701005/9- -4- to prevent if from falling down.





– Push assembly thrust piece - VAS 701005/7- -1- upwards. Allow it to engage in bonded rubber bush when doing this.

• -Arrow- on assembly thrust piece - VAS 701005/7- must be aligned to opening on subframe.

– Push hydraulic press - VAS 6178- -2- against assembly thrust piece - VAS 701005/7- and secure with nut - VAS 701005/9- -3-.

– Check that assembly tool is mounted correctly.

Mounting not correct:

– Loosen nut - VAS 701005/9- and rectify seat of assembly tool.

Mounting correct:

– Tighten nut - VAS 701005/9- by hand.

– Fit hose of foot pump - VAS 6179- on pressure gauge with connection line - VAS 6179/1- .

– Connect pressure gauge with connection line - VAS 6179/1- to pressure fitting of hydraulic press - VAS 6178- .

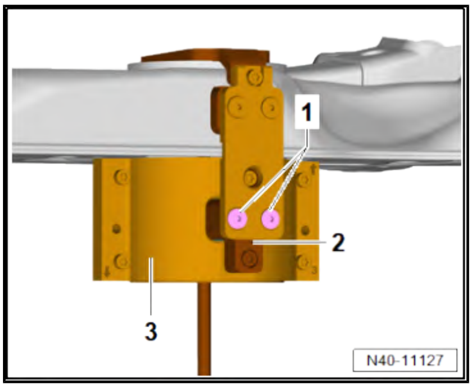
– Start pressing in sequence.

– Continue the pressing in sequence until a pressure increase of approx. 100-120 bar is reached on the pressure gauge with connection line - VAS 6179/1- .

– Relieve pressure.

– Retract piston rod from hydraulic press - VAS 6178- and release hose from pressure fitting.

– Loosen nut - VAS 701005/9- . When doing this, remove hydraulic press - VAS 6178- with assembly thrust piece - VAS 701005/7- .



– Unscrew bolts -1- from jaw element - VAS 701005/4- -2-. Hold jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- -3- when doing this.

If jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- became jammed during pressing in, proceed as follows:

– Tap lightly against jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- with plastic hammer.

– Unscrew positioning plate with subframe element - VAS 701005/5a- .

– Unscrew threaded spindle - VAS 701005/6- from counterhold tool - VAS 701005/10- . Remove both items when doing this.

– Install pendulum support ⇒ Rep. gr. 10 ; Assembly mountings; Removing and installing pendulum support .

Vehicles with electric drive

WARNING

Danger to life from high voltage.

Risk of severe or fatal injury due to electric shock.

– Have a qualified technician re-energise the high-voltage system.

– Re-energise high-voltage system ⇒ Electric motor (270, LS2); Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

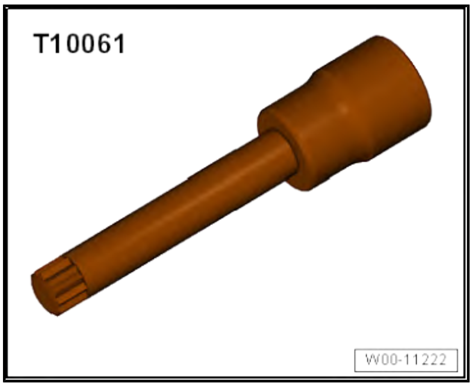
Specified torques

Component	Specified torque
Tighten all bolts in connection with assembly tool - VAS 701005-	5 Nm
Threaded connection of both bonded rubber bushes to each other with bolt from pendulum support	5 Nm
Alternate threaded connection jaw 1 - VAS 701005/2- and jaw 2 - VAS 701005/3- to new bonded rubber bushes	18 Nm

2.4 Removing and installing anti-roll bar

Special tools and workshop equipment required

♦ Insert tool - T10061-



- Torque wrench - V.A.G 1332-

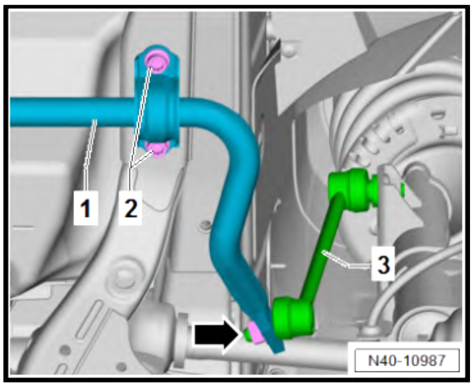
Removing

– If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

– Unscrew nut -arrow-.

– Pull coupling rod -3- out of anti-roll bar -1-.

– Unscrew bolts -2-.

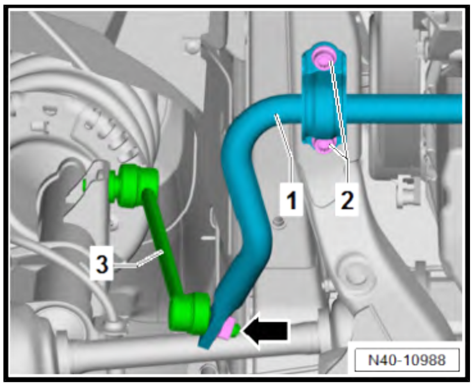


– Unscrew nut -arrow-.

– Pull coupling rod -3- out of anti-roll bar -1-.

– Unscrew bolts -2-.

– Remove anti-roll bar -1-.

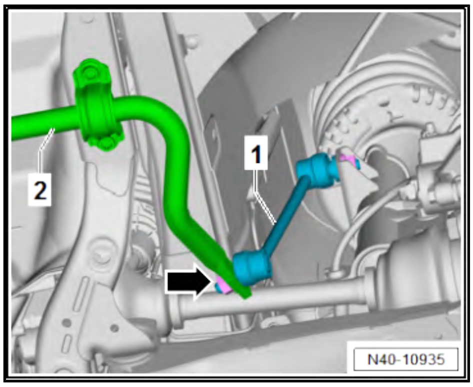


Installing

Install in the reverse order of removal, observing the following:

Note

- *Tighten bolted connections to the anti-roll bar -2- only with the vehicle standing on its wheels and unladen or with the vehicle raised and in the unladen position (normal position)* ⇒ page 8 .
- *When tightening the nut -arrow-, counterhold on the internal Torx of the joint pin of the coupling rod -1-.*



Specified torques

- ⇒ “2.1 Assembly overview - subframe”, page 23
- Assembly overview - noise insulation ⇒ Rep. gr. 66 ; Exterior equipment; Noise insulation; Assembly overview - noise insulation

2.5 Renewing anti-roll bar bushes

Anti-roll bar bushes cannot be replaced separately ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

In the event of a complaint the entire anti-roll bar must be replaced ⇒ page 49 .

2.6 Removing and installing coupling rod

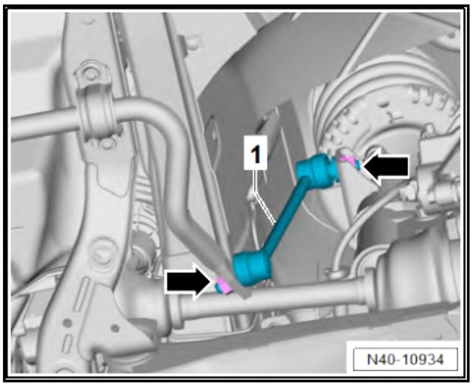
Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-

Removing

– Remove nuts -arrows-

– Remove coupling rod -1-.



Installing

Install in the reverse order of removal, observing the following:

Note

When tightening the nuts -arrows-, counterhold on the multipoint socket of the joint pin of the coupling rod.

Specified torques

♦ ⇒ “2.1 Assembly overview - subframe”, page 23

2.7 Fixing position of subframe

Special tools and workshop equipment required

- Tensioning strap - T10038-
- Locating pins - T10227-
- Torque wrench - V.A.G 1331-
- Torque wrench - V.A.G 1332-
- Engine and gearbox jack - VAS 6931-
- Rear axle support - T10552-

Assembly and preparation of engine and gearbox jack - VAS 6931- with rear axle support - T10552-

– Adjust both mounting pins, support and mounting block on rear axle support - T10552- to specified dimensions as follows.

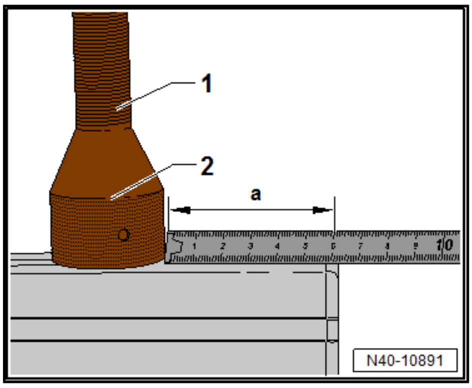
Note

Please use the dimensions specified in the text and never use the values shown on the folding rule in the illustrations.

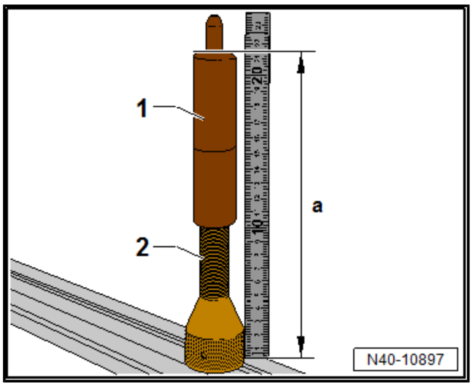
– Adjust mounting pin -T10552/1- -1- on left to dimension -a- (9 mm) with union cap -2- screwed on.

– Adjust mounting pin -T10552/1- -1- on right to dimension -a- (30 mm) with union cap -2- screwed on.

– Tighten union caps of mounting pins.

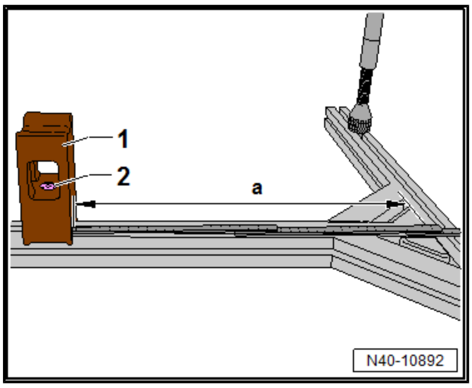


– Adjust height of mounting pins -2- by hand on left and right at threaded sleeve -1- to dimension -a- (183 mm).



– Adjust support -T10552/2- -1- to dimension -a- (600 mm).

– Tighten bolt -2- to specified torque ⇒ page 58 .

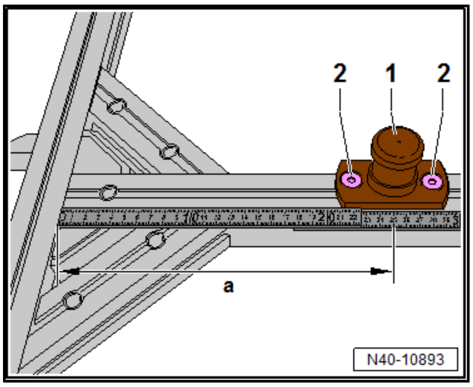


– Adjust mounting block -1- to dimension -a- (530 mm).

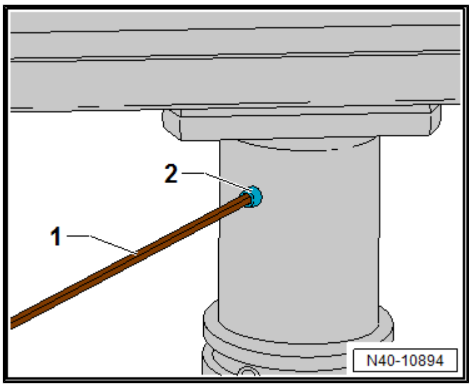
– Tighten bolts -2- to specified torque ⇒ page 58 .

– Align rear axle support - T10552- above engine and gearbox jack - VAS 6931- in such a way that mounting block points downwards.

– Place rear axle support - T10552- with mounting block on engine and gearbox jack - VAS 6931- .



– Tighten bolt -2- with hexagon socket -1- to secure rear axle support - T10552- on engine and gearbox jack - VAS 6931- to specified torque ⇒ page 58 .



Further steps on vehicle

Vehicles with electric drive

DANGER

Danger to life from high voltage.

Severe or fatal injury from electric shock.

– The high-voltage system must be de-energised by a suitably qualified technician.

– De-energise high-voltage system ⇒ Rep. gr. 93 ; De-energising high-voltage system .

– Raise vehicle. ⇒ Maintenance ; Booklet 10.4

Continued for all vehicles

– Raise vehicle. ⇒ Maintenance ; Booklet 10.3

– If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

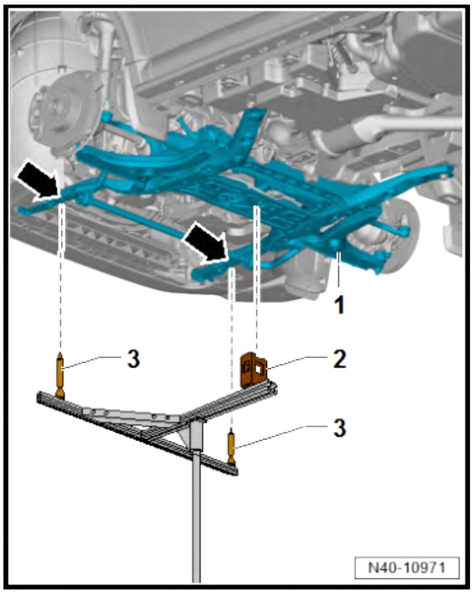
– If fitted, remove underbody cover ⇒ General body repairs, exterior; Rep. gr. 66 ; Underbody cover; Removing and installing underbody cladding .

Installing locating pins, vehicles with combustion engine

– Position engine and gearbox jack - VAS 6931- with attached rear axle support - T10552- beneath subframe.

– Allow rear axle support - T10552- to move upwards with engine and gearbox jack - VAS 6931- .

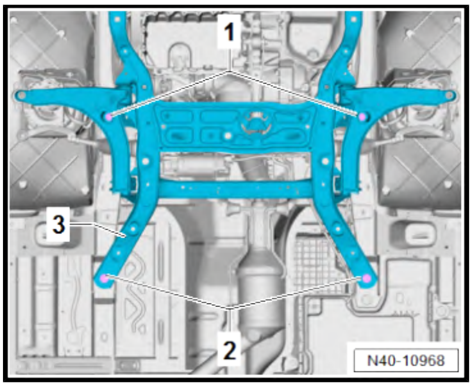
– Insert front mountings -3- in mounting holes -arrows- of subframe -1- as far as stop free of stress. When doing this, place support -2- beneath lateral reinforcement of subframe -1-.



Note

When positioning the rear axle support - T10552- in the subframe with the engine and gearbox jack - VAS 6931- , make sure that the vehicle is not raised as well under any circumstances.

– Substitute bolts -1- of subframe -3- with locating pins - T10227- one after the other.

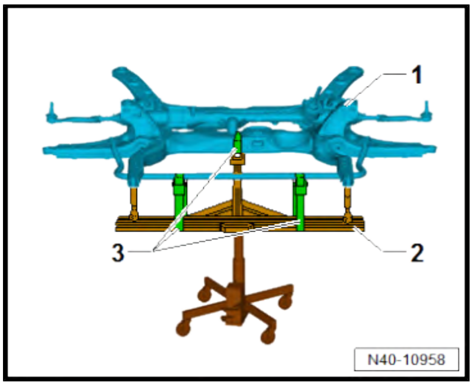


– Unscrew bolts -2-.

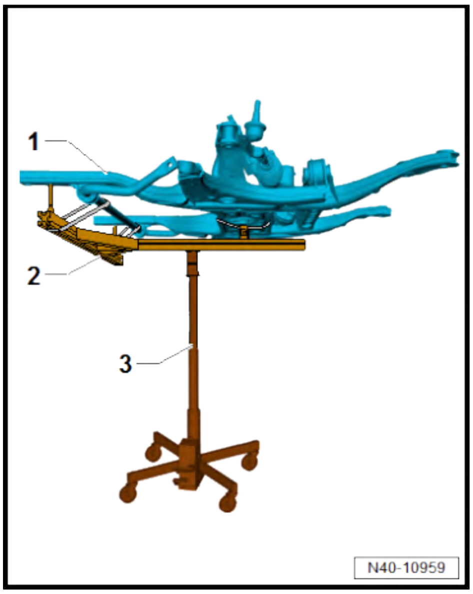
Note

- *Adhere to specified torque of locating pins - T10227- ⇒ page 58 .*
- *Fixing of the subframe is complete when the 2 bolts of the subframe have been substituted one after the other with locating pins - T10227- .*
- *The position of the subframe is now fixed.*

– Use tensioning straps - T10038- or -T10552/3- -3- to locate and secure subframe -1- on rear axle support - T10552- -2- as shown in figure.



– Lower subframe -1- placed on rear axle support - T10552- -2- with engine and gearbox jack - VAS 6931- -3-.

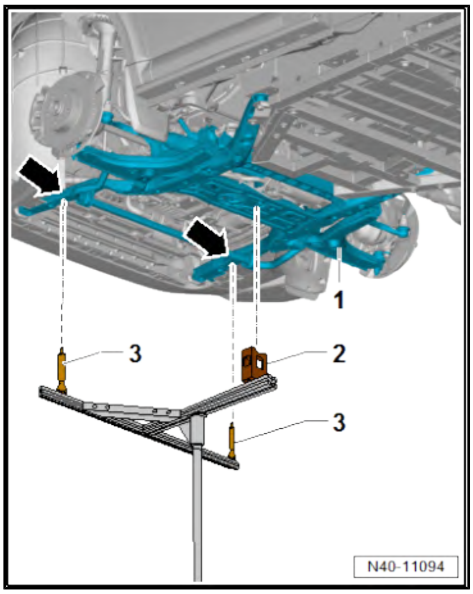


Installing locating pins, vehicles with electric drive

– Position engine and gearbox jack - VAS 6931- with attached rear axle support - T10552- beneath subframe.

– Allow rear axle support - T10552- to move upwards with engine and gearbox jack - VAS 6931- .

– Insert front mountings -3- in mounting holes -arrows- of subframe -1- as far as stop free of stress. When doing this, place support -2- beneath lateral reinforcement of subframe -1-.

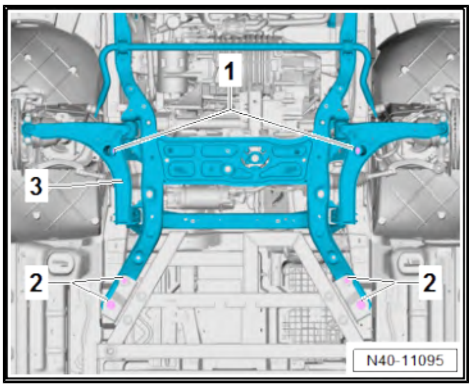


Note

When positioning the rear axle support - T10552- in the subframe with the engine and gearbox jack - VAS 6931- , make sure that the vehicle is not raised as well under any circumstances.

– Substitute bolts -1- of subframe -3- with locating pins - T10227- one after the other.

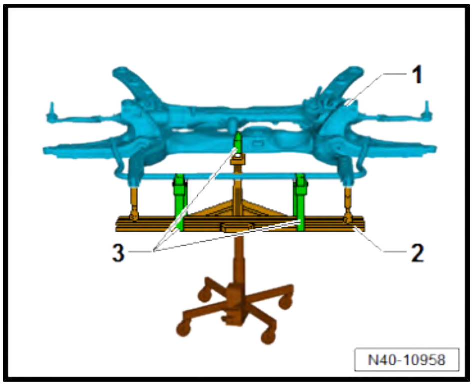
– Unscrew bolts -2-.



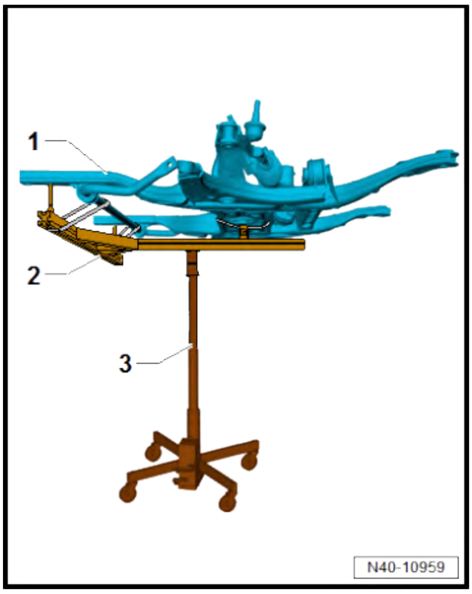
Note

- *Adhere to specified torque of locating pins - T10227-* *⇒ page 58* *.*
- *Fixing of the subframe is complete when the 2 bolts of the subframe have been substituted one after the other with locating pins - T10227- .*
- *The position of the subframe is now fixed.*

– Use tensioning straps - T10038- or -T10552/3- -3- to locate and secure subframe -1- on rear axle support - T10552- -2- as shown in figure.



– Lower subframe -1- placed on rear axle support - T10552- -2- with engine and gearbox jack - VAS 6931- -3-.



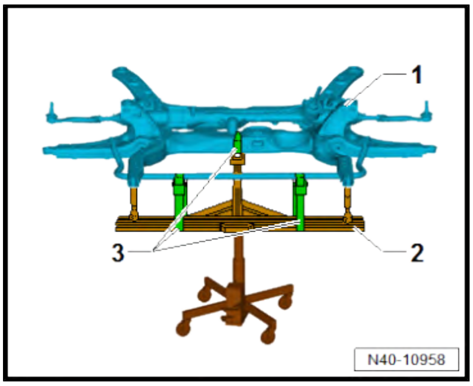
Removing locating pins, all vehicles

Remove in reverse order, observing the following:

Note

The engine and gearbox jack - VAS 6931- may only be moved with the subframe on the rear axle support - T10552- in the lowest position.

– Use tensioning straps - T10038- or -T10552/3- -3- to locate and secure subframe -1- on rear axle support - T10552- -2- as shown in figure.

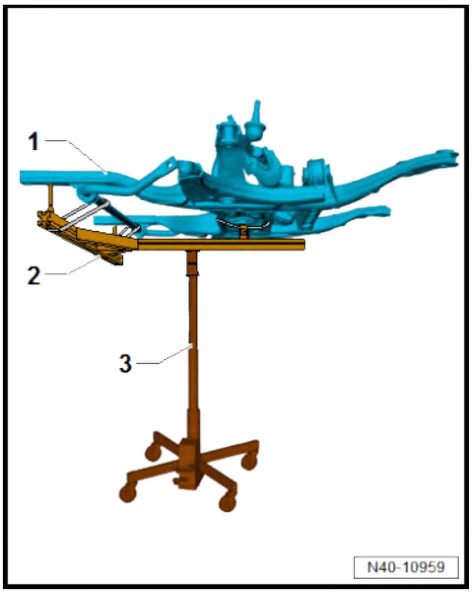


Note

When positioning the rear axle support - T10552- in the subframe with the engine and gearbox jack - VAS 6931- , make sure that the vehicle is not raised as well under any circumstances.

– Carefully allow subframe -1- placed on rear axle support - T10552- -2- to move upwards against underbody with engine and gearbox jack - VAS 6931- -3-.

– One after the other, replace locating pins - T10227- with 2 new bolts.



To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

If the steering wheel is found to be off centre during the road test despite using the locating pins - T10227- , wheel alignment is necessary. In this case the wheel alignment test results must be archived in the vehicle files.

Vehicles with electric drive

WARNING

Danger to life from high voltage.

Risk of severe or fatal injury due to electric shock.

– **Have a qualified technician re-energise the high-voltage system.**

– Re-energise high-voltage system ⇒ Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

Specified torques

♦ ⇒ “2.1 Assembly overview - subframe”, page 23

Component	Specified torque
Locating pins - T10227-	20 Nm
Bolt for securing rear axle support - T10552- to engine and gearbox jack - VAS 6931-	M6: 5 Nm M8: 10 Nm
Bolt securing support to rear axle support - T10552-	10 Nm

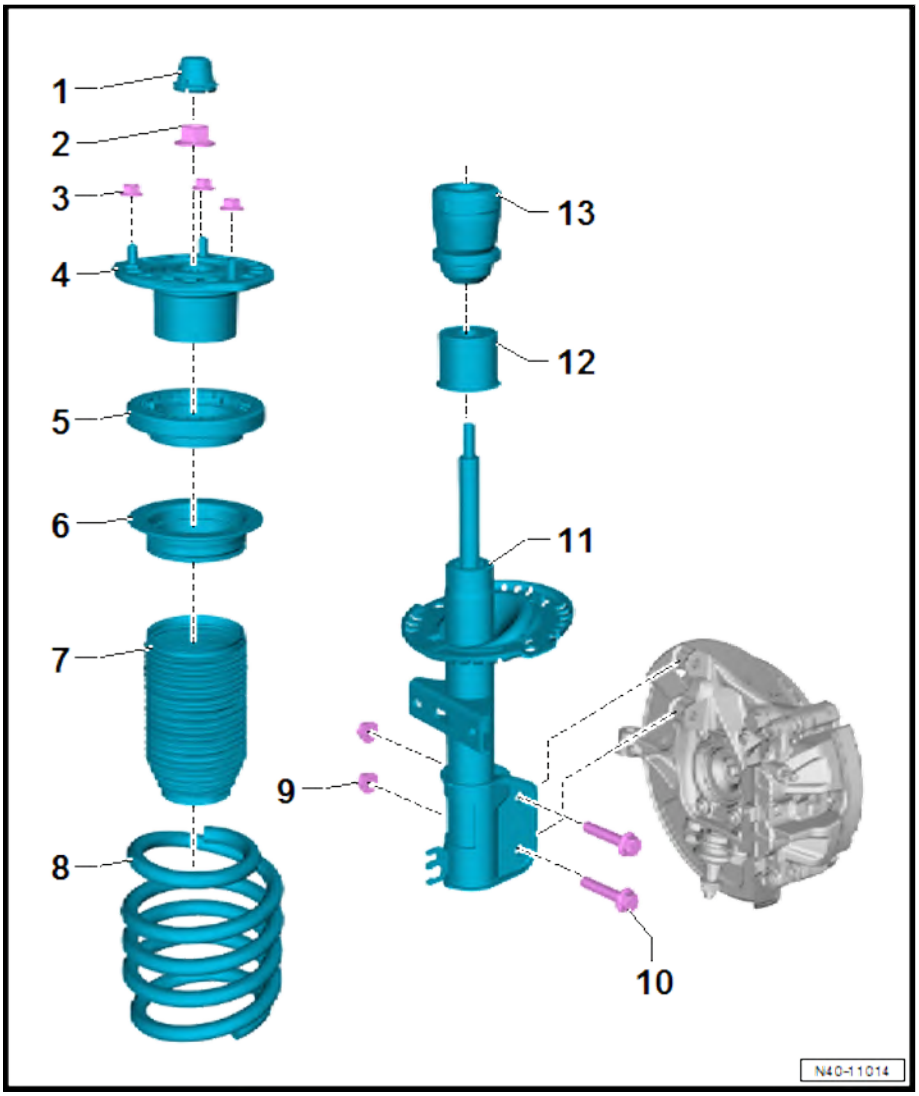
3 Suspension strut, upper suspension link

⇒ “3.1 Assembly overview - suspension strut, upper suspension link”, page 59

⇒ “3.2 Removing and installing suspension strut”, page 59

⇒ “3.3 Repairing suspension strut”, page 62

3.1 Assembly overview - suspension strut, upper suspension link



1 - Cap

2 - Nut
- ❑ Renew after removal
- ❑ M14: 80 Nm
- ❑ M16: 110 Nm

3 - Nuts
- ❑ Renew after removal
- ❑ Qty. 3
- ❑ 50 Nm +45°

4 - Suspension strut mounting
- ❑ Removing and installing ⇒ page 62

5 - Deep groove ball thrust bearing

6 - Spring seat

7 - Protective cover

8 - Coil spring
- ❑ Removing and installing ⇒ page 62

9 - Nuts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 130 Nm +270°

10 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2

11 - Suspension strut
- ❑ Removing and installing ⇒ page 59

12 - Cap

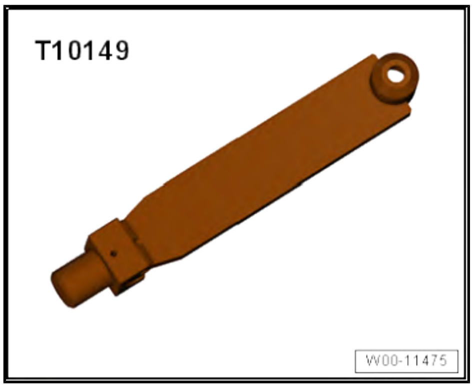
13 - Buffer stop

3.2 Removing and installing suspension strut

Special tools and workshop equipment required

- Engine and gearbox jack - VAS 6931-

- Support - T10149-



- Torque wrench - V.A.G 1332-

Removing

– Remove front wheel ⇒ page 150 .

Left side of vehicle:

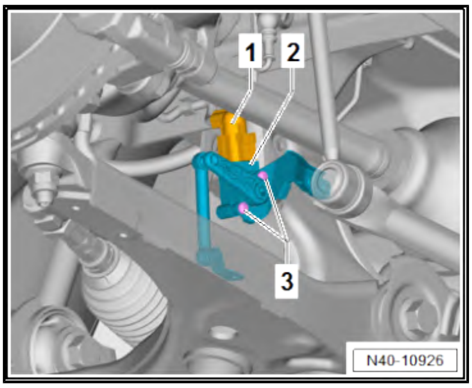
– Remove plenum chamber cover ⇒ General body repairs, exterior; Rep. gr. 50 ; Bulkhead; Removing and installing plenum chamber cover .

Vehicles with right front vehicle level sender - G289-

– Separate electrical connector -1-.

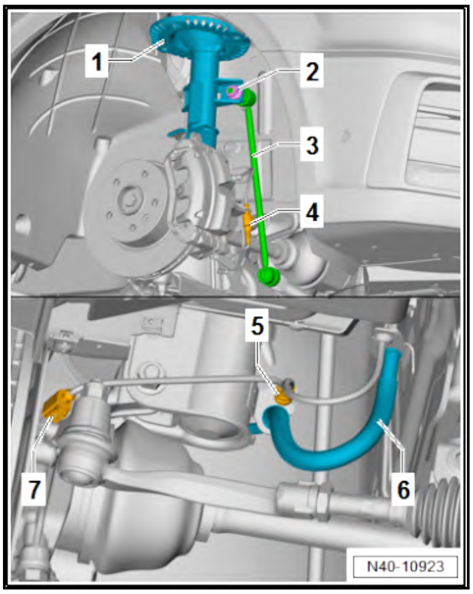
– Unscrew bolts -3-.

– Lay right front vehicle level sender - G289- -2- aside.



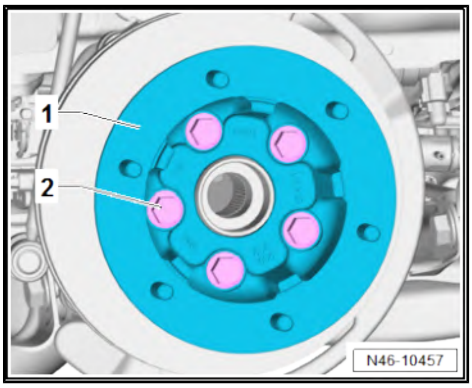
Continued for all vehicles

– Unscrew nut -2- from suspension strut -1-.
– Swing coupling rod -3- towards rear.
– If fitted, disconnect connector -4- of brake pad wear indicator.
– Separate electrical connector -7-.
– Press lines -5- and -6- out of retainer on suspension strut.



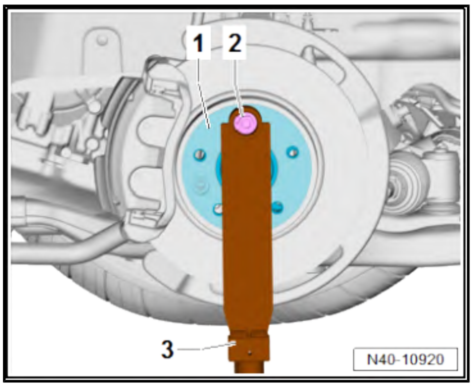
Vehicles with twin wheels

– Unscrew bolts -2-.
– Remove spacer -1-.



Continued for all vehicles

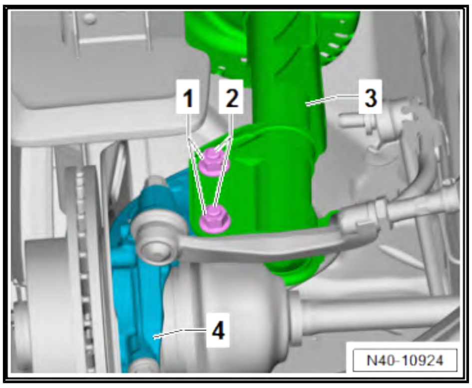
– Turn wheel hub until one of the wheel bolt holes is at the top.
– Attach support - T10149- -3- to wheel hub -1- using wheel bolt -2-.
– Support wheel bearing housing -1- using engine and gearbox jack - VAS 6931- .



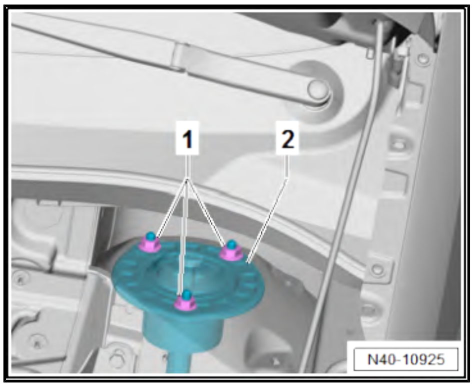
Note

♦ *Never raise or lower the vehicle while the engine and gearbox jack is positioned beneath the vehicle.*
♦ *Do not leave engine and gearbox jack - VAS 6931- under vehicle longer than necessary.*

– Unscrew nuts -1-.
– Pull out bolts -2-.
– Pull suspension strut -3- out of wheel bearing housing -4-.



– Unscrew nuts -1- from suspension strut -2-.
– Lower engine and gearbox jack - VAS 6931- as far as necessary and remove suspension strut.



Installing

Install in the reverse order of removal, observing the following:

If components such as damper have been renewed, it may be necessary to carry out wheel alignment.

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

Specified torques

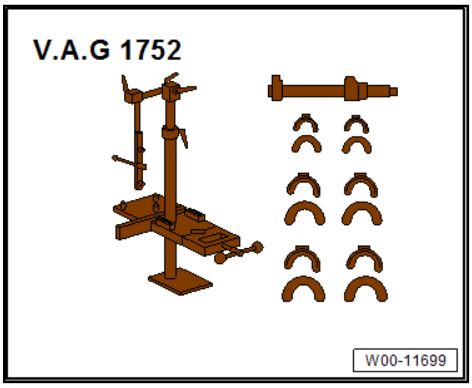
- ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145
- ⇒ “2.1 Assembly overview - subframe”, page 23
- ⇒ “3.1 Assembly overview - suspension strut, upper suspension link”, page 59
- Assembly overview – front brake ⇒ Brake systems; Rep. gr. 46 ; Front brake; Assembly overview – front brake

3.3 Repairing suspension strut

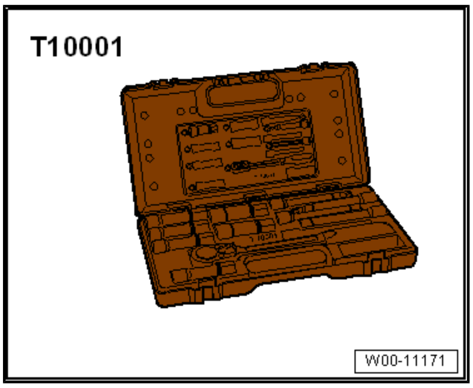
Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Suspension strut clamp - V.A.G 1752-



- Shock absorber tool set - T10001-



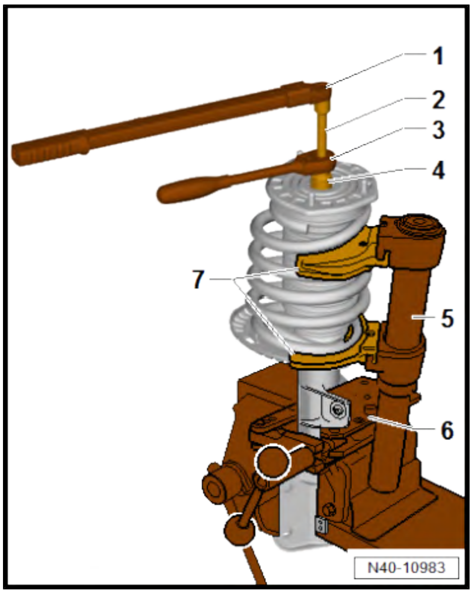
Removing

– Remove coil spring strut ⇒ page 59 .

– Clamp suspension strut support clamp - V.A.G 1752/20- -6- in a vice.

– Clamp suspension strut in suspension strut support clamp - V.A.G 1752/20- -6-.

– Pre-tension coil spring with spring compressor - V.A.G 1752/1- -5- until deep groove ball thrust bearing is free at top.



1 - Torque wrench - V.A.G 1332-

2 - Hexagon bit, long reach - T10001/8-

3 - Ratchet wrench - T10001/11-

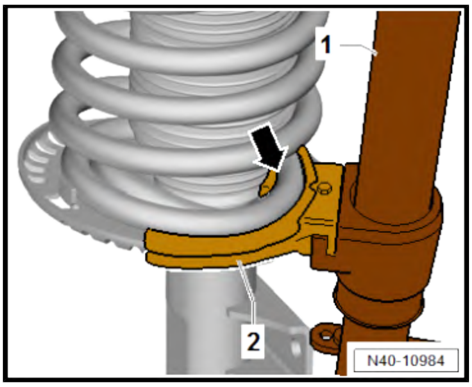
4 - Hexagon bit, long reach - T10001/5-

5 - Spring compressor - V.A.G 1752/1-

6 - Strut support clamp - V.A.G 1752/20-

7 - Spring retainer - V.A.G 1752/5-

– Ensure coil spring -arrow- is seated correctly in spring retainer - V.A.G 1752/4- -2- on spring compressor - V.A.G 1752/1- -1-.



Note

Keep limbs away from top of springs.

– Unscrew nut from piston rod.

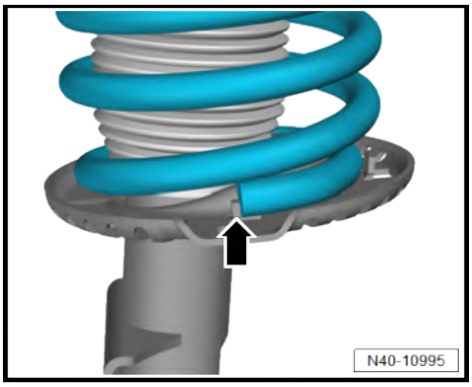
– Remove suspension strut mounting, upper spring plate with bump stop and deep groove ball thrust bearing with protective boot from piston rod.

– Remove coil spring with spring compressor from shock absorber.

Installing

Install in the reverse order of removal, observing the following:

– Fit coil spring with spring compressor - V.A.G 1752/1- onto lower spring support.



The end of the coil spring must lie against the stop -arrow-.

– Ensure that bump stop is correctly seated in upper spring plate.

– Pull lower end of protective boot down far enough that the locking lugs of the protective cap engage in the groove of the protective boot.

– Assemble all other parts and use new nut on piston rod.

– Relieve tension on spring compressor - V.A.G 1752/1- and remove from coil spring.

– Remove suspension strut from suspension strut support clamp - V.A.G 1752/20- .

Specified torques

♦ ⇒ “3.1 Assembly overview - suspension strut, upper suspension link”, page 59

4 Lower suspension link, swivel joint

⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65

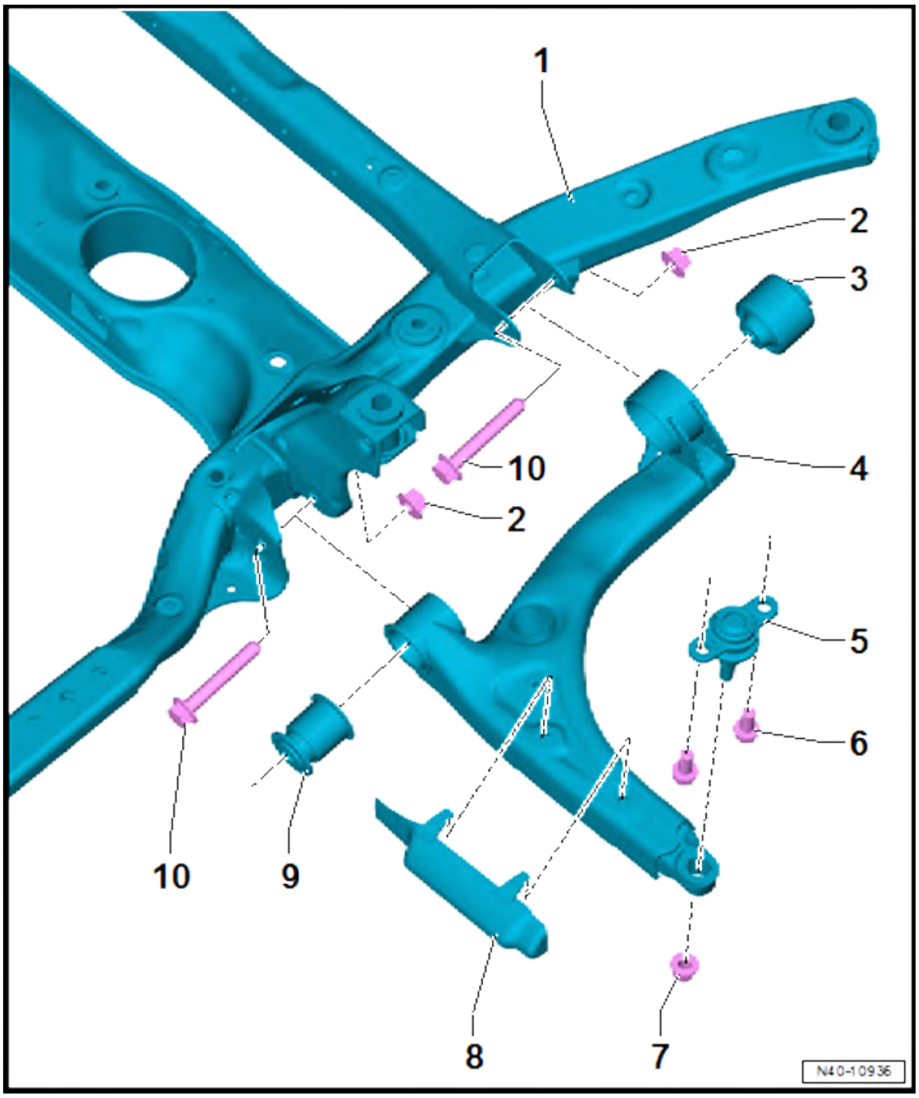
⇒ “4.2 Removing and installing lower suspension link”, page 65

⇒ “4.3 Checking swivel joint”, page 67

⇒ “4.4 Removing and installing swivel joint”, page 67

⇒ “4.5 Renewing bonded rubber bush for lower suspension link”, page 69

4.1 Assembly overview - lower suspension link, swivel joint



1 - Subframe
- ❑ Removing and installing ⇒ page 23

2 - Nuts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 180 Nm +180°

3 - Bonded rubber bush
- ❑ Removing and installing ⇒ page 69

4 - Lower suspension link
- ❑ Removing and installing ⇒ page 65

5 - Swivel joint
- ❑ Removing and installing ⇒ page 67

6 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 130 Nm +45°

7 - Nut
- ❑ Renew after removal
- ❑ 60 Nm +90°

8 - Shield

9 - Bonded rubber bush
- ❑ Removing and installing ⇒ page 69

10 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 180 Nm +180°

4.2 Removing and installing lower suspension link

Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

Removing

– Remove front wheel ⇒ page 150 .

Note

If the suspension link is deformed, the swivel joint must be re- newed together with the suspension link.

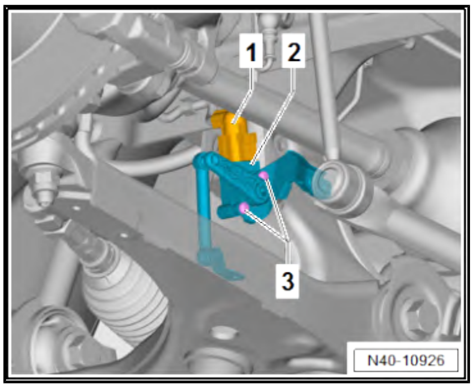
– If fitted, remove noise insulation ⇒ General body repairs, ex- terior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

Vehicles with right front vehicle level sender - G289-

– Separate electrical connector -1-.

– Unscrew bolts -3-.

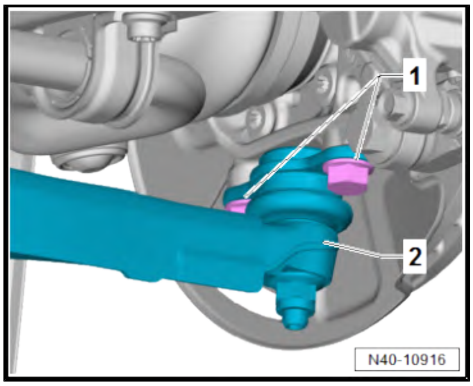
– Lay right front vehicle level sender - G289- -2- aside.

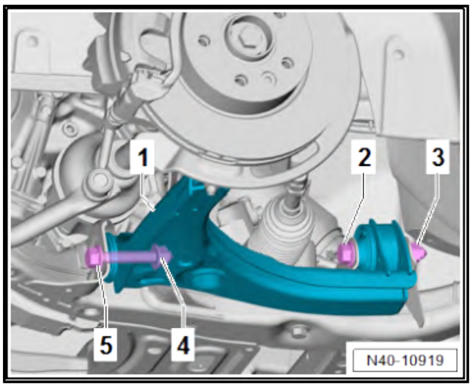


Continued for all vehicles

– Unscrew bolts -1-.

– Pull off suspension link -2-.





– Unscrew nuts -3- and -4-.

– Remove bolts -2- and -5-.

– Remove suspension link -1- from subframe.

Installing

Install in the reverse order of removal, observing the following:

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

Note

Tighten the nuts for the suspension link in unladen position *⇒ page 8* *.*

Specified torques

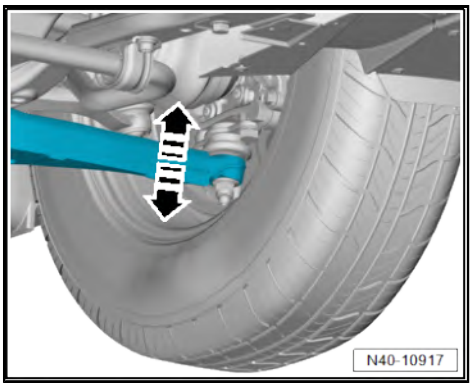
- ♦ ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65
- ♦ ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145
- ♦ Assembly overview - noise insulation ⇒ Rep. gr. 66 ; Exterior equipment; Noise insulation; Assembly overview - noise insulation

4.3 Checking swivel joint

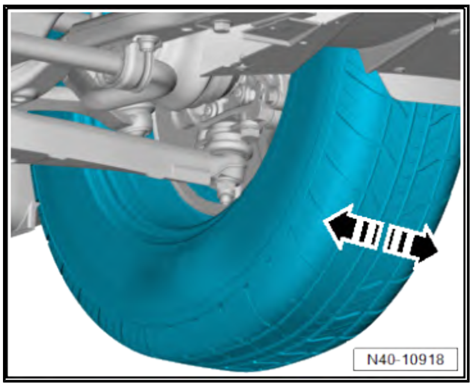
Checking axial play

– Forcefully pull suspension link down and press up again.

Checking radial play



– Press lower part of wheel forcefully outwards and inwards.



Note

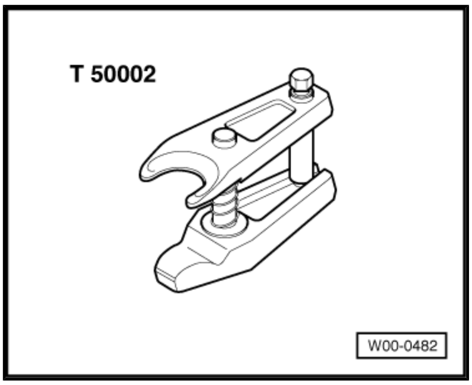
- ♦ *There should be no tangible or visible “play” during both tests.*
- ♦ *Check rubber boot for damage and renew swivel joint if necessary.*

4.4 Removing and installing swivel joint

Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Ball joint puller - T50002-



Removing

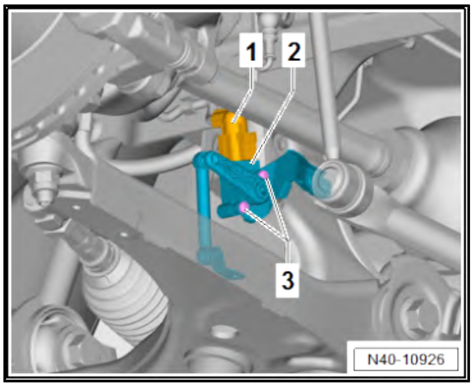
– Remove front wheel ⇒ page 150 .

Vehicles with right front vehicle level sender - G289-

– Separate electrical connector -1-.

– Unscrew bolts -3-.

– Lay right front vehicle level sender - G289- -2- aside.



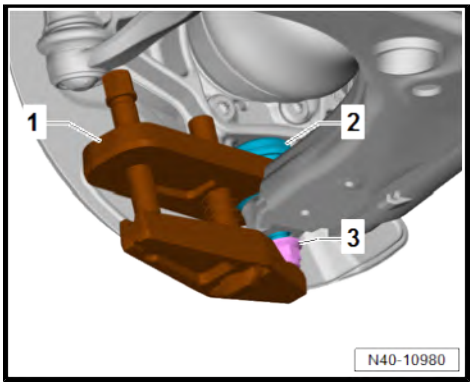
Continued for all vehicles

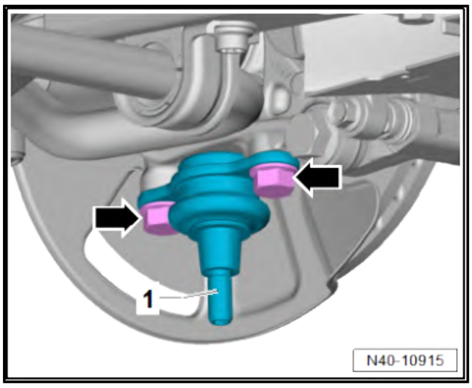
– Loosen nut -3-.

– Press suspension link off swivel joint -2- using ball joint puller - T50002- -1-.

– Unscrew nut -3-.

– Pull suspension link off swivel joint -2-.





– Remove bolts -arrows-

– Remove swivel joint -1-.

Installing

Install in the reverse order of removal, observing the following:

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

Specified torques

- ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65
- ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145

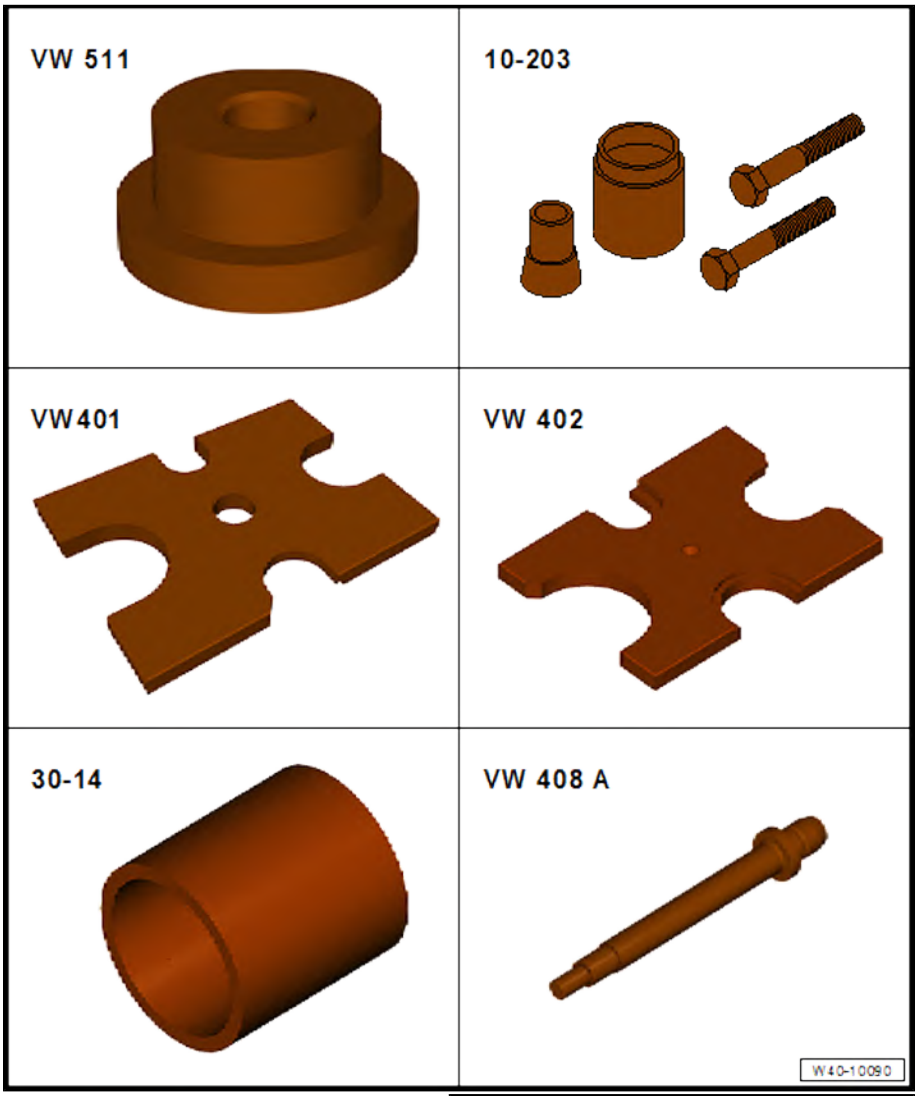
4.5 Renewing bonded rubber bush for lower suspension link

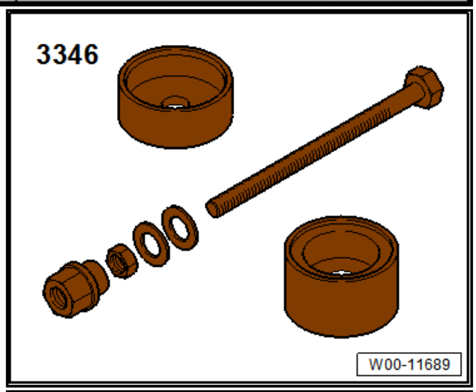
⇒ “4.5.1 Renewing front bonded rubber bush for suspension link”, page 69

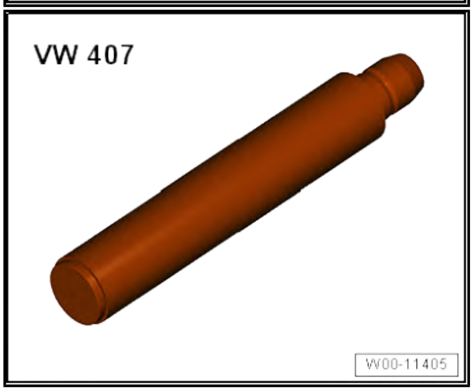
⇒ “4.5.2 Renewing rear bonded rubber bush for suspension link”, page 72

4.5.1 Renewing front bonded rubber bush for suspension link

Special tools and workshop equipment required







- Thrust plate - VW 401-
- Thrust plate - VW 402-
- Press tool - VW 408 A-
- Thrust plate - VW511-
- Bolt from fitting tool - 10 - 203-
- Tube - 30-14-
- Assembly tool - 3346-
- Press tool - VW407-

Removing

- Suspension link removed ⇒ page 65 .

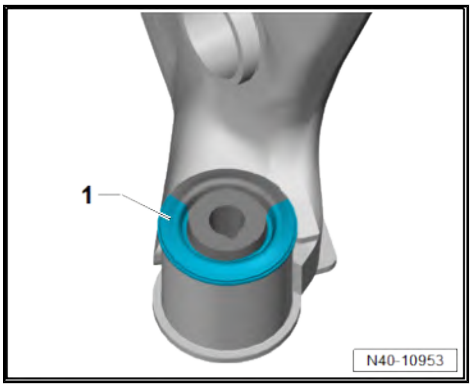
CAUTION

Risk of injury from swarf being flung into air.
Risk of irritation and injury to skin and eyes.
– Wear safety goggles.
– Wear protective gloves.

Note

When bending up metal collar, do not damage suspension link.

– To support the suspension link, use a sharp -chisel- to bend up metal collar -1-.



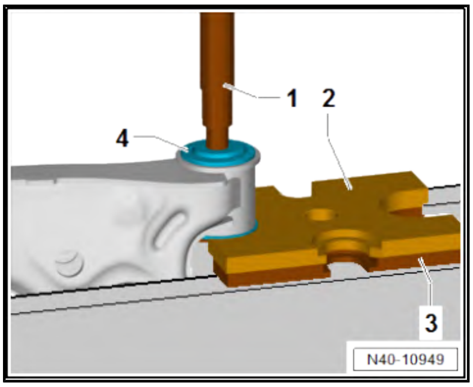
– Position tools as shown in illustration.

1 - Press tool - VW 408 A-

2 - Pressure plate - VW 401-

3 - Pressure plate - VW 402-

– Press out bonded rubber bush -4-.

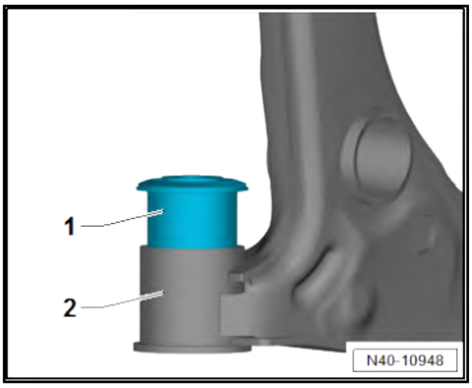


Installing

Install in the reverse order of removal, observing the following:

– Coat rubber bead of bonded rubber bush with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Press bonded rubber bush -1- into suspension link -2- by hand as far as possible.



– Press in bonded rubber bush -A- onto stop.

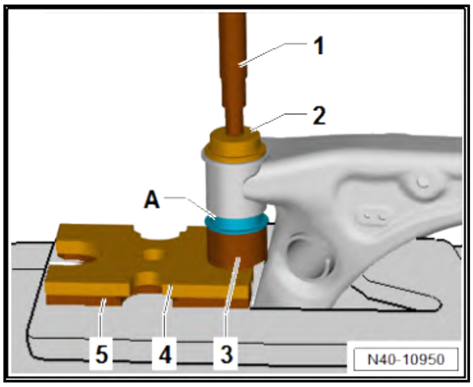
1 - Press tool - VW 408 A-

2 - Thrust pad - VW511-

3 - Assembly tool - 3346-

4 - Pressure plate - VW 401-

5 - Pressure plate - VW 402-



– Press in bonded rubber bush further with bolt -A- from fitting tool - 10 - 203- .

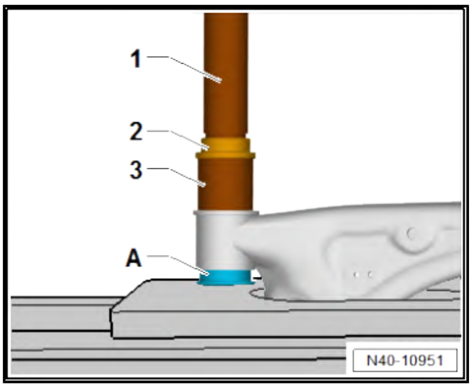
1 - Press tool - VW407-

2 - Thrust pad - VW511-

3 - Tube - 30-14-

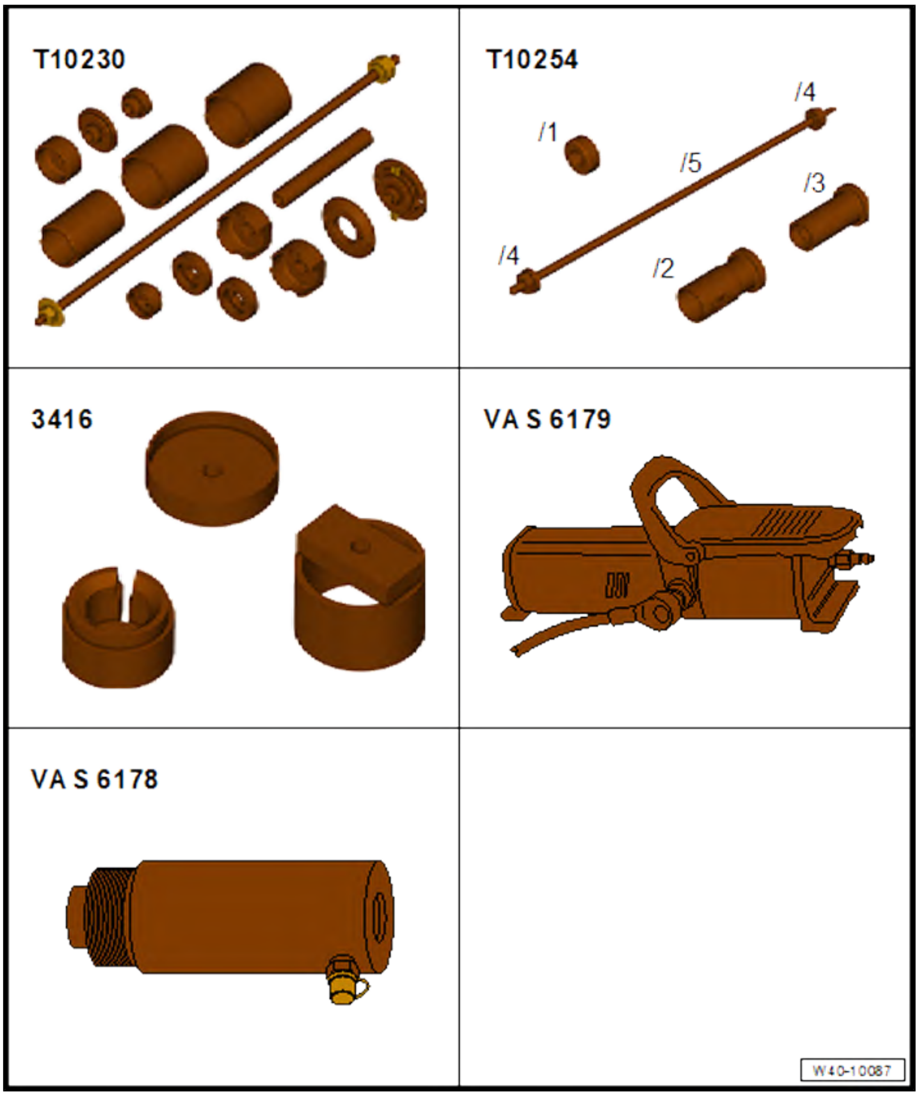
Note

This causes the rubber bead to »pop« out of the suspension link.



4.5.2 Renewing rear bonded rubber bush for suspension link

Special tools and workshop equipment required

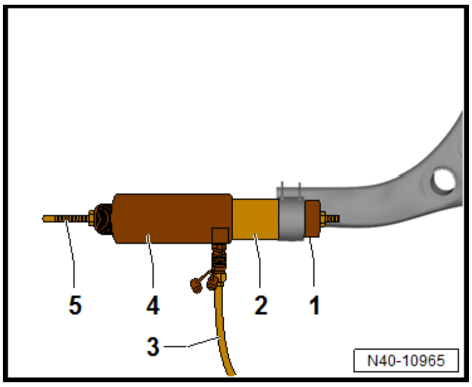


- ♦ Foot pump - VAS 6179-
- ♦ Hydraulic cylinder - VAS 6178-
- ♦ Assembly tool - T10254-
- ♦ Assembly tool - T10230-
- ♦ Assembly tool - 3416-

Removing

Note

Before pressing out the bonded rubber bush with the »kidney-shaped« recess, mark its installation position on the suspension link.



– Position tools as shown in illustration.

1 - Assembly tool - 3416-

2 - Assembly tool - T10230-

3 - Foot pump - VAS 6179-

4 - Hydraulic press - VAS 6178-

5 - Assembly tool - T10254-

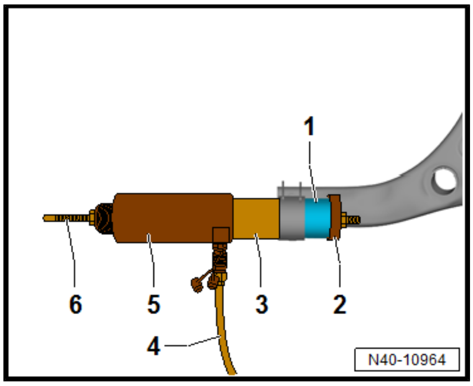
– Press out bonded rubber bush.

Installing

Install in the reverse order of removal, observing the following:

– Coat rubber bead of bonded rubber bush with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Align installation position of bonded rubber bush -1- to suspension link -2- as shown in diagram.



– Position tools as shown in illustration.

1 - Rear bonded rubber bush for suspension link

2 - Assembly tool - T10230-

3 - Assembly tool - T10230-

4 - Foot pump - VAS 6179-

5 - Hydraulic press - VAS 6178-

6 - Assembly tool - T10254-

– Pressing in bonded rubber bush.

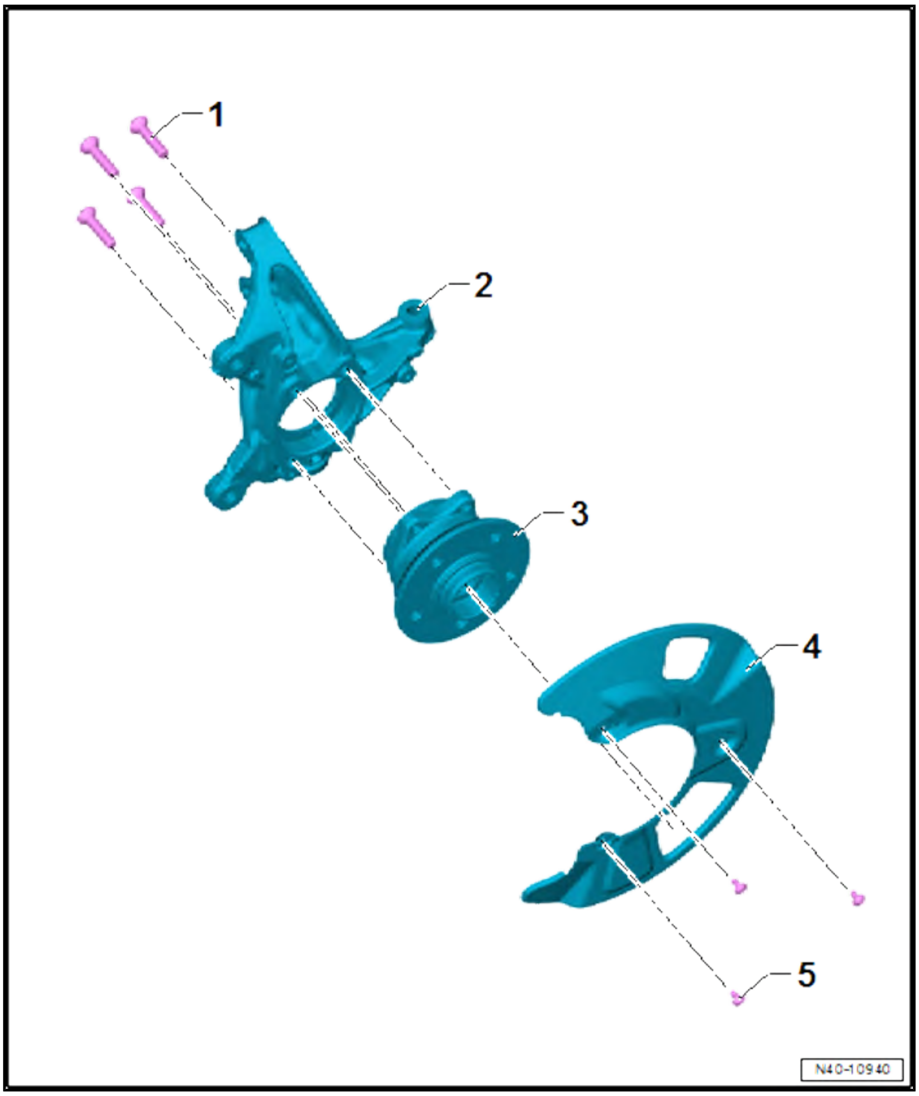
5 Wheel bearing

⇒ “5.1 Assembly overview - wheel bearing”, page 75

⇒ “5.2 Removing and installing wheel bearing housing”, page 75

⇒ “5.3 Removing and installing wheel bearing unit”, page 79

5.1 Assembly overview - wheel bearing



1 - Bolts

- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 70 Nm +90°

2 - Wheel bearing housing

- ❑ Removing and installing ⇒ page 75

3 - Wheel bearing unit

- ❑ Removing and installing ⇒ page 79

4 - Cover plate

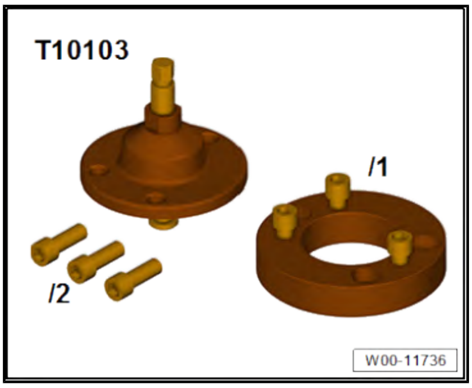
5 - Bolts

- ❑ Number of bolts may differ
- ❑ Qty. 3
- ❑ 9.5 Nm

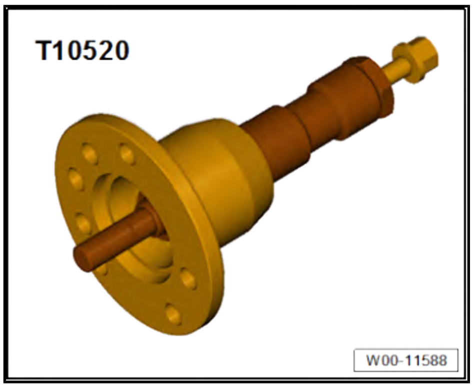
5.2 Removing and installing wheel bearing housing

Special tools and workshop equipment required

- Ejector - T10103-

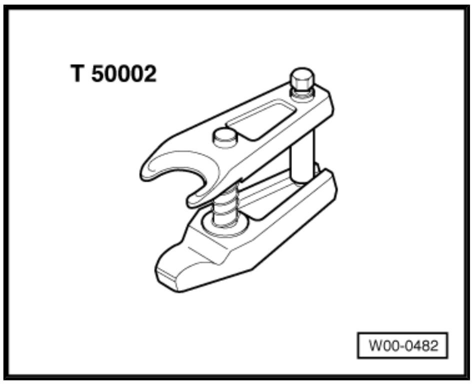


- Ejector - T10520-



- Torque wrench - V.A.G 1332-

- Ball joint puller - T50002-



Removing

Front-wheel or all-wheel drive vehicles

– Unscrew bolt for drive shaft ⇒ page 95 .

– Unscrew nut for drive shaft ⇒ page 95 .

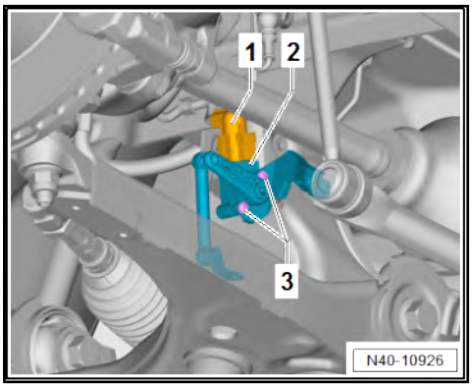
Continued for all vehicles

– Remove front wheel ⇒ page 150 .

– Remove brake disc ⇒ Brake system; Rep. gr. 46 ; Front brake; Removing and installing brake disc .

– Disconnect connector for front left speed sensor - G47- / front right speed sensor - G45- ⇒ Brake system; Rep. gr. 45 ; Sensors; Removing and installing speed sensors on front axle -G45- / -G47- .

Vehicles with right front vehicle level sender - G289-

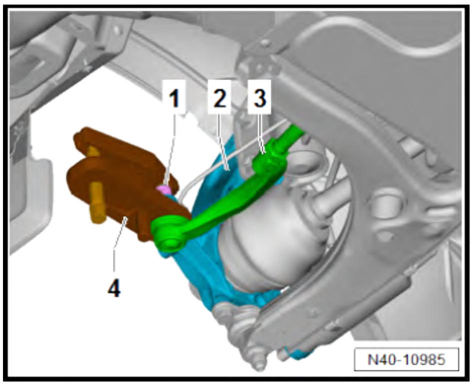


– Separate electrical connector -1-.

– Unscrew bolts -3-.

– Lay right front vehicle level sender - G289- -2- aside.

Continued for all vehicles

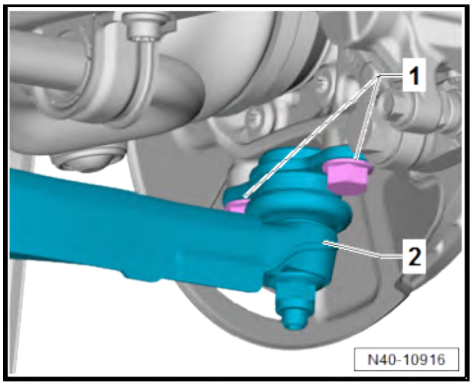


– Loosen nut -1-.

– Press track rod ball joint -3- off wheel bearing housing -2- using ball joint puller - T50002- -4-.

– Unscrew nut -1-.

– Pull track rod ball joint -3- off wheel bearing housing -2-.

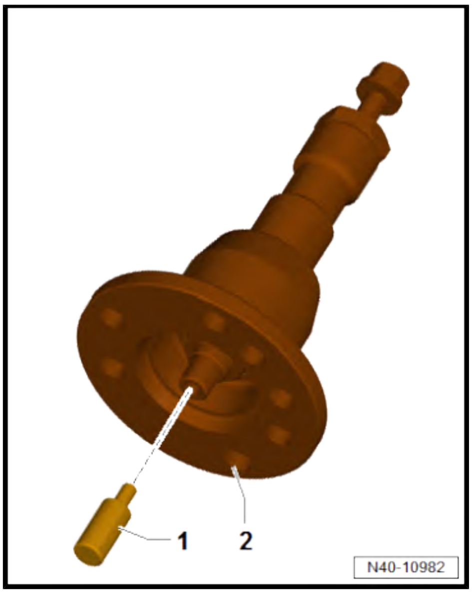


– Unscrew bolts -1-.

– Pull swivel joint -2- together with suspension link off wheel bearing housing.

Front-wheel or all-wheel drive vehicles

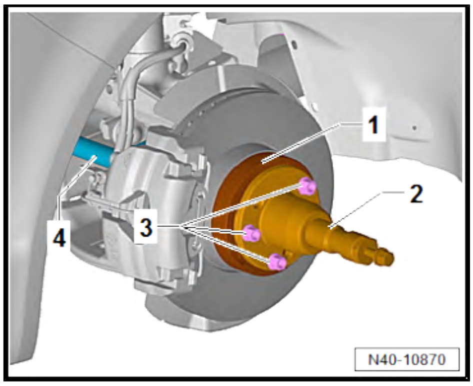
– Insert thrust piece -1- in press tool - T10520- -2-.



How to use press tool - T10520-

– Use bolts - T 10103/1- to secure adapter - T10103/2- -1- to wheel bearing.

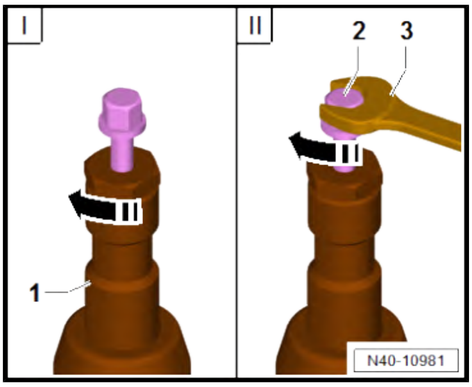
– To press out drive shaft -3-, secure ejector - T10520- -2- to wheel hub using 4 wheel bolts -3-.



– It is essential to follow the specified sequence.

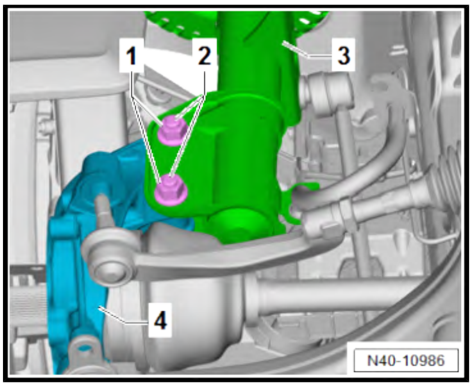
I - Tighten knurled nut in direction of -arrow- hand-tight.

II - Turn only bolt -2- using a wrench -3- in order to press out drive shaft with press tool - T10520- -1-.



Note

At the end of the procedure or for pressing out drive shaft further, the spindle must be moved to its original position in order to apply the hydraulic force.



Continued for all vehicles

– Unscrew nuts -1-.

– Pull out bolts -2-.

– Pull wheel bearing housing -4- out of suspension strut -3-.

Installing

Install in the reverse order of removal, observing the following:

Front-wheel or all-wheel drive vehicles

– Tighten bolt of drive shaft ⇒ page 95 .

– Tighten nut for drive shaft ⇒ page 95 .

Continued for all vehicles

Specified torques

- ⇒ “5.1 Assembly overview - wheel bearing”, page 75
- ⇒ “3.1 Assembly overview - suspension strut, upper suspension link”, page 59
- ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65
- ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145
- Front brake; Assembly overview – front brake ⇒ Brake system; Rep. gr. 46 ; Front brake; Assembly overview – front brake

5.3 Removing and installing wheel bearing unit

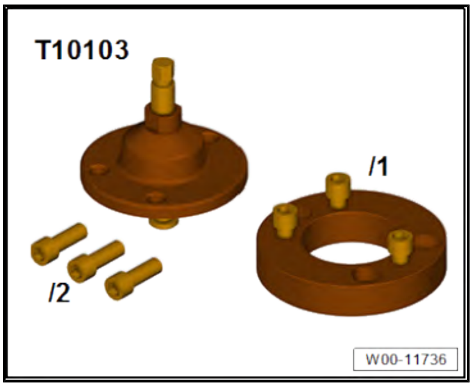
⇒ “5.3.1 Removing and installing wheel bearing unit, front-wheel and four-wheel drive”, page 79

⇒ “5.3.2 Removing and installing wheel bearing unit, rear-wheel drive”, page 82

5.3.1 Removing and installing wheel bearing unit, front-wheel and four-wheel drive

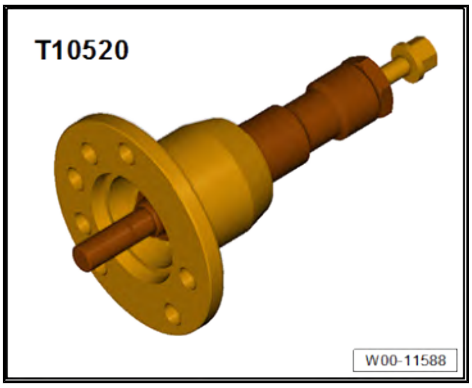
Special tools and workshop equipment required

- Adapter - T10103/1-



- Bolts - T10103/2-

- ♦ Ejector - T10520-



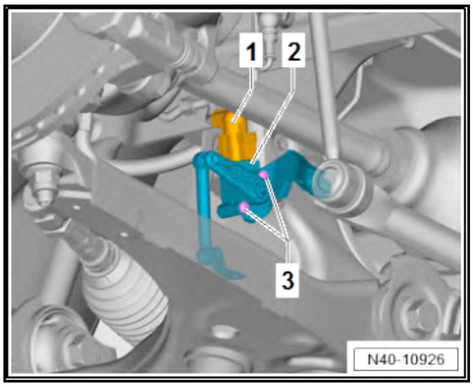
- ♦ Torque wrench - V.A.G 1332-

- ♦ Welding wire

Removing

- – Unscrew bolt for drive shaft ⇒ page 95 .
- – Unscrew nut for drive shaft ⇒ page 95 .
- – Remove brake disc ⇒ Brake system; Rep. gr. 46 ; Front brake; Removing and installing brake disc .
- – Remove front left speed sensor - G47- / front right speed sensor - G45- ⇒ Brake system; Rep. gr. 45 ; Sensors; Removing and installing speed sensors on front axle -G45- / -G47- .

Vehicles with right front vehicle level sender - G289-

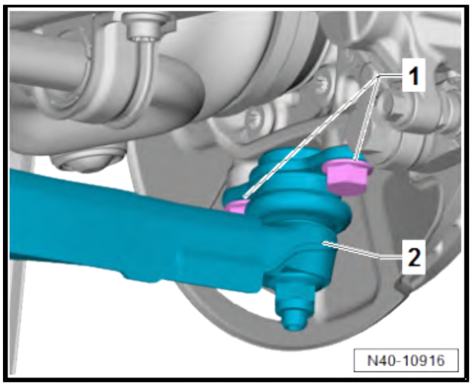


- – Separate electrical connector -1-.
- – Unscrew bolts -3-.
- – Lay right front vehicle level sender - G289- -2- aside.

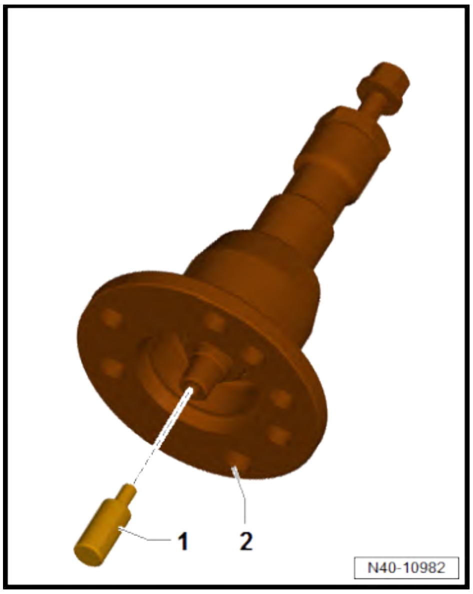
Continued for all vehicles

– Unscrew bolts -1-.

– Pull swivel joint -2- together with suspension link off wheel bearing housing.



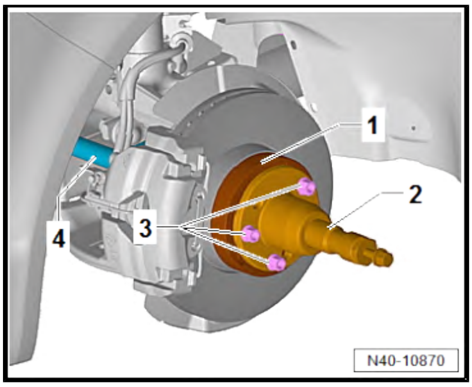
Before using press tool - T10520- -2-, ensure that thrust piece -1- is inserted.



How to use press tool - T10520-

– Use bolts - T 10103/1- to secure adapter - T10103/2- -1- to wheel bearing.

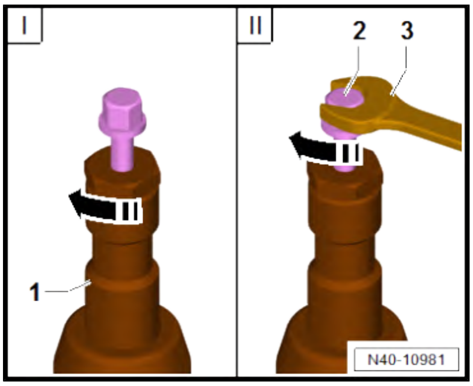
– To press out drive shaft -3-, secure ejector - T10520- -2- to wheel hub using 4 wheel bolts -3-.



– It is essential to follow the specified sequence.

I - Tighten knurled nut in direction of -arrow- hand-tight.

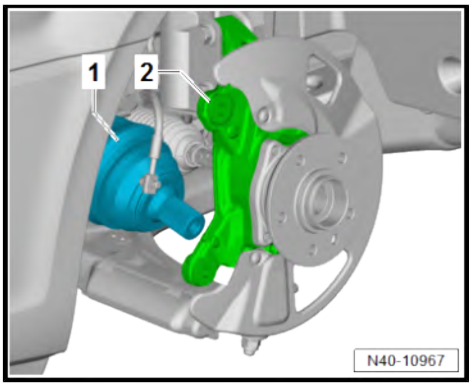
II - Turn only bolt -2- using a wrench -3- in order to press out drive shaft with press tool - T10520- -1-.



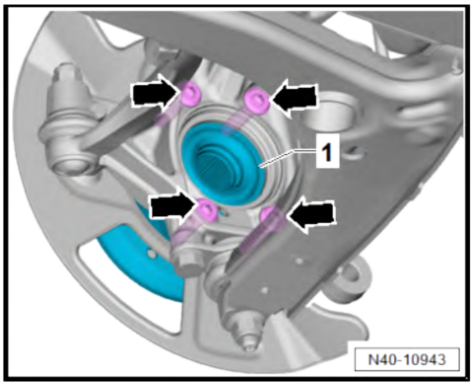
Note

At the end of the procedure or for pressing out drive shaft further, the spindle must be moved to its original position in order to apply the hydraulic force.

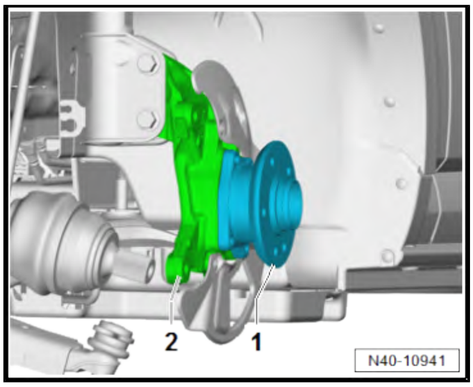
– Pull drive shaft -1- out of wheel bearing housing -2-.
– Secure drive shaft to body with welding wire.
– To stabilise the wheel bearing housing, insert swivel joint back in wheel bearing housing, and tighten swivel joint bolts hand-tight.



– Unscrew bolts -arrows- from wheel bearing unit -1-.



– Pull wheel bearing unit -1- out of wheel bearing housing -2-.



Installing

Install in the reverse order of removal, observing the following:

– Tighten bolt of drive shaft ⇒ page 95 .
– Tighten nut for drive shaft ⇒ page 95 .

Specified torques

♦ ⇒ “5.1 Assembly overview - wheel bearing”, page 75
♦ ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65
♦ ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145
♦ Front brake; Assembly overview – front brake ⇒ Brake system; Rep. gr. 46 ; Front brake; Assembly overview – front brake

5.3.2 Removing and installing wheel bearing unit, rear-wheel drive

Special tools and workshop equipment required

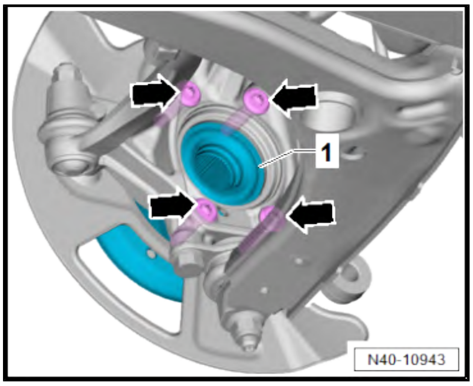
- Torque wrench - V.A.G 1332-

Removing

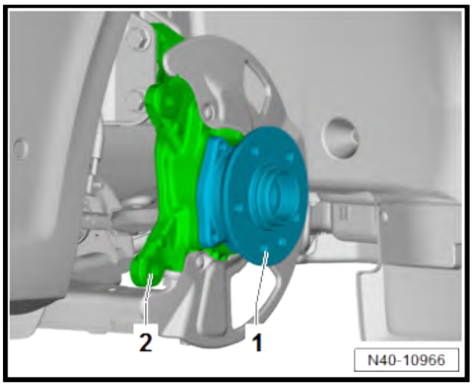
– Remove brake disc ⇒ Brake system; Rep. gr. 46 ; Front brake; Removing and installing brake disc .

– Remove front left speed sensor - G47- / front right speed sensor - G45- ⇒ Brake system; Rep. gr. 45 ; Sensors; Removing and installing speed sensors on front axle -G45- / -G47- .

– Unscrew bolts -arrows- from wheel bearing unit -1-.



– Pull wheel bearing unit -1- out of wheel bearing housing -2-.



Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ⇒ “5.1 Assembly overview - wheel bearing”, page 75
- ⇒ “1.1 Assembly overview - front vehicle level senders”, page 145
- Front brake; Assembly overview – front brake ⇒ Brake system; Rep. gr. 46 ; Front brake; Assembly overview – front brake

6 Drive shaft

⇒ “6.1 Assembly overview - drive shaft”, page 84

⇒ “6.2 Removing and installing drive shaft”, page 85

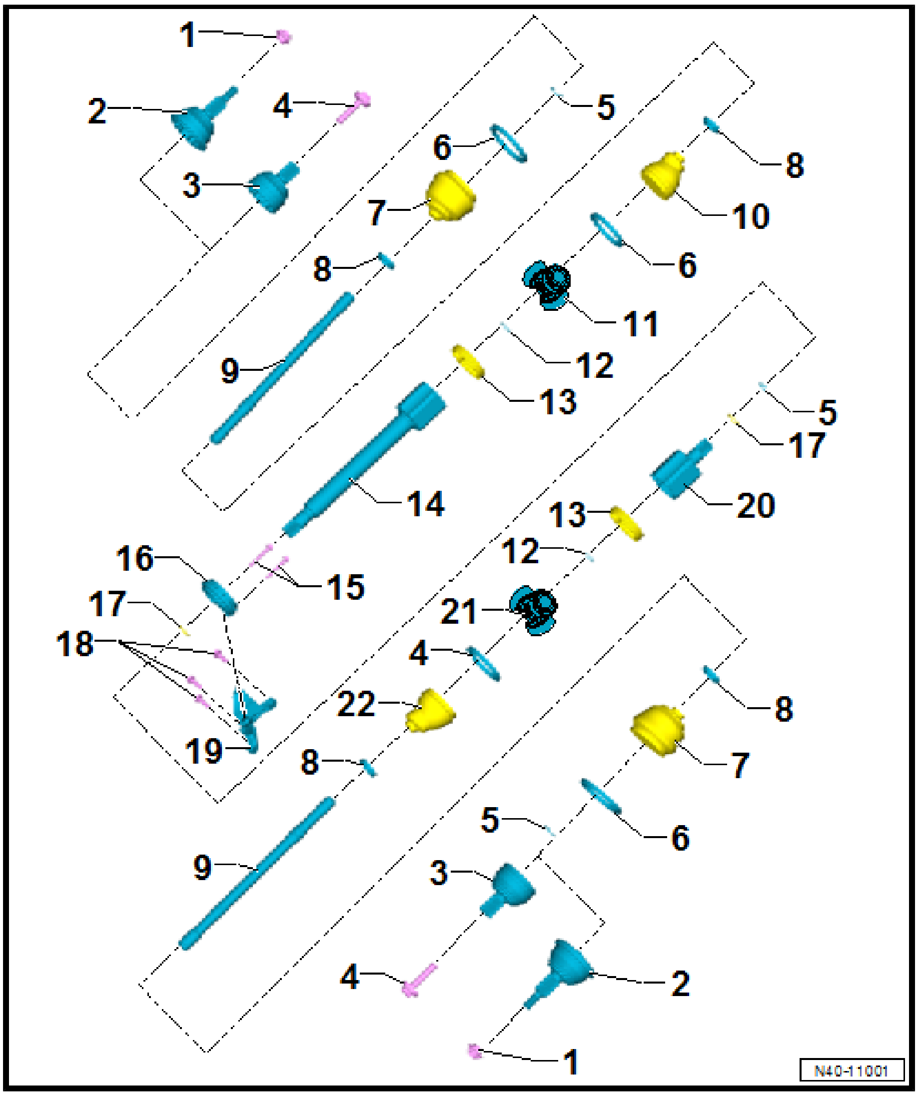
⇒ “6.3 Dismantling and assembling drive shaft”, page 90

⇒ “6.4 Loosening and tightening threaded connections of drive shaft”, page 95

⇒ “6.5 Checking outer constant velocity joint”, page 97

⇒ “6.6 Repairing intermediate shaft”, page 98

6.1 Assembly overview - drive shaft



1 - Nuts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ page 95

2 - Constant velocity joint
- ❑ On outside
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 90

3 - Constant velocity joint
- ❑ On outside
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 90

4 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ Specified torque ⇒ page 95

5 - Retaining rings
- ❑ Renew after removal
- ❑ Qty. 3

6 - Clips
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 25 Nm

7 - Boots
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 90

8 - Clips
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 25 Nm

9 - Drive shafts
- ❑ Qty. 2

- ❑ Removing and installing ⇒ page 85

10 - Boot

- ❑ Removing and installing ⇒ page 90

11 - Triple roller joint

- ❑ Removing and installing ⇒ page 90

12 - Retaining ring

13 - Boot adapter

- ❑ Depending on equipment
- ❑ Qty. 2

14 - Triple roller shaft

- ❑ For right drive shaft

15 - Bolts

- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 20 Nm +90°

16 - Support mounting

- ❑ For right drive shaft
- ❑ Removing and installing ⇒ page 90

17 - O-rings

- ❑ Renew after removal
- ❑ Qty. 2

18 - Bolts

- ❑ Renew after removal
- ❑ Qty. 3
- ❑ 50 Nm +90°

19 - Console

20 - Triple roller joint

- ❑ On inside
- ❑ For left drive shaft

21 - Triple roller joint

- ❑ Removing and installing ⇒ page 90

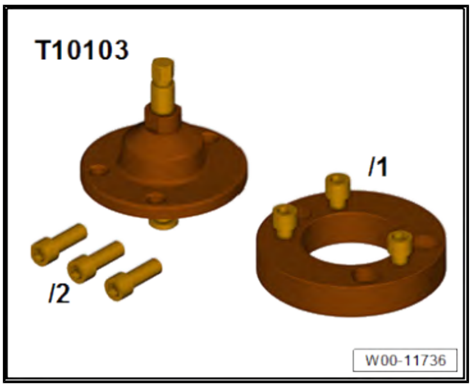
22 - Boot

- ❑ Removing and installing ⇒ page 90

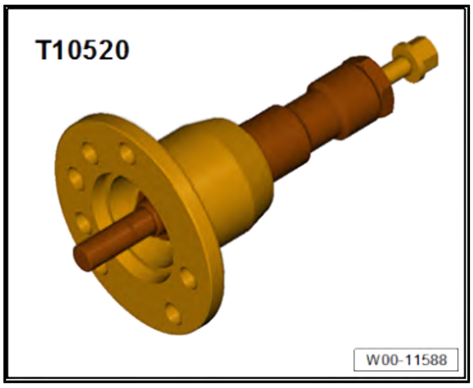
6.2 Removing and installing drive shaft

Special tools and workshop equipment required

- Adapter - T10103/1-



- Bolts - T10103/2-
- Ejector - T10520-



- Torque wrench - V.A.G 1332-

- Lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE

Removing

Vehicles with electric drive

DANGER

Danger to life from high voltage.

Severe or fatal injury from electric shock.

– The high-voltage system must be de-energised by a suitably qualified technician.

– De-energise high-voltage system ⇒ Rep. gr. 93 ; De-energising high-voltage system .

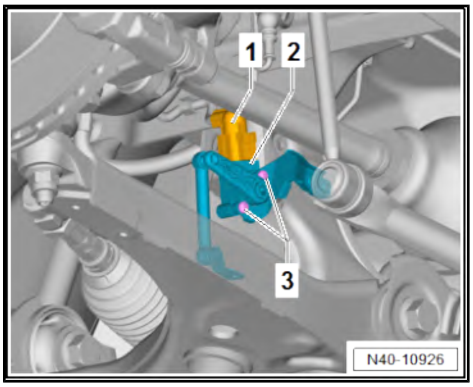
Continued for all vehicles

Procedure for left and right

– Remove front wheels ⇒ page 150 .
– Unscrew bolt for drive shaft ⇒ page 95 .
– Unscrew nut for drive shaft ⇒ page 95 .
– If fitted, remove noise insulation ⇒ General body repairs, exterior; Rep. gr. 66 ; Noise insulation; Removing and installing noise insulation .

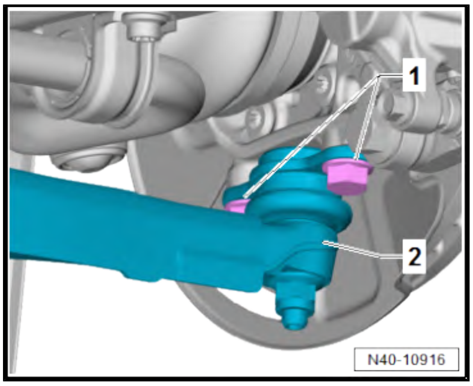
Vehicles with right front vehicle level sender - G289-

– Separate electrical connector -1-.
– Unscrew bolts -3-.
– Lay right front vehicle level sender - G289- -2- aside.

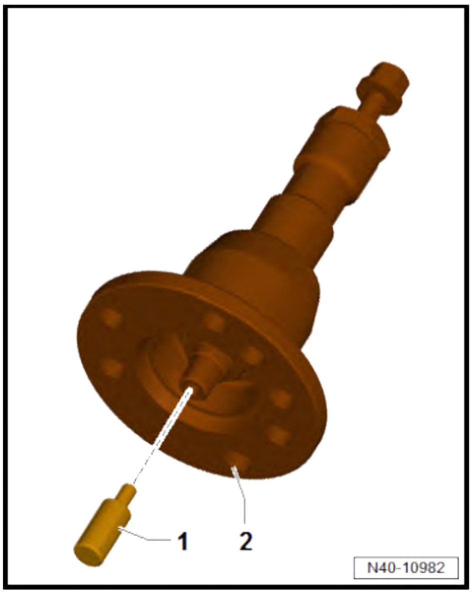


Continued for all vehicles

– Unscrew bolts -1-.
– Pull swivel joint -2- together with suspension link off wheel bearing housing.



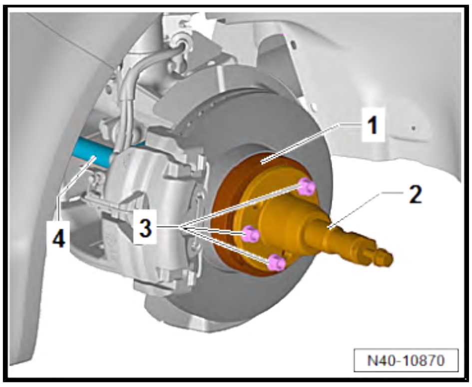
Before using press tool - T10520- -2-, ensure that thrust piece -1- is inserted.



How to use press tool - T10520-

– Use bolts - T 10103/1- to secure adapter - T10103/2- -1- to wheel bearing.

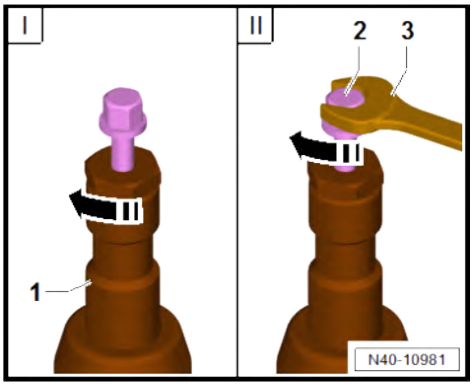
– To press out drive shaft -3-, secure ejector - T10520- -2- to wheel hub using 4 wheel bolts -3-.



– It is essential to follow the specified sequence.

I - Tighten knurled nut in direction of -arrow- hand-tight.

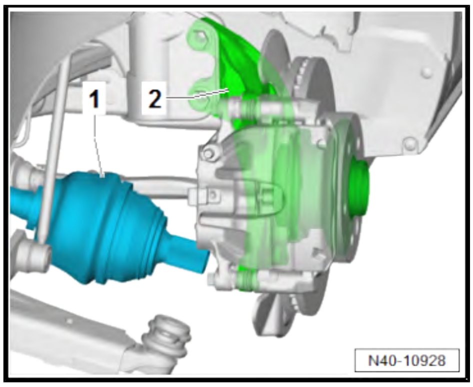
II - Turn only bolt -2- using a wrench -3- in order to press out drive shaft with press tool - T10520- -1-.



Note

At the end of the procedure or for pressing out drive shaft further, the spindle must be moved to its original position in order to apply the hydraulic force.

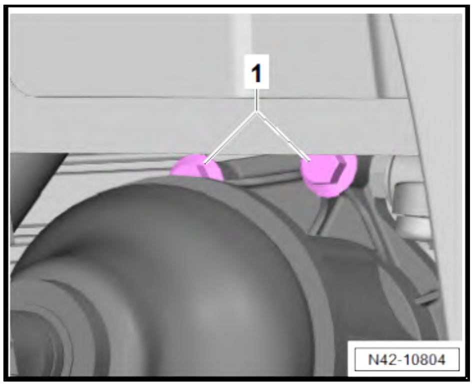
– Pull drive shaft -1- out of wheel bearing unit -2-.



Procedure for right side, front-wheel drive

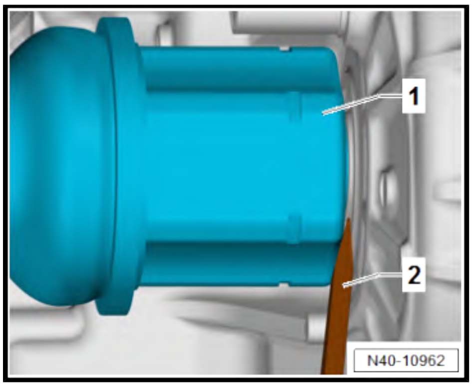
– Unscrew bolts -1-.

– Remove drive shaft from gearbox splines and pull out complete drive shaft.



Continuation of procedure for left and right side

– Fit assembly lever -2- between gearbox housing and triple roller joint -1-.
– Use assembly lever -2- to push inner joint out of gearbox.
– Remove drive shaft from gearbox splines and pull out complete drive shaft.



Note

- ♦ *When removing drive shafts, make sure that the triple roller spider with rollers is not pulled out of the triple roller joint.*
- ♦ *If triple roller star with rollers has slipped out, joint protective boot must be opened and triple roller star with rollers must be re-inserted in triple roller joint.*

Installing

Install in the reverse order of removal, observing the following:

Procedure for both sides

Note

Before installing the drive shafts, ensure that the O-rings on both sides of the gearbox splines are not damaged. Renew them if necessary.

Installing left drive shaft

Note

Insert new retaining ring directly into groove of drive shaft. It is not permitted for retaining ring to be pushed over splines. This will overstretch retaining ring and render it unusable.

– Coat final drive on gearbox »generously« with lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Place drive shaft in splines of final drive.
– With a hefty »jolt«, push drive shaft as far as possible in direction of gearbox.

Installing right drive shaft

– Coat final drive on gearbox »generously« with lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Insert stub shaft on greased splines of final drive.
– Tighten bolt of drive shaft ⇒ page 95 .
– Tighten nut for drive shaft ⇒ page 95 .

Vehicles with electric drive

WARNING

Danger to life from high voltage.

Risk of severe or fatal injury due to electric shock.

– Have a qualified technician re-energise the high-voltage system.

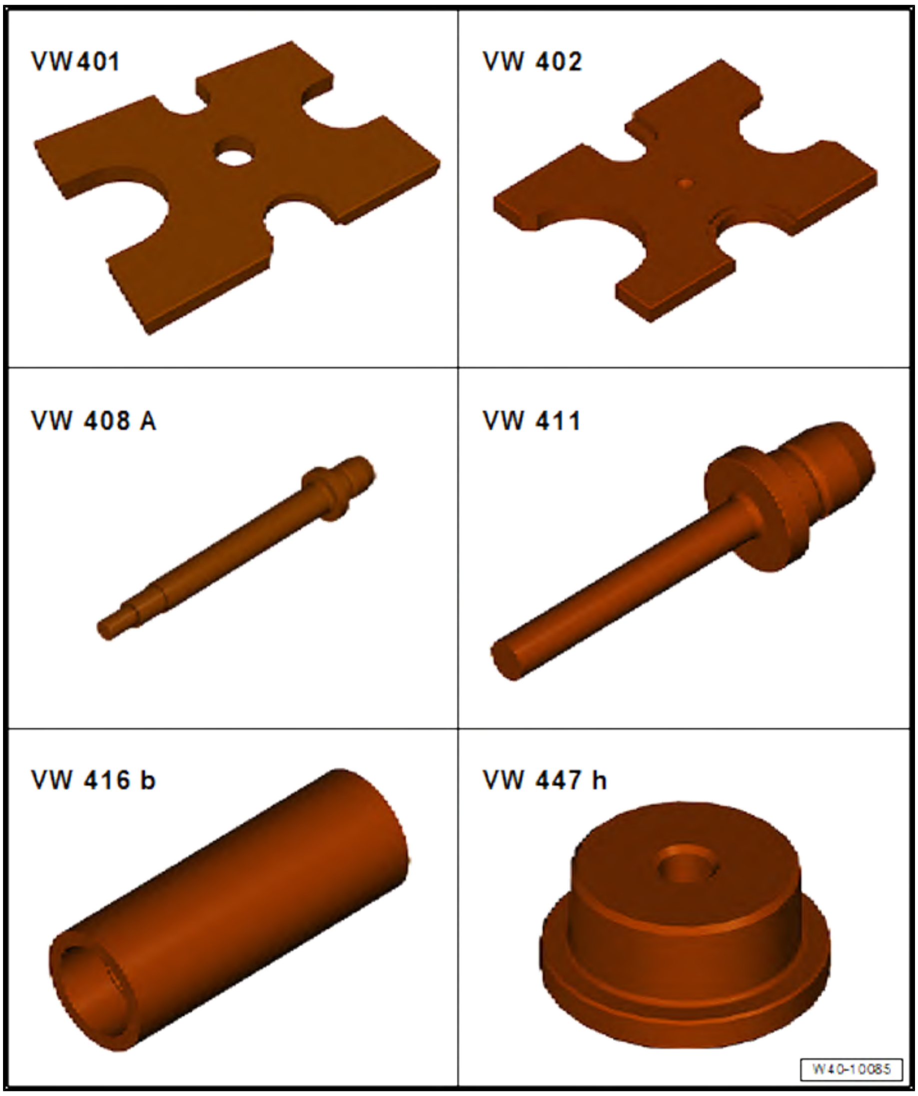
– Re-energise high-voltage system ⇒ Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

Specified torques

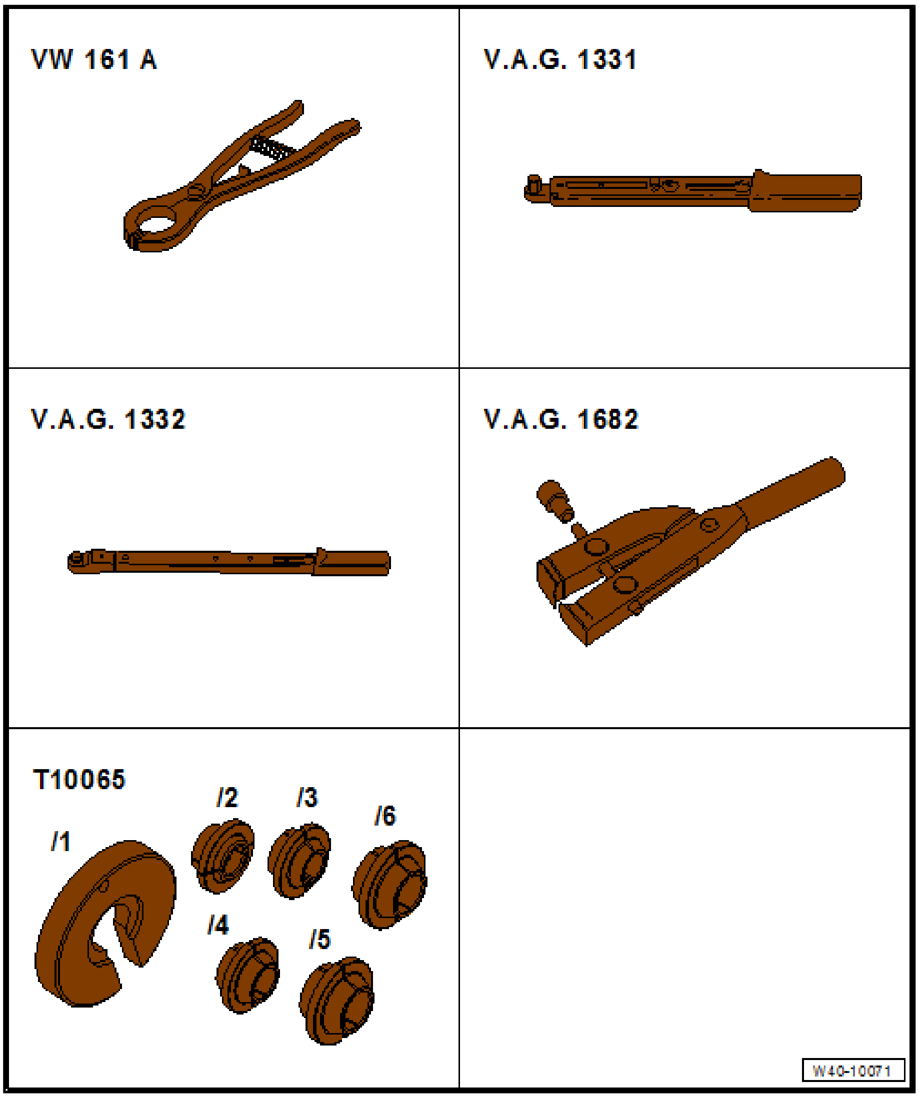
- ⇒ “6.1 Assembly overview - drive shaft”, page 84
- ⇒ “4.1 Assembly overview - lower suspension link, swivel joint”, page 65
- Assembly overview - noise insulation ⇒ Rep. gr. 66 ; Exterior equipment; Noise insulation; Assembly overview - noise insulation

6.3 Dismantling and assembling drive shaft

Special tools and workshop equipment required

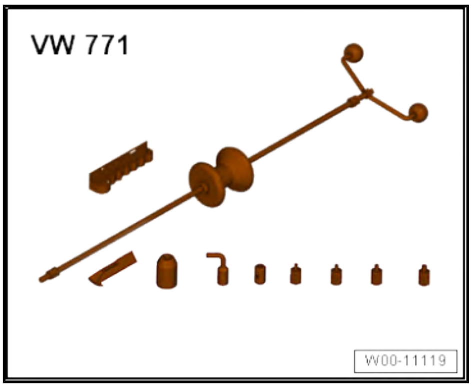


- Thrust plate - VW 401-
- Thrust plate - VW 402-
- Press tool - VW 408 A-
- Press tool - VW 411-
- Tube - VW 416 B-
- Thrust plate - VW 447 H-



- Snap ring pliers - VW 161 A-
- Torque wrench - V.A.G 1331-
- Torque wrench - V.A.G 1332-
- Clamp tensioner - V.A.G 1682-
- Assembly tool - T10065-

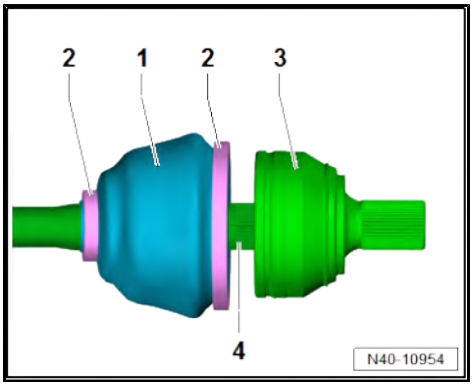
- ♦ Multipurpose tool - VW 771-



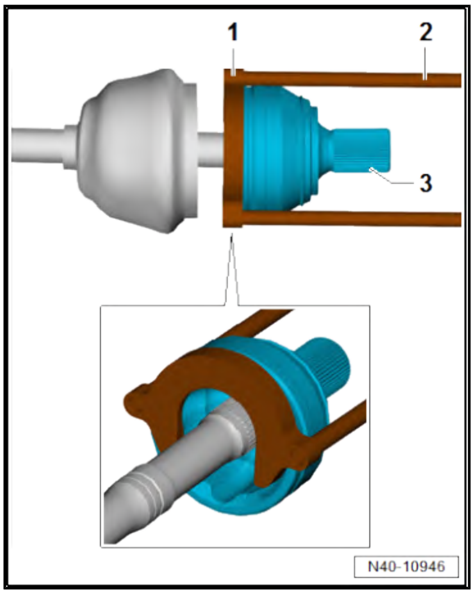
- ♦ Puller - T10382-

Removing outer constant velocity joint

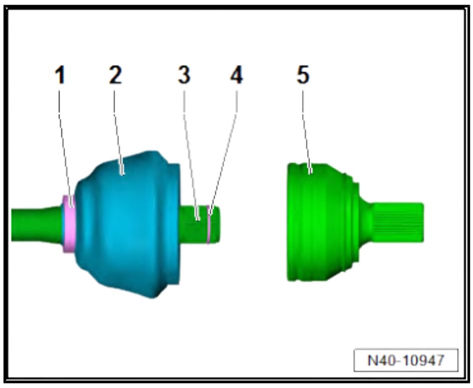
- Drive shaft removed ⇒ page 85
- – Clamp drive shaft in vice using jaw protectors.
- – Release clamps -2- of boot -1-.
- – Push boot -1- of outer constant velocity joint -3- back onto drive shaft -4-.



– Set up puller - T10382- -1- so that smooth side of puller plate - T10382/1- points to spindles - T10382/2- -2-.
– Assemble puller - T10382- complete with multi-purpose tool - VW 771- .
– Pull constant velocity joint -3- off drive shaft using puller - T10382- and multi-purpose tool - VW 771- .



Driving on outer constant velocity joint

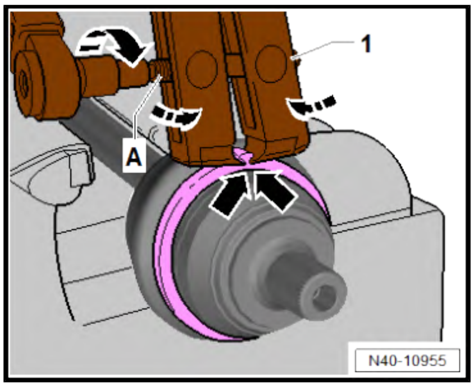


– Insert retaining ring -4- in groove of drive shaft -3-.
– Push clamp -1- onto drive shaft -3-.
– Drive clean outer constant velocity joint -5- onto drive shaft -3- using a plastic hammer until circlip -4- engages.
– Inject 100 grammes of lubricating grease ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE into outer constant velocity joint through ball tracks.
– Pack drive shaft boot -2- with 135 grammes of lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.
– Fit boot -2- to constant velocity joint -5-.

Tension clamps on constant velocity joint

Note

The clamp tensioner - V.A.G 1682- must not be twisted or canted as it is tightened.

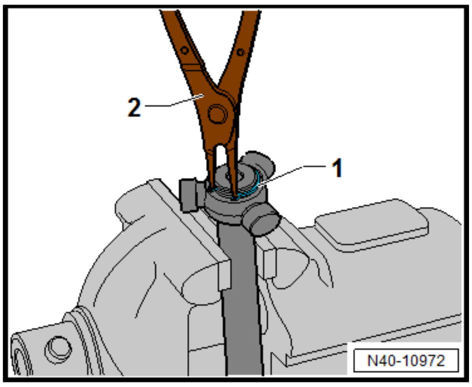


– Position clamp tensioner - V.A.G 1682- -1- as shown in illustration.
– Ensure jaws of tensioner lie in corners -arrows- of ear on O-type clip.
– Tension clamp by turning spindle -A- in direction of -arrow-.

Dismantling triple roller joint

• Drive shaft removed ⇒ page 85
– Open both clamps on triple roller joint and push back boot.
– Pull joint body off drive shaft.

– Remove retaining ring -1- with circlip pliers - VW 161 A- -2-.



– Insert drive shaft in assembly tool - T10065- -3- and -4-.

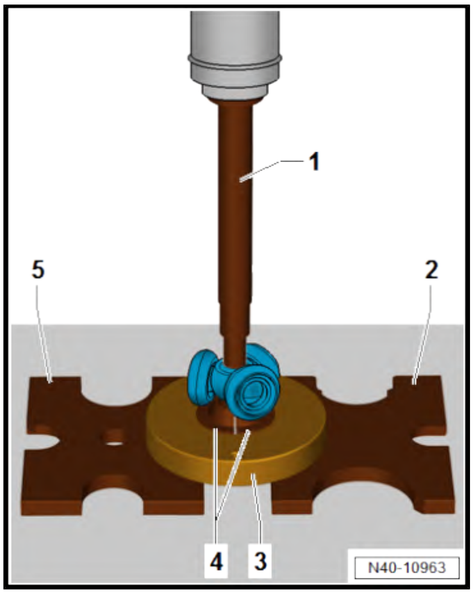
1 - Press tool - VW408A-

2 - Pressure plate - VW402-

3 - Assembly tool - T10065-

4 - Assembly tool - T10065-

5 - Pressure plate - VW401-



– Press drive shaft off triple roller star.

– Pull boot off shaft.

– Clean shaft, joint body and groove for seal.

Installing

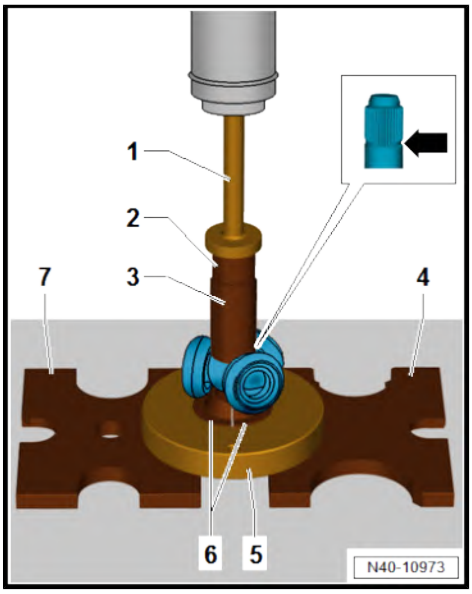
– Push clamp for boot onto shaft.

– Push joint boot onto shaft.

– Push joint body onto shaft.

Fit triple roller star

Chamfer on shaft -arrow- points towards triple roller star and serves as an assembly aid.



– Fit triple roller star onto shaft and press on to stop.

1 - Press tool - VW411-

2 - Thrust pad - VW447H-

3 - Tube - VW416B-

4 - Pressure plate - VW402-

5 - Assembly tool - T10065-

6 - Assembly tool - T10065-

7 - Pressure plate - VW401-

– Ensure that pressure does not exceed 3.0 t.

– If necessary, coat splines of drive shaft and triple roller star with lubricating paste ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Insert retaining ring, ensuring that it is seated correctly.

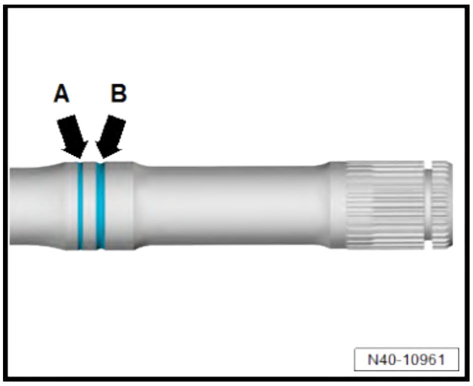
– Pack 80 grammes of lubricating grease ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE into triple roller joint.

– Slide joint body over rollers and hold.

– Pack 80 grammes of grease ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE into back of triple roller joint.

– Install boot for joint.

Groove for detecting that joint protective boot has been installed correctly



– Position boot in groove -B-.

Groove -A- must remain visible.

– Install clamps ⇒ page 93 .

Specified torques

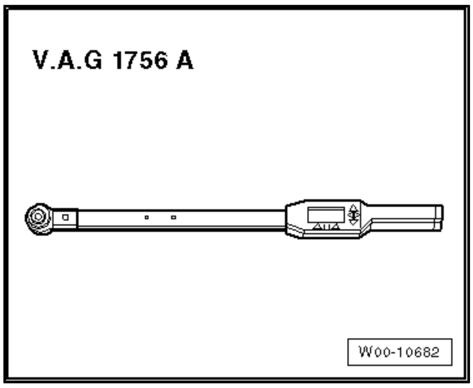
♦ ⇒ “6.1 Assembly overview - drive shaft”, page 84

6.4 Loosening and tightening threaded connections of drive shaft

Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Torque angle wrench - V.A.G 1756 A-



Note

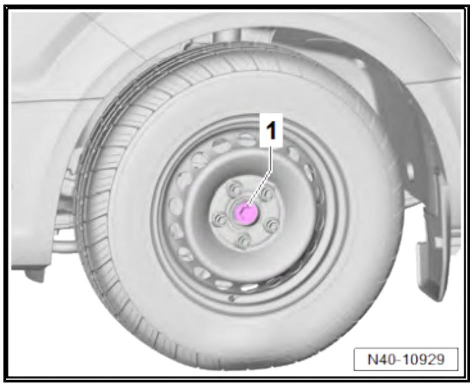
- *Wheel bearings must not be subjected to load after bolt securing drive shaft to wheel hub has been loosened.*
- *If wheel bearings are loaded with weight of vehicle, bearing will be damaged.*
- *Avoid moving vehicle without installed drive shaft!*
- *Instead of a drive shaft, an outer joint can be installed for moving the vehicle.*

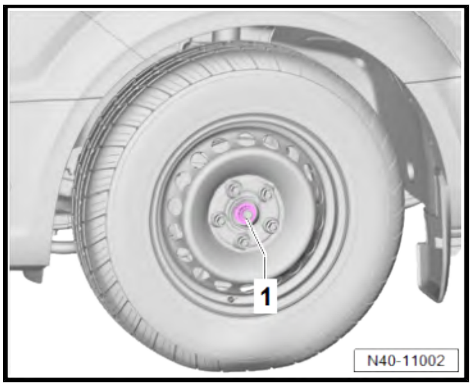
Loosen nut / bolt securing drive shaft.

- If vehicle is still standing on wheels, loosen threaded connection securing drive shaft by maximum 90°.
- Raise vehicle so that wheels are off the ground.
- Have a second mechanic depress the brake pedal.

Vehicles with bolt

- Unscrew bolt -1-.





Vehicles with nut

– Unscrew nut -1-.

Tighten nut / bolt securing drive shaft.

– Renew bolt -1-.

– Renew nut -1-.

Note

It must be ensured that the wheels do not yet touch the floor when the drive shaft is being tightened. Otherwise the wheel bearing may be damaged.

– Have a second mechanic depress the brake pedal.

Specified torques

Vehicles with bolt

Step	Component	Specified torque
1.	Bolt	120 Nm
2.	Bolt	90°

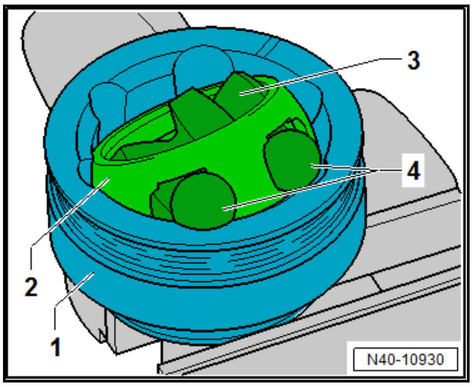
Vehicles with nut

Step	Component	Specified torque
1.	Nut	200 Nm
2.	Nut	Loosen 180°
3.	Nut	200 Nm

6.5 Checking outer constant velocity joint

The joint is to be dismantled to renew the grease if it is heavily soiled, or to check the running surfaces of the balls for wear and damage.

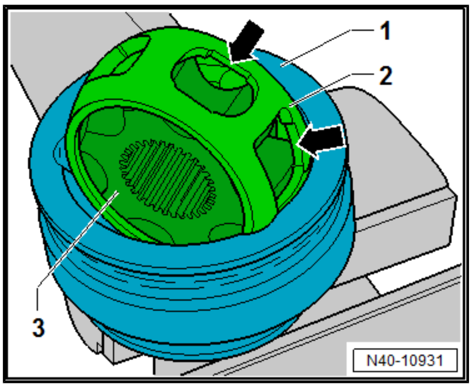
Removing



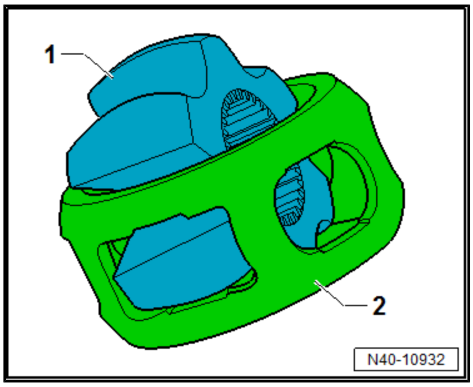
– Before dismantling, mark position of ball hub -3- in relation to ball cage -2- and joint body -1- with an electric scriber or oil stone.

– Swing ball hub -3- and ball cage -2-.

– Remove balls -4- one at a time.



- Turn ball cage -2- until the two rectangular windows -arrows- rest against joint body.
- Take out cage -2- with hub -3-.



- Swing segment of hub into square cage window.
- Tip hub -1- out of cage -2-.

The 6 balls for each joint belong to a tolerance group. Check stub axle, hub, cage and balls for small indentations (pitting) and traces of seizing. Excessive circumferential backlash in the joint is noticeable during load change jolts. In this case the joint must be replaced. Smoothing and traces of wear of the balls are no reason to change the joint.

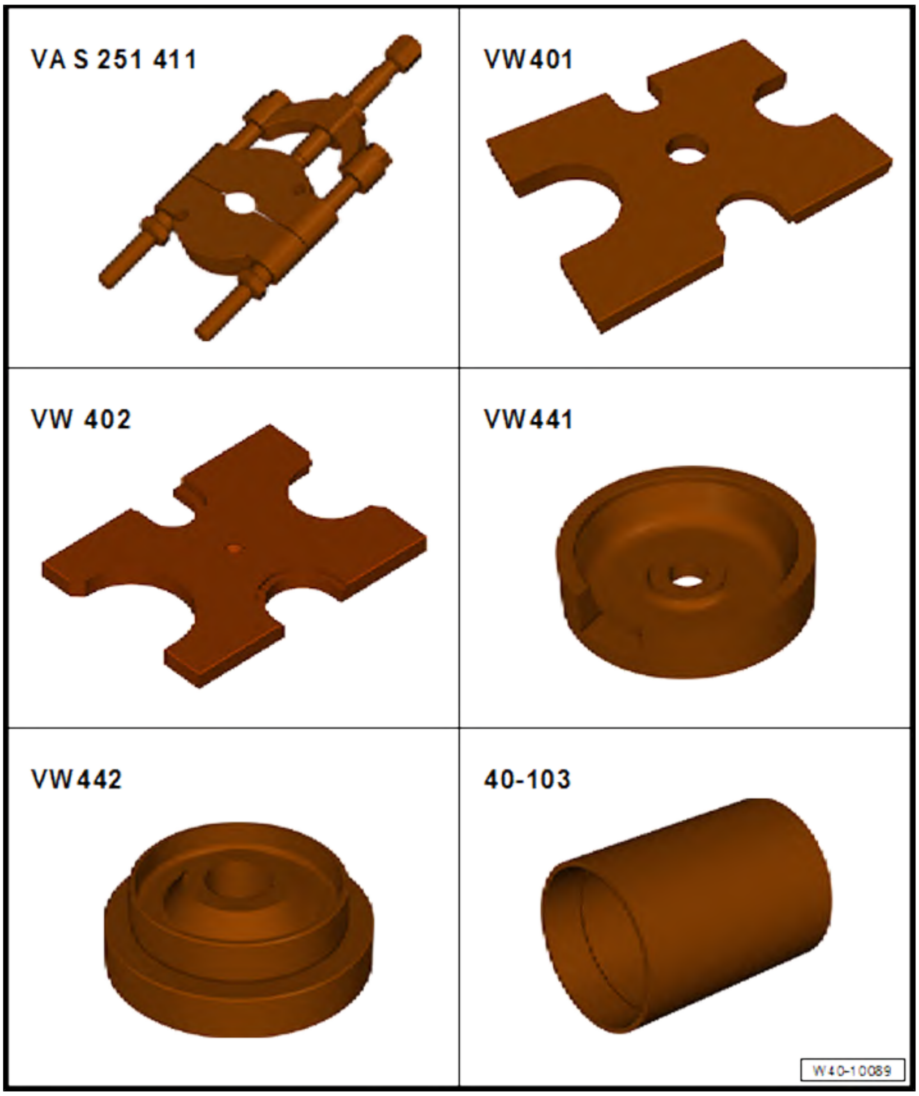
Installing

Install in the reverse order of removal, observing the following:

- Pack half of total grease quantity (40 g) into joint body.
- Fit cage with hub into joint body.
- Press in opposing balls one after the other; the original position of the hub relative to the cage and joint body must be restored.
- Fit new retaining ring into hub.
- Distribute remaining grease in boot.

6.6 Repairing intermediate shaft

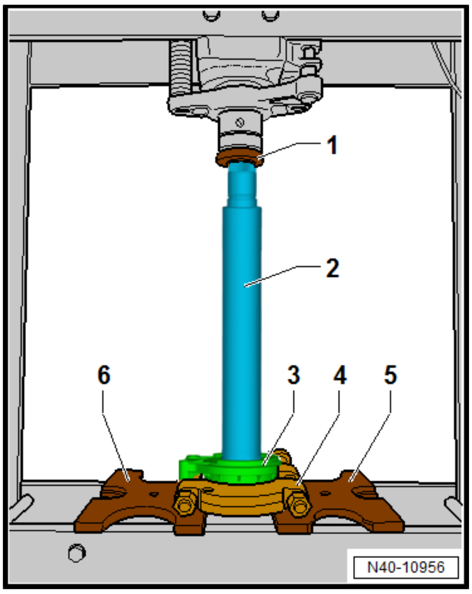
Special tools and workshop equipment required



- ♦ Splitter KUKKO 17-2 - VAS 251 411-
- ♦ Thrust plate - VW401-
- ♦ Thrust plate - VW402-
- ♦ Support - VW441-
- ♦ Thrust piece - VW442-
- ♦ Support - 40-103-

Press off support bearing

- • Right drive shaft removed ⇒ page 85
- – Separate drive shaft at inner joint ⇒ page 90 .



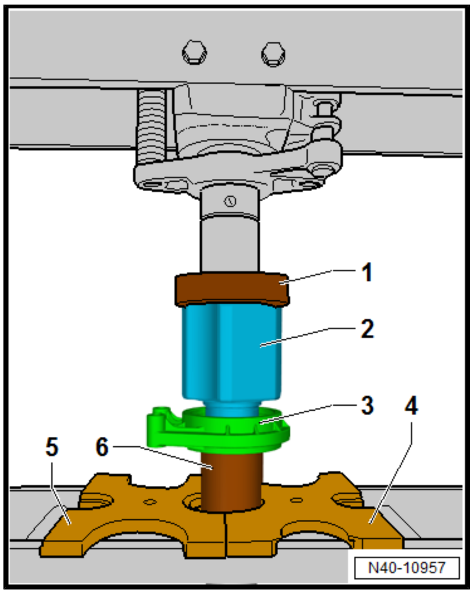
- Fit splitter KUKKO 17-2 - VAS 251 411- -4- between inner joint and support bearing -3-.
- Fit thrust plate - VW 401- -5- and thrust plate - VW 402- -6- onto press as shown in illustration.
- Fit drive shaft -2- on thrust plates with splitter KUKKO 17-2 - VAS 251 411- -4-.
- Set thrust piece - VW 442- -1- on upper end of drive shaft -2-.

Note

Hold drive shaft at inner joint from below to prevent it from falling to the ground.

- Press drive shaft -1- downwards out of support bearing -2-.
- Remove drive shaft from below.

Pressing on support bearing



- Fit thrust plate - VW 401- -4- and thrust plate - VW 402- -5- onto press as shown in illustration.
- Push support bearing -3- onto drive shaft.

Note

Observe installation position of support bearing. Push support bearing onto drive shaft as shown in figure.

- Push support - 40 - 103- -6- onto drive shaft.
- Set drive shaft wit support bearing -3- and support - 40 - 103- onto press plates.
- Set support - VW 441- -1- on inner joint -2-.
- Press drive shaft downwards onto support bearing -3- to stop.
- Remove drive shaft from press and check that support bearing moves freely.
- Assembling drive shaft ⇒ page 90 .

42 – Rear suspension

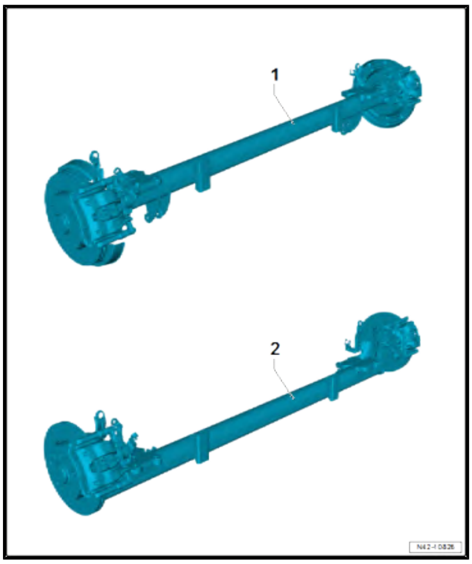
1 Rear axle

⇒ “1.1 Overview - rear axle”, page 101
⇒ “1.2 Removing and installing rear axle”, page 102

1.1 Overview - rear axle

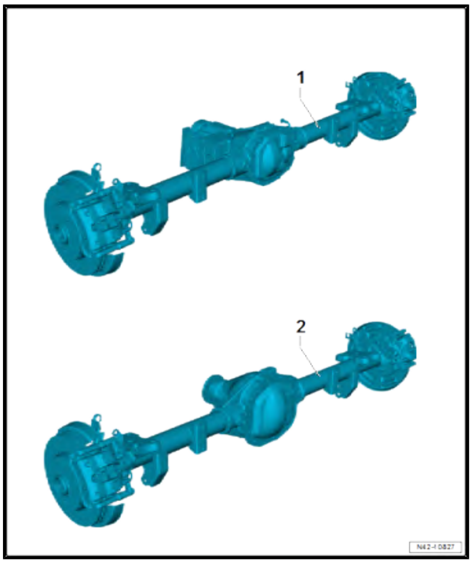
1 - Rear axle, high

2 - Rear axle, low

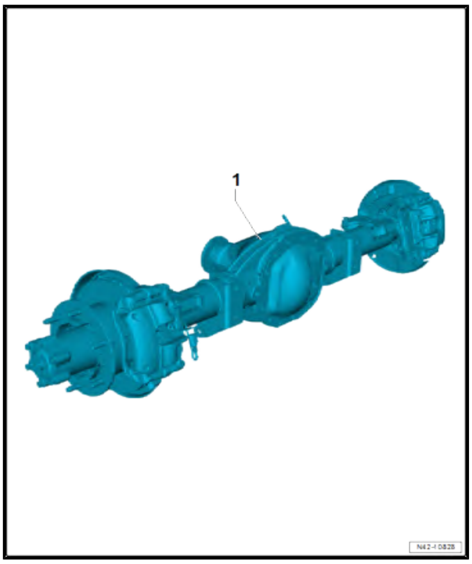


1 - Rear axle, all-wheel drive, open/lockable

2 - Rear axle, rear-wheel drive



1 - Rear axle, twin wheels



1.2 Removing and installing rear axle

⇒ “1.2.1 Removing and installing rear axle, front-wheel drive”, page 102

⇒ “1.2.2 Removing and installing rear axle, all-wheel drive/rear-wheel drive”, page 104

1.2.1 Removing and installing rear axle, front-wheel drive

Special tools and workshop equipment required

- ♦ Engine and gearbox jack - VAS 6931-

- ♦ Torque wrench - V.A.G 1332-

◆ Tensioning strap - T10038-

Removing

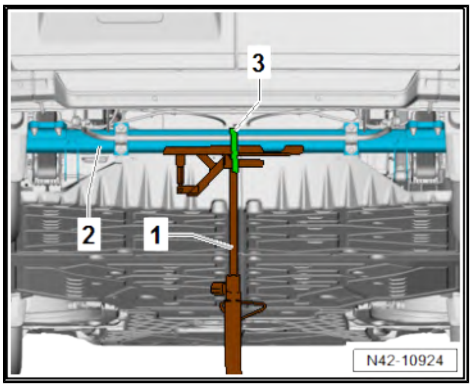
– Remove brake caliper ⇒ Brake system; Rep. gr. 46 ; Rear brake; Removing and installing brake pads .

Vehicles with electric drive

– Position engine and gearbox jack - VAS 6931- -1- under rear axle -2-.

– Secure rear axle -2- using tensioning strap - T10038- -3-.

– Raise rear axle -2- slightly.

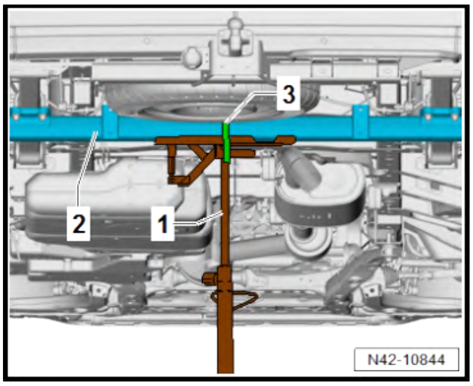


Continued for all vehicles

– Position engine and gearbox jack - VAS 6931- -1- under rear axle -2-.

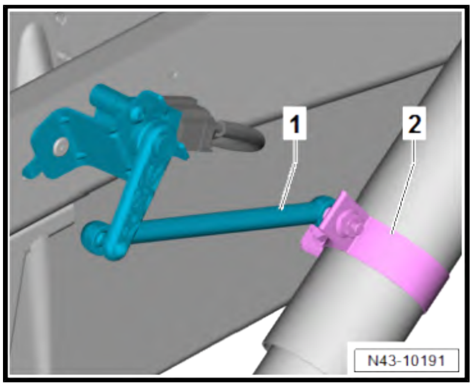
– Secure rear axle -2- using tensioning strap - T10038- -3-.

– Raise rear axle -2- slightly.



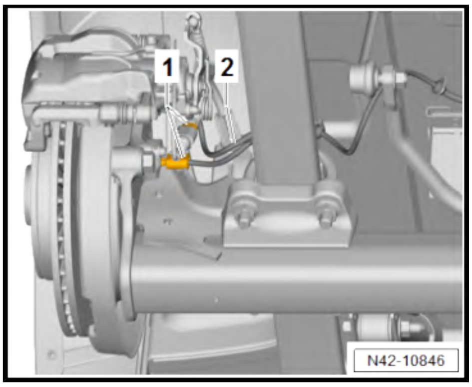
Vehicles with rear right vehicle level sender - G77-

– Detach rear right vehicle level sender - G77- -1- from connecting tab -2-.

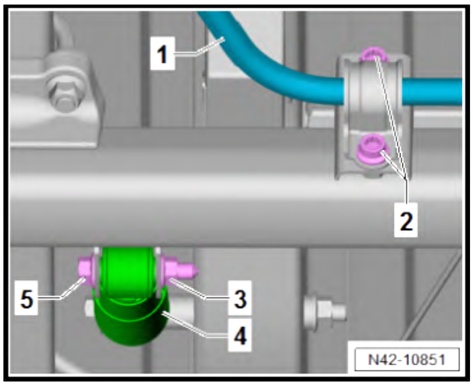


Continued for all vehicles

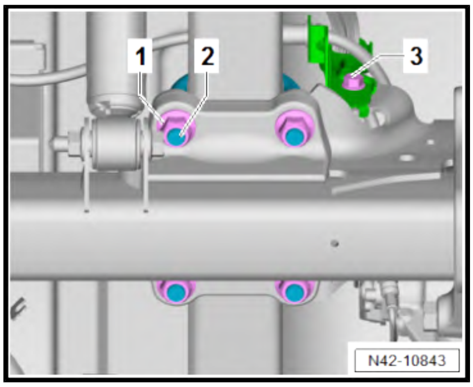
– Separate electrical connectors -1-.
– Unclip wiring harness -2- from retainers.



– Unscrew bolts -2- from anti-roll bar -1-.
– Unscrew bolt -5-. When doing this, counterhold nut -3-.
– Take shock absorber -4- out of mounting.



– Unscrew bolt -3-.
– Unscrew nuts -1-.
– Remove spring clip -2-.
– Lower rear axle with aid of second mechanic.



Installing

Install in the reverse order of removal, observing the following:

Specified torques

♦ ⇒ “2.1 Assembly overview - anti-roll bar”, page 108
♦ ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115
♦ Rear brake; Assembly overview – rear brake ⇒ Brake system; Rep. gr. 46 ; Rear brake; Assembly overview – rear brake

1.2.2 Removing and installing rear axle, all-wheel drive/rear-wheel drive

Special tools and workshop equipment required

♦ Engine and gearbox jack - VAS 6931-

♦ Torque wrench - V.A.G 1332-

♦ Tensioning strap - T10038-

♦ Welding wire

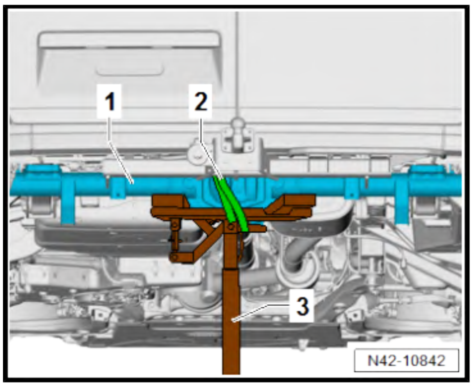
Removing

– Remove brake caliper ⇒ Brake system; Rep. gr. 46 ; Rear brake; Removing and installing brake pads .

– Position engine and gearbox jack - VAS 6931- -3- under rear axle -1-.

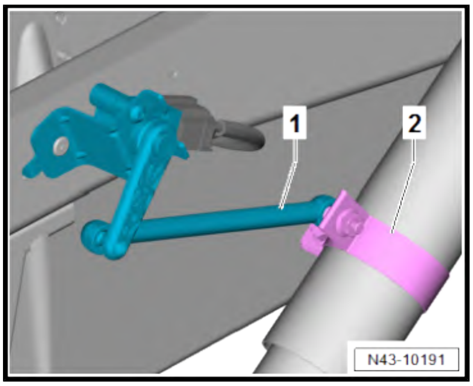
– Secure rear axle -1- using tensioning strap - T10038- -2-.

– Raise rear axle -1- slightly.



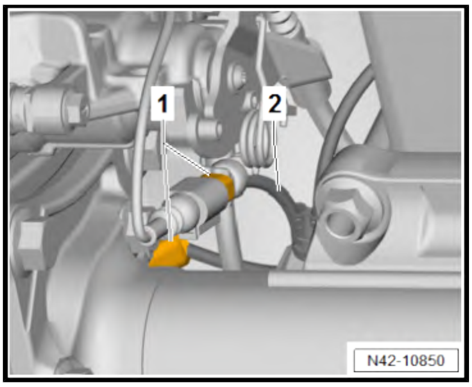
Vehicles with rear right vehicle level sender - G77-

– Detach rear right vehicle level sender - G77- -1- from connecting tab -2-.

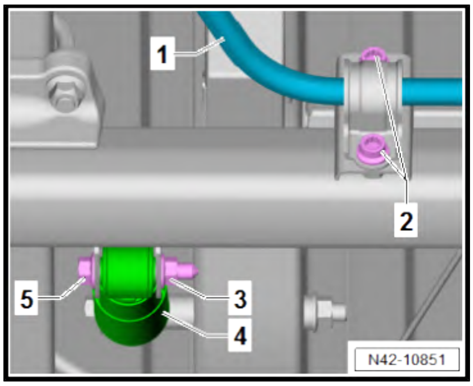


Continued for all vehicles

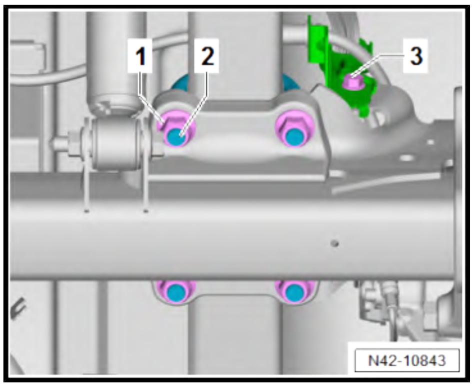
– Separate electrical connectors -1-.
– Unclip wiring harness -2- from retainer.



– Unscrew bolts -2- from anti-roll bar -1-.
– Unscrew bolt -5-. When doing this, counterhold nut -3-.
– Take shock absorber -4- out of mounting.



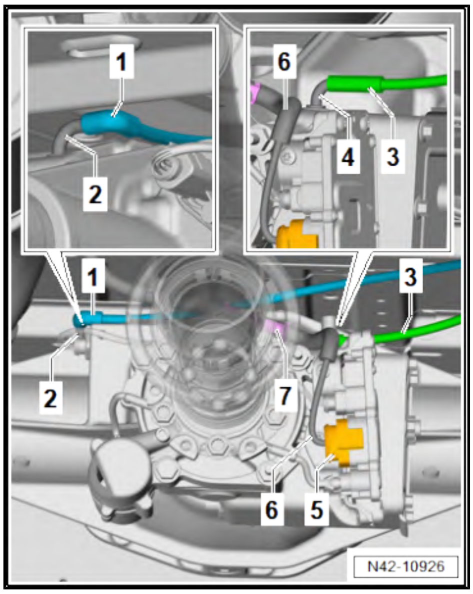
– Unscrew bolt -3-.
– Unscrew nuts -1-.
– Remove spring clip -2-.

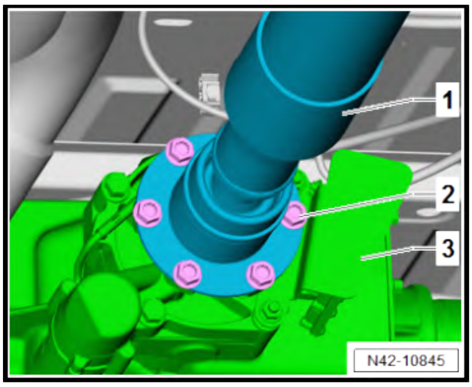


Note

- *Mark position of parts in relation to each other before removing all of them.*
- *Installation must be restored to original condition.*

– Pull off breather hose -1- from breather connection -2-.
– Pull off breather hose -3- from breather connection -4-.
– Separate electrical connector -5-.
– Unclip wiring harness -6- from bracket -7-.





– Mark position of propshaft -1- relative to propshaft flange -3-.
– Unscrew bolts -2-.
– Pull off propshaft -1- from propshaft flange -3-.
– Support propshaft -1- on underbody.
– Lower rear axle with aid of second mechanic.

Installing

Install in the reverse order of removal, observing the following:

Specified torques

- ⇒ “2.1 Assembly overview - anti-roll bar”, page 108
- ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115
- Propshaft, Assembly overview - propshaft ⇒ Rep. gr. 39 ; Propshaft; Assembly overview - propshaft
- Rear brake; Assembly overview – rear brake ⇒ Brake system; Rep. gr. 46 ; Rear brake; Assembly overview – rear brake

2 Anti-roll bar

2.1 Assembly overview - anti-roll bar

Assembly overview - anti-roll bar, single wheels ⇒ page 108

Assembly overview - anti-roll bar, twin wheels ⇒ page 108

Assembly overview - anti-roll bar, single wheels

1 - Rear axle
- ❑ Removing and installing ⇒ page 102

2 - Coupling rod
- ❑ Removing and installing ⇒ page 113

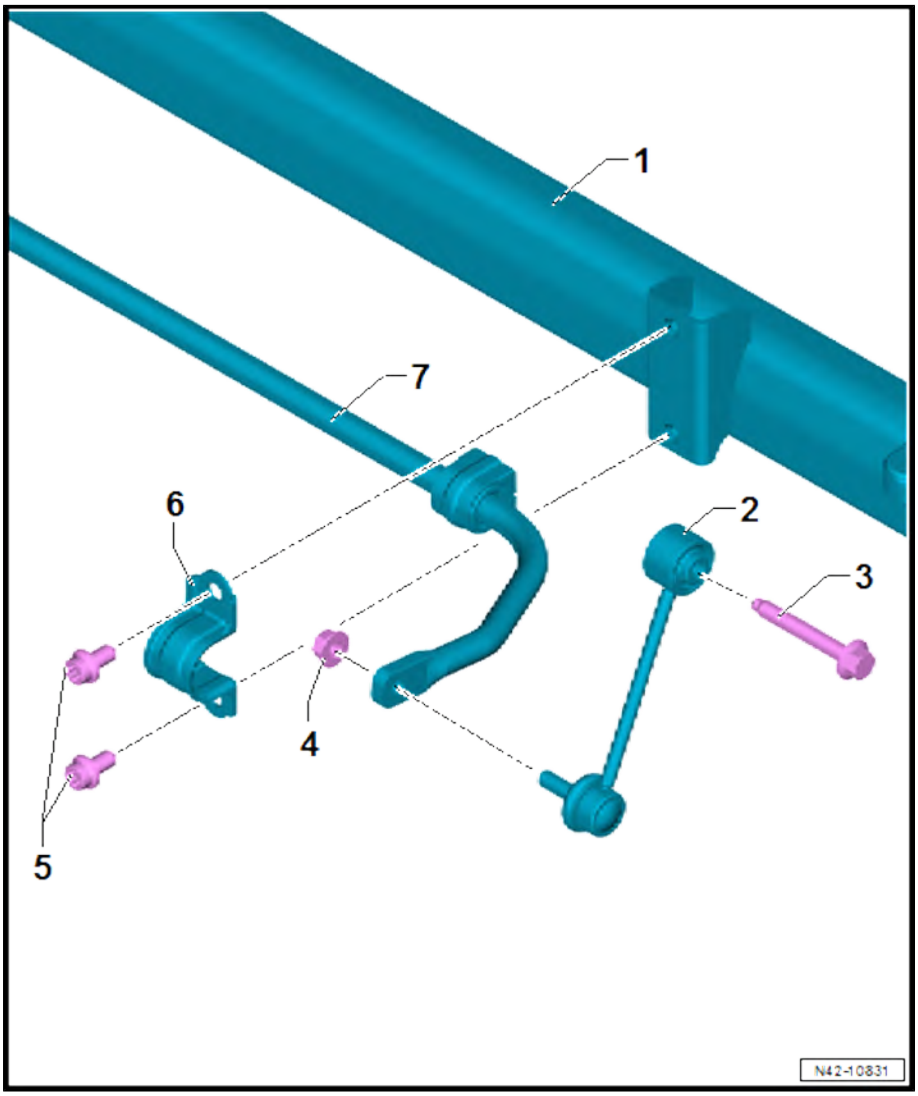
3 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +90°

4 - Nut
- ❑ Renew after removal
- ❑ M12: 60 Nm +45°
- ❑ M14: 100 Nm +45°

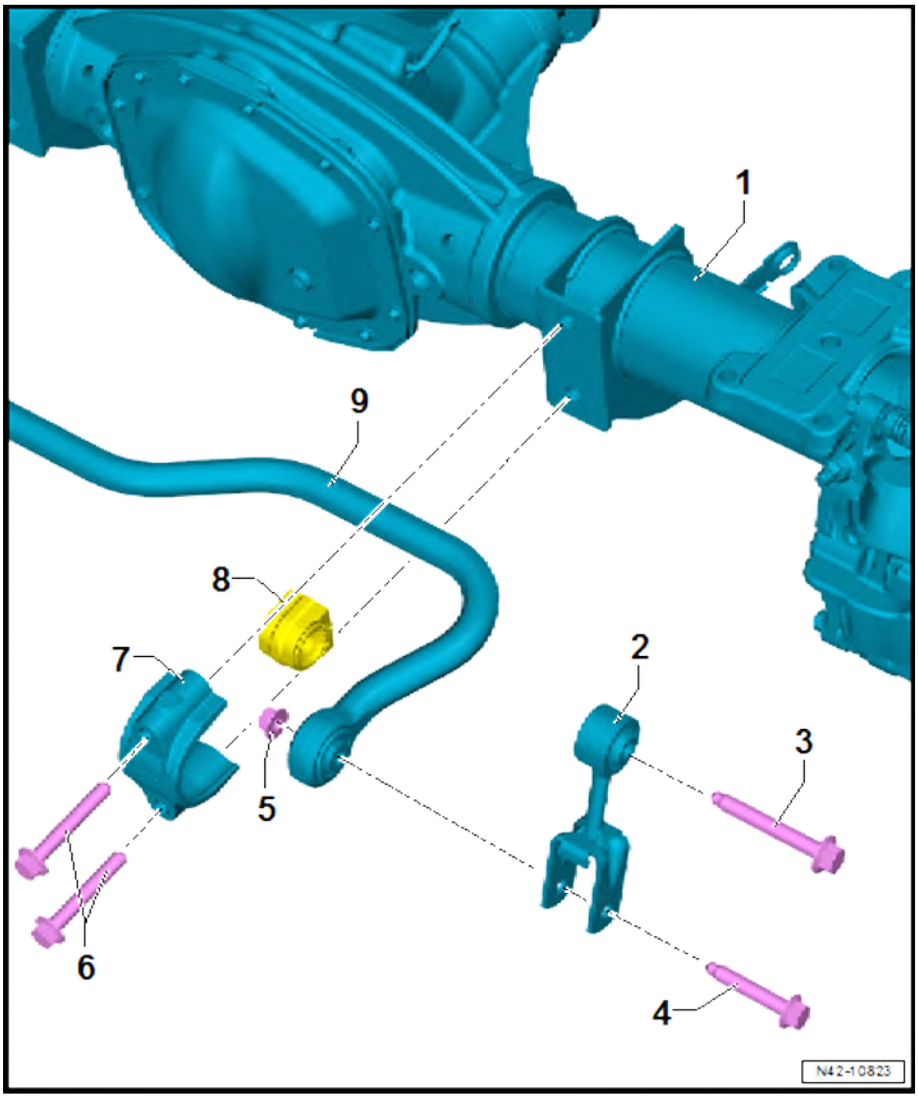
5 - Bolts
- ❑ Qty. 2
- ❑ 100 Nm

6 - Clip

7 - Anti-roll bar
- ❑ Removing and installing ⇒ page 109



Assembly overview - anti-roll bar, twin wheels



1 - Rear axle
- ❑ Removing and installing ⇒ page 102

2 - Coupling rods
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 113

3 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +90°

4 - Bolts
- ❑ Renew after removal

5 - Nut
- ❑ Renew after removal
- ❑ 70 Nm +90°

6 - Bolts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 70 Nm +180°

7 - Clip

8 - Anti-roll bar bush
- ❑ Removing and installing ⇒ page 112

9 - Anti-roll bar
- ❑ Removing and installing ⇒ page 109

2.2 Removing and installing anti-roll bar

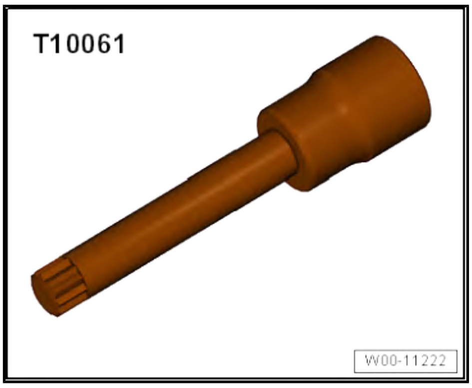
⇒ “2.2.1 Removing and installing anti-roll bar, single tyres”, page 109

⇒ “2.2.2 Removing and installing anti-roll bar, twin tyres”, page 111

2.2.1 Removing and installing anti-roll bar, single tyres

Special tools and workshop equipment required

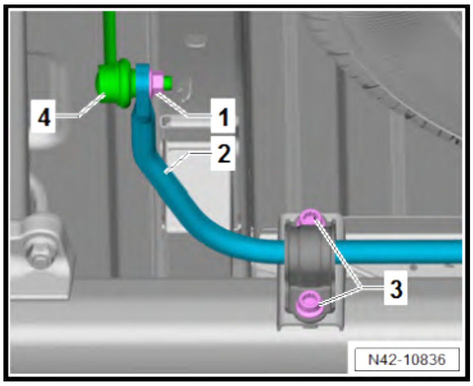
♦ Insert tool - T10061-

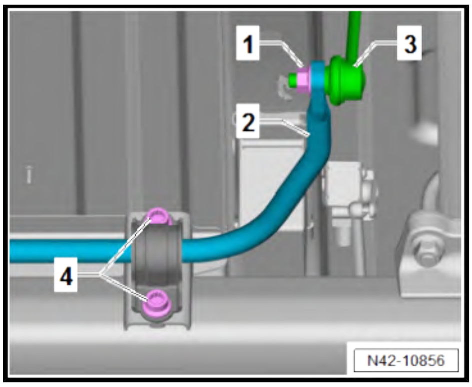


♦ Torque wrench - V.A.G 1332-

Removing

– Unscrew nut -1-.
– Pull coupling rod -4- out of anti-roll bar -2-.
– Unscrew bolts -3-.





– Unscrew nut -1-.
– Pull coupling rod -3- out of anti-roll bar -2-.
– Unscrew bolts -4-.
– Remove anti-roll bar -2-.

Installing

Install in the reverse order of removal, observing the following:

Note

- *Tighten bolted connections to the anti-roll bar only if the vehicle is standing on its wheels and is unladen or if the vehicle is raised and is in the unladen position (normal position)* *⇒ page 8* *.*
- *When tightening the nut -1-, counterhold on the internal Torx of the joint pin of the coupling rod.*

Specified torques

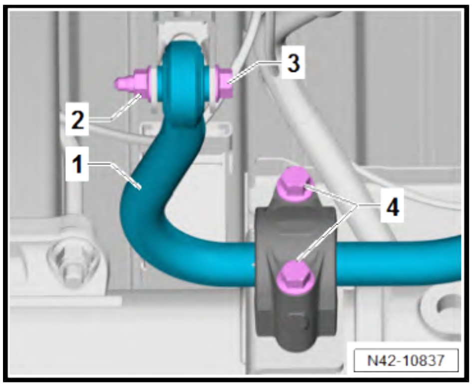
- ⇒ “2.1 Assembly overview - anti-roll bar”, page 108

2.2.2 Removing and installing anti-roll bar, twin tyres

Special tools and workshop equipment required

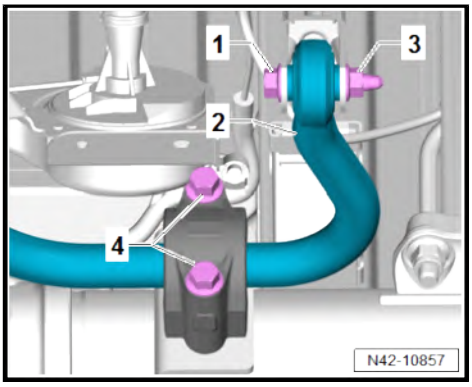
- Torque wrench - V.A.G 1332-

Removing



– Unscrew bolt -3-. When doing this, counterhold nut -2-.
– Unscrew bolts -4-.
– Secure anti-roll bar -1-.

– Unscrew bolt -1-. When doing this, counterhold nut -3-.
– Unscrew bolts -4-.
– Remove anti-roll bar -2-.



Installing

Install in the reverse order of removal, observing the following:

Note

Tighten bolted connections to the anti-roll bar only if the vehicle is standing on its wheels and is unladen or if the vehicle is raised and is in the unladen position (normal position) *⇒ page 8* .

Specified torques

- ⇒ “2.1 Assembly overview - anti-roll bar”, page 108

2.3 Renewing anti-roll bar bushes

⇒ “2.3.1 Renewing anti-roll bar bushes, single tyres”, page 112
⇒ “2.3.2 Renewing anti-roll bar bushes, twin tyres”, page 112

2.3.1 Renewing anti-roll bar bushes, single tyres

Anti-roll bar bushes cannot be replaced separately ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

In the event of a complaint the entire anti-roll bar must be replaced ⇒ page 109 .

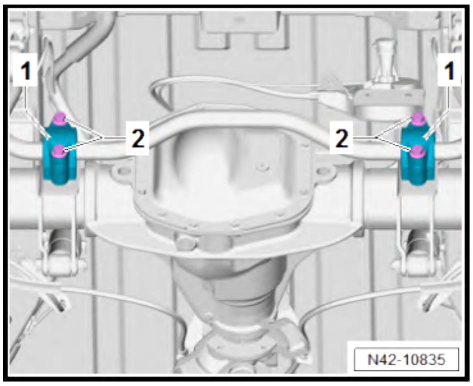
2.3.2 Renewing anti-roll bar bushes, twin tyres

Special tools and workshop equipment required

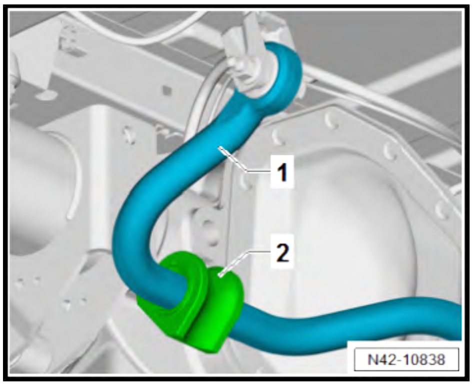
- Torque wrench - V.A.G 1332-

Removing

– Unscrew bolts -2-.
– Remove clamps -1-.



– Lower anti-roll bar -2-.
– Remove anti-roll bar bushes -1-.



Installing

Install in the reverse order of removal, observing the following:

Note

Tighten bolted connections to the anti-roll bar only if the vehicle is standing on its wheels and is unladen or if the vehicle is raised and is in the unladen position (normal position) ⇒ page 8 .

♦ ⇒ “2.1 Assembly overview - anti-roll bar”, page 108

2.4 Removing and installing coupling rod

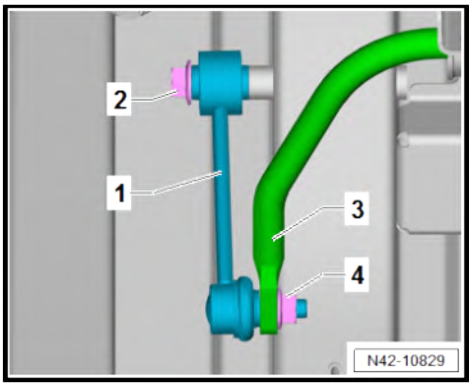
⇒ “2.4.1 Removing and installing coupling rod, single tyres”, page 113

⇒ “2.4.2 Removing and installing coupling rod, twin tyres”, page 114

2.4.1 Removing and installing coupling rod, single tyres

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-



Removing

– Unscrew nut -4-.
– Pull coupling rod -1- out of anti-roll bar -3-.
– Unscrew bolt -2-. Remove coupling rod -1-.

Installing

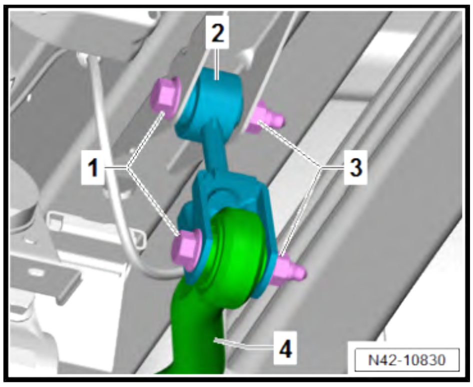
Install in the reverse order of removal, observing the following:

♦ ⇒ “2.1 Assembly overview - anti-roll bar”, page 108

2.4.2 Removing and installing coupling rod, twin tyres

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-



Removing

– Unscrew bolts -1-. When doing this, counterhold nuts -3-.
– Pull coupling rod -2- out of anti-roll bar -4-.
– Remove coupling rod -2-.

Installing

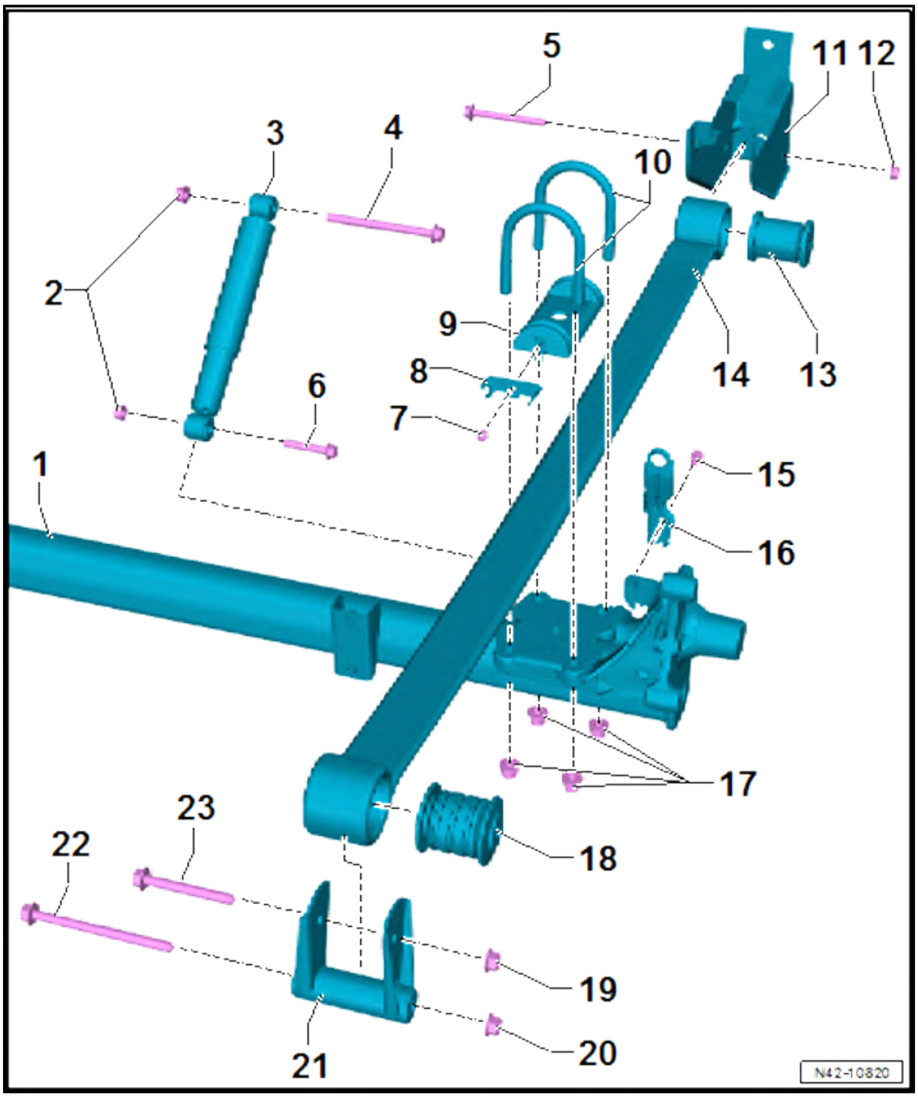
Install in the reverse order of removal, observing the following:

♦ ⇒ “2.1 Assembly overview - anti-roll bar”, page 108

3 Suspension strut, shock absorber, spring

3.1 Assembly overview - suspension strut, shock absorber, spring

3.1.1 Assembly overview - suspension strut/shock absorber, spring, single tyres, panel van and Kombi



1 - Rear axle
- ❑ Removing and installing ⇒ page 102

2 - Nuts
- ❑ Renew after removal
- ❑ Qty. 2

3 - Shock absorbers
- ❑ Removing and installing ⇒ page 120

4 - Bolt
- ❑ Renew after removal
- ❑ 130 Nm +180°

5 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +180°

6 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +90°

7 - Bolts
- ❑ Qty. 2
- ❑ 8 Nm

8 - Bracket

9 - Saddle plate
- ❑ Observe installation position

10 - U bolt
- ❑ Renew after removal
- ❑ Qty. 2

11 - Support

12 - Nut
- ❑ Renew after removal

13 - Bonded rubber bush
- ❑ Removing and installing ⇒ page 125

14 - Leaf spring
- ❑ Removing and installing ⇒ page 121

15 - Bolt
- ❑ 20 Nm

16 - Bracket

17 - Nuts
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 130 Nm +180°

18 - Bonded rubber bush
- ❑ Removing and installing ⇒ page 125

19 - Nut
- ❑ Renew after removal

20 - Nut

- ❑ Renew after removal

21 - Shackle

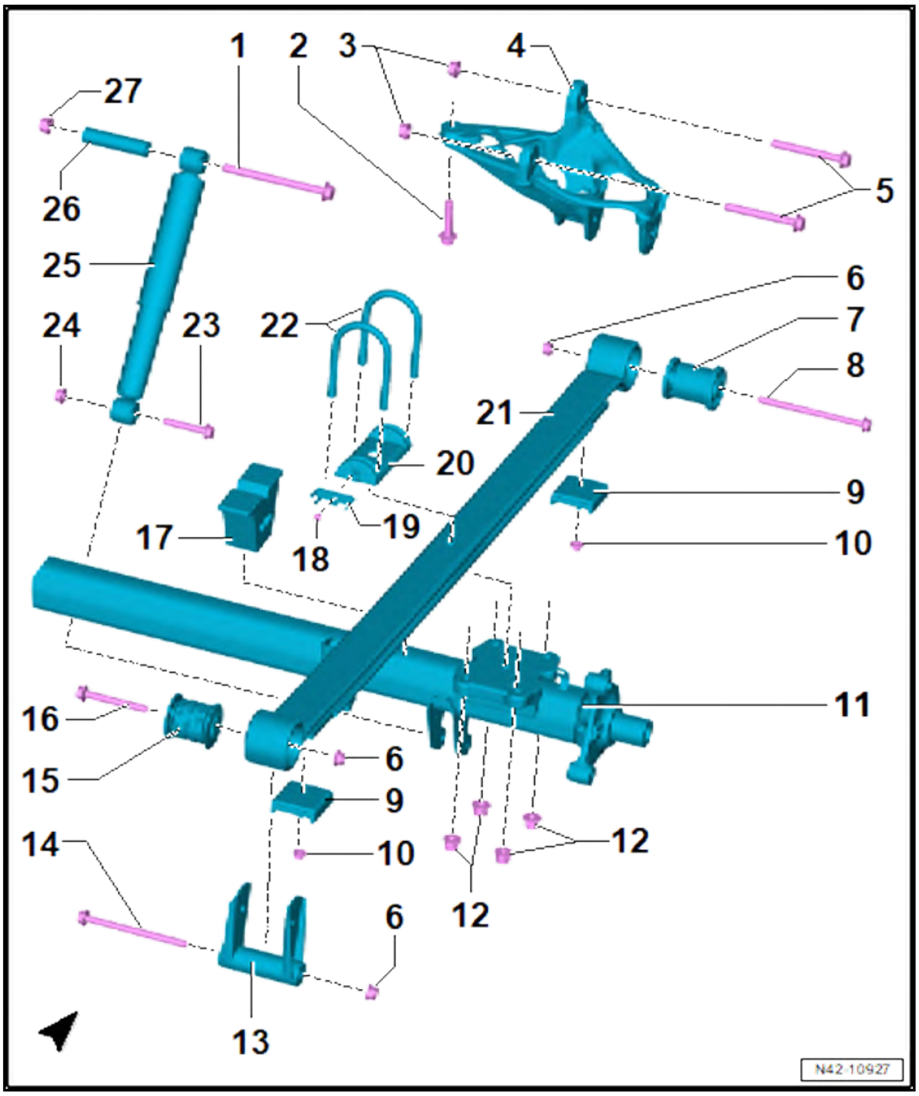
22 - Bolt

- ❑ Renew after removal
- ❑ 70 Nm +180°

23 - Bolt

- ❑ Renew after removal
- ❑ 70 Nm +180°

3.1.2 Assembly overview - suspension strut/shock absorber, spring, single tyres, drop-side



1 - Bolt

- ❑ Renew after removal
- ❑ 130 Nm +180°

2 - Bolt

- ❑ Renew after removal
- ❑ 130 Nm +90°

3 - Nuts

- ❑ Renew after removal
- ❑ Qty. 2

4 - Spring bracket

- ❑ For leaf spring
- ❑ Removing and installing ⇒ page 131

5 - Bolts

- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 130 Nm +180°

6 - Nut

- ❑ Renew after removal

7 - Bonded rubber bush

- ❑ Front
- ❑ Removing and installing ⇒ page 125

8 - Bolt

- ❑ Renew after removal
- ❑ 70 Nm +180°

9 - Support spring

- ❑ Removing and installing ⇒ page 132

10 - Nut

- ❑ Renew after removal
- ❑ 40 Nm

11 - Rear axle

- ❑ Removing and installing ⇒ page 102

12 - Nuts
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 130 Nm +180°

13 - Shackle

14 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +180°

15 - Bonded rubber bush
- ❑ Rear
- ❑ Removing and installing ⇒ page 125

16 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +180°

17 - Buffer stop
- ❑ Removing and installing ⇒ page 131

18 - Bolt
- ❑ Renew after removal
- ❑ 8 Nm

19 - Bracket

20 - Saddle plate
- ❑ Observe installation position

21 - Leaf spring
- ❑ Removing and installing ⇒ page 121

22 - Spring clip
- ❑ Renew after removal
- ❑ Qty. 2

23 - Bolt
- ❑ Renew after removal
- ❑ 70 Nm +90°

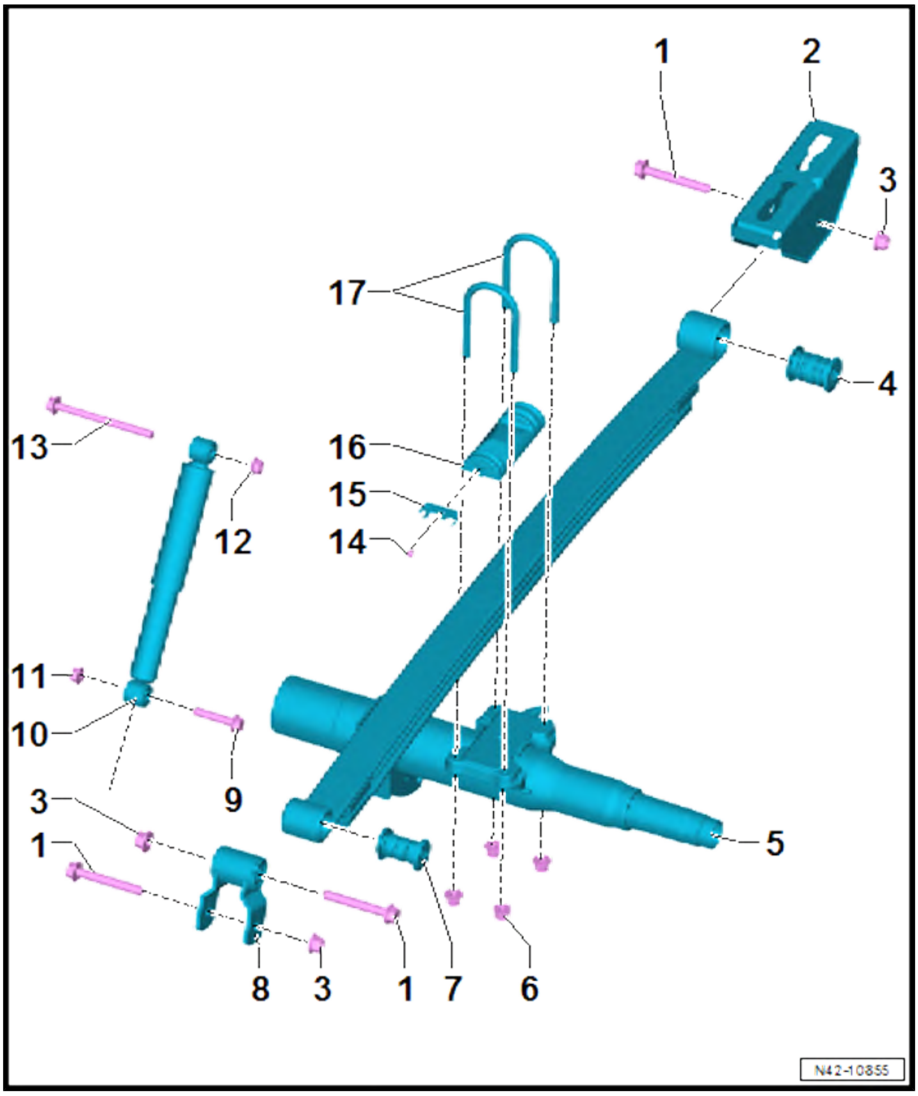
24 - Nut
- ❑ Renew after removal

25 - Shock absorbers
- ❑ Removing and installing ⇒ page 120

26 - Sleeve

27 - Nut
- ❑ Renew after removal

3.1.3 Assembly overview - suspension strut/shock absorber, spring, twin tyres



1 - Bolts
- ❑ Renew after removal
- ❑ Qty. 3
- ❑ 180 Nm +180°

2 - Support

3 - Nuts
- ❑ Renew after removal
- ❑ Qty. 3

4 - Bonded rubber bush
- ❑ Removing and installing ⇒ page 125

5 - Rear axle
- ❑ Removing and installing ⇒ page 102

6 - Nuts
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 130 Nm +180°

7 - Bonded rubber bush
- ❑ Removing and installing ⇒ page 125

8 - Shackle

9 - Bolt
- ❑ Renew after removal
- ❑ 130 Nm +90°

10 - Shock absorbers
- ❑ Removing and installing ⇒ page 120

11 - Nut
- ❑ Renew after removal

12 - Nut
- ❑ Renew after removal

13 - Bolt
- ❑ Renew after removal
- ❑ 130 Nm +180°

14 - Bolt
- ❑ 8 Nm

15 - Bracket

16 - Saddle plate
- ❑ Observe installation position

17 - U bolt
- ❑ Renew after removal
- ❑ Qty. 2

3.2 Removing and installing shock absorbers

Special tools and workshop equipment required

- Engine and gearbox jack - VAS 6931-

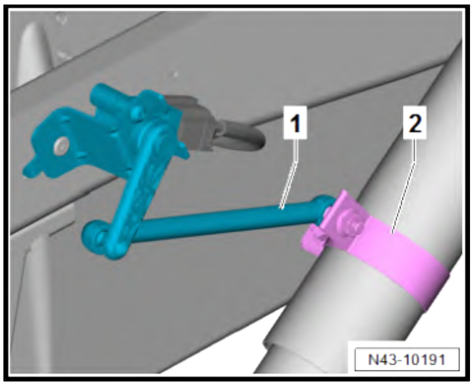
- Torque wrench - V.A.G 1332-

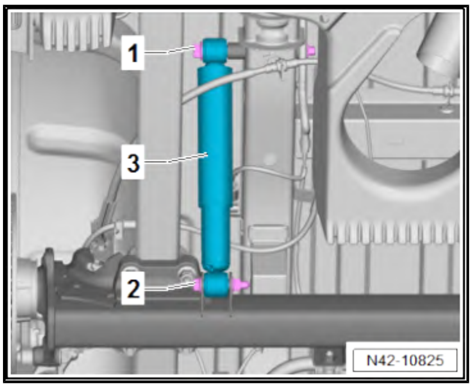
Removing

– Position engine and gearbox jack - VAS 6931- under rear axle and slightly raise.

Vehicles with rear right vehicle level sender - G77-

– Detach rear right vehicle level sender - G77- -1- from connecting tab -2-.





Continued for all vehicles

– Unscrew bolt -2-. When doing this, counterhold nut.
– Pull shock absorber -3- out of rear axle mounting.
– Unscrew bolt -1-, and, at the same time, remove shock absorber -3-.

Installing

Install in the reverse order of removal, observing the following:

Vehicles with rear right vehicle level sender - G77-

- Fish plate for rear right vehicle level sender - G77- must face towards rear.

Note

Note installation position of fish plate *⇒ page 146* *.*

Continued for all vehicles

Note

- *If the shock absorbers are renewed, the plastic stone guard strip on some shock absorbers must be transferred to the new units.*
- *The plastic stone guard strip must be renewed if damaged.*

Specified torques

- ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115
- ⇒ “1.2 Assembly overview - rear vehicle level senders”, page 146

3.3 Removing and installing leaf spring

Special tools and workshop equipment required

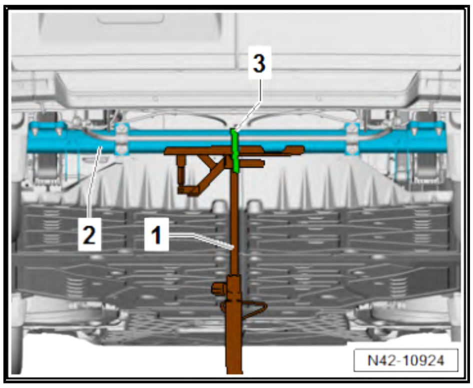
- Engine and gearbox jack - VAS 6931-

- Torque wrench - V.A.G 1332-

- Tensioning strap - T10038-

Removing

Vehicles with electric drive



– Position engine and gearbox jack - VAS 6931- -1- under rear axle -2-.

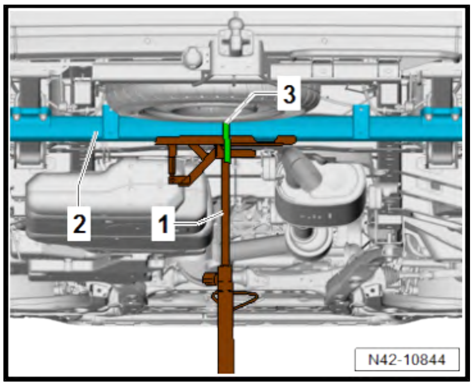
– Secure rear axle -2- using tensioning strap - T10038- -3-.

– Raise rear axle -2- slightly.

Continued for all vehicles

– Remove rear wheel ⇒ page 150 .

Continued for vehicles with combustion engine



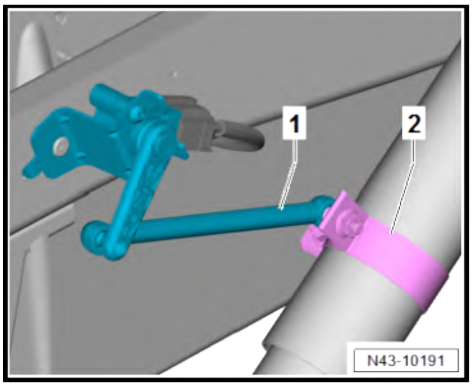
– Position engine and gearbox jack - VAS 6931- -1- under rear axle -2-.

– Secure rear axle -2- using tensioning strap - T10038- -3-.

– Raise rear axle -2- slightly.

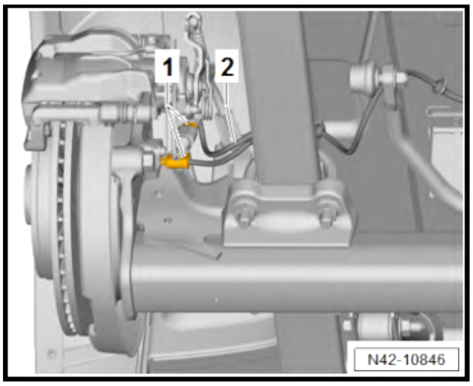
Vehicles with rear right vehicle level sender - G77-

– Detach rear right vehicle level sender - G77- -1- from connecting tab -2-.

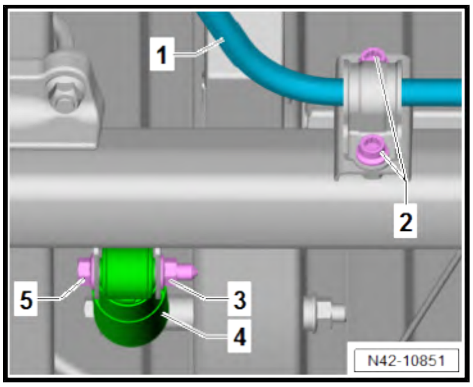


Continued for all vehicles

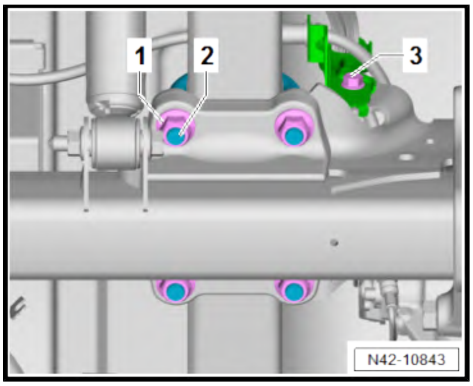
– Separate electrical connectors -1-.
– Unclip wiring harness -2- from retainer.



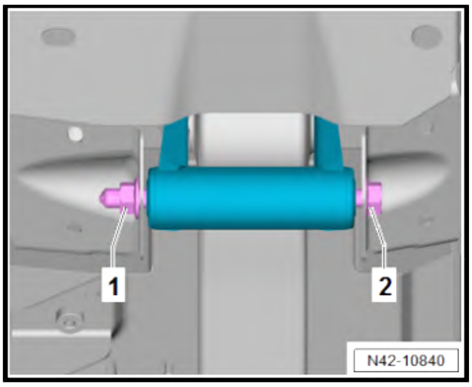
– Unscrew bolts -2- from anti-roll bar -1-.
– Unscrew bolt -5-. When doing this, counterhold nut -3-.
– Take shock absorber -4- out of mounting.



– Unscrew bolt -3-.
– Unscrew nuts -1-.
– Remove spring clip -2-.
– Remove spring plate.

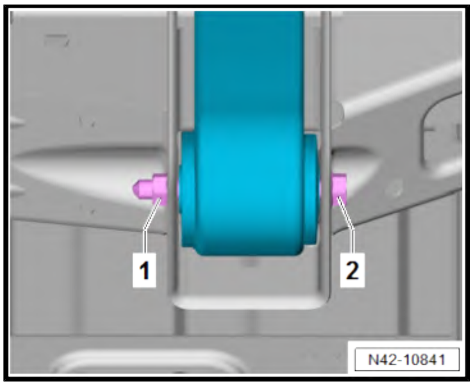


– Unscrew bolt -2-. When doing this, counterhold nut -1-.



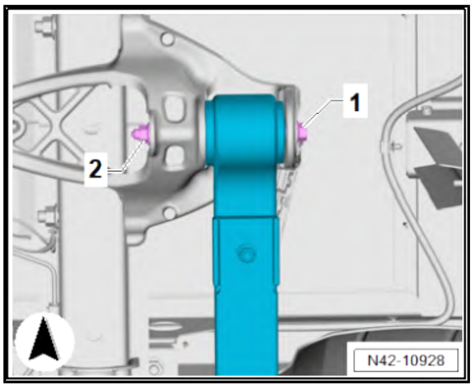
Panel van or Kombi vehicles

– Unscrew bolt -2-. When doing this, counterhold nut -1-.



Drop-side vehicles

– Unscrew bolt -1-. When doing this, counterhold nut -2-.



Continued for all vehicles

Note

Remove leaf spring with aid of second mechanic.

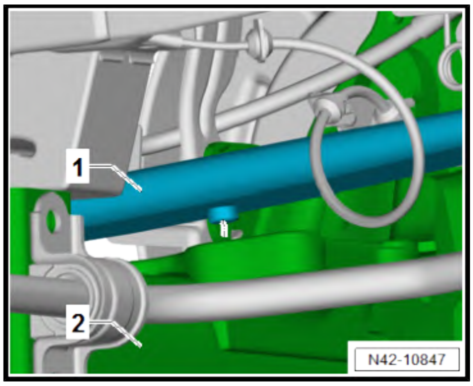
– Lower rear axle enough so that leaf spring can be removed.

Installing

Install in the reverse order of removal, observing the following:

– Install leaf spring -1- with aid of second mechanic.

– Insert leaf spring -1- in rear axle. Observe installation position in relation to rear axle -2- when doing this.



Note

Tighten bolts in unladen state ⇒ page 8 .

Specified torques

- ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115
- ⇒ “2.1 Assembly overview - anti-roll bar”, page 108

3.4 Removing and installing bonded rubber bushes for leaf spring

⇒ “3.4.1 Removing and installing bonded rubber bushes for leaf spring, front, single wheels”, page 125

⇒ “3.4.2 Removing and installing bonded rubber bushes for leaf spring, front, twin wheels”, page 127

⇒ “3.4.3 Removing and installing bonded rubber bushes for leaf spring, rear, single wheels”, page 128

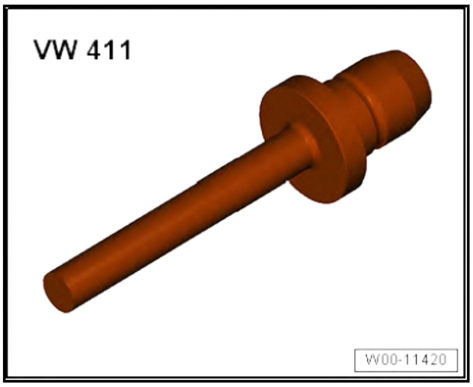
⇒ “3.4.4 Removing and installing bonded rubber bushes for leaf spring, rear, twin wheels”, page 130

3.4.1 Removing and installing bonded rubber bushes for leaf spring, front, single wheels

Special tools and workshop equipment required

- ♦ Thrust plate - VW402-

- ♦ Press tool - VW411-



- ♦ Assembly tool - T10230-

♦ Tube - 40-203/1-

Removing

• Leaf spring removed ⇒ page 121

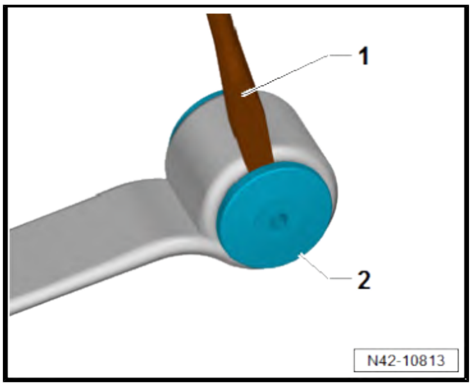
CAUTION

Risk of injury from swarf being flung into air.

Risk of irritation and injury to skin and eyes.

– Wear safety goggles.
– Wear protective gloves.

– Use a sharp chisel -1- to remove metal collar from bonded rubber bush -2-.



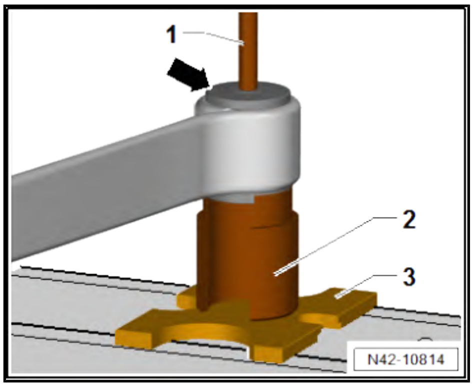
– Position tools as shown in illustration.

1 - Press tool - VW411-

2 - Tube - 40-203/1-

3 - Pressure plate - VW402-

– Press out bonded rubber bush -arrow-.



Installing

Install in the reverse order of removal, observing the following:

– Coat new bonded rubber bush with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

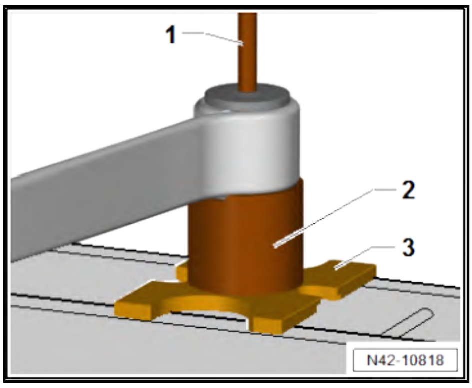
– Position tools as shown in illustration.

1 - Press tool - VW411-

2 - Assembly tool - T10230-

3 - Pressure plate - VW402-

– Pressing in bonded rubber bush.

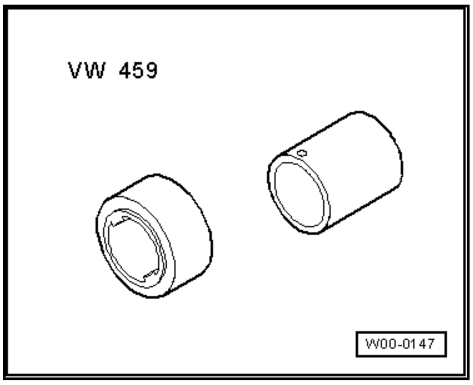


3.4.2 Removing and installing bonded rubber bushes for leaf spring, front, twin wheels

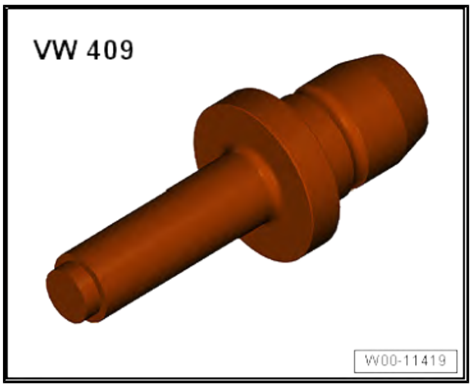
Special tools and workshop equipment required

- Thrust plate - VW 401-

- Removing and installing tool - VW459-

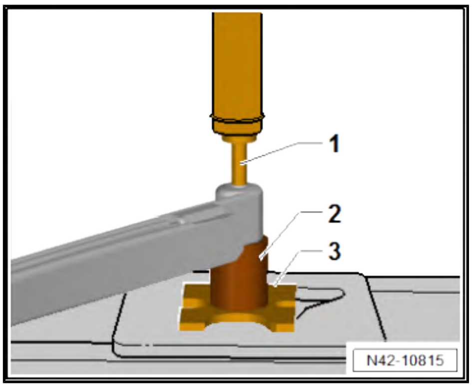


- Press tool - VW409-



Removing

- Leaf spring removed ⇒ page 121



– Position tools as shown in illustration.

1 - Press tool - VW409-

2 - Removing and installing tool - VW459-

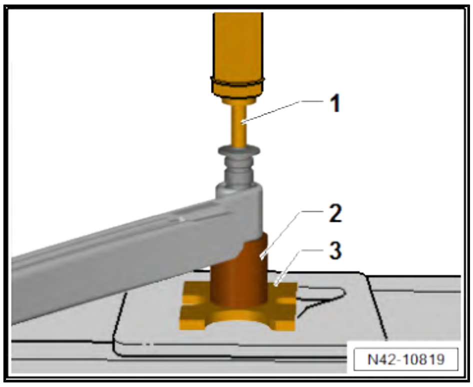
3 - Pressure plate - VW 401-

– Press out bonded rubber bush.

Installing

Install in the reverse order of removal, observing the following:

– Coat new bonded rubber bush with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.



– Position tools as shown in illustration.

1 - Press tool - VW409-

2 - Removing and installing tool - VW459-

3 - Pressure plate - VW 401-

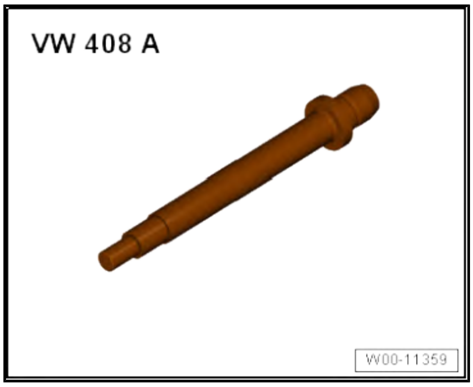
– Press in bonded rubber bush until shoulders project evenly on both sides.

3.4.3 Removing and installing bonded rubber bushes for leaf spring, rear, single wheels

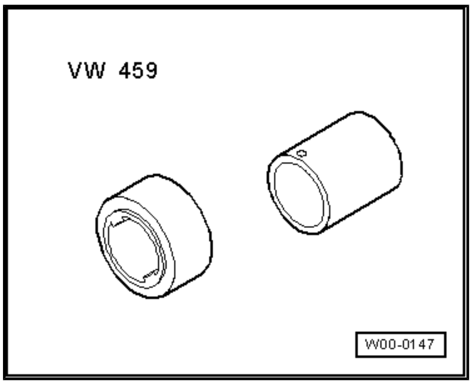
Special tools and workshop equipment required

♦ Thrust plate - VW402-

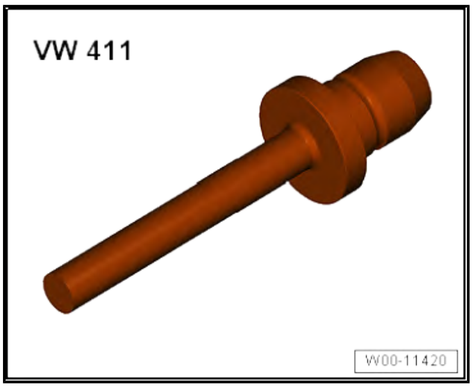
♦ Press tool - VW408A-



♦ Removing and installing tool - VW459-



♦ Press tool - VW411-

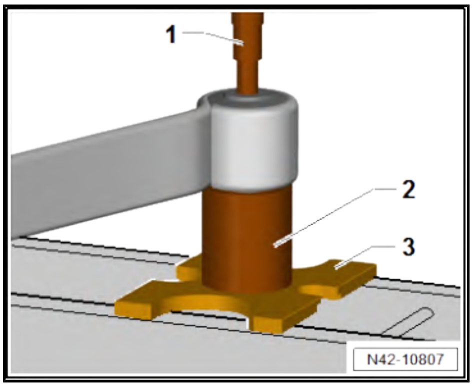


Removing

- Leaf spring removed ⇒ page 121

– Remove spring shackle.

– Position tools as shown in illustration.



1 - Press tool - VW408A-

2 - Removing and installing tool - VW459-

3 - Pressure plate - VW402-

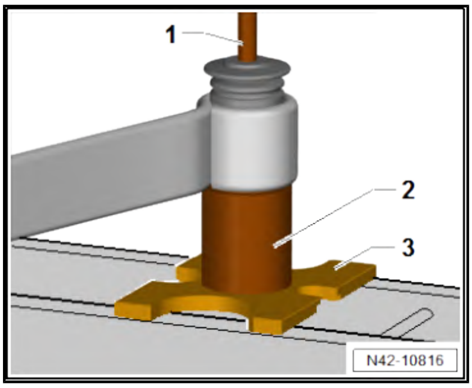
– Press out bonded rubber bush.

Installing

Install in the reverse order of removal, observing the following:

– Coat new bonded rubber bush with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Position tools as shown in illustration.



1 - Press tool - VW411-

2 - Removing and installing tool - VW459-

3 - Pressure plate - VW402-

– Pressing in bonded rubber bush.

Specified torques

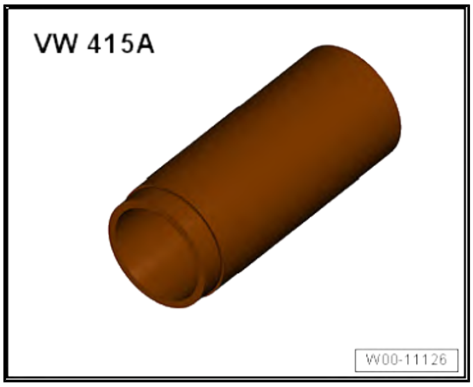
♦ ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115

3.4.4 Removing and installing bonded rubber bushes for leaf spring, rear, twin wheels

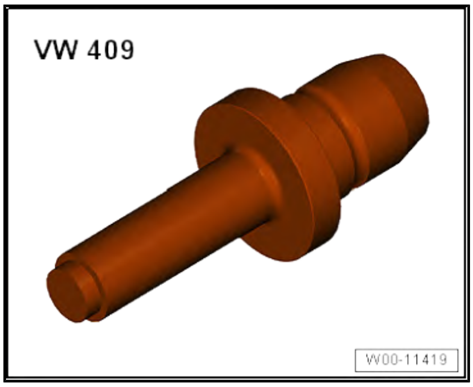
Special tools and workshop equipment required

- ♦ Thrust plate - VW 401-

- ♦ Tube - VW 415 A-



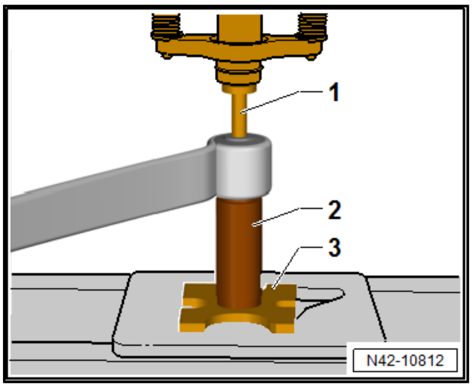
- ♦ Press tool - VW409-



Removing

- • Leaf spring removed ⇒ page 121
- – Remove spring shackle.

– Position tools as shown in illustration.



1 - Press tool - VW409-

2 - Tube - VW 415 A-

3 - Pressure plate - VW 401-

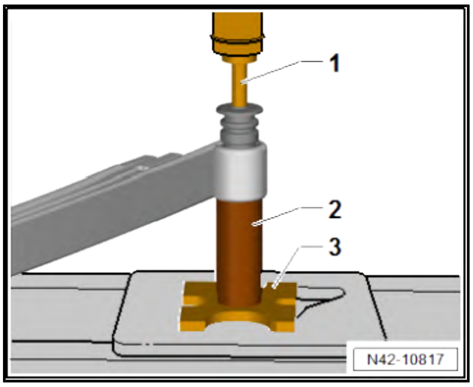
– Press out bonded rubber bush.

Installing

Install in the reverse order of removal, observing the following:

– Coat new bonded rubber bush with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Position tools as shown in illustration.



1 - Press tool - VW409-

2 - Tube - VW 415 A-

3 - Pressure plate - VW 401-

– Press in bonded rubber bush until shoulders project evenly on both sides.

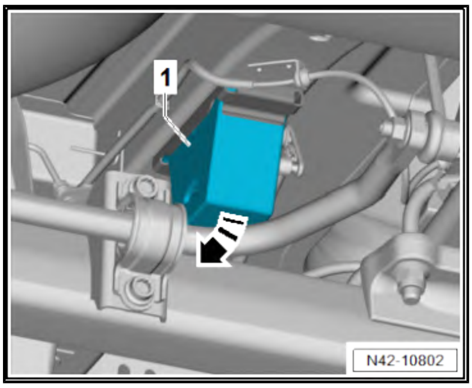
Specified torques

♦ ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115

3.5 Removing and installing axle beam stop buffer

Removing

– Use a suitable tool to lever stop buffer -1- in -direction of arrow- out of mounting.



Installing

Install in reverse order.

3.6 Removing and installing spring hanger for leaf spring

Special tools and workshop equipment required

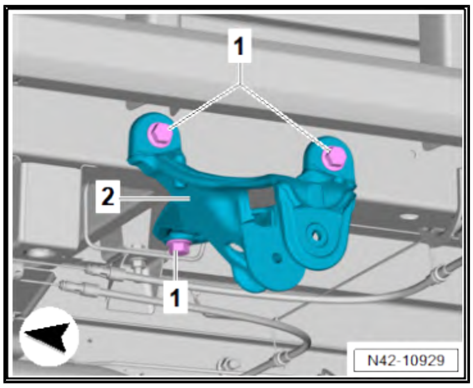
♦ Torque wrench - V.A.G 1332-

Removing

– Remove leaf spring ⇒ page 121 .

– Unscrew bolts -1-.

– Remove spring hanger for leaf spring -2-.



Installing

Install in the reverse order of removal, observing the following:

Specified torques

♦ ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115

3.7 Removing and installing support spring

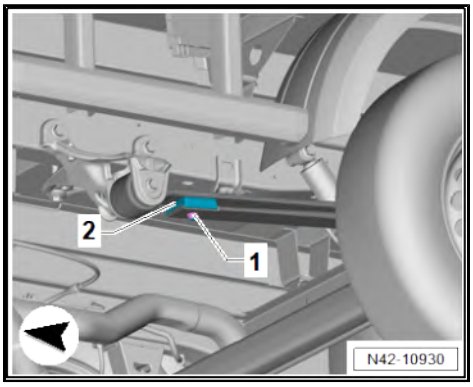
Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-

Removing

– Unscrew nut -1-.

– Remove support spring -2-.



Installing

Install in the reverse order of removal, observing the following:

Specified torques

♦ ⇒ “3.1 Assembly overview - suspension strut, shock absorber, spring”, page 115

4 Wheel bearing assembly, trailing arm

⇒ “4.1 Assembly overview - wheel bearing”, page 133

⇒ “4.2 Removing and installing wheel bearing housing”, page 136

⇒ “4.3 Removing and installing wheel bearing unit”, page 137

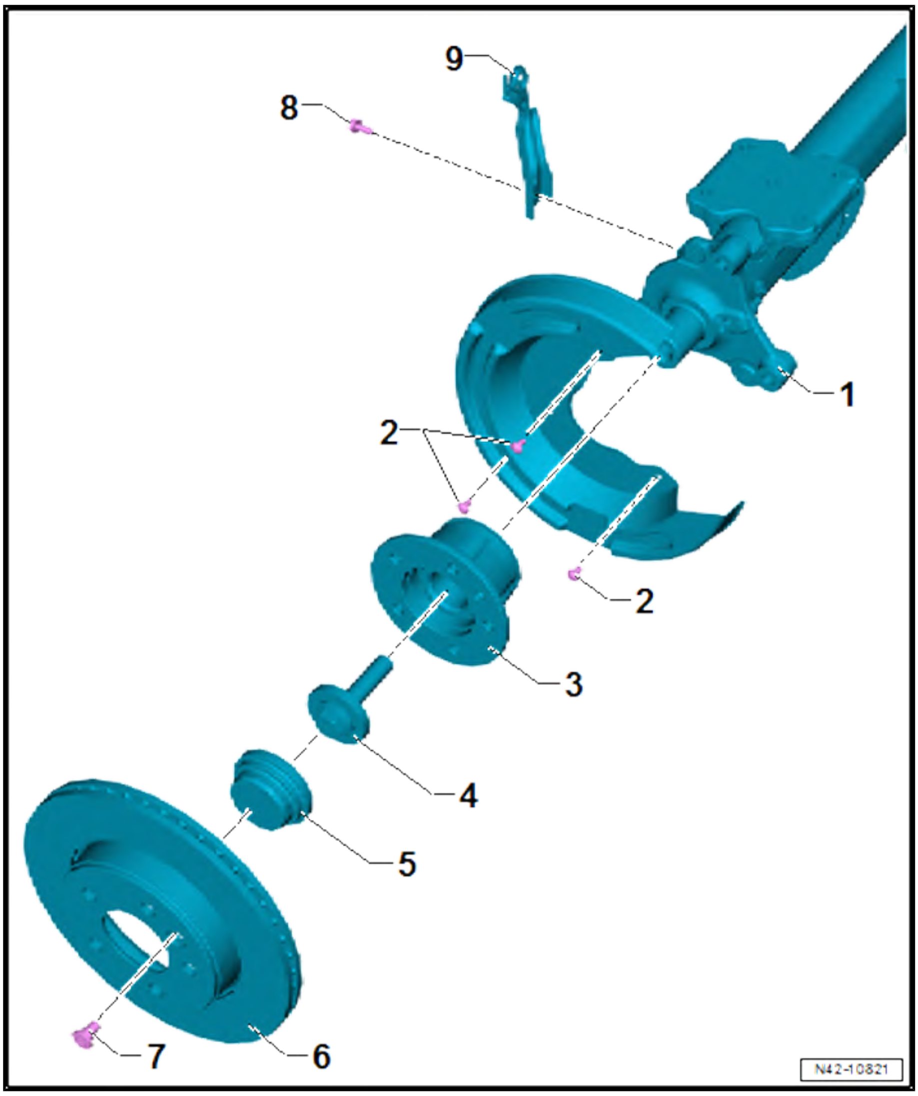
4.1 Assembly overview - wheel bearing

⇒ “4.1.1 Assembly overview - wheel bearing, single wheels”, page 133

⇒ “4.1.2 Assembly overview - wheel bearing, all-wheel/rear-wheel drive”, page 134

⇒ “4.1.3 Assembly overview - wheel bearing, twin wheels”, page 135

4.1.1 Assembly overview - wheel bearing, single wheels



1 - Rear axle
- ❑ Removing and installing ⇒ page 102

2 - Bolts
- ❑ Qty. 3
- ❑ 9.5 Nm

3 - Wheel bearing unit
- ❑ Removing and installing ⇒ page 137

4 - Bolt
- ❑ Renew after removal
- ❑ 250 Nm +180°

5 - Wheel centring seat

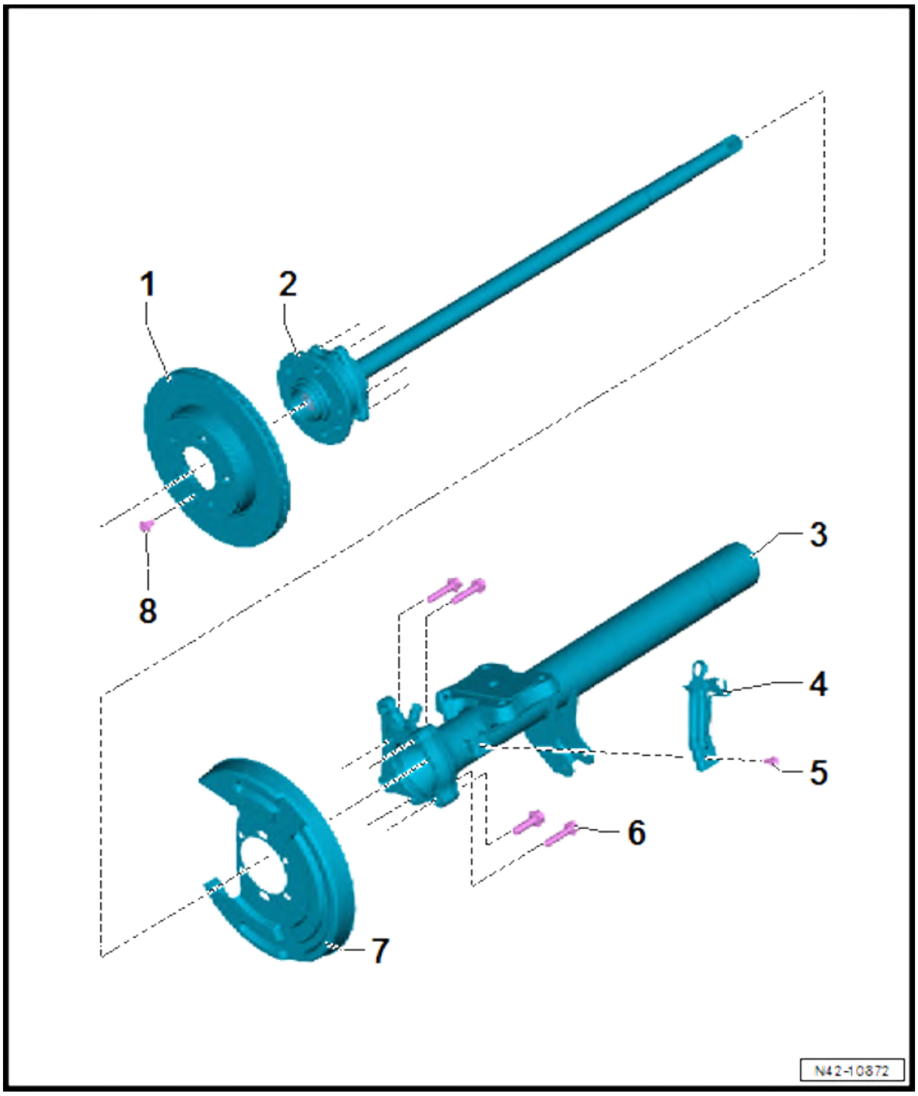
6 - Brake disc
- ❑ Removing and installing ⇒ Brake system; Rep. gr. 46 ; Rear brake; Removing and installing brake disc .

7 - Bolt
- ❑ 20 Nm

8 - Bolt
- ❑ 20 Nm

9 - Bracket

4.1.2 Assembly overview - wheel bearing, all-wheel/rear-wheel drive



1 - Brake disc

- ❑ Removing and installing ⇒ Brake system; Rep. gr. 46 ; Rear brake; Removing and installing brake disc .

2 - Stub shaft

- ❑ Together with wheel bearing unit
- ❑ Removing and installing ⇒ page 137

3 - Rear axle

- ❑ Removing and installing ⇒ page 102

4 - Bracket

5 - Bolt

- ❑ Specified torque ⇒ Item 8 (page 133)

6 - Bolts

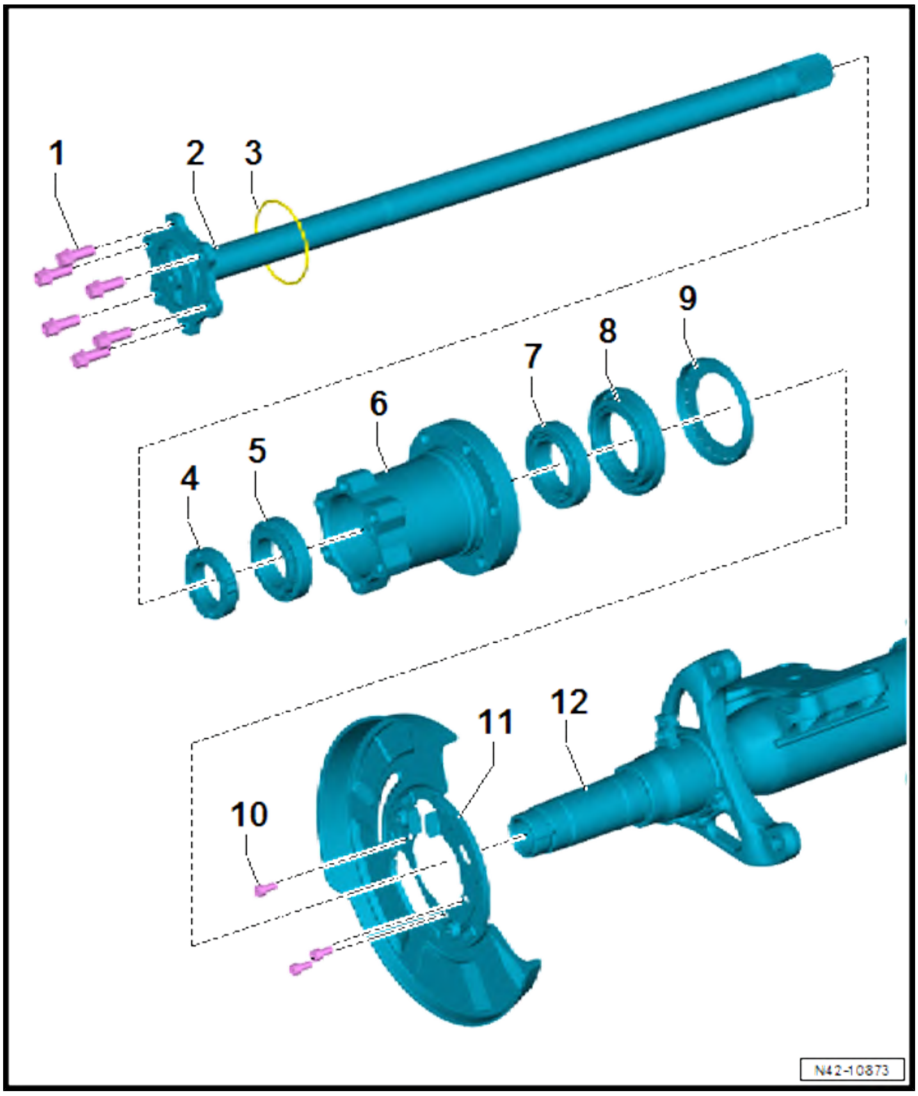
- ❑ Renew after removal
- ❑ Qty. 4
- ❑ 70 Nm +90°

7 - Cover plate

8 - Bolt

- ❑ Specified torque ⇒ Item 7 (page 133)

4.1.3 Assembly overview - wheel bearing, twin wheels



1 - Bolts
- ❑ Renew after removal
- ❑ Qty. 6
- ❑ 70 Nm +45°

2 - Stub shaft
- ❑ Different lengths on left and right
- ❑ Removing and installing ⇒ Propshaft and rear final drive; Rep. gr. 39 ; Final drive; Differential; Removing and installing stub shaft

3 - Seal
- ❑ Renew after removal

4 - Slotted nut
- ❑ Renew after removal
- ❑ Specified torque and tightening sequence ⇒ page 135

5 - Wheel bearing
- ❑ Outer
- ❑ With outer race
- ❑ Removing and installing ⇒ page 137

6 - Wheel bearing housing
- ❑ Removing and installing ⇒ page 136

7 - Wheel bearing
- ❑ Inner
- ❑ With outer race
- ❑ Removing and installing ⇒ page 137

8 - Seal
- ❑ Renew after removal

9 - Sender wheel of speed sensor
- ❑ Renew after removal
- ❑ Removing and installing ⇒ page 137

10 - Bolts
- ❑ Qty. 3
- ❑ 15 Nm

11 - Heat shield

12 - Rear axle
- ❑ Removing and installing ⇒ page 102

Grooved nut - specified torque and tightening sequence

Stage	Specified torque/turning further angle
1.	100 Nm

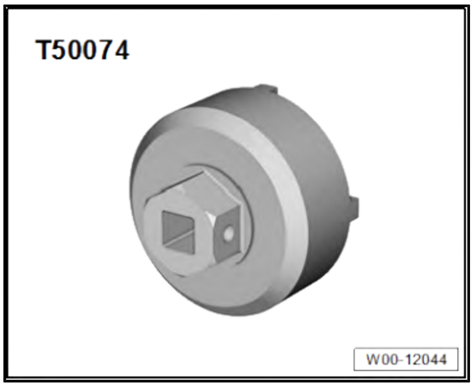
Stage	Specified torque/turning further angle
2.	Loosen 180°
3.	27 Nm

4.2 Removing and installing wheel bearing housing

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1332-

- ♦ Special wrench - T50074-



- ♦ Drip tray for workshop hoist - VAS 6208-

Removing

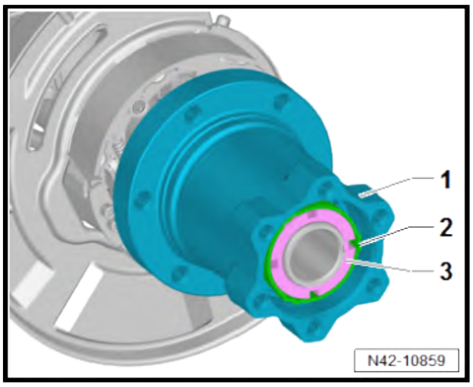
– Remove stub shaft ⇒ Propshaft and final drive; Rep. gr. 39 ; Differential; Removing and installing stub shaft, twin tyres .

– Remove rear brake disc ⇒ Brake system; Rep. gr. 46 ; Rear brake; Removing and installing brake disc .

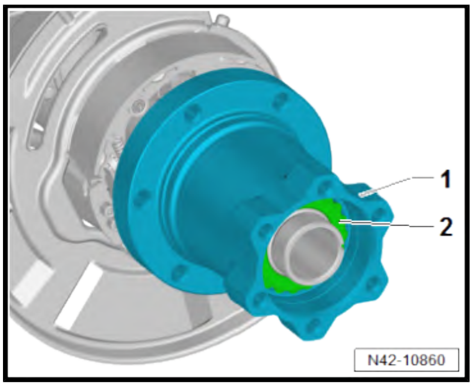
Note

Slotted nut has a left-hand thread on the left side and a right-hand thread on the right side.

– Unscrew slotted nut -3- using special wrench - T50074- .
– Remove slotted nut -3- with securing clip -2- from wheel bearing housing -1-.



– Remove tapered roller bearing -2- from wheel bearing housing -1-.
– Pull off wheel bearing housing -1- from axle tube.



Installing

Install in the reverse order of removal, observing the following:

Note

Continually turn wheel bearing housing while tightening to torque to prevent tension.

– Check axle oil level ⇒ Propshaft and rear final drive; Rep. gr. 39 ; Axle oil; Checking axle oil level .

Specified torques

♦ ⇒ “4.1 Assembly overview - wheel bearing”, page 133
♦ Rear brake; Assembly overview – rear brake ⇒ Brake system; Rep. gr. 46 ; Rear brake; Assembly overview – rear brake

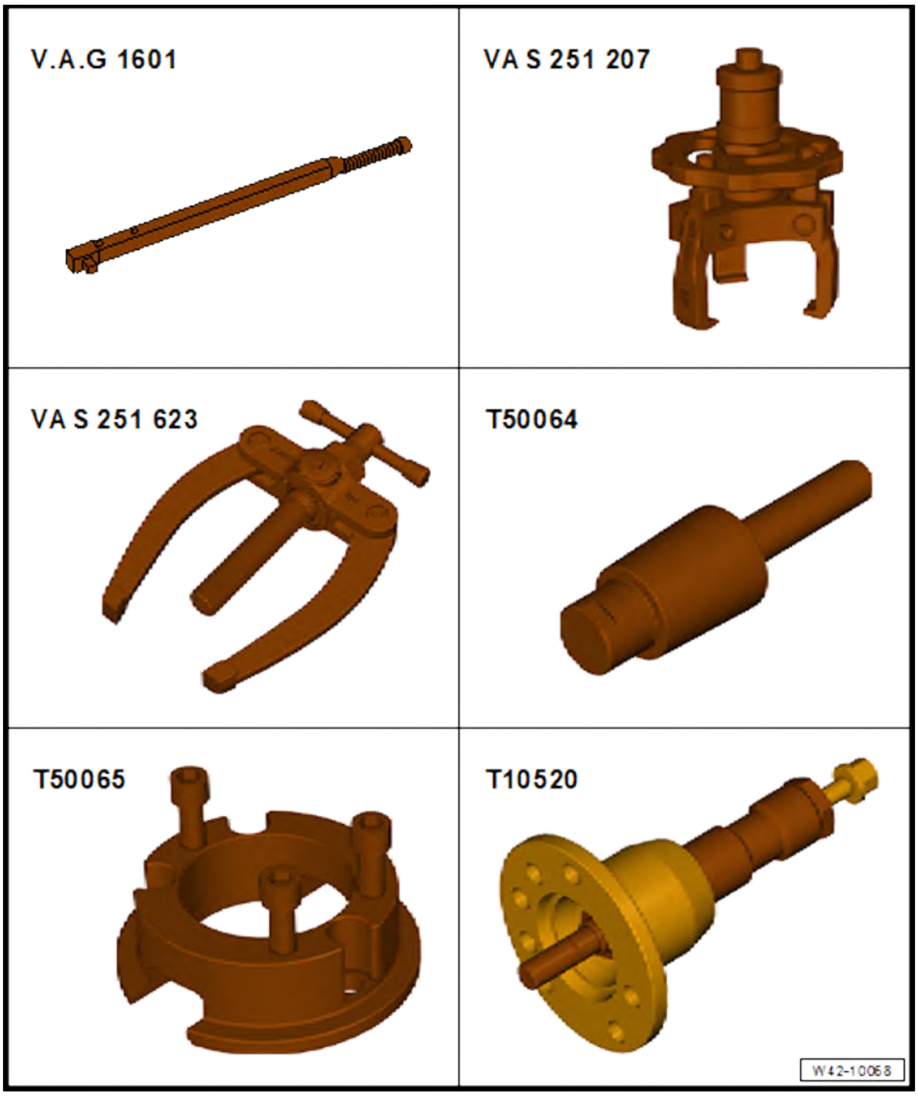
4.3 Removing and installing wheel bearing unit

⇒ “4.3.1 Removing and installing wheel bearing unit, single wheels”, page 137

⇒ “4.3.2 Removing and installing wheel bearing unit, rear-wheel drive/four-wheel drive”, page 140

⇒ “4.3.3 Removing and installing wheel bearing unit, twin wheels”, page 141

4.3.1 Removing and installing wheel bearing unit, single wheels



Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1601-
- ♦ Three-arm puller - VAS 251 207-
- ♦ Counter support KUKKO 22-2 - VAS 251 623-
- ♦ Centring mandrel - T50064-
- ♦ Assembly tool - T50065-
- ♦ Ejector - T10520-

Removing

Vehicles with electric drive

– Observe safety precautions when working on the high-voltage system ⇒ page 1 .

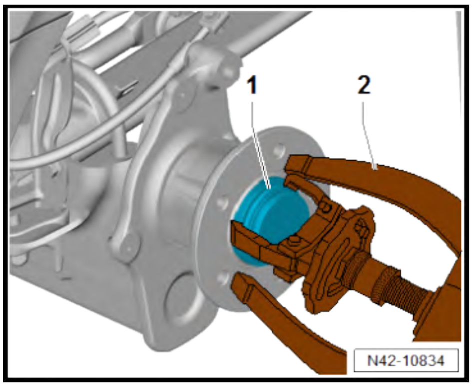
– Observe safety precautions when working in the vicinity of high-voltage components ⇒ page 2 .

– Observe classification of dangers of the high-voltage system ⇒ page 15 .

Continued for all vehicles

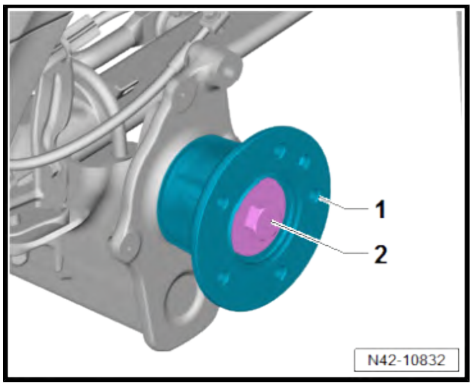
– Remove rear brake disc ⇒ Brake system; Rep. gr. 46 ; Rear brake; Removing and installing brake disc .

– Pull off wheel centring seat -1- with counter support KUKKO 22-2 - VAS 251 623- and three-arm puller - VAS 251 207- -2-.



– Unscrew bolt -2-.

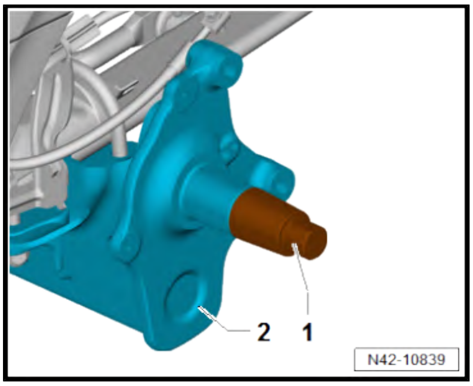
– Remove wheel bearing unit -1-.



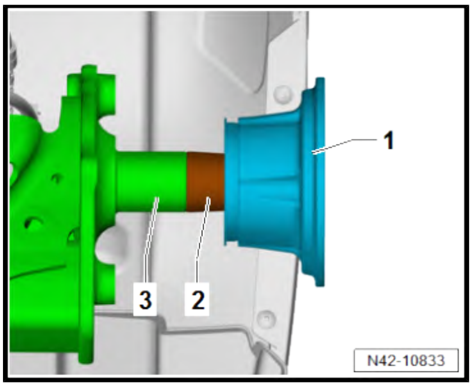
Installing

Install in the reverse order of removal, observing the following:

– Tighten centring mandrel - T50064- -1- in stub axle -2- by hand.



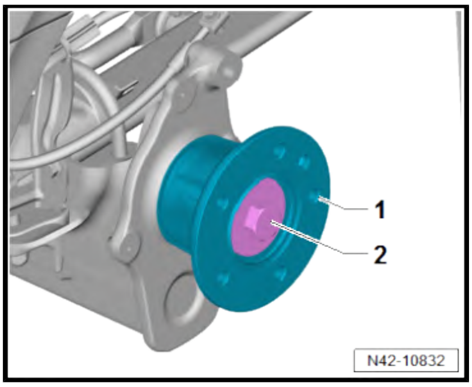
– Carefully push wheel bearing unit -1- over centring mandrel - T50064- -2- onto stub axle -3-.



Note

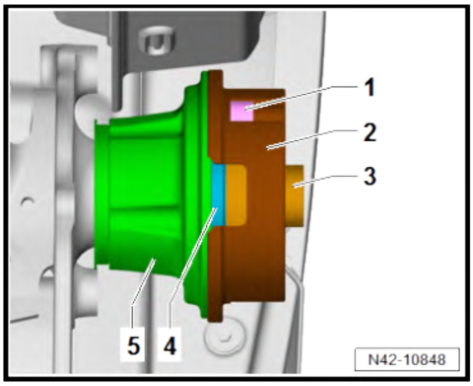
Continually turn wheel bearing unit -1- while tightening to torque to prevent tension.

– Tighten bolt -2-.



– Tighten assembly tool - T50065- -2- on wheel bearing unit -5- hand-tight with bolts -1-.

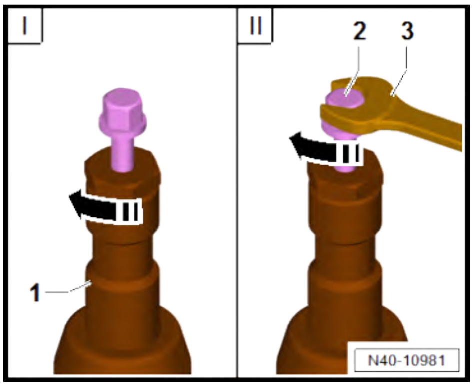
– Insert assembly tool - T50065- -3- with wheel centring seat -4- in assembly tool - T50065- -2-.



– It is essential to follow the specified sequence.

I - Tighten knurled nut in direction of -arrow- hand-tight.

II - Using spanner -3-, turn only bolt -2-, not the press tool itself.

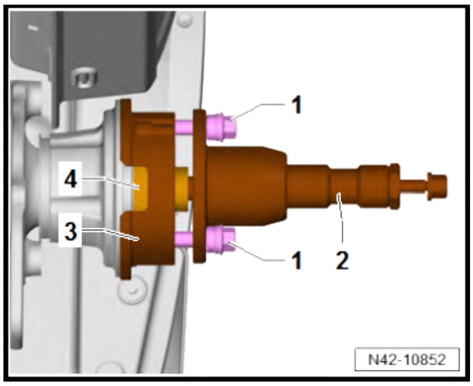


Note

At the end of the procedure or for pressing out drive shaft further, the spindle must be moved to its original position in order to apply the hydraulic force.

– Screw press tool - T10520- -2- onto assembly tool - T50065- -3- hand-tight using wheel bolts -1-.

– Press in wheel centring seat -4- with press tool - T10520- -2-.



Specified torques

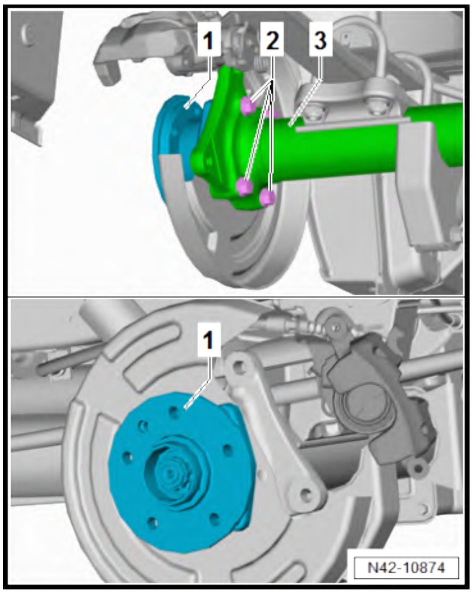
- ⇒ “4.1 Assembly overview - wheel bearing”, page 133
- Rear brake; Assembly overview – rear brake ⇒ Brake system; Rep. gr. 46 ; Rear brake; Assembly overview – rear brake

4.3.2 Removing and installing wheel bearing unit, rear-wheel drive/four-wheel drive

Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Drip tray for workshop hoist - VAS 6208-



Removing

– Remove brake disc ⇒ Brake systems; Rep. gr. 46 ; Rear brake; Removing and installing brake disc .

– Remove speed sensor ⇒ Brake systems; Rep. gr. 45 ; Sensors; Removing and installing speed sensor on rear axle G44/ G46 .

– Unscrew bolts -2-.

– Place drip tray for workshop hoist - VAS 6208- underneath.

– Pull stub shaft together with wheel bearing unit -1- out of axle beam -3-.

Installing

Install in the reverse order of removal, observing the following:

– Check axle oil level ⇒ Propshaft and rear final drive; Rep. gr. 39 ; Axle oil; Checking axle oil level .

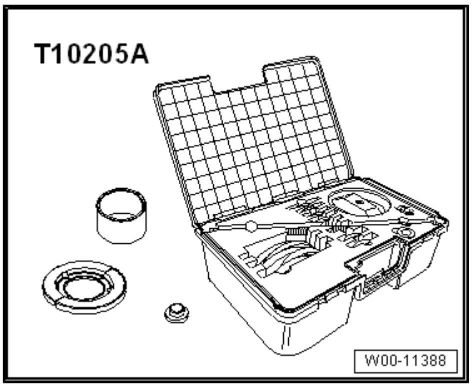
Specified torques

- ⇒ “4.1 Assembly overview - wheel bearing”, page 133
- Sensors; Assembly overview - speed sensor on rear axle ⇒ Brake systems; Rep. gr. 45 ; Sensors; Assembly overview - speed sensor on rear axle
- Rear brake; Assembly overview – rear brake ⇒ Brake system; Rep. gr. 46 ; Rear brake; Assembly overview – rear brake

4.3.3 Removing and installing wheel bearing unit, twin wheels

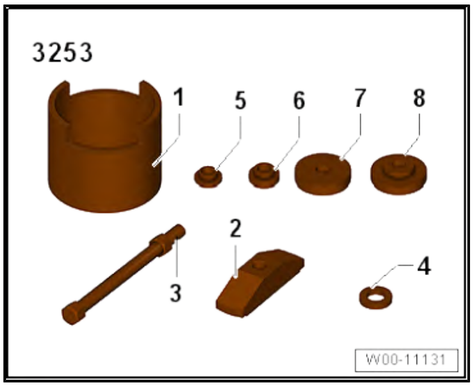
Special tools and workshop equipment required

♦ Assembly tool - T10205A-

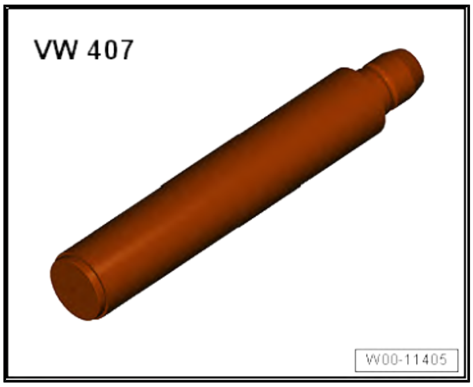


♦ Thrust plate - VW401-

♦ Assembly tool - 3253/1-



♦ Press tool - VW407-



Removing

• Wheel bearing housing removed ⇒ page 136

– Clamp wheel bearing housing in vice.

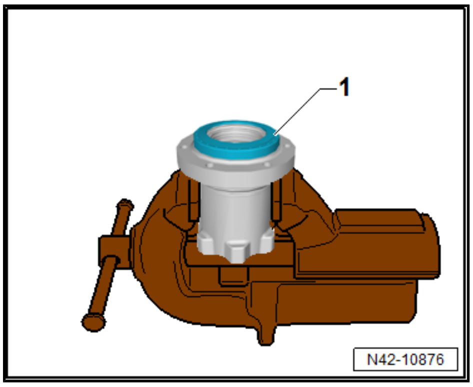
CAUTION

Risk of injury from swarf being flung into air.

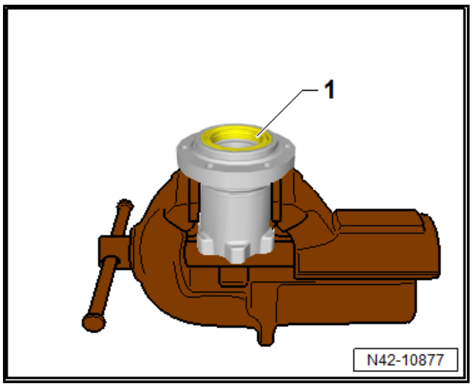
Risk of irritation and injury to skin and eyes.

– Wear safety goggles.
– Wear protective gloves.

– Lever of sender wheel of speed sensor -1- using a suitable tool.



– Lever out seal -1- using a suitable tool.

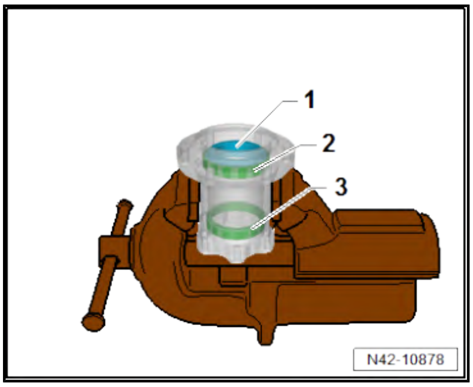


– Remove inner wheel bearing -1-.

– Drive out outer race on inside -2- and outside -3- using a drift.

Installing

Install in the reverse order of removal, observing the following:

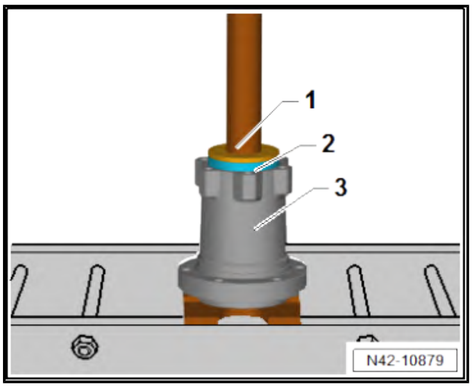


– Position tools as shown in illustration.

1 - Assembly tool - T10205A/9-

3 - Wheel bearing housing

– Press in outer race -2- for outer wheel bearing.



– Position tools as shown in illustration.

1 - Press tool - VW407-

2 - Assembly tool - T10205A/7-

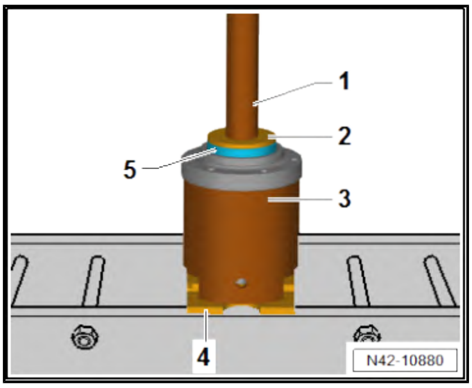
3 - Assembly tool - 3253/1-

4 - Pressure plate - VW 401-

– Press in outer race -5- for inner wheel bearing.

– Lubricate inner wheel bearing with lubricating paste ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

– Place inner wheel bearing in outer race.



– Position tools as shown in illustration.

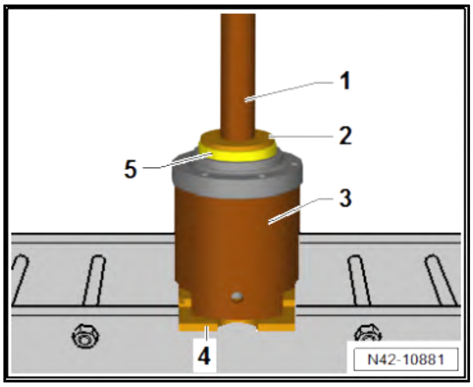
1 - Press tool - VW407-

2 - Assembly tool - T10205A/4-

3 - Assembly tool - 3253/1-

4 - Pressure plate - VW 401-

– Press in seal -5-.



– Position tools as shown in illustration.

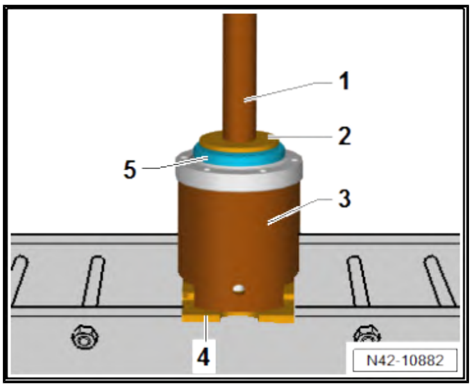
1 - Press tool - VW407-

2 - Assembly tool - T10205A/4-

3 - Assembly tool - 3253/1-

4 - Pressure plate - VW 401-

– Press on sender wheel for speed sensor -5-.



43 – Self-levelling suspension

1 Vehicle level sender

⇒ “1.1 Assembly overview - front vehicle level senders”, page 145

⇒ “1.2 Assembly overview - rear vehicle level senders”, page 146

⇒ “1.3 Removing and installing front right vehicle level sender G289 ”, page 147

⇒ “1.4 Removing and installing rear right vehicle level sender G77 ”, page 147

1.1 Assembly overview - front vehicle level senders

1 - Subframe
- ❑ Removing and installing ⇒ page 23

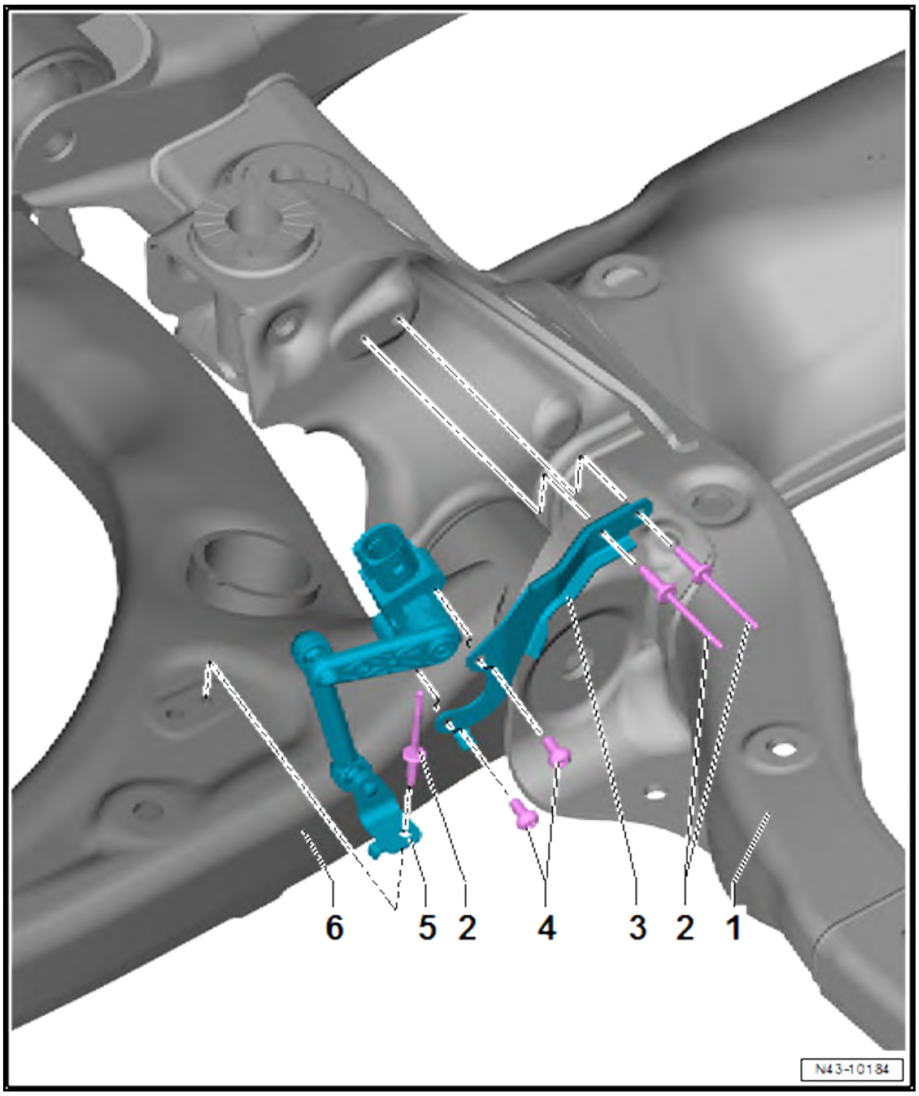
2 - Pop rivet
- ❑ Qty. 3

3 - Retainer

4 - Bolts
- ❑ Qty. 2
- ❑ 4 Nm

5 - Front right vehicle level sender - G289-
- ❑ Removing and installing ⇒ page 147

6 - Lower suspension link
- ❑ Removing and installing ⇒ page 65



1.2 Assembly overview - rear vehicle level senders

1 - Rear right vehicle level sender - G77-

❑ Removing and installing ⇒ page 147

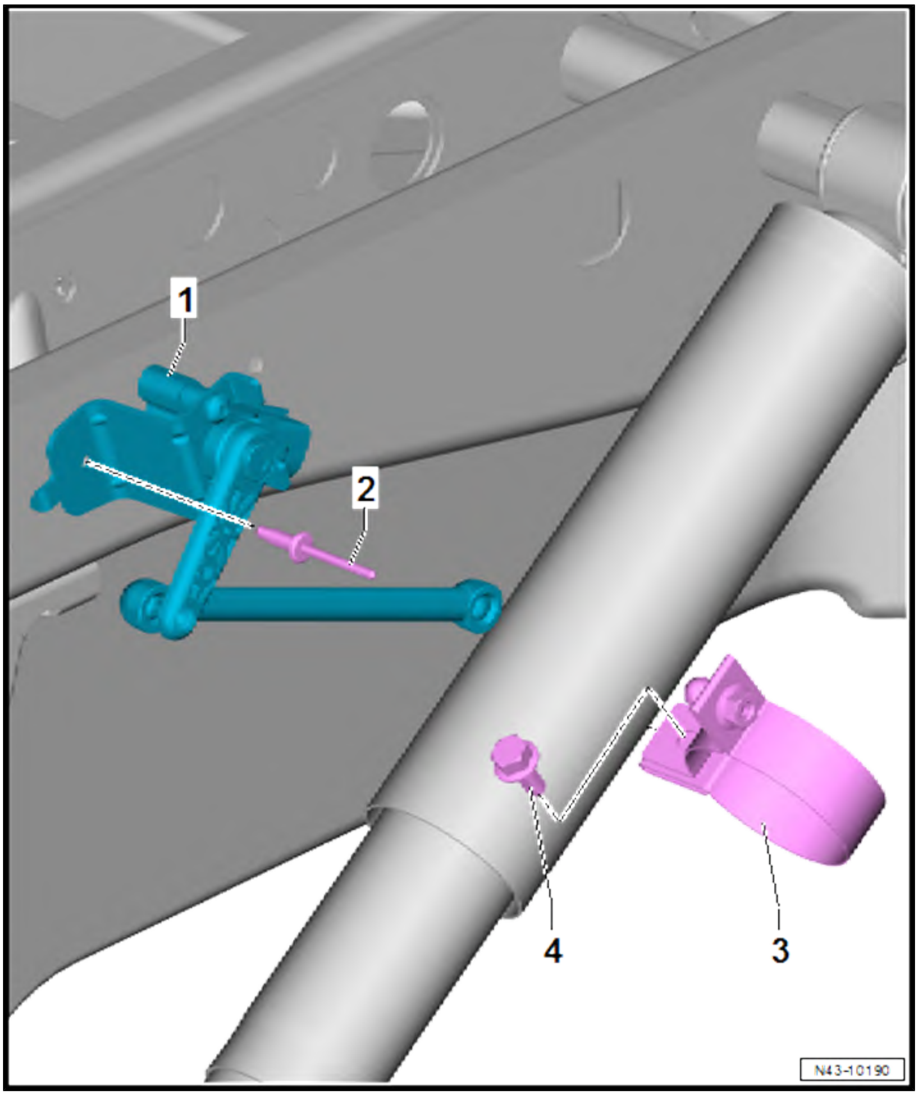
2 - Pop rivet

3 - Connecting tab

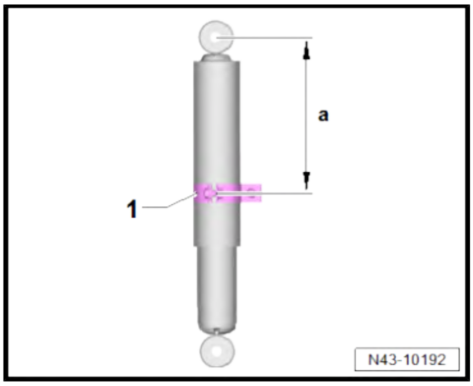
❑ Note installation position ⇒ page 146

4 - Bolt

❑ 8 Nm



Installation position of connecting tab -1-



PR number	Height -a- in mm
K4A	183
K4B, K4F, K4G, 1X1, 1X4	205

Explanation of PR numbers ⇒ page 187 .

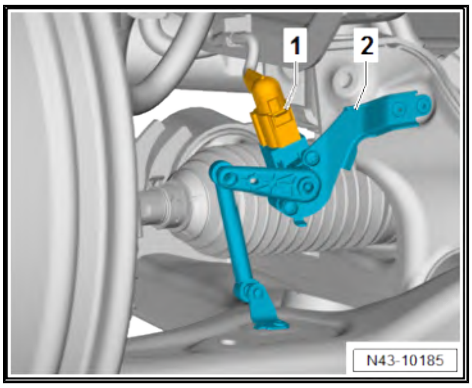
1.3 Removing and installing front right vehicle level sender - G289-

Special tools and workshop equipment required

- ♦ Hand drill
- ♦ Pop rivet pliers

Removing

– Disconnect connector -1- from front right vehicle level sender - G289- -2-.

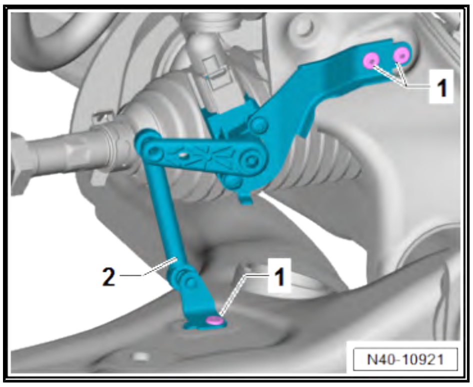


CAUTION

Risk of injury from swarf being flung into air.
Risk of irritation and injury to skin and eyes.
– Wear safety goggles.
– Wear protective gloves.

– Drill out pop rivets -1-.

– Remove front right vehicle level sender - G289- -2-.



Installing

Install in the reverse order of removal, observing the following:

– Secure front right vehicle level sender - G289- -2- with special pop rivets ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

1.4 Removing and installing rear right vehicle level sender - G77-

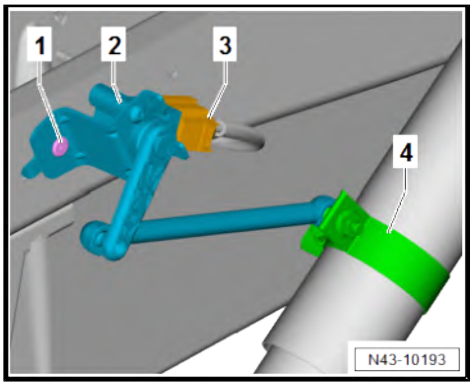
Special tools and workshop equipment required

- ♦ Hand drill
- ♦ Pop rivet pliers

Removing

– Separate electrical connector -3-.

– Detach rear right vehicle level sender - G77- -2- from connecting tab -4-.



CAUTION

Risk of injury from swarf being flung into air.

Risk of irritation and injury to skin and eyes.

– Wear safety goggles.

– Wear protective gloves.

– Drill out pop rivet -1-.

– Remove rear right vehicle level sender - G77- -2-.

Installing

Install in the reverse order of removal, observing the following:

– Secure rear right vehicle level sender - G77- -2- with special pop rivet ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE.

Specified torques

♦ ⇒ “1.2 Assembly overview - rear vehicle level senders”, page 146

44 – Wheels, tyres, vehicle geometry

1 Wheels, tyres

1.1 Removing tyre from wheel

Tyres sealed with tyre sealant

Note

- *Avoid eye and skin contact with tyre sealant.*
- *It is damaging to health and can cause eye irritation and allergies.*
- *During this work, wear protective gloves and safety goggles.*

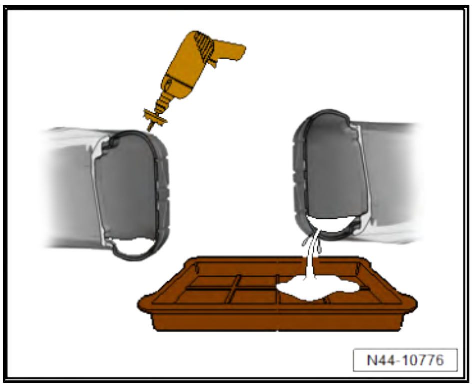
– Place wheel on an even surface.

– Unscrew and remove valve insert of tyre valve.

– Use a suitable drill or mill to carefully drill a hole in shoulder area of tyre.

– Hold wheel over a suitable container and drain sealant.

– Remove tyre from wheel rim.



Note

For tyres filled or sealed with tyre sealant, the entire sensor for tyre pressure monitoring system (TPMS) has to be replaced.

– Clean wheel rim with, for example, a moist cloth.

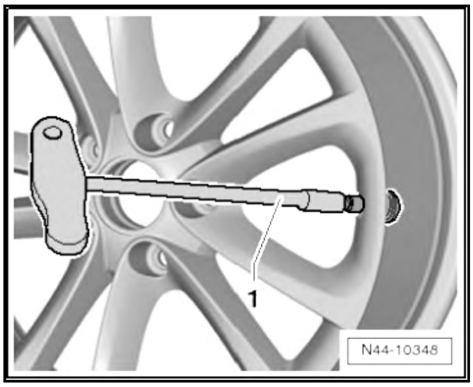
1.2 Fitting tyres

• Ensure that the wheel rim is clean.

Vehicles with tyre pressure control

– Install tyre pressure sensor ⇒ page 160 .

Continued for all vehicles



– Insert new tyre valve with valve fitting tool - VAS 6459- -1-.
– Unscrew valve insert.
– Inflate tyre to approx. 3…4 bar. The bead of the tyre must slip audibly over the hump of the rim.
– Screw in valve insert.
– Check tyre pressure to make sure prescribed pressure has been reached.
– Balance wheel.

1.3 Wheel change

⇒ “1.3.1 Wheel change, protecting wheel centring seat against corrosion”, page 150

⇒ “1.3.2 Wheel change, installation requirements”, page 151

⇒ “1.3.3 Wheel change, installation instructions for wheel change”, page 152

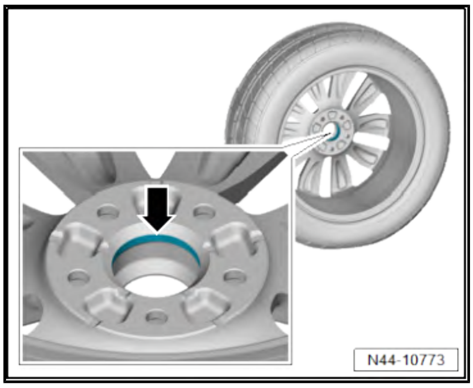
⇒ “1.3.4 Changing wheel, mounting wheel on vehicles with single wheels”, page 153

⇒ “1.3.5 Changing wheel, mounting wheel on vehicles with twin wheels”, page 154

1.3.1 Wheel change, protecting wheel centring seat against corrosion

Applies to alloy and steel wheels

When changing a wheel, wheel centring seat should be waxed using wax spray ⇒ Electronic parts catalogue (ETKA) or ⇒ web-MANTIS for MAN TGE to prevent corrosion between wheel centring seat and wheel.



– Remove wheel.
– Thoroughly clean wheel centring seat on wheel hub and centring of rim.
– Apply wax with a brush in the area of centring ring -arrow-.

Note

Wheel bolts, contact surfaces of wheel hub and wheel, brake discs and threads of wheel hub must not be waxed. Never treat wheel securing bolts with lubricant or corrosion protection materials!

Observe notes for revised wheel bolts ⇒ Wheels and Tyres Guide - General information; Rep. gr. 44 ; Useful information about wheels/tyres (Commercial vehicles); Fitting wheels .

– Install wheel and tighten bolts or nuts ⇒ page 153 .

1.3.2 Wheel change, installation requirements

Warming cold tyres to minimum fitting temperature

Note

- *The minimum fitting temperature of a tyre should be between 15°C and 30°C in the core of the tyre.*
- *This instruction also applies to ultra-high performance tyres (aspect ratio less than or equal to 45% and speed symbol V or higher).*

- To install tyres without damage, it is especially important to warm the upper part of the sidewall and the inside of the upper bead to at least 15°C.
- This internal temperature is referred to as the core temperature.
- Rubber is a poor conductor of heat. Therefore, a cold tyre must be left in an area with the correct temperature for a sufficiently long period so the inner rubber layers can warm up to at least 15°C.
- The surface temperature of the tyre during the warming up phase is no indication of its internal temperature.
- To enable cold tyres to absorb heat from the ambient air as quickly as possible, they should not be stacked on top of one another but instead stored individually in order to allow the warm air to “circulate” around them effectively.
- Tyres must never be placed in front of a radiator or hot air blower for warming, since this can very quickly lead to critical surface temperatures.
- When cold tyres (below 0°C) are transferred to a warm environment (above 0°C), a layer of condensation immediately forms on the surface of the tyre. This layer of condensation indicates that the tyre is intensively absorbing heat from its environment through the process of water vapour in the air condensing out on the tyre surface.
- Thawing of the frost layer leads to formation of condensation water on the surface. The condensation should be dried with a cloth so that the further warming process is not delayed due to a temperature drop caused by evaporation.

Warming times:

- Assuming a minimum room temperature of 19°C and a tyre temperature of 0°C or more, a tyre must be kept at a minimum of 19°C for at least 2 hours.
- Assuming a minimum room temperature of 19°C and a tyre temperature of below 0°C, a tyre must be kept at a minimum of 19°C for at least 2.5 hours.

Warming recommendations:

- If possible, tyres should be stored in the workshop 1 day before they are fitted (order preparation).
- Store on an insulated base, pallet or the like, as high up as possible.
- Position the tyres individually to allow the warm air to “circulate” around them effectively.
- Wipe off perspiration water.

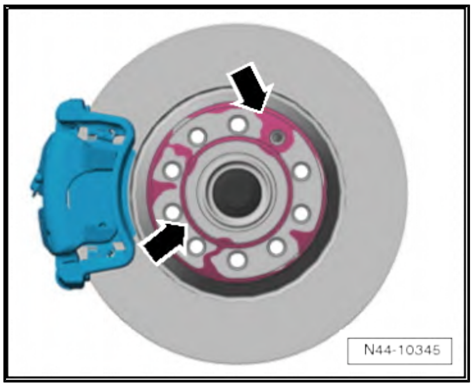
♦ Never heat with a radiator or hot air blower!

1.3.3 Wheel change, installation instructions for wheel change

Note

Perform the checks and follow the instructions listed below. This is important to ensure that the wheel bolts and the wheels are properly secured.

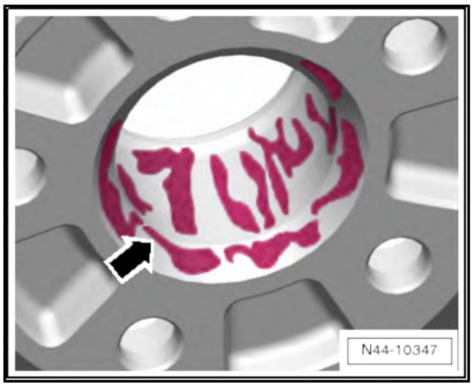
– Check to ensure that contact surfaces -arrows- on brake disc are free of corrosion and dirt.



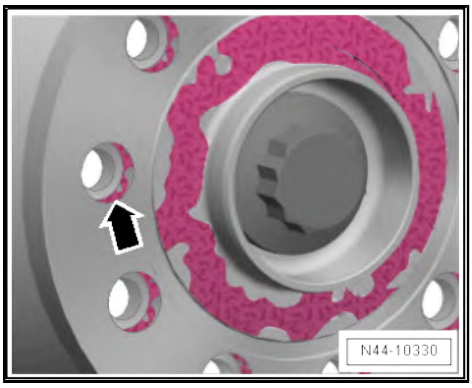
– Check to ensure that contact surface -arrow- on centring seat of brake disc are free of corrosion and dirt.

– Check to ensure that contact surface -arrow- on inner side of wheel (rim) and also centring seat of rim are free of corrosion and dirt.

– The lug seats* in the holes for the wheel bolts and the threads of the wheel bolts must also be free of corrosion and dirt, oil or grease.



– Check whether the wheel bolts can be easily screwed in by hand. The thread of the wheel bolts must not come into contact with the bore in the brake disc -arrow-.



* The lug seat is the curved surface of a section of a sphere cut by a plane.

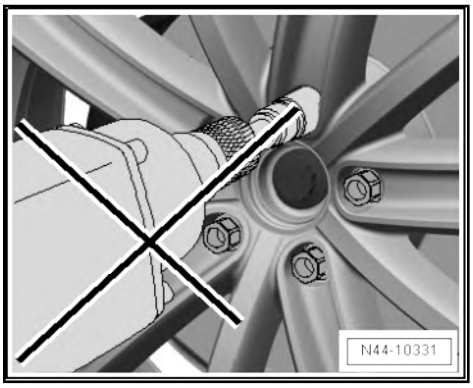
If the thread of the wheel bolt touches the hole -arrow-, turn the brake disc relative to the wheel hub accordingly.

Note

- *If necessary, remove dirt and corrosion, oil or grease from the contact surfaces, threads in the wheel hub and/or wheel bolts as necessary.*
- *Damaged, badly corroded or difficult to remove wheel bolts must be renewed.*

1.3.4 Changing wheel, mounting wheel on vehicles with single wheels

– Preserve wheel centring seat ⇒ page 150 .



1 - When fitting the wheel, pull in all wheel bolts uniformly by hand.

2 - Tighten bolts diagonally and evenly.

3 - Lower vehicle onto the floor and tighten all wheel bolts in diagonal sequence to specified torque, using torque wrench.

Note

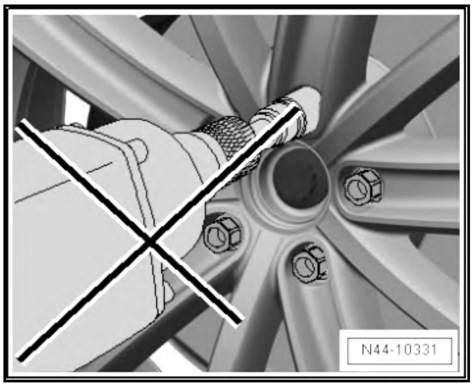
Do not use an impact wrench to screw in the wheel nuts.

Observe notes for revised wheel bolts ⇒ Wheels and Tyres Guide - General information; Rep. gr. 44 ; Useful information about wheels/tyres (Commercial vehicles); Fitting wheels .

Specified torques

- ⇒ “1.4 Specified torques for wheel bolts”, page 154

1.3.5 Changing wheel, mounting wheel on vehicles with twin wheels



– Preserve wheel centring seat ⇒ page 150 .

1 - Position inner wheel.

2 - Fit valve extension, screw only hand-tight.

3 - Position outer wheel.

4 - Make sure that the valves are in correct position. They must be mounted offset to one another by 180°.

5 - Tighten all wheel nuts uniformly by hand.

6 - Tighten wheel nuts diagonally and evenly.

7 - Lower vehicle onto the floor and tighten all wheel nuts in diagonal sequence to specified torque, using torque wrench.

Note

Do not use an impact wrench to screw in the wheel nuts.

Observe notes for revised wheel bolts ⇒ Wheels and Tyres Guide - General information; Rep. gr. 44 ; Useful information about wheels/tyres (Commercial vehicles); Fitting wheels .

Specified torques

♦ ⇒ “1.4 Specified torques for wheel bolts”, page 154

1.4 Specified torques for wheel bolts

Specified torques

Component	Specified torque
Wheel bolt to wheel hub for all vehicles with single wheels	200 Nm
Wheel bolt to wheel hub for all vehicles with Super single wheels	200 Nm
Wheel nut to wheel hub for all vehicles with twin wheels	180 Nm

– Before fitting wheel, protect wheel centring seat against corrosion ⇒ page 150 .

Observe notes for revised wheel bolts ⇒ Wheels and Tyres Guide - General information; Rep. gr. 44 ; Useful information about wheels/tyres (Commercial vehicles); Fitting wheels .

1.5 Pressing tyre off wheel rim

The tyre fitting unit must be furnished with the tyre fitting head designed for these rims.

Note

Otherwise there is a danger of the rim becoming damaged.

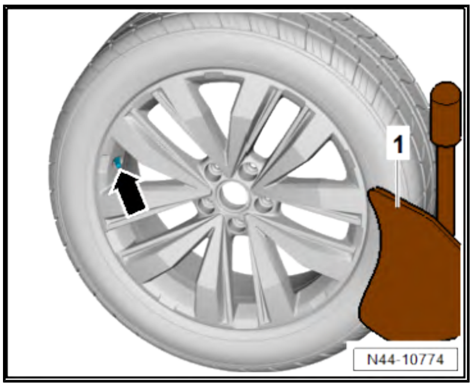
If your tyre fitting unit has not yet been modified, please contact the manufacturer of the unit.

Notes on safety and conditions for removing and fitting tyres (wheels with tyre pressure monitoring)

- It is extremely important to adhere to the instructions and warnings in the following descriptions.
- Check whether the tyre pressure sensor should also be replaced ⇒ Vehicle diagnostic tester.

– Let air out of tyre. To do this, unscrew nickel-plated valve core.

– When pressing off a tyre using tyre fitting equipment with a press-off plate, always ensure that the tyre valve/ tyre pressure sensor -arrow- is directly opposite the bead breaker -1-.



The bead breaker must be positioned no more than 2 cm away from the rim flange.

– Remove balancing weights and dirt from wheel rim.

– Press both tyre beads off all round and liberally coat tyre and wheel rim flange with tyre assembly paste -arrow-.

1.6 Tyre sealant disposal

♦ Tyre sealant or residues must not be mixed with other waste/ fluids.

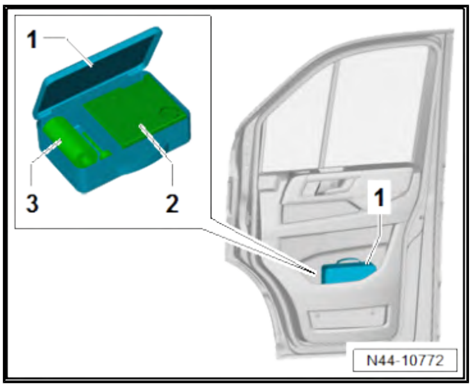
♦ Excess tyre sealant must be collected and stored in a plastic container. The plastic container can be disposed of through the disposal system along with the breakdown set (when the expiry date is exceeded).

♦ The items can be returned or disposed of through the existing workshop disposal system.

♦ Consult with the waste disposal representative of the distribution centre or importer.

1.7 Models with breakdown set

Depending vehicle equipment, the vehicles are equipped with a tyre mobility set.



The breakdown set -1- is in the front passenger door.

The breakdown set -1- includes a bottle of tyre sealant -3- and a compressor -2-.

The tyre sealant in the bottle has a limited shelf life.

Therefore the expiry date is indicated on the bottle -3-.

Renew tyre sealant if the expiry date is reached (the tyre sealant must not be older than 4 years).

If the bottle was opened e.g. at a "flat tyre", it must also be renewed.

Observe regulations for disposal ⇒ page 155 .

2 Tyre Pressure Monitoring System

⇒ “2.1 System description – Tyre Pressure Monitoring System”, page 157

⇒ “2.2 Overview of fitting locations - tyre pressure monitoring system”, page 157

⇒ “2.3 Assembly overview - tyre pressure sensor”, page 159

⇒ “2.4 Removing and installing tyre pressure monitoring system control unit J502 ”, page 160

⇒ “2.5 Removing and installing tyre pressure sensors G222 / G223 / G224 / G225 ”, page 160

2.1 System description – Tyre Pressure Monitoring System

- As previously the system is based on sensors in the tyres, the sensors send the inflation pressure data to the Tyre Pressure Monitoring System control unit - J502- .
- The function “autolocation” allows the system to react over a relatively short distance, sending inflation pressures and warnings correctly to the display in the dash panel.
- The “intelligent aerial” is a combined tyre pressure monitor control unit and a central receiver aerial.
- Manual teaching is not necessary after renewing tyre pressure sensors or changing tyre sets. The Tyre Pressure Monitoring System automatically identifies new tyre pressure sensors and programs these as soon as driving commences.

2.2 Overview of fitting locations - tyre pressure monitoring system

1 - Dash panel insert - KX2-

- ❑ Removing and installing ⇒ Electrical system; Rep. gr. 90 ; Dash panel insert; Removing and installing dash panel insert KX2

2 - Tyre Pressure Monitoring System control unit - J502-

- ❑ Removing and installing ⇒ page 160

3 - Front left tyre pressure sensor - G222-

- ❑ Removing and installing ⇒ page 160

4 - Front right tyre pressure sensor - G223-

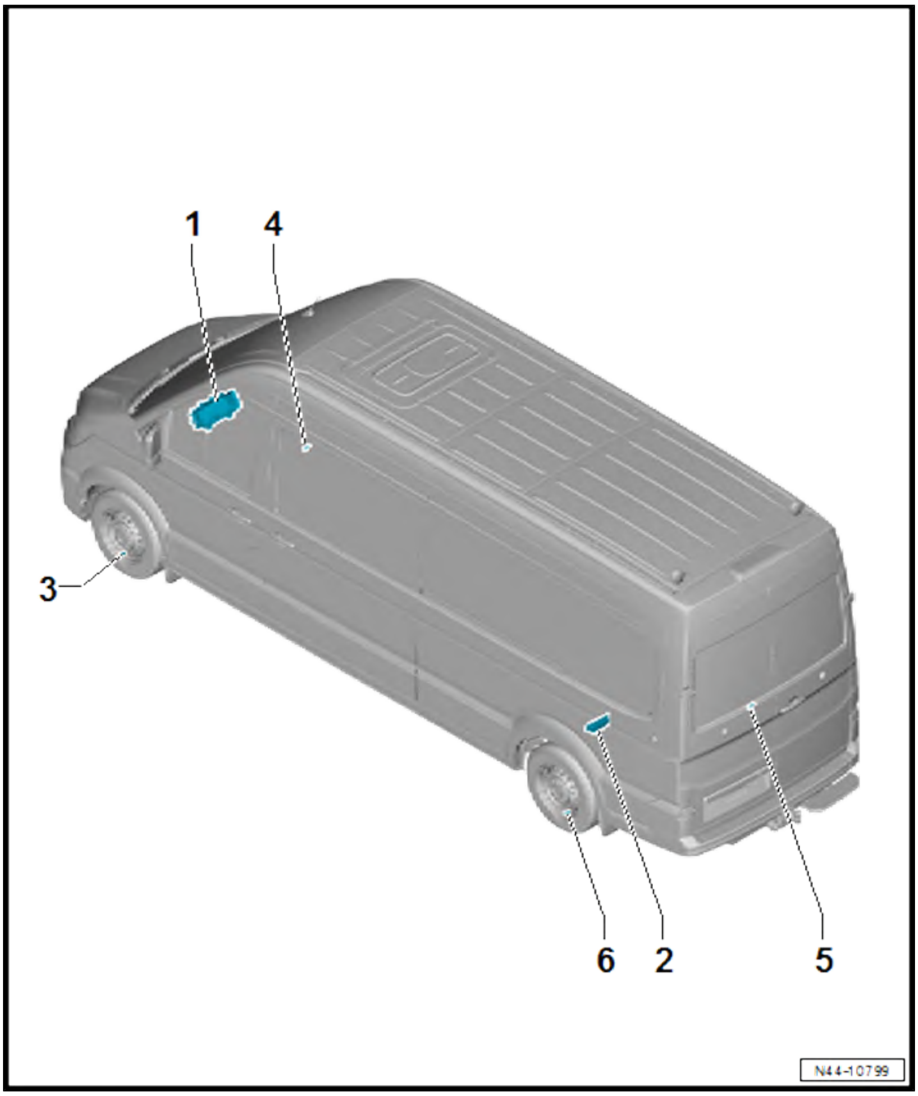
- ❑ Removing and installing ⇒ page 160

5 - Rear right tyre pressure sensor - G225-

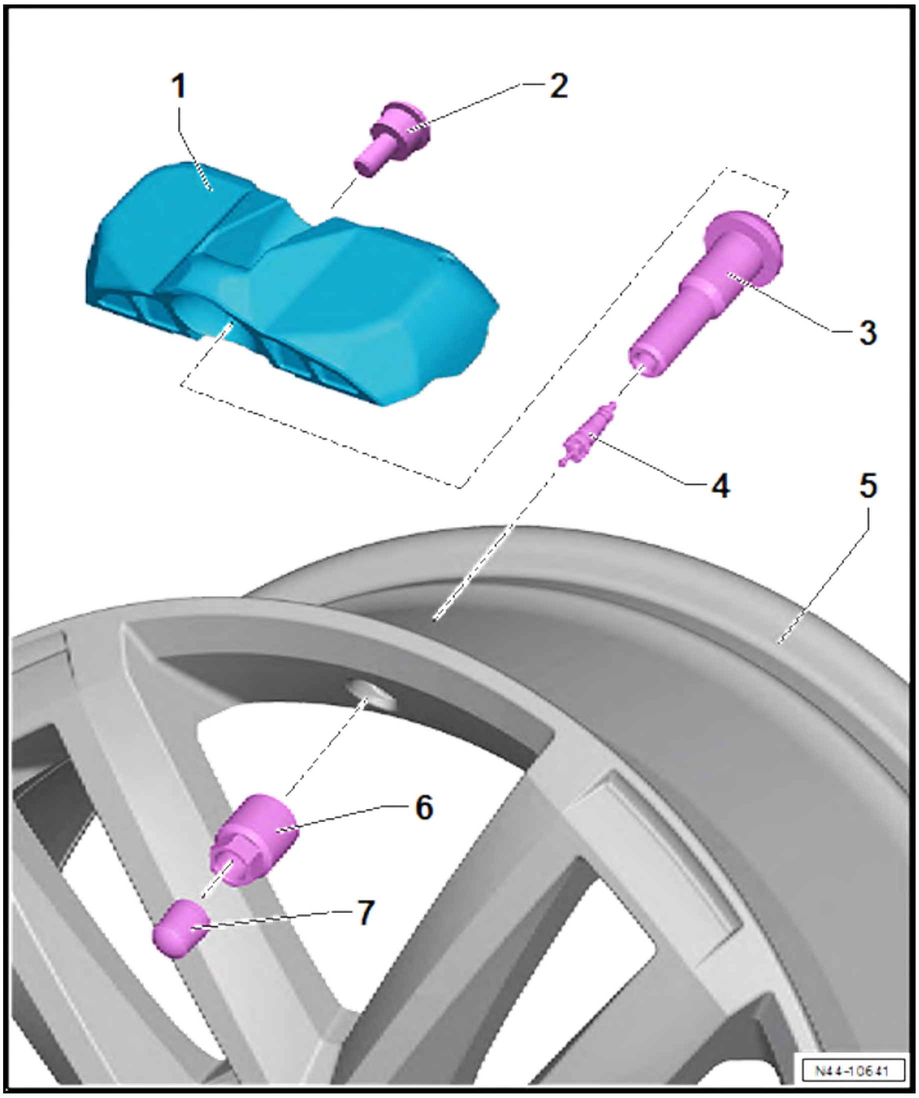
- ❑ Removing and installing ⇒ page 160

6 - Rear left tyre pressure sensor - G224-

- ❑ Removing and installing ⇒ page 160



2.3 Assembly overview - tyre pressure sensor



1 - Tyre pressure sensor

- ❑ Front left tyre pressure sensor - G222-
- ❑ Front right tyre pressure sensor - G223-
- ❑ Rear left tyre pressure sensor - G224-
- ❑ Rear right tyre pressure sensor - G225-
- ❑ Removing and installing ⇒ page 160

2 - Bolt

- ❑ Square-head bolt with flat head
- ❑ Is supplied as a replacement part complete with the tyre pressure sensor ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

3 - Metal valve

- ❑ Is supplied as a replacement part complete with the tyre pressure sensor ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE

4 - Valve insert

- ❑ Always renew when changing tyre

5 - Wheel rim

6 - Union nut

- ❑ Always renew after removing

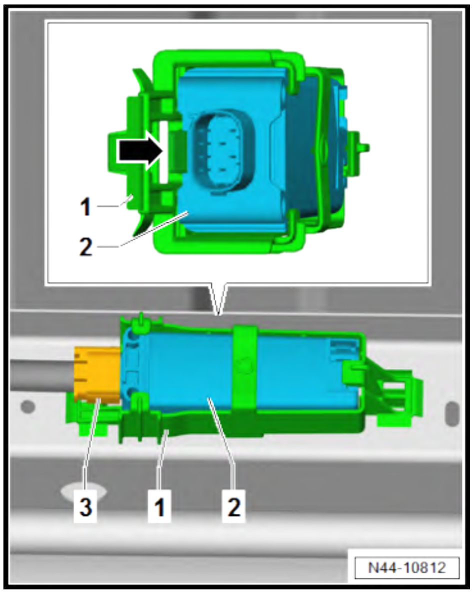
♦ When tightening, do not use metal valve as a counterhold.

♦ The result of a disc being placed in the union nut is that the metal valve is first screwed onto the tyre pressure sensor when the nut is tightened. The tyre pressure sensor is screwed onto the rim after the disc breaks.

- ❑ 4 Nm

7 - Valve cap

2.4 Removing and installing tyre pressure monitoring system control unit - J502-



Removing

– Switch off ignition.

– Raise vehicle ⇒ Maintenance ; Booklet 10.3 ; Description of work; Raising vehicle with lifting platform and trolley jack .

– Separate electrical connector -3-.

– Slightly push detent lug -arrow- outwards.

– Pull out Tyre Pressure Monitoring System control unit - J502- -2- from bracket -1-.

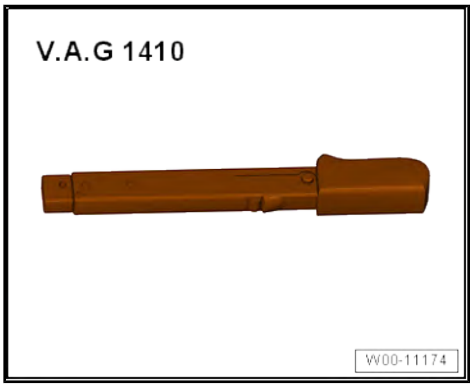
Installing

Install in reverse order.

2.5 Removing and installing tyre pressure sensors - G222- / -G223- / -G224- / -G225-

Special tools and workshop equipment required

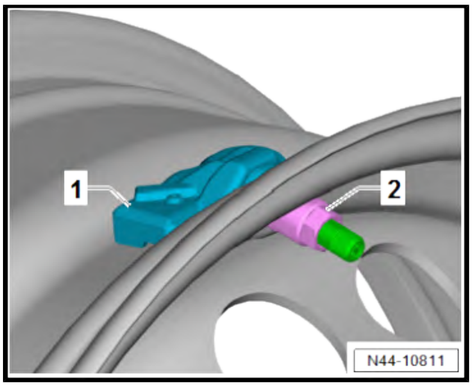
♦ Torque wrench - V.A.G 1410-



Removing

– Press tyre off wheel rim ⇒ page 154 .

– Turn nut -2- anti-clockwise until tyre pressure sensor -1- can be removed.



Note

When the nut -2- is turned, the valve turns as well.

Installing

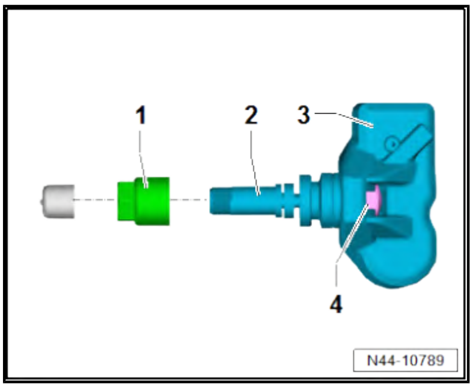
– Before installing tyre pressure sensor , clean valve hole.

Note

♦ *Always renew the metal valve completely.*

♦ *Before installing tyre pressure sensor , clean valve hole.*

– Using square-head bolt -4-, secure tyre pressure sensor -3- loosely to metal valve -2-.



Note

When doing this, ensure the metal valve is not bolted tight on the sensor.

– Insert metal valve into valve hole in rim.

Note

♦ *When tightening, do not use metal valve as a counterhold.*

♦ *The result of a washer being placed in the union nut -1- is that the metal valve is first screwed onto the tyre pressure sensor when the nut is tightened. The tyre pressure sensor is screwed onto the rim after the disc breaks.*

– Bolt new union nut onto metal valve. While doing so, the metal valve will turn and is bolted to tyre pressure sensor .

– Tighten union nut until rupture disc breaks.

– Fit tyres ⇒ page 149 .

Specified torques

♦ ⇒ “2.3 Assembly overview - tyre pressure sensor”, page 159

3 Wheel alignment

3.1 Notes for wheel alignment

Wheel alignment may only be performed using a manufacturer-approved wheel alignment unit.

Whenever wheels are aligned, both the front and rear axles must be measured.

– Carry out alignment with the wheel alignment computer.

All the information on wheel alignment can be found in the wheel alignment computer.

Note

- *Wheel alignment should not be checked before the vehicle has completed 1000 to 2000 km after the coil springs have settled.*
- *When making adjustments, adhere to the relevant specifications as closely as possible.*
- *Whenever adjustments are made to the running gear, always check whether any driver assist systems must be recalibrated following the adjustments.*

Crabbing and accident damaged vehicles

This may be due to the rack in the steering rack not being exactly centred when vehicle is driven in a straight line.

Power steering assistance is thereby slightly one sided, rather than central. The vehicle will pull to one side as a consequence.

When checking the alignment of a "vehicle which pulls to one side", always ensure that the steering rack is centred.

3.2 Conditions for testing

⇒ "3.2.1 Conditions for testing, Crafter", page 163

⇒ "3.2.2 Conditions for testing, MAN TGE", page 163

3.2.1 Conditions for testing, Crafter

- Check wheel suspension, wheel bearing assembly and steering for excessive play and damage.
- Tread depth difference of no more than 2 mm on one axle.
- Tyres inflated to correct pressures.
- The payload must equate to the standard load as per everyday use.
- Fuel tank must be full.
- Spare wheel and vehicle tools are stowed in correct locations.
- Windscreen washer fluid reservoir must be full.
- When checking wheel alignment, ensure that sliding plates and turn tables are not touching end stop.

Please note the following:

- The test equipment must be properly adjusted and attached to the vehicle; observe device manufacturer's operating instructions.

If necessary, ask the manufacturer to show you how to use the wheel alignment machine.

Wheel alignment platforms and wheel alignment computers can lose their original calibration over a period of time.

Wheel alignment platforms and computers should be checked and adjusted as necessary during inspection and maintenance at least once per year!

– Handle these highly sensitive devices carefully and conscientiously.

3.2.2 Conditions for testing, MAN TGE

- Check wheel suspension, wheel bearing assembly and steering for excessive play and damage.
- Tread depth difference of no more than 2 mm on one axle.
- Tyres inflated to correct pressures.
- The payload must equate to the standard load as per everyday use.
- Fuel tank must be full.
- Spare wheel and vehicle tools are stowed in correct locations.
- Windscreen washer fluid reservoir must be full.
- Perform wheel alignment on even ground. Vehicle must be laden for measurement and steered wheels must be on turntables.
- When checking wheel alignment, ensure that sliding plates and turn tables are not touching end stop.

Please note the following:

- The test equipment must be properly adjusted and attached to the vehicle; observe device manufacturer's operating instructions.
- Handle highly sensitive devices carefully and conscientiously.

If necessary, ask the manufacturer to show you how to use the wheel alignment machine.

3.3 Test preparations

⇒ "3.3.1 Test preparations, wheel alignment with no driver assist systems, Crafter", page 164

⇒ "3.3.2 Test preparations, wheel alignment with no driver assist systems, wheel alignment system AXIS 500, MAN TGE", page 165

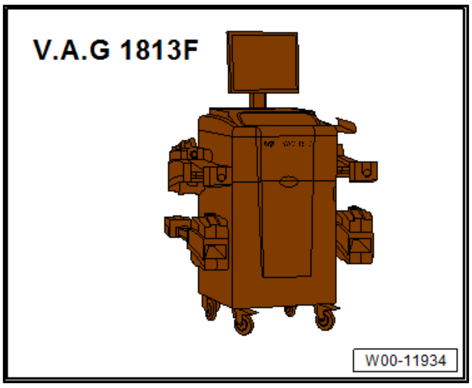
⇒ "3.3.3 Test preparations, wheel alignment with no driver assist systems, wheel alignment system HD-30 Easy Touch, MAN TGE", page 167

⇒ "3.3.4 Test preparations, wheel alignment with driver assist systems, Crafter", page 169

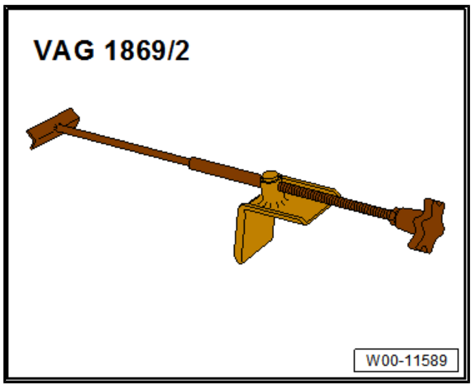
3.3.1 Test preparations, wheel alignment with no driver assist systems, Crafter

Special tools and workshop equipment required

- ♦ Wheel alignment computer - V.A.G 1813F-



- ♦ Brake pedal depressor - V.A.G 1869/2-

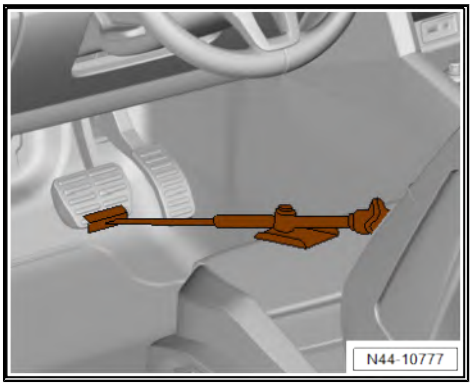


The existing lateral runout of the wheel must be compensated for, otherwise the result will be falsified.

If runout compensation is not performed, it is not possible to adjust toe-in correctly!

Please follow the instructions of the manufacturer of the wheel alignment equipment.

– Carry out wheel runout compensation.
– Apply brake pedal depressor - V.A.G 1869/2- .
– Use brake pedal depressor to depress brake pedal.

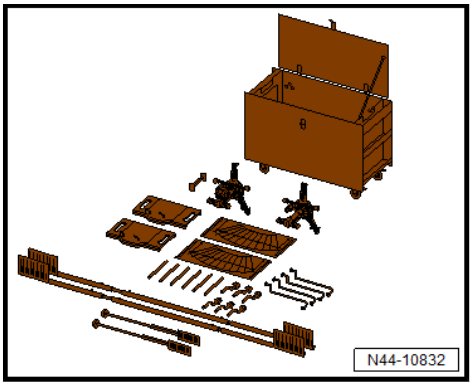


Preparations required before calibration of driver assist systems

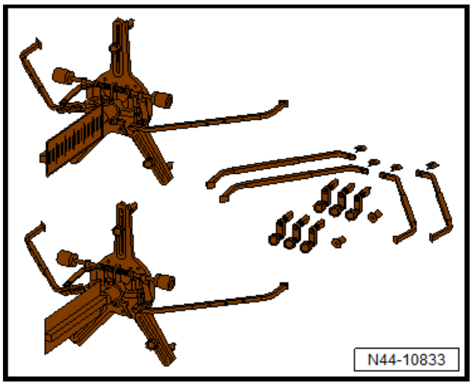
3.3.2 Test preparations, wheel alignment with no driver assist systems, wheel alignment system AXIS 500, MAN TGE

Special tools and workshop equipment required

♦ Wheel alignment system AXIS 500 - 80.99605-6039-



♦ Upgrade kit Transporter - 80.99605-6049-



Overview of laser measuring head of wheel alignment system AXIS 500

1 - Centring pins
- ❑ Qty. 3

2 - Gripper arms
- ❑ Qty. 2

3 - Inclinometer

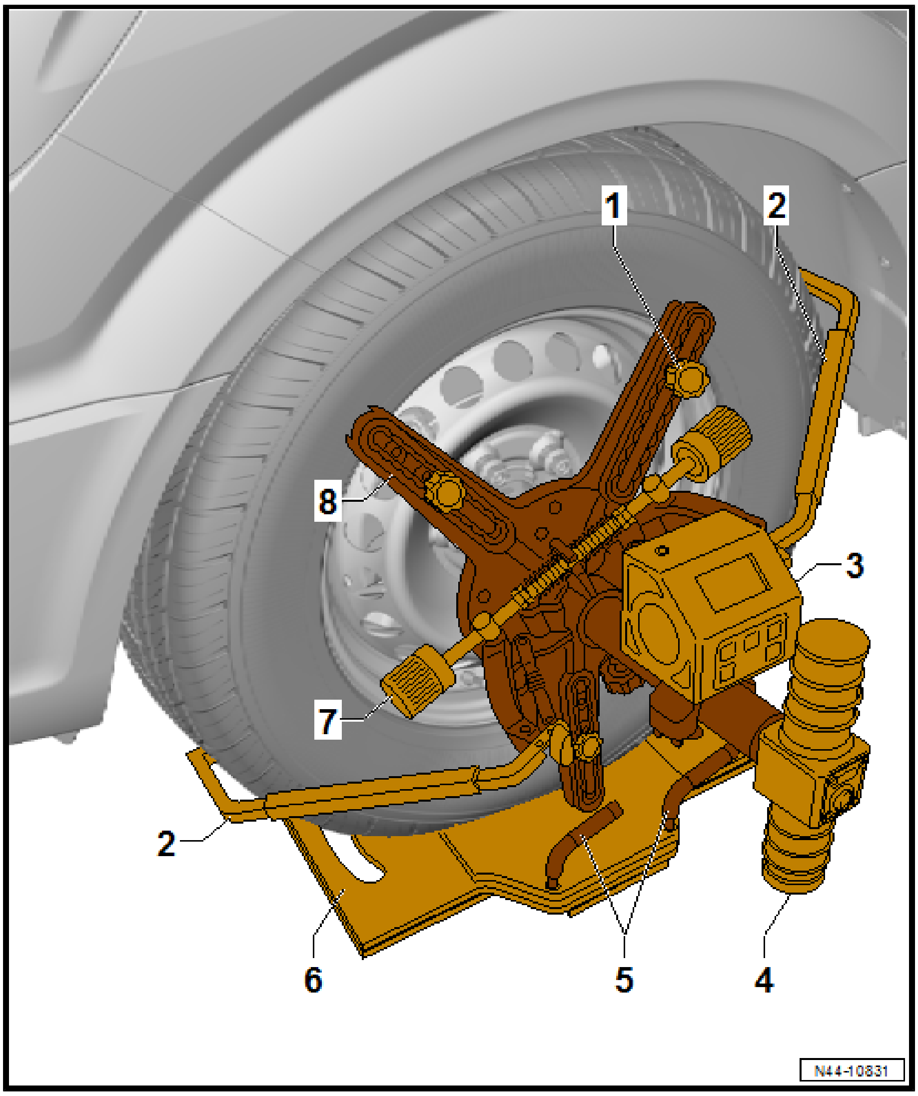
4 - Laser

5 - Grab handles
- ❑ Qty. 2

6 - Turntable

7 - Clamping device

8 - Laser measuring head



Procedure

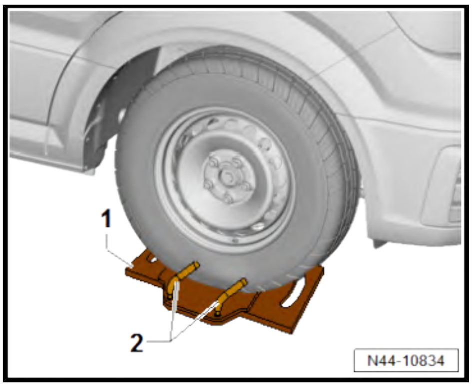
– Insert grab handles -2- in turntables -1-.

– Bring turntables into position at steered wheels with arrow pointing in direction of travel.

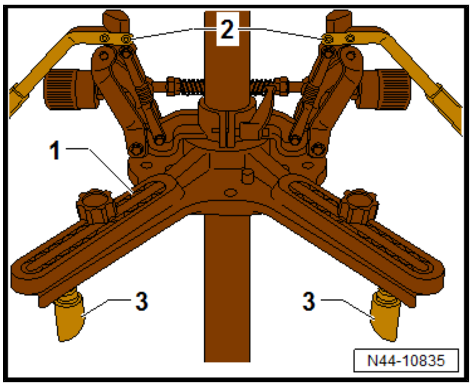
– Drive vehicle with steered wheels onto turntables.

– Secure vehicle to prevent it from rolling away ⇒ Owner's Manual .

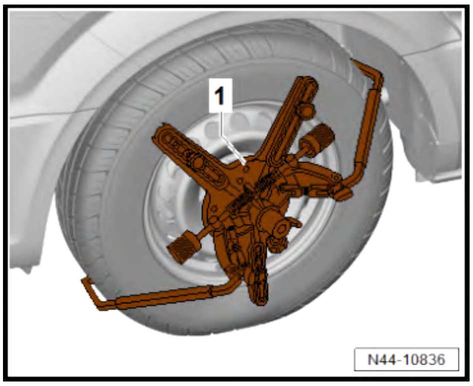
– Pull out grab handles from turntables -1-.



Converting laser measuring heads



– Remove magnetic holder from laser measuring head -1-.
– Mount centring pins -3- (qty. 3) and gripper arms -2- on laser measuring head.
– Align centring pins to rim diameter.
– Repeat procedure for all laser measuring heads.



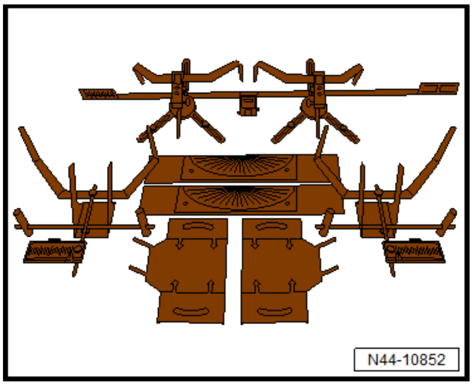
– Mount laser measuring heads -1- on all wheels.

Procedure for wheel alignment, wheel alignment system AXIS 500, MAN TGE ⇒ page 177 .

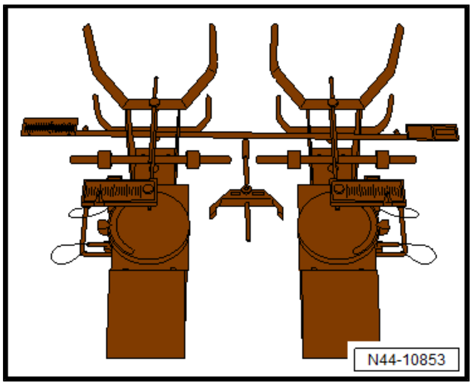
3.3.3 Test preparations, wheel alignment with no driver assist systems, wheel alignment system HD-30 Easy Touch, MAN TGE

Special tools and workshop equipment required

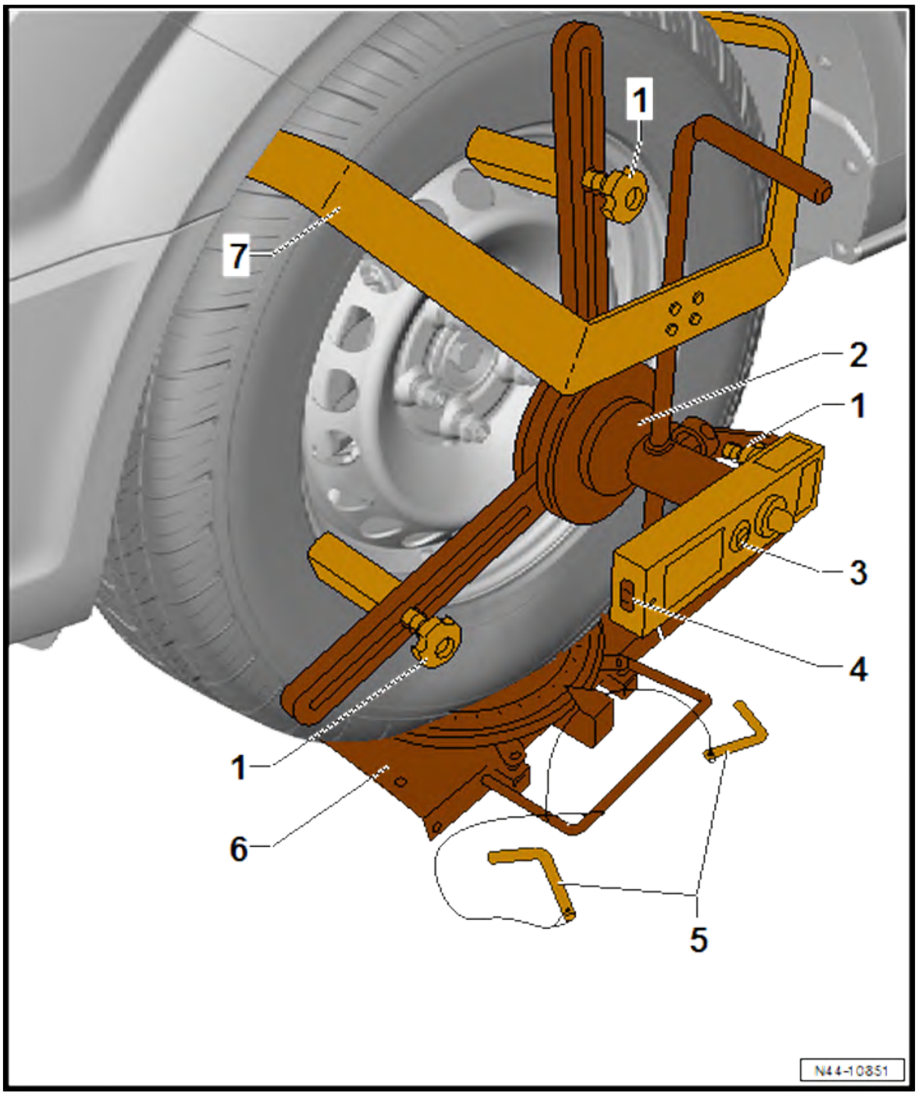
♦ Wheel alignment system HD-30 Easy Touch - 80.99607-6080-



♦ Passenger vehicle adapter - 80.99607-6089-



Overview of laser measuring head of wheel alignment system HD 30 Easy Touch



1 - Measuring sensor
- ❑ Qty. 3

2 - Laser measuring head

3 - Laser

4 - Spirit level
- ❑ Integrated in laser housing

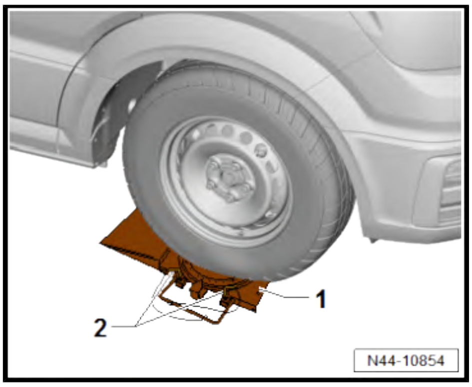
5 - Locating pins
- ❑ Qty. 2

6 - Turntable

7 - Retaining bracket
- ❑ Qty. 2

Procedure

Continued for all vehicles



– Insert locating pins -2- in turntables -1-.

– Bring turntables into position at steered wheels with arrow pointing in direction of travel.

– Drive vehicle with steered wheels onto turntables.

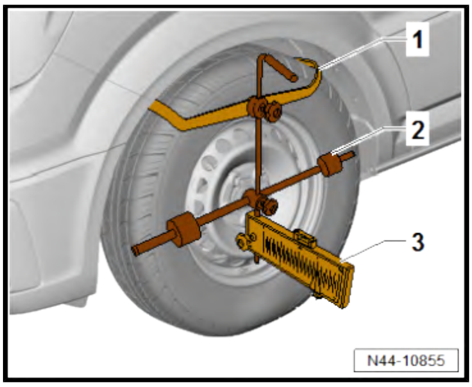
– Secure vehicle to prevent it from rolling away ⇒ Owner's Manual .

– Pull out locating pins -2-.

Mounting hanging scale

– Fit hanging scale -3- with retaining bracket -1- loosely on rear wheel.

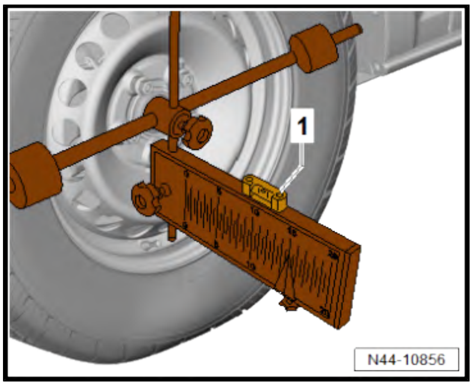
– Loosen sensor rollers -2- on left and right and mount on rim flange.



Aligning hanging scale

– Align hanging scale to retaining brackets until spirit level -1- is in centre.

– Repeat procedure on other side of vehicle.



Mounting laser measuring head

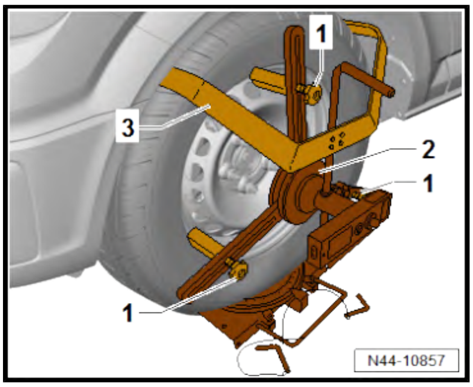
– Fit laser measuring head -2- with retaining bracket -3- loosely on front wheel.

– Loosen measuring sensor -1-, relocate and mount on rim flange.

– Tighten threaded connection on measuring sensors.

– Adjust retaining brackets until all measuring sensors are in contact with rim.

– Repeat procedure on other side of vehicle.



Procedure for wheel alignment, wheel alignment system HD-30 Easy Touch, MAN TGE ⇒ page 181 .

3.3.4 Test preparations, wheel alignment with driver assist systems, Crafter

The following steps are required if one or more driver assist systems on the vehicle are to be calibrated via the “Quick-start” procedure (i.e. without first checking and adjusting the wheel alignment):

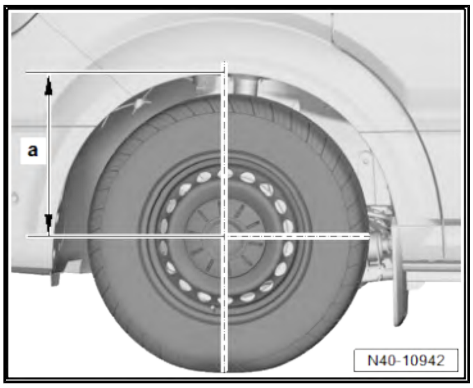
- Before driving the vehicle onto the alignment platform, make sure there is sufficient space between the vehicle and the calibration unit. The distance between the calibrating device and the vehicle must be 120 cm ± 2.5 cm.
- If the available space is not adequate, move the vehicle backwards on the alignment platform as required. This allows the use of the respective floor area.
- Before starting calibration, read out event memory and rectify any faults.
- Vehicle accurately aligned, suspension bounced and rocked several times.

- When checking wheel alignment, ensure that sliding plates and turn tables are not touching end stop.
- Connect battery charger ⇒ Vehicle electrics; Rep. gr. 27 ; Battery; Charging battery .
- Move front wheels to straight-ahead position.
- Connect ⇒ Vehicle diagnostic tester to vehicle, route diagnostic cable through open door window.
- Vehicle exterior lighting switched off.
- All doors of vehicle are closed.
- Press button to select required calibration on wheel alignment computer.

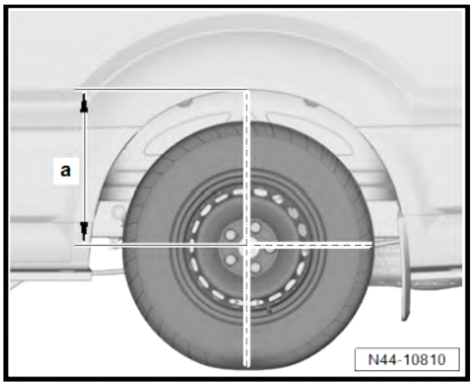
3.4 Specifications for wheel alignment

The ride heights shown in the table refer to dimension -a-.

Front axle



Rear axle, closed body type

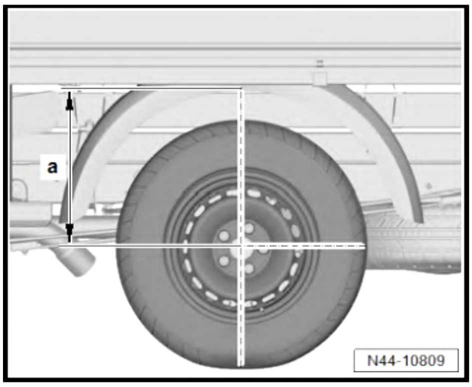


Rear axle, open body type

Single tyres

Front axle			
PR numbers	K4A, 0WQ/0WP, 1UA/1UD/1UV, 1X0 *)	K4A, 0WQ/0WP, 1UA/1UD/1UV, 1X1, 1X4 *)	K4B, 0WQ, 1UA, 1X0 *)
Toe-in per wheel (not pressed)	0° 5′ (± 10′)	0° 5′ (± 10′)	0° 5′ (± 10′)
Total toe (not pressed)	0° 10′ (± 10′)	0° 10′ (± 10′)	0° 10′ (± 10′)
Camber (in straight-ahead position) per wheel	-0.7° (± 45')	-0.7° (± 45')	-0.7° (± 45')
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Caster per wheel	4° 10′ (± 40′)	4° 10′ (± 40′)	4° 10′ (± 40′)
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Ride height dimension -a-	466 ± 17 mm	466 ± 17 mm	466 ± 17 mm

Front axle			
PR numbers	K4B, 0WQ, 1UA, 1X1 *)	K4F/K4G, 0WQ/0WP, 1UA/1UD/1UV, 1X0 *)	K4F/K4G, 0WQ/0WP, 1UA/1UD/1UV, 1X1, 1X4 *)
Toe-in per wheel (not pressed)	0° 5′ (± 10′)	0° 5′ (± 10′)	0° 5′ (± 10′)
Total toe (not pressed)	0° 10′ (± 10′)	0° 10′ (± 10′)	0° 10′ (± 10′)
Camber (in straight-ahead position) per wheel	-0.7° (± 45')	-0.7° (± 45')	-0.7° (± 45')
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Caster per wheel	4° 10′ (± 40′)	4° 10′ (± 40′)	4° 10′ (± 40′)
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Ride height dimension -a-	466 ± 17 mm	466 ± 17 mm	466 ± 17 mm



Front axle			
PR numbers	**K4A, 0WQ, 1UA, 1US, 1X0 *)**		
Toe-in per wheel (not pressed)	0° 5′ (± 10′)		
Total toe (not pressed)	0° 10′ (± 10′)		
Camber (in straight-ahead position) per wheel	-0.7° (± 45')		
Maximum permissible difference between sides	Max. 1°		
Caster per wheel	4° 10′ (± 40′)		
Maximum permissible difference between sides	Max. 1°		
Ride height dimension -a-	466 ± 17 mm		

Rear axle			
PR numbers	**K4A, 0WQ/ 0WP, 1UA/ 1UD/1UV, 1X0 *)**	**K4A, 0WQ/ 0WP, 1UA/ 1UD/1UV, 1X1, 1X4 *)**	**K4B, 0WQ, 1UA, 1X0 *)**
Total toe (not pressed)	0° 10′ (+15′ -10′)	0° 0′ (± 30′)	0° 10′ (+15′ -10′)
Camber (in straight-ahead position) per wheel	-0° 40′ (± 15′)	0° 0′ (± 30′)	-0° 40′ (± 15′)
Maximum permissible difference between sides	-	-	-
Caster per wheel	-	-	-
Maximum permissible difference between sides	-	-	-
Ride height dimension -a-	526/ -39; +12 mm	526/ -40; +11 mm	526/ -46; +18 mm

Rear axle			
PR numbers	K4B, 0WQ, 1UA, 1X1 *)	K4F/K4G, 0WQ/0WP, 1UA/1UD/ 1UV, 1X0 *)	K4F/K4G, 0WQ/0WP, 1UA/1UD/ 1UV, 1X1, 1X4 *)
Total toe (not pressed)	0° 0′ (± 30′)	0° 10′ (+15′ -10′)	0° 0′ (± 30′)
Camber (in straight-ahead position) per wheel	0° 0′ (± 30′)	-0° 40′ (± 15′)	0° 0′ (± 30′)
Maximum permissible difference between sides	-	-	-
Caster per wheel	-	-	-
Maximum permissible difference between sides	-	-	-
Ride height dimension -a-	526/ -47; +16 mm	525/ -25; +33 mm	525/ -27; +32 mm

Rear axle			
PR numbers	K4A, 0WQ, 1UA, 1US, 1X0 *)		
Total toe (not pressed)	0° 10′ (+15′ -10′)		
Camber (in straight-ahead position) per wheel	-0° 40′ (± 15′)		
Maximum permissible difference between sides	-		
Caster per wheel	-		
Maximum permissible difference between sides	-		
Ride height dimension -a-	515/ -19; +10 mm		

Twin tyres

Front axle			
PR numbers	**K4A, 0WS, 1UA, 1X2 *)**	**K4A, 0WS, 1UI, 1X2 *)**	**K4A, 0WS, 1UH/1UF/ 1UE, 1X2 *)**
Toe-in per wheel (not pressed)	0° 5′ (± 10′)	0° 5′ (± 10′)	0° 5′ (± 10′)
Total toe (not pressed)	0° 10′ (± 10′)	0° 10′ (± 10′)	0° 10′ (± 10′)
Camber (in straight-ahead position) per wheel	-0.7° (± 45')	-0.7° (± 45')	-0.7° (± 45')
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Caster per wheel	4° 10′ (± 40′)	4° 10′ (± 40′)	4° 10′ (± 40′)
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Ride height dimension -a-	466 ± 17 mm	466 ± 17 mm	466 ± 17 mm

Front axle			
PR numbers	**K4F/K4G, 0WS, 1UA, 1X2 *)**	**K4F/K4G, 0WS, 1UI, 1X2 *)**	**K4F/K4G, 0WS, 1UH/ 1UF/1UE, 1X2 *)**
Toe-in per wheel (not pressed)	0° 5′ (± 10′)	0° 5′ (± 10′)	0° 5′ (± 10′)
Total toe (not pressed)	0° 10′ (± 10′)	0° 10′ (± 10′)	0° 10′ (± 10′)
Camber (in straight-ahead position) per wheel	-0.7° (± 45')	-0.7° (± 45')	-0.7° (± 45')
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Caster per wheel	4° 10′ (± 40′)	4° 10′ (± 40′)	4° 10′ (± 40′)
Maximum permissible difference between sides	Max. 1°	Max. 1°	Max. 1°
Ride height dimension -a-	466 ± 17 mm	466 ± 17 mm	466 ± 17 mm

Rear axle			
PR numbers	K4A, 0WS, 1UA, 1X2 *)	K4A, 0WS, 1UI, 1X2 *)	K4A, 0WS, 1UH/1UF/1UE, 1X2 *)
Total toe (not pressed)	0° 0′ (± 30′)	0° 0′ (± 30′)	0° 0′ (± 30′)
Camber (in straight-ahead position) per wheel	0° 0′ (± 30′)	0° 0′ (± 30′)	0° 0′ (± 30′)
Maximum permissible difference between sides	-	-	-
Caster per wheel	-	-	-
Maximum permissible difference between sides	-	-	-
Ride height dimension -a-	549.5 ± 24 mm	548.5 ± 24 mm	544.5 ± 24 mm

Rear axle			
PR numbers	K4F/K4G, 0WS, 1UA, 1X2 *)	K4F/K4G, 0WS, 1UI, 1X2 *)	K4A/K4G, 0WS, 1UH/1UF/1UE, 1X2 *)
Total toe (not pressed)	0° 0′ (± 30′)	0° 0′ (± 30′)	0° 0′ (± 30′)
Camber (in straight-ahead position) per wheel	0° 0′ (± 30′)	0° 0′ (± 30′)	0° 0′ (± 30′)
Maximum permissible difference between sides	-	-	-
Caster per wheel	-	-	-
Maximum permissible difference between sides	-	-	-
Ride height dimension -a-	553 ± 17 mm	552 ± 17 mm	548 ± 17 mm

1) The value given in brackets represents the tolerance.

3.5 Wheel alignment procedure

⇒ “3.5.1 Wheel alignment procedure, Crafter”, page 175

⇒ “3.5.2 Procedure for wheel alignment, wheel alignment system AXIS 500, MAN TGE”, page 177

⇒ “3.5.3 Procedure for wheel alignment, wheel alignment system HD-30 Easy Touch, MAN TGE”, page 181

3.5.1 Wheel alignment procedure, Crafter

Always adhere to the following procedure!

Note

The vehicle must always be in unladen condition when measuring wheel alignment ⇒ page 8 .

– Take note of the information in the wheel alignment unit.

Alignment procedure

Start	→	Carry out wheel runout compensation.	→	Bounce springs.
				↓
Turn steering wheel to straight-ahead position and fix in place [2)].	←	Measure vehicle height.	←	Fit brake pedal depressor - V.A.G 1869/2-
↓				
Check front axle camber	→	Is actual value within tolerance?	→	no
		↓		↓
↓ ←	←	yes		Adjusting ⇒ page 188
↓				↓
↓ ←	←	←	←	←
↓				
Check rear axle camber	→	Is actual value within tolerance?	→	no
		↓		↓
↓ ←	←	yes		Adjusting ⇒ page 188
↓				↓
↓ ←	←	←	←	←
↓				
Check rear axle toe	→	Is actual value within tolerance?	→	no
		↓		↓
↓ ←	←	yes		Adjusting ⇒ page 188
↓				↓
↓ ←	←	←	←	←
↓				
Check front axle caster	→	Is actual value within tolerance?	→	no
		↓		↓
↓ ←	←	yes		Inspect axle components and body
↓				↓
↓ ←	←	←	←	←
↓				
Check front axle toe	→	Is actual value within tolerance?	→	no
		↓		↓

End	←	yes		Adjusting ⇒ page 188
↑				↓
↑	← ←	←	←	←

2) If steering wheel is not centred, it must be straightened after wheel alignment is finished. Then carry out basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.

Note

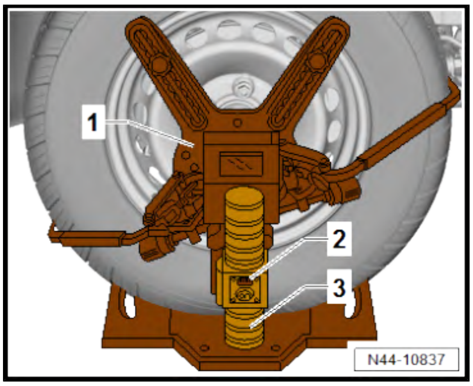
If adjustments have been made to the suspension after a wheel alignment check on vehicles with ESP or ABS, the steering angle sender - G85- must be calibrated using the ⇒ Vehicle diagnostic tester.

3.5.2 Procedure for wheel alignment, wheel alignment system AXIS 500, MAN TGE

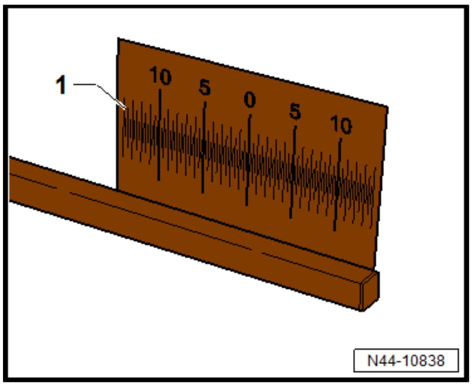
Procedure

– Perform preparations for measurement ⇒ page 165 .

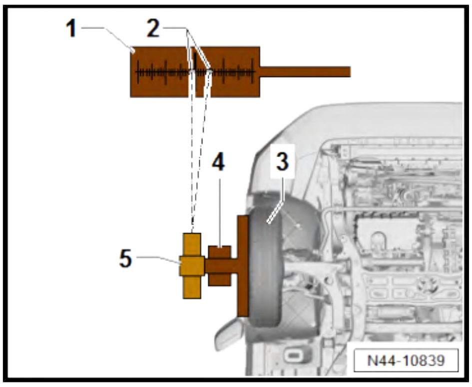
Wheel runout compensation



– Mount laser -3- on laser measuring head -1- on front wheel.
– Swivel laser downwards until spirit level -2- is exactly between both markings.
– Mark laser measuring point on ground.
– Repeat procedure on other side of vehicle.
– Determine rim diameter with tape measure.
– Multiply rim diameter in cm by 10 and divide by 2.
– Mark determined toe setting (e.g. 240 cm) at front of and behind measuring points on ground.
– Bring toe scale into position at marked measuring points.

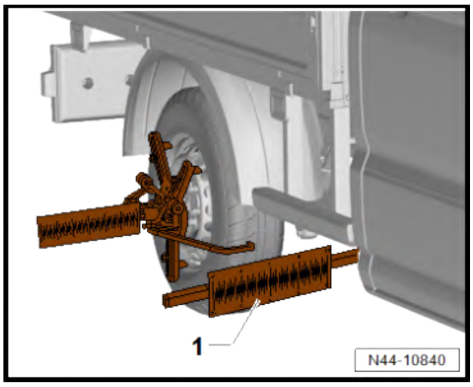


– Swivel laser forwards onto target -1- and align scale to zero.
– Repeat procedure on other side of vehicle.
– Raise vehicle on one side until one front wheel is free and other is still on turntable ⇒ Maintenance ; Booklet 10.3 ; Work descriptions; Raising vehicle with lifting platform and trolley jack .



– Swivel laser -5- forwards onto target -1-.
– Take measurement reading and record.
– Remove laser measuring head -4-.
– Turn wheel -3- 120° further.
– Bring laser measuring head into position and swivel laser onto target.
– Take measurement reading and record.
– Repeat work procedure for last wheel rotation of 120°.
– Continue to turn wheel until laser is aligned centrally between total tolerance values -2-.
– Repeat procedure on other side of vehicle.
– Lower vehicle.

Adjusting straight-ahead position

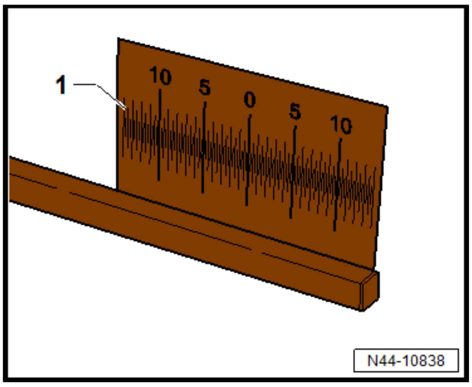


– Mount magnetic scale -1- on chassis approx. right of vehicle centre.

Note

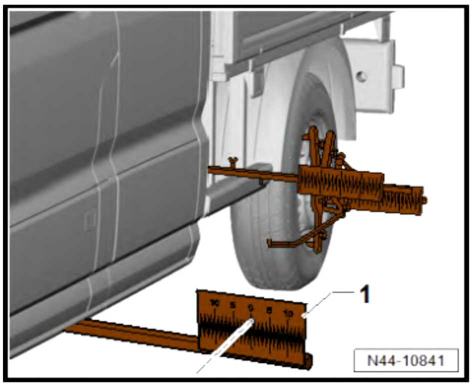
If, owing to the type of vehicle, the magnetic scale -1- cannot be attached to the chassis in the centre of the vehicle, the magnetic scale must be inserted above the rear wheel and secured to the chassis there.

– Swivel laser backwards onto target and align to zero.
– Repeat procedure on second target.
– Mount one target -1- on chassis on left-hand side of vehicle approx. in centre of vehicle.
– Swivel laser backwards onto target and take reading.
– If value is not zero, turn steering wheel until calculated value is halved.



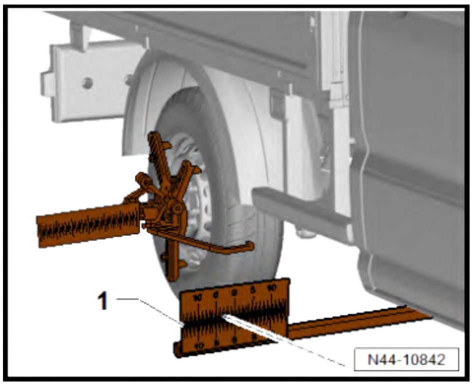
– Position second scale at front where measuring points are marked on floor.
– Swivel laser forwards onto target -1- and align to zero.
– Repeat procedure on other side of vehicle.
– Block steering wheel in straight-ahead position.
– Determine height between centre of hub and edge of wing ⇒ page 170 .

– Position second scale -1- at rear where measuring points are marked on floor.
– Swivel left laser backwards onto target -1- and align scale to zero.
– Repeat procedure on other side of vehicle.



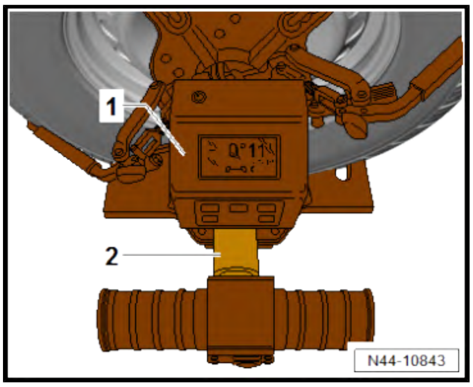
Determining toe setting

– Swivel right laser backwards onto target -1- and take toe reading.
– Convert toe setting using conversion table.
– Determine toe setting depending on height; adjusting toe if necessary ⇒ page 188 .
– Read measured value and enter it measurement log.
– Repeat procedure on other side of vehicle.



Measuring camber

– Turn steering wheel to straight-ahead position.
– Bring inclinometer -1- into position on axle shaft -2- of laser measuring head and secure with knurled screw.
– Switch on inclinometer.
– Read measured value and enter it measurement log.
– Repeat procedure on other side of vehicle.

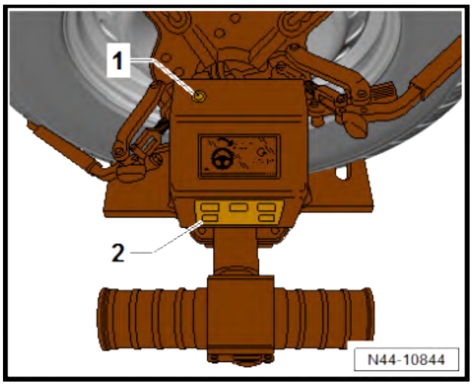


Determining turning angles, caster, steering axis inclination and toe-out on turns

– Switch on turning angle assist with button -2-.

Command appears in display to turn steering wheel 20°.

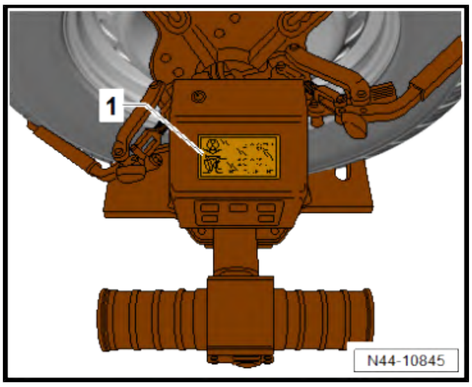
– Turn steering wheel clockwise 20° until LED -1- flashes and confirms steering wheel angle.
– After LED goes out, turn steering wheel in opposite direction by 20°.
– Hold steering wheel in this position until LED light is steady.



Once measured values have been recorded, caster (top) and steering axis inclination (bottom) are shown in display -1-.

– Read measured values and enter in measurement log.

– Repeat procedure on other side of vehicle.

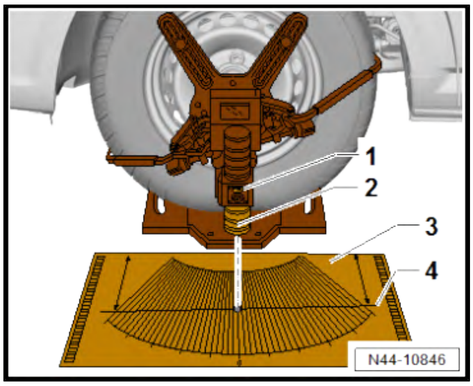


– Swivel laser -2- downwards until spirit level -1- is exactly between both markings.

– Place turning angle scale -3- parallel to front wheel on floor.

– Align turning angle scale until laser dot swivels along zero line -4-.

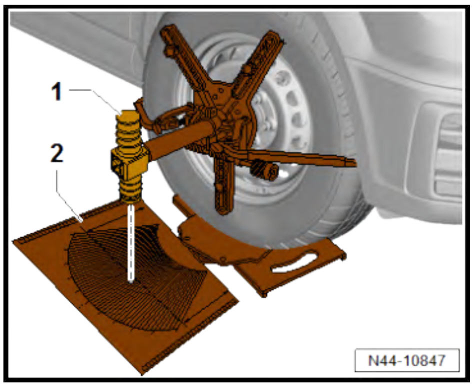
– Repeat procedure on other side of vehicle.

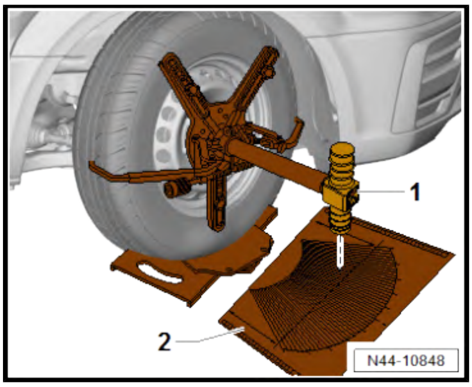


– Turn steering wheel approx. 1.5 turns in anti-clockwise direction until dot of laser -1- on turning angle scale -2- hits 20° marking on left.

– Align laser vertically until spirit level -1- is exactly between both markings.

– Hold steering wheel in this position.





– Read off turning angle from scale -2- on other side of vehicle.

– Align laser vertically until spirit level -1- is exactly between both markings.

The difference between the measured value on the right-hand side and 20° on the left-hand side is the toe-out on turns.

– Enter toe-out on turns in measurement log.

– Repeat procedure on other side of vehicle.

Final checks

– If necessary, adjust toe ⇒ page 188 .

– Remove wheel alignment system.

– Drive vehicle off turntables.

– Carry out road test.

Note

- *If steering wheel is not centred, it must be straightened after wheel alignment is finished. Then carry out basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.*
- *If adjustments have been made to the suspension after a wheel alignment check on vehicles with ESP or ABS, the steering angle sender - G85- must be calibrated using the ⇒ Vehicle diagnostic tester.*

3.5.3 Procedure for wheel alignment, wheel alignment system HD-30 Easy Touch, MAN TGE

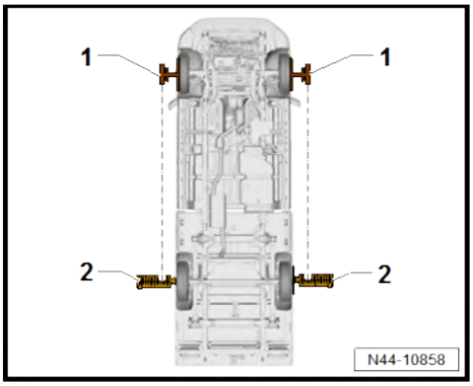
Note

Always adhere to the following procedure.

Procedure

– Perform preparations for measurement ⇒ page 165 .

Aligning laser measuring head

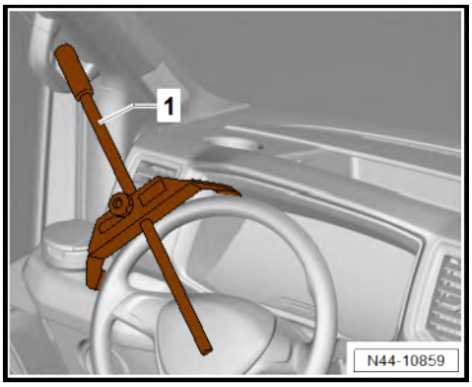


– Swivel laser -1- backwards onto targets -2- of hanging scales.

– Turn steering wheel until same value is displayed on both targets.

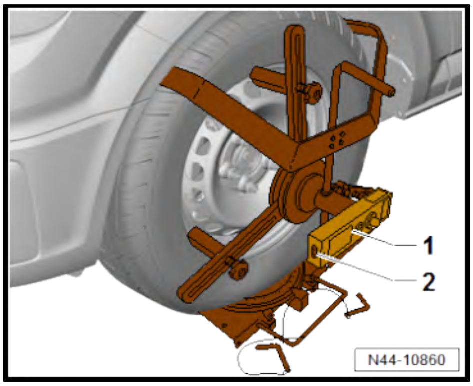
The value 6.5, for example, is shown on both targets. Wheels are in straight-ahead position.

Locking steering wheel



– Lock steering wheel in straight-ahead position with locking device -1-.

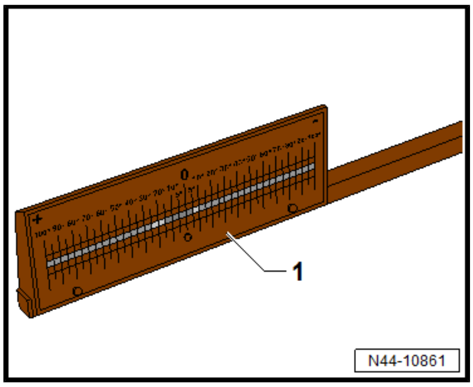
Aligning laser



– Swivel laser -1- downwards until spirit level -2- in laser housing is exactly between both markings.

– Mark measuring point on ground.

Bringing laser measuring bream into position at front

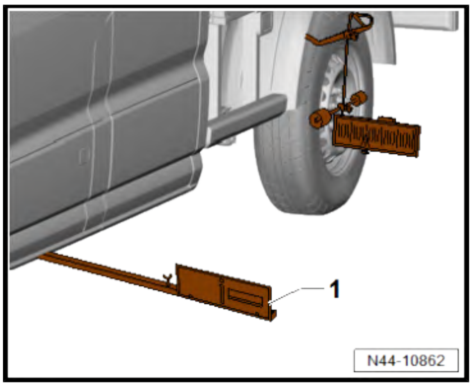


– Bring laser measuring beam -1- into position on both sides 1.70 metres in front of marked measuring points.

– Swivel laser forwards onto laser measuring beam on both sides and align laser measuring beam in such a way that laser dot hits zero mark on driver side.

– Secure laser measuring beam with wing bolt to prevent position being lost.

Checking toe



– Bring laser measuring beam -1- into position on both sides 1.70 metres behind marked measuring points.
– Swivel laser on driver side backwards onto laser measuring beam.
– Align laser measuring beam in such a way that laser dot hits zero mark on driver side.
– Swivel laser on other side of vehicle backwards onto laser measuring beam and read off measured value.

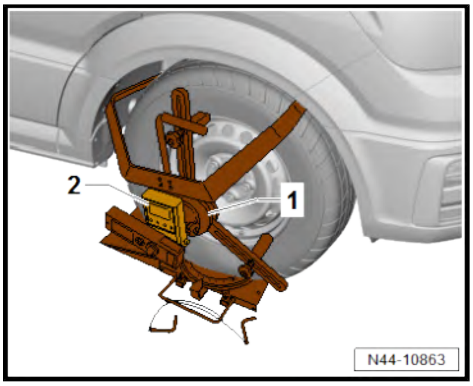
• Laser dot at zero = toe is zero
• Laser dot points inwards from zero = toe-out
• Laser dot points outwards from zero = toe-in

– If necessary, adjust toe ⇒ page 188 .

Determining toe setting

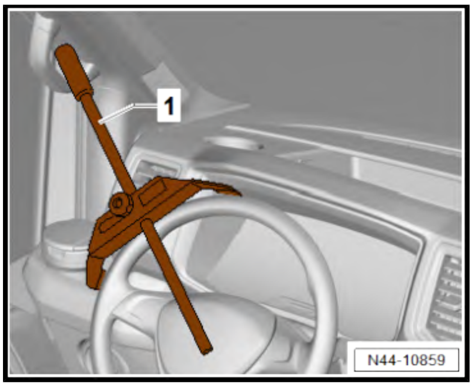
– Determine height between centre of hub and edge of wing ⇒ page 170 .
– Convert toe setting using conversion table.
– Determine toe setting depending on height; adjusting toe if necessary ⇒ page 188 .

Measuring camber



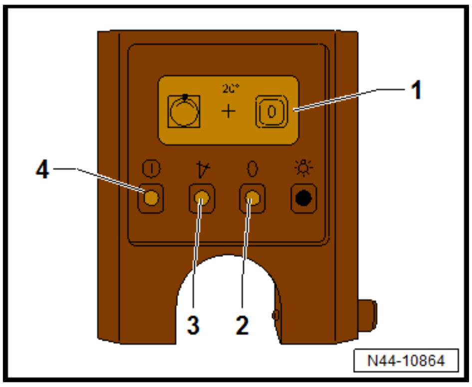
– Turn steering wheel to straight-ahead position.
– Bring inclinometer -2- into position on axle shaft of laser measuring head -1- and secure with knurled screw.
– Switch on inclinometer.
– Read measured value and enter it measurement log.
– Repeat procedure on other side of vehicle.

Removing locking device



– Release locking device -1- and remove from steering wheel.

Switching on turning angle assist

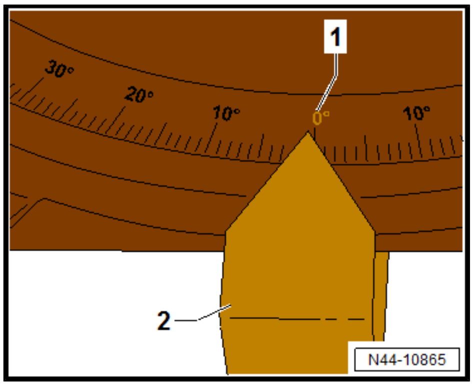


- Switch on inclinometer with button -4-.
- Press button -3- to switch on turning angle assist.
- Press button -2- on inclinometer.

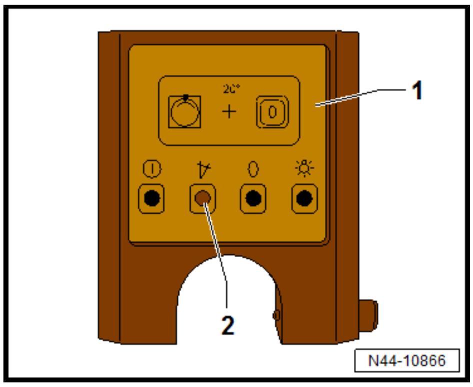
Turning angle assist is switched on.

Display -1- is shown on inclinometer.

Setting and saving straight-ahead position

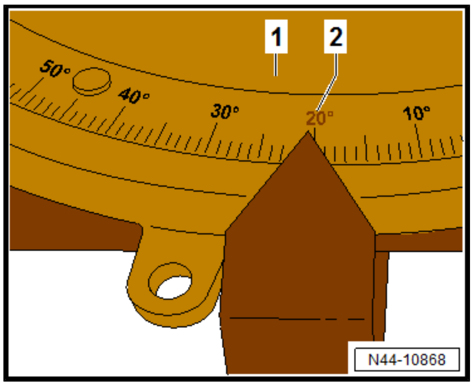


- Set wheels to straight-ahead position.
- Set indicator -2- on turntables to 0° marking -1-.



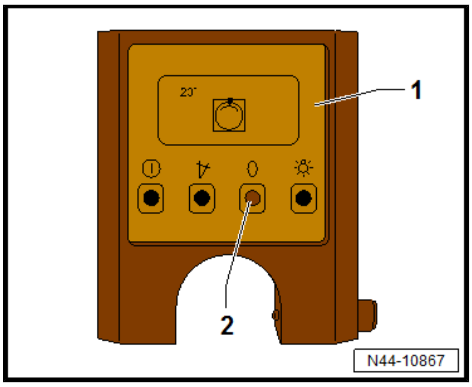
- Press button -2- on inclinometer -1-.

Preparation for measurement of caster



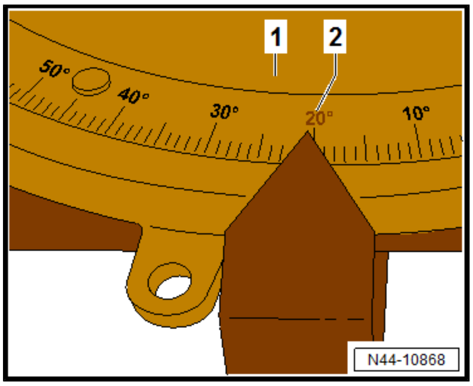
- Turn wheel on turntable -1- forwards into wheel housing to 20° marking -2-.

– Press button -2- on inclinometer -1-.

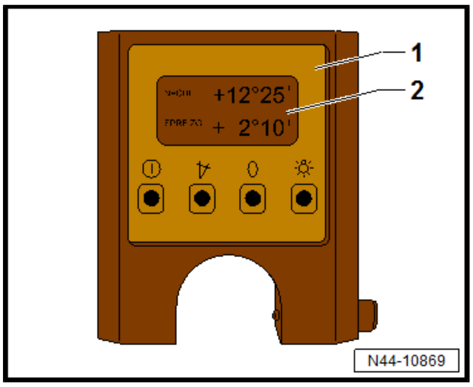


Measuring caster

– Turn wheel on turntable -1- in opposite direction to 20° marking -2-.



– Read off caster value in display -2- of inclinometer -1- and enter.



Measuring toe-out on turns

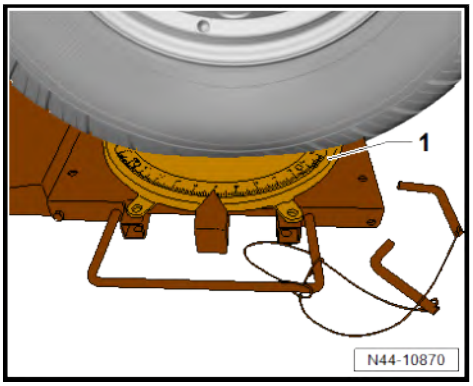
– Turn steering wheel approx. 1.5 turns in anti-clockwise direction until arrow on turntable -1- points to 20°.

– Read off turning angle from turntable on other side.

The difference between the measured value on the right-hand side and 20° on the left-hand side is the toe-out on turns.

– Enter toe-out on turns in measurement log.

– Repeat procedure on other side of vehicle.

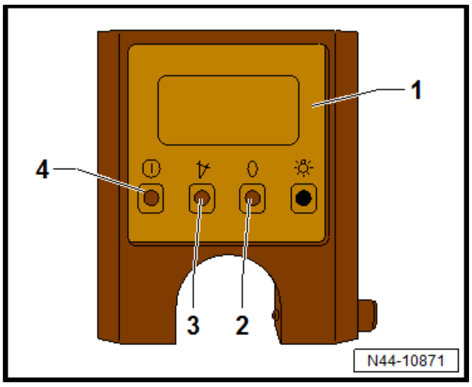


Switching on toe assist function

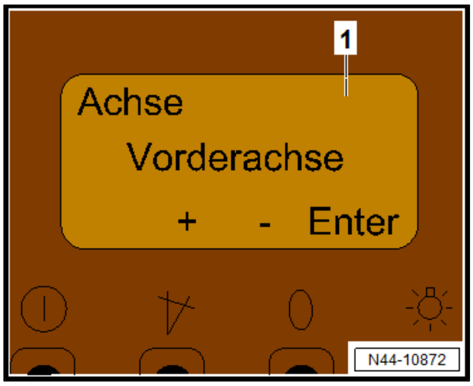
– Switch on inclinometer -1- with button -4-.

– Press buttons -2- and -3- at same time and hold for approx. 3 seconds.

Toe assist function of inclinometer is switched on.



Entering measured values in inclinometer



– Use + or - button in display -1- to select front axle and confirm with Enter button.

– Use + or - button in display -1- to enter measured values for straight-ahead position shown on hanging scales and confirm with Enter button.

– Use + or - button in display -1- to enter desired toe-in and confirm with Enter button.

– Use + or - button in display -1- to enter present toe value and confirm with Enter button.

– Use + or - button in display -1- to enter wheelbase (distance between front axle and rear axle) and confirm with Enter button.

Display -1- of inclinometer shows value to be set on rear axle scales.

Final checks

– If necessary, adjust toe ⇒ page 188 .

– Remove wheel alignment system.

– Drive vehicle off turntables.

– Carry out road test.

Note

♦ *If steering wheel is not centred, it must be straightened after wheel alignment is finished. Then carry out basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.*

♦ *If adjustments have been made to the suspension after a wheel alignment check on vehicles with ESP or ABS, the steering angle sender - G85- must be calibrated using the ⇒ Vehicle diagnostic tester.*

3.6 Necessity of wheel alignment

If:

♦ Vehicle does not handle properly

♦ Vehicle has been involved in an accident and components have been renewed

♦ Axle components have been removed or renewed

♦ Tyres wear on one side.

Front axle component was renewed.	Alignment necessary	
	yes	no
Lower suspension link	X	
Bonded rubber bush for suspension link		X
Wheel bearing housing	X	
Track rod/track rod ball joint	X	
Steering rack	X	
Subframe	X	
Suspension strut	X	
Swivel joint	X	

Front axle component removed and reinstalled	Alignment necessary	
	yes	no
Lower suspension link		X
Wheel bearing housing	X	
Track rod/track rod ball joint	X	
Steering rack	X	
Subframe		X
Suspension strut		X
Swivel joint	X	

Rear axle

All camber and toe values on the rear axle are prescribed by the design. If measured values exceed tolerances, they can only be corrected/restored by replacing defective/damaged components.

3.7 Explanatory notes on production control numbers (PR no.)

Note

- *Various types of running gear are installed depending on engine and equipment level. These are identified by production control numbers.*
- *Production control numbers are relevant for the wheel alignment specifications.*
- *The appropriate production control numbers on vehicle data sticker or in "Vehicle-specific notes" indicate which running gear is fitted to a vehicle ⇒ Vehicle-specific notes; Vehicle data; PR numbers .*

PR number table

PR number	Definition
K4A	Van
K4B	Kombi
K4F	Chassis with cab (single cab)
K4G	Chassis with double cab
0WP	Gross vehicle weight rating 3.0 t
0WQ	Gross vehicle weight rating 3.5 t

PR number	Definition
0WS	Gross vehicle weight rating 5.5 t
1UA	Without uprating/downrating
1UD	Uprating gross vehicle weight from 3.5 t to 4.0 t
1UE	Downrating gross vehicle weight from 5.5 t to 3.5 t
1UF	Downrating gross vehicle weight from 5.5 t to 3.88 t
1UH	Downrating gross vehicle weight from 5.5 t to 4.0 t
1UI	Downrating gross vehicle weight from 5.5 t to 5.0 t
1US	Uprating gross vehicle weight from 3.5 t to 4.25 t
1UV	Uprating gross vehicle weight from 3.5 t to 3.88 t
1X0	Front-wheel drive
1X1	All-wheel drive
1X2	Rear-wheel drive (twin tyres)
1X4	Rear-wheel drive (single tyres)
5HJ	Without dropside bed attachments
5HN	Preparation for 3-sided tipper

3.8 Adjusting camber at front wheels

Note

- *It is not possible to adjust camber on front axle.*
- *If the measured values are outside the allowed tolerance, camber can only be corrected by renewing the affected axle components.*

3.9 Adjusting camber on rear axle

Note

- *It is not possible to adjust camber on rear axle.*
- *If the measured values are outside the allowed tolerance, camber can only be corrected by renewing the affected axle components.*

3.10 Adjusting toe at rear axle

Note

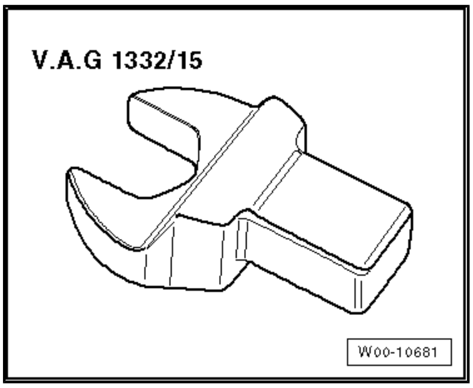
- *It is not possible to adjust toe on rear axle.*
- *If the measured values are outside the allowed tolerance, toe can only be corrected by renewing the affected axle components.*

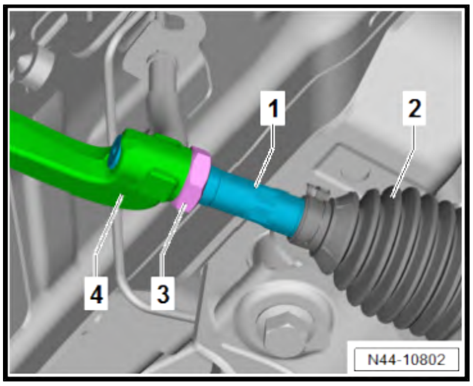
3.11 Adjusting front axle toe

Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Open jaw socket insert AF 30 - V.A.G 1332/15-





Procedure

– Set steering rack to zero position ⇒ page 190 .

– Loosen nut -3-. Counterhold on track rod ball joint -4-.

– Adjust toe by turning the track rod -1-.

– Hold boot -2- while turning track rod -1-.

Specified torques

- ⇒ “3.8 Repairing steering rack”, page 242

3.12 Wheel runout compensation

The existing lateral runout of the wheel must be compensated for. Otherwise, the result of the measurement will be incorrect.

The specified toe-in tolerance may already be exceeded by the permitted amount of axial runout at the wheel rims. In such cases it is not possible to set the toe-in correctly without first compensating for the wheel runout.

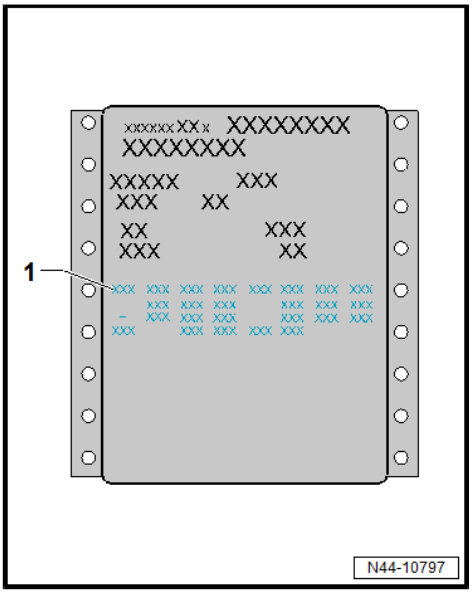
Please follow the instructions of the manufacturer of the wheel alignment equipment.

3.13 Vehicle data sticker

Vehicle data sticker is located in vehicle on side of open glove compartment and in service schedule/maintenance record.

Example of a vehicle data sticker

1 - PR numbers



3.14 Checking maximum wheel lock

3.14.1 Checking maximum wheel lock

The maximum wheel lock is determined with the aid of the wheel alignment computer or wheel alignment system.

– When the values for the maximum wheel lock lie outside the tolerances, then check following:

♦ Are steering or suspension components damaged or deformed?

♦ Are track rods visually OK.?

♦ Is the symmetry of the track rods OK.?

Damaged components must be renewed.

– If the steering wheel is found to be off centre, check the following:

♦ Check steering components for damage and deformation. If necessary, renew damaged components.

♦ Check suspension components for damage and deformation. If necessary, renew damaged components.

♦ Check symmetry of track rods also.

– When returning steering to central position, “swivel” steering wheel with even movements.

3.15 Setting steering rack to zero position

Procedure

– Lock turntable in position on left and right; roll vehicle off turntable, if necessary.

– Turn steering wheel to left and to right.

• No power steering

• Steering angle of 75°

– Repeat steering cycle at least 7 times. When doing this, lower steering angle for each steering cycle until it is 0°.

– Roll vehicle onto turntable, if necessary.

– Unlock turntable on left and right.

4 Vibration due to wheels/tyres - causes and rectification

4.1 Causes of rough running

Rough running can have a number of different causes. It can also be caused by tyre wear. Tyre wear caused by driving is not always evenly spread across the entire running surface of the tyre. This causes slight imbalances which affect the smooth running of a wheel which was previously exactly balanced.

Minor imbalances will not be felt at the steering wheel, but that does not mean that they are not there. They increase wear on the tyre and thus reduce the tyre service life.

Recommendation

To ensure

- optimal safety,
- smoothest possible running and
- even wear

throughout a tyre's service life, we recommend having the wheels and tyres balanced at least twice during the tyre's service life.

4.2 Conducting a road test before balancing wheels

If a customer brings a vehicle to the workshop complaining about “vibration”, a road test is essential prior to balancing the wheels.

♦ This will give you information about the nature of the rough running.

♦ You will be able to determine the speed range in which rough running occurs.

– Raise the vehicle on a lifting platform immediately after the road test.

– Mark installation position on tyre.

Tyre position	Marked with …
Front left tyre	FL
Front right tyre	FR
Rear left tyre	RL
Rear right tyre	RR

– Remove wheels.

– Balance wheels.

4.3 Balancing wheel

⇒ “4.3.1 Balancing wheel, conditions”, page 193

⇒ “4.3.2 Balancing wheel, balancing wheel on stationary wheel balancer”, page 193

⇒ “4.3.3 Balancing wheel, balancing wheel with finish balancer”, page 196

4.3.1 Balancing wheel, conditions

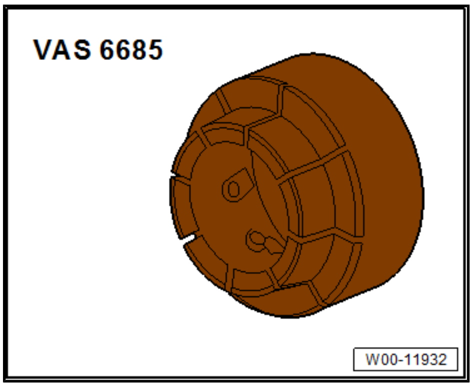
Before you start balancing the wheels, the following requirements must be met.

- Tyre pressure must be OK.
- Tread must not be worn on one side. Tread depth should be at least 4 mm.
- The tyre must not show any signs of damage, for example cuts, piercing, foreign bodies, etc.
- Wheel suspension and steering, including shock absorbers, must be in perfect condition.
- You must have conducted a road test.

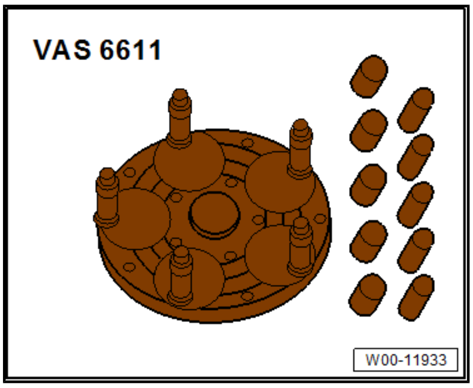
4.3.2 Balancing wheel, balancing wheel on stationary wheel balancer

Special tools and workshop equipment required

- ♦ Wheel balancing machine centring system - VAS 6685-



- ♦ 5-hole clamp plate - VAS 6611-



- ♦ Wheel balancing machine
- Road test has been carried out ⇒ page 192 .

Clamp wheel onto wheel balancing machine .

Note

When balancing tyres, please remember that cleanliness is absolutely essential, as is the case in any other repair work you carry out. Only then can you attain a flawless result!

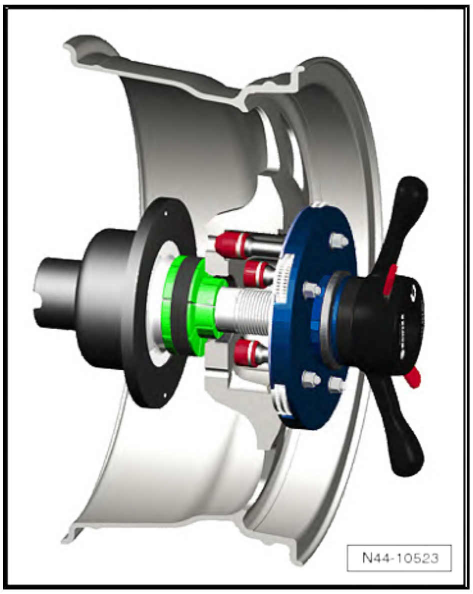
Dirt and rust in the area of the contact surfaces and centre of the wheel distort the result.

– Clean contact surface, centring element and inside of wheel using e.g. pneumatic brush grinding set - VAS 6446- before clamping wheel on wheel balancing machine .

Note

It is very important that the correct tools for centring and clamping the wheels are used on the wheel balancing machine . Before starting any work, find out about the respective centring system for wheel balancing machines .

– Attach the wheel (together with tyre) to the wheel balancing machine .



Note

- *To clamp the wheel in place, use centring system for wheel balancing machines - VAS 6685- , for example.*
- *This ensures that the wheel is 100% centred and that the wheel is clamped without its being damaged.*
- *The wheel cannot be centred 100% with conical clamping elements on the wheel balancing machine .*
- *A deviation of 0.1 mm from the centre results in an imbalance of 10 g at the wheel/rim.*

Procedure for balancing wheels and tyres

– Allow the wheel (together with tyre) to rotate on the wheel balancing machine .

– Check that the indicator lines on the sidewalls of the tyre near the wheel rim flange run evenly.

– Check tyre profile with wheel/tyre rotating.

Note

If one-sided wear, flat spots from braking or severely washed out spots are apparent, balancing cannot achieve smooth running. In this case, the tyres should be renewed in pairs.

– Check true running of the wheel/tyre. If the wheel and tyre do not run true although there are no flat spots, radial or lateral runout may be the cause.

– Check the wheel for radial or lateral run-out ⇒ page 197 .

– If radial and lateral runout are within the specified tolerance, balance the wheel and tyre.

Note

- *More than 60 g of weight per tyre should not be used.*
- *If more weight is required, you may be able to achieve smoother running by "matching" the tyre and rim. Matching tyres ⇒ page 199 .*
- *The wheel balancing machine display should indicate 0 g.*
- *The wheel balancing machine can be used as an alternative to matching ⇒ page 197 .*

– Bolt wheel onto vehicle.

– First, hand-tighten the lowest wheel bolt to about 30 Nm.

– Then, tighten the remaining wheel bolts diagonally to about 30 Nm. This process centres the wheel on the hub.

– Lower vehicle onto its wheels.

– Now use the torque wrench to firmly tighten the wheel bolts diagonally to the prescribed torque.

Carrying out road test

– After balancing the wheel/tyre, carry out a road test.

If you detect vibration during the road test, it may be due to wheel centring tolerances.

In unfavourable circumstances, the component tolerances of wheels and hubs could cumulate. This too can lead to vibration. This can be alleviated using a finish balancer. ⇒ page 196

4.3.3 Balancing wheel, balancing wheel with finish balancer

Note

- ♦ *Before working with a finish balancer , the mechanic needs to have been instructed by the manufacturer of the balancer.*
- ♦ *To balance the wheels, set the wheels of the driven axle on the sensor platforms. On front-wheel drive models, front wheels must be set. On four-wheel drive models, all 4 wheels must be set.*

If you determine a residual imbalance greater than 20 grams when balancing the wheels, you should rotate the mounting position of the wheel on the hub.

– Mark the point at which the imbalance is indicated.

– Then, unbolt the wheel and rotate its position on the hub so that the marking points downwards.

Note

The hub must not rotate during this procedure.

– First, hand-tighten the lowest wheel bolt to about 30 Nm.

– Then, tighten the remaining wheel bolts diagonally to about 30 Nm. This process ensures that the wheel is centred properly on the hub.

– Check whether the imbalance is less than 20 g using the finish balancer.

Note

The imbalance should always be less than 20 grams before you change the balance weight.

– Loosen the wheel bolts again if necessary.

– Rotate wheel again by 1 or 2 wheel bolt holes in relation to wheel hub.

– Tighten wheels using the method described above.

Note

Do not try to reduce the imbalance using balancing weights unless the imbalance is less than 20 grams.

– Balance wheels if imbalance is less than 5 grams.

– Tighten wheel bolts to specified torque if you have not already done so.

Note

Always tighten wheel bolts to specified tightening torque using a torque wrench!

4.4 Vibration control system

More functions are available on the wheel balancing machine than encountered with conventional stationary balancing.

A special feature of this system is the testing of the radial force of the wheel and tyre while rolling.

A roller presses against the wheel with a force of about 635 kg. This simulates the vertical tyre force against the road surface during travel.

Radial and lateral runout in the wheel and tyre and differences in the stiffness of the tyre cause the vertical force of the wheel to vary.

The wheel balancing machine detects and stores the position of the maximum measured radial force in the tyre. Then the position of the smallest distance between the wheel rim flange and the centre of the rim is measured.

4.5 Checking radial and lateral runout of wheels and tyres

Radial and lateral runout occur when the wheel and tyre do not run absolutely true.

For technical reasons, 100% true running is not possible.

Therefore, the manufacturers of these components allow a precisely determined tolerance.

Mounting the tyre in an unfavourable position on the wheel can cause the maximum allowed tolerance for wheel with tyre to be exceeded.

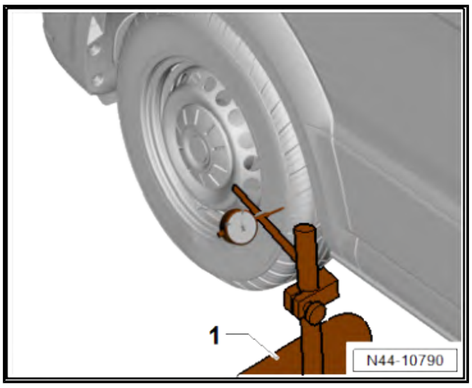
The table shows the maximum permissible tolerances for a wheel with mounted tyre.

Tolerances for radial and lateral runout of wheels with tyres

Wheel with tyre	Radial runout (mm)	Lateral runout (mm)
Commercial vehicle	0.9	1.1 (1.3 in vicinity of lettering)

Checking lateral run-out

– Preload tyre gauge -1- approx. 2 mm.



– Fit tyre gauge -1- to side wall of tyre.
– Turn wheel slowly.
– Make a note of smallest and largest deflection of the indicator needle.

Note

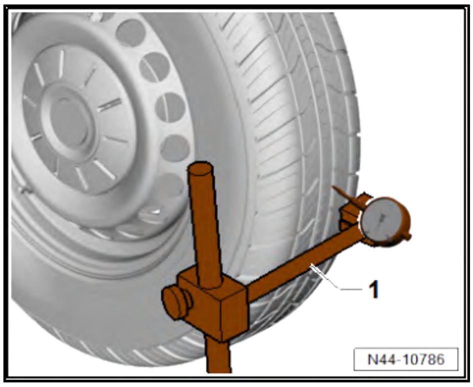
If the difference is greater than 1.3 mm, the lateral runout is too great.

In this case, the lateral runout can be reduced by match mounting the tyre ⇒ page 199 .

Extreme values on the tyre gauge due to small irregularities in the rubber may be disregarded.

Checking radial runout

– Preload tyre gauge -1- approx. 2 mm.



– Fit tyre gauge -1- to tread of tyre.
– Turn wheel slowly.
– Make a note of smallest and largest deflection of the indicator needle.

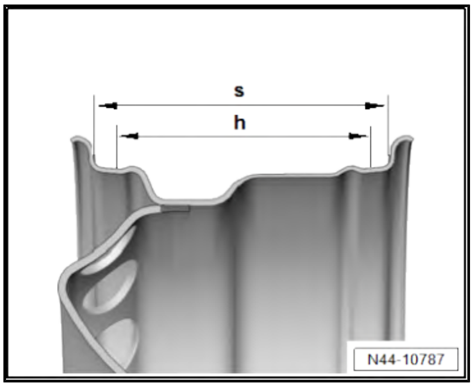
Note

If the difference is greater than 1 mm, the radial runout is too great.

In this case, the lateral runout can be reduced by match mounting the tyre ⇒ page 199 .

4.6 Checking radial and lateral run-out on wheel rim

– Mount rim on wheel balancing machine .
– Use the centring system for wheel balancing machines - VAS 6685- .
– Preload the tyre gauge approx. 2 mm.
– Turn wheel rim slowly.
– Make a note of smallest and largest deflection of indicator needle.



S - Lateral runout

H - Radial runout

– Compare measured value with the specifications in the table ⇒ page 199 .

Note

Extreme values on the tyre gauge due to small irregularities may be disregarded.

Specifications for radial and lateral runout on wheel

Wheel rim	Radial runout (mm)	Lateral runout (mm)
Steel wheel	0.5	0.5
Alloy wheel	0.5	0.8

Note

If the measured value exceeds the specification, acceptably smooth running cannot be attained.

4.7 Matching wheels and tyres

General information

When radial or lateral runout of the wheel and tyre coincide, the imbalance of the wheel is amplified by the tyre.

For technical reasons, 100% true running is not possible ⇒ page 197 .

Before match mounting the used wheels which are fitted on the vehicle, run the tyres warm. This will eliminate any flat spots caused by storage or handling, ⇒ page 200 .

Procedure for match mounting

– Let air out of tyre.
– Press tyre beads off wheel rim flanges.
– Apply tyre assembly paste all round the tyre beads.
– Rotate tyre by 180° in relation to wheel rim.
– Inflate tyre to approx. 4 bar.
– Clamp wheel (together with tyre) on wheel balancing machine .
– Check wheel for true running and for radial and lateral runout .

Note

- ♦ *If the specified values for radial and lateral runout are not exceeded, the wheel can be balanced to 0 g. Specified values appear on* *⇒ page 199* *.*
- ♦ *If the radial and lateral runout is not within the specifications, the tyre must be rotated again.*

– Release air, and press tyre beads off wheel flanges.
– Rotate tyre by 90° in relation to wheel rim (quarter of a rotation).
– Inflate tyre again to 4 bar and check true running.

Note

- ♦ *If the specified values for radial and lateral runout are not exceeded, the wheel can be balanced to 0 g.*
- ♦ *If the radial and lateral runout are not within the specified values, the tyre must be rotated again.*

– Press tyre off wheel rim flange again as described above.

– Rotate tyre by 180° in relation to wheel rim (half a rotation).

If the radial and/or lateral runout are still not within the specified values, check the wheel for radial and/or lateral runout ⇒ page 199 .

If the measured values for radial and lateral runout of the wheel are within the specified values, the tyre has unacceptably high radial or lateral runout. In this case, the tyre must be renewed.

Note

- *After fitting the tyres there will be fitting lubricant between the tyres and the rim flanges.*
- *You should therefore avoid severe braking and acceleration manoeuvres for the first 100 or 200 km driven. The tyres may otherwise rotate on the rims and the work will have been in vain.*

4.8 Determining flat spots on tyres

What is a flat spot?

The terms flat area and flats are also used for the term flat spot.

Flat spots caused by storage or handling also cause vibration in the same way as incorrectly balanced wheels do. It is important that flat spots on the running surface of the tyre are identified as such.

Flat spots caused by storage or handling cannot be balanced and they can reoccur at any time due to various circumstances. Flat spots caused by storage or handling can be eliminated without complicated special tools. Assuming it is not a flat spot caused by full braking ⇒ Wheels and Tyres Guide - General Information; Rep. gr. 44 ; Rolling noises due to tyres, locked-brake flat spots .

Note

Flat spots caused by hard braking cannot be repaired. Such tyres must be renewed.

Reasons for flat spots caused by storage or handling:

- The vehicle has been left standing in one place without being moved for several weeks.
- The tyre pressure is too low.
- The vehicle was placed in a paint shop drying booth after being painted.
- The vehicle was parked with warm tyres in a cool garage or similar for a long period of time. In this case, a standing flat spot may even occur overnight.

Eliminating flat spots caused by storage or handling

- Flat spots caused by storage or handling cannot be eliminated from the tyre using workshop equipment.
- Flat spots caused by storage or handling can be removed only by running the tyres warm.
- The method described below is not recommended in cold and wintry weather.

Requirements and conditions:

– If necessary, check tyre pressure and correct it.

– If possible, drive the vehicle onto a motorway.

– Traffic and road conditions permitting, drive a 20 to 30 km stretch at a speed of 120 to 150 km/h (where legally permissible).

Note

♦ *Do not endanger yourself or other road users during this road test.*

♦ *Observe the highway code and speed limitations in force when performing the road test.*

– Jack up the vehicle immediately after the trip.

– Remove wheels.

– Balancing wheels on stationary wheel balancer ⇒ page 193 .

5 Adaptive cruise control

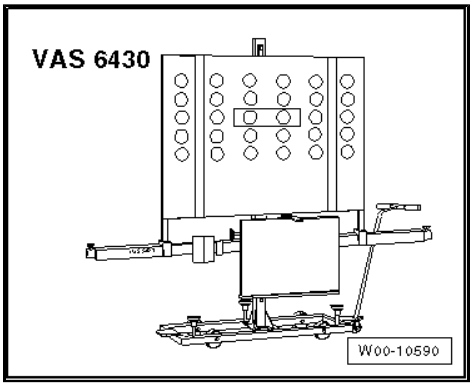
⇒ “5.1 Calibrating adaptive cruise control, Crafter”, page 202

⇒ “5.2 Calibrating adaptive cruise control, MAN TGE”, page 205

5.1 Calibrating adaptive cruise control, Crafter

Special tools and workshop equipment required

♦ Setting device - VAS 6430- or setting device, basic kit - VAS 6430/1-



♦ ACC reflector mirror - VAS 6430/10-

♦ Wheel alignment computer

Before adjusting the front radar, check the sensor, its bracket and its securing elements for damage, external influence and secure seating. Repair any damaged parts if necessary.

Before adjusting the front radar, read out the event memory and rectify any faults found.

The measured misalignment-angle value of the front radar control unit indicates whether or not the sensor is out of adjustment.

Use only a wheel alignment machine and adjusting devices approved by the manufacturer to adjust the front radar.

Correct adjustment is essential for proper front assist operation.

The adaptive cruise control unit - J428- needs to be calibrated under the following conditions:

♦ Adaptive cruise control unit - J428- has been renewed

♦ Bumper cover was detached or removed and installed

♦ Lock carrier has been moved to service position

♦ Lock carrier has been removed and installed

♦ Lock carrier has been renewed

♦ Rear axle has been loosened

♦ Vertical misalignment angle from tolerance range is ±4°

♦ Horizontal misalignment angle from tolerance range is ±6°

Note

- *Before driving vehicle onto wheel alignment platform, make sure there is sufficient space between vehicle and setting device - VAS 6430- . The distance between ACC reflecting mirror - VAS 6430- and the sensor must be 120 cm ± 2.5 cm.*
- *If available space is not adequate, move vehicle backwards onto alignment platform as required to achieve space required.*
- *If the position of the ACC reflecting mirror - VAS 6430/10- on alignment beam is changed during the adjustment work, the setting of adjustment device - VAS 6430- must always be checked (e.g. spirit levels, individual toe values on adjusting beam, etc.).*

- Before starting adjustment, read out event memory and rectify any faults.

Adjustment procedure described here is based on adjustment device - VAS 6430- .

The following setting sequence is to be adhered to:

1 - Establish a distance of 120 cm ± 2.5 cm between the ACC reflecting mirror - VAS 6430/10- fitted in the middle and the sensor in the radiator grille.

2 - Install ACC reflector mirror - VAS 6430/10- centred on alignment beam.

If wheel alignment has just been carried out, the steps described under “Setting without preceding wheel alignment” do not have to be carried out.

Setting without preceding wheel alignment

– Press button to select front radar calibration in wheel alignment computer.

– Note testing preconditions for wheel alignment ⇒ page 163 .

– Drive car onto wheel alignment platform.

– Connect battery charger ⇒ Electrical system; General information; Rep. gr. 27 ; Charging battery .

– Connect ⇒ Vehicle diagnostic tester to vehicle, route diagnostic cable through open door window.

Note

During the setting procedure, make sure that all the car doors remain closed and that the car's exterior lighting is switched off.

– Move front wheels to straight-ahead position.

– Mount quick-release clamps on rear wheels.

– Mount measurement transducers on rear wheels.

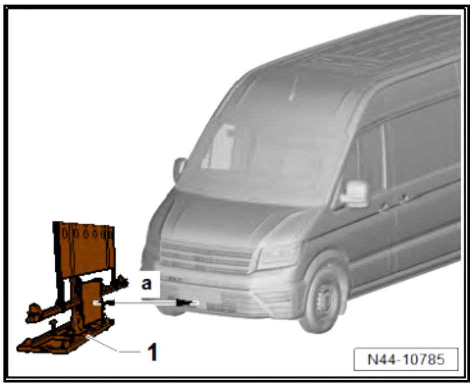
– Carry out wheel run-out compensation at rear wheels.

Setting with or without preceding wheel alignment

– Position ACC setting device - VAS 6430- -1- centrally and parallel at distance -a- from centrally fitted ACC reflector mirror - VAS 6430/10- to adaptive cruise control unit - J428- .

Maß a - 120 cm ± 2.5 cm

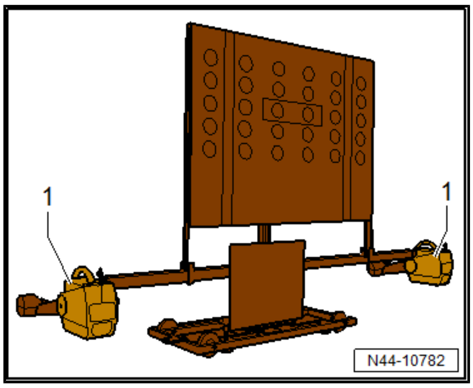
– Remove soiling on adaptive cruise control unit - J428-



Note

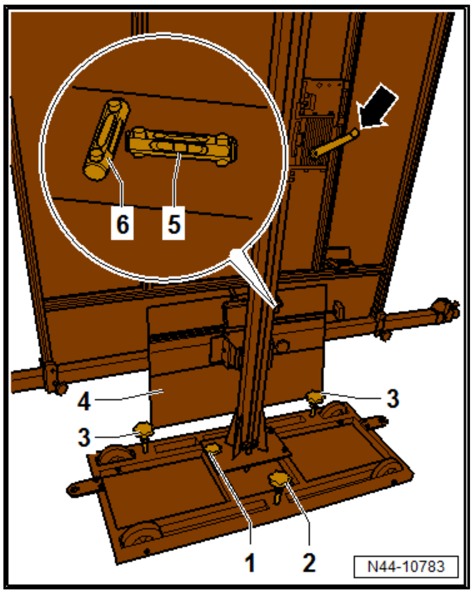
ACC setting device - VAS 6430- is not allowed to be moved on alignment beam.

– Mount measurement transducers -1- of front wheels on alignment beam.

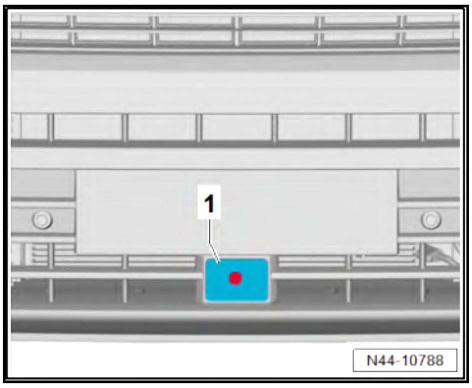


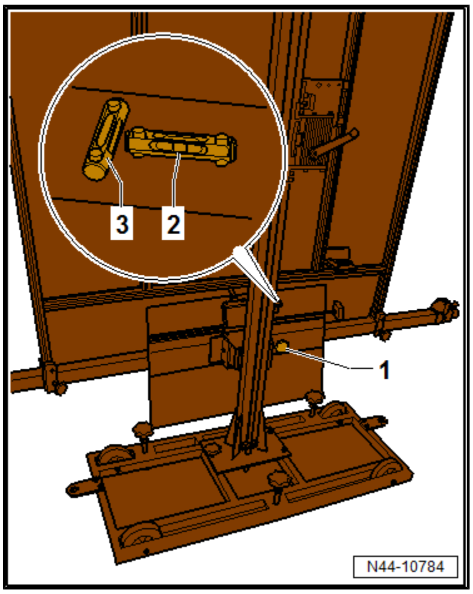
– Use adjusting screws -1, 2, 3- to balance spirit levels -5- and -6- on - VAS 6430/10- .

– Adjust mirror -4- by means of crank handle -arrow- in such a way that the laser beam contacts the centre of the sensor lens.



– Position mirror on side of alignment beam so that the laser beam contacts the centre of the adaptive cruise control unit - J428- -1-.





– Turn fine adjuster screw -1- until display on wheel alignment computer is within tolerance.

– Level spirit levels -2- and -3- of wheel alignment sensor.

– Now, use laser beam of -VAS 6430/10- to check again whether spirit levels are horizontal and laser beam hits centre of sensor lens.

Note

If the laser does not point at the sensor lens, - VAS 6430/10- must be pushed on the setting device - VAS 6430- .

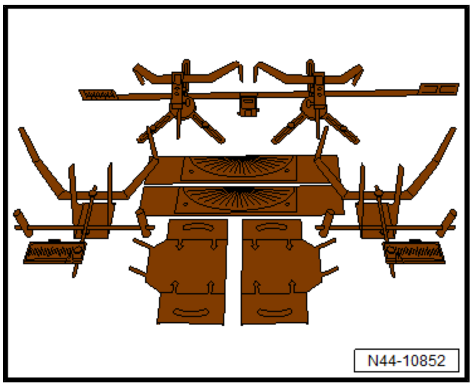
– Press ⇒ Vehicle diagnostic tester `GoTo` button and select `Select function/component` function.

– Follow further instructions displayed on ⇒ Vehicle diagnostic tester.

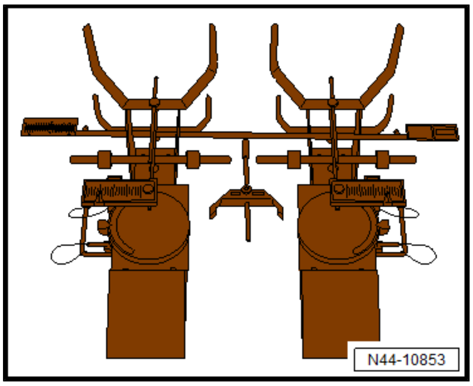
5.2 Calibrating adaptive cruise control, MAN TGE

Special tools and workshop equipment required

♦ Wheel alignment system HD-30 Easy Touch - 80.99607-6080-

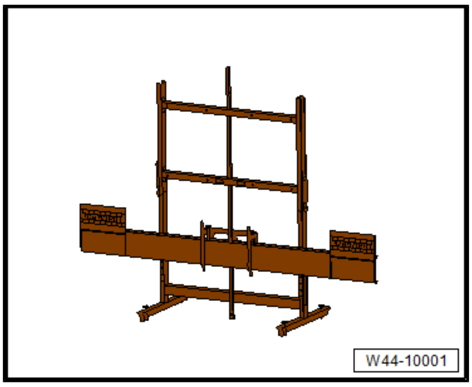


♦ Upgrade kit Transporter - 80.99607-6125-

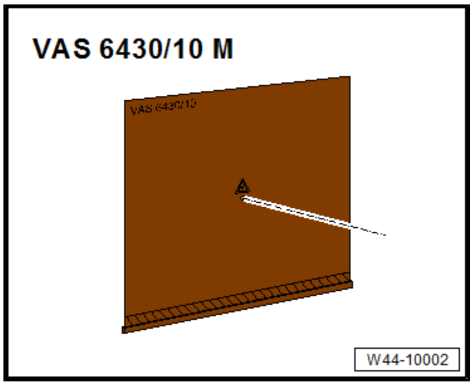


♦ HD-10 light TGE Transporter - 80.99607-6126- can be used for different wheel alignment systems as an alternative to upgrade kit Transporter - 80.99607-6125- .

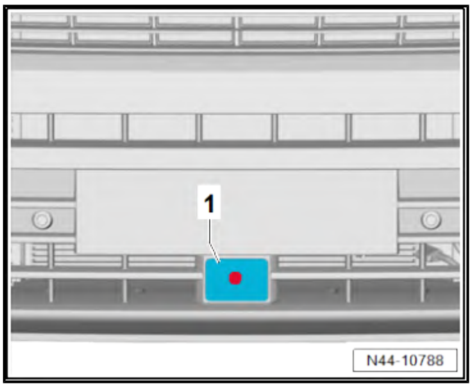
- Adjusting device CLS TGE - 80.99607-6120-



- ACC reflector mirror VAS 6430/10 M - 80.99607-6123-



- Vehicle diagnostic tester
- Battery charger



Before adjusting the front radar, check the sensor, its bracket and its securing elements for damage, external influence and secure seating. Repair any damaged parts if necessary. Remove soiling on sensor -1-.

Before adjusting the front radar, read out the event memory and rectify any faults found.

The measured misalignment-angle value of the front radar control unit indicates whether or not the sensor is out of adjustment.

The adaptive cruise control unit - J428- needs to be calibrated under the following conditions:

- ♦ Adaptive cruise control unit - J428- has been renewed
- ♦ Bumper cover was detached or removed and installed
- ♦ Lock carrier has been moved to service position
- ♦ Lock carrier has been removed and installed
- ♦ Lock carrier has been renewed
- ♦ Rear axle has been loosened
- ♦ Vertical misalignment angle from tolerance range is ±4°
- ♦ Horizontal misalignment angle from tolerance range is ±6°

Note

- ♦ *Park the vehicle on a flat hard standing. The distance between setting device CLS TGE - 80.99607-6120- and the sensor must be 120 cm ± 2.5 cm.*
- ♦ *Incorrect wheels or tyres and tyre inflation pressures can lead to false measuring and calibrating procedures.*
- ♦ *If the position of the ACC reflecting mirror VAS 6430/10 M - 80.99607-6123- on alignment beam is changed during the adjustment work, the setting of adjustment device CLS TGE - 80.99607-6120- must always be checked (e.g. spirit levels, individual toe values on adjusting beam, etc.).*
- ♦ *Observe the test prerequisites ⇒ page 163 .*

The following setting sequence is to be adhered to:

1 - Install ACC reflector mirror VAS 6430/10 M - 80.99607-6123- centred on alignment beam.

2 - Drive vehicle centrally in front of ACC reflector mirror VAS 6430/10 M - 80.99607-6123- at a distance of 120 cm ± 2.5 cm.

3 - Position setting device CLS TGE - 80.99607-6120- -1- parallel to sensor with distance -a- from centrally attached ACC reflector mirror VAS 6430/10 M - 80.99607-6123- .

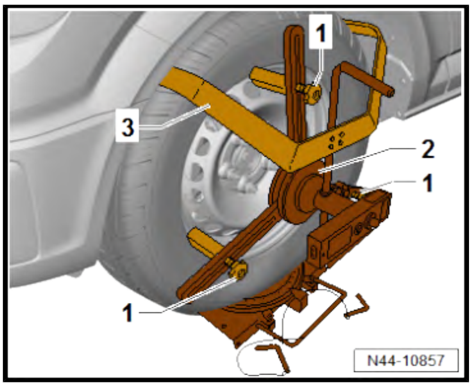
-Dimension a- 120 cm ± 2.5 cm

– Switch on laser on ACC reflector mirror VAS 6430/10 M - 80.99607-6123- .

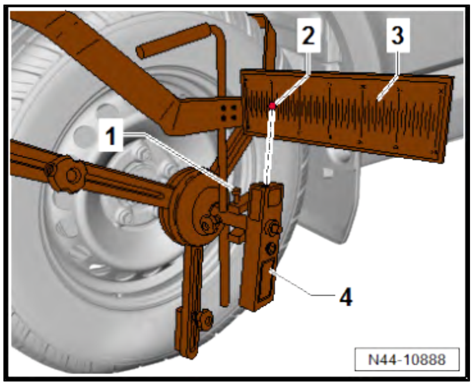
– Direct laser beam so that it hits centre of adaptive cruise control unit - J428- -1-.

Mounting laser measuring head on rear axle

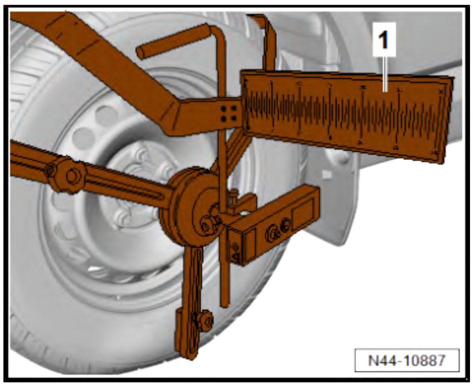
– Attach laser measuring head -2- with retaining bracket -3- loosely on rear wheel.
– Loosen measuring sensor -1-, relocate and mount on rim flange or tyre.
– Tighten threaded connection on measuring sensors.
– Adjust retaining brackets until all measuring sensors are in contact with rim. If necessary, choose contact surface of tyre if rims are aluminium.



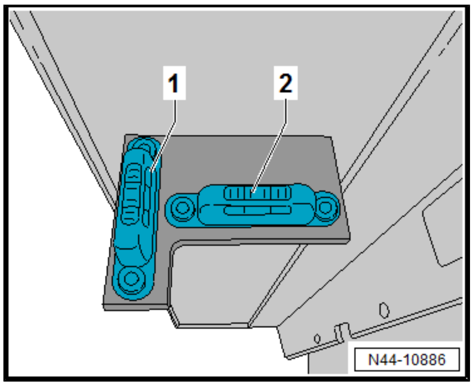
– Switch on laser measuring head -4- and turn upwards until point of laser -2- is on measuring scale -3-.
– Loosen locking screw -1- and move measuring scale -3- to value 5.
– Tighten locking screw -1-.
– Position laser measuring head in such a way that front mirror on setting device CLS TGE - 80.99607-6120- is hit and laser beam is reflected onto rear mirror.
– Repeat procedure on other side of vehicle.



– To align setting device parallel to front of vehicle, move setting device CLS TGE - 80.99607-6120- so that same value is displayed on left and right rear axle mirror -1-.



– Use four setting screws on setting device CLS TGE - 80.99607-6120- with aid of upper levels -1- and -2- to set vertical position.
– Fine adjust via screws on ACC reflector mirror VAS 6430/10 M - 80.99607-6123- to adjust board horizontally.



Note

During the setting procedure, make sure that all the car doors remain closed and that the car's exterior lighting is switched off.

– Connect battery charger ⇒ Electrical system; General information; Rep. gr. 27 ; Charging battery .
– Connect ⇒ Vehicle diagnostic tester to vehicle, route diagnostic cable through open door window.
– On ⇒ Vehicle diagnostic tester, press `GoTo` button and choose `Select function/component` function.
– Follow further instructions displayed on ⇒ Vehicle diagnostic tester.

6 Front camera for assist systems

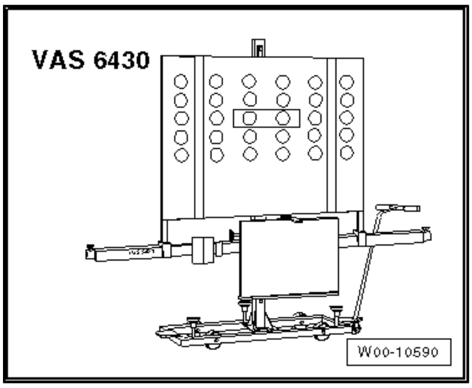
⇒ “6.1 Calibrating front camera for assist systems, Crafter”, page 209

⇒ “6.2 Calibrating front camera for assist systems, MAN TGE”, page 214

6.1 Calibrating front camera for assist systems, Crafter

Special tools and workshop equipment required

- Setting device - VAS 6430-



- Calibration unit - VAS 6350A-

- Wheel alignment computer
- Vehicle diagnostic tester
- Battery charger

Note

The following situations can cause the camera function to be impaired by sustained poor visibility of the lane marker lines:

- The field of view of the camera is soiled or iced over. In this case, clean the affected area.
- The field of view of the camera is fogged over.

The camera’s viewing window must be cleaned manually if there is heavy soiling on the inside of the windscreen in the field of view of the camera. To do this, the control unit and lens hood must be removed and the windscreen cleaned using cleaning solution. Remove control unit and lens hood ⇒ Electrical system; Rep. gr. 96 ; Front camera for driver assist systems .

Correct calibration is a basic precondition for the front camera for driver assist systems - R242- to function without faults.

The front camera for driver assist systems - R242- must be recalibrated if:

- ♦ Entry “no or incorrect basic setting / adaptation” is stored in event memory.
- ♦ Front camera for driver assist systems - R242- has been renewed.
- ♦ Windscreen has been renewed or removed.
- ♦ Rear axle has been loosened
- ♦ Modifications having an effect on the body height have been made to the car's running gear.

Note

- ♦ *Before calibrating the front camera for driver assist systems, read out event memory and rectify faults as necessary.*
- ♦ *Calibration of the front camera for driver assist systems must be performed using only manufacturer-approved wheel alignment equipment!*
- ♦ *Calibration of the driver assistant system's front camera is permitted only with the setting device - VAS 6430- !*

Note

- ♦ *The front camera for driver assist systems - R242- must be correctly located in the holder.*
- ♦ *The field of view of the camera must be clean and clear.*
- ♦ *Before driving the vehicle onto the wheel alignment platform, make sure there is sufficient space between the midpoint of the hub of the front wheels and the setting device - VAS 6430- .*
- ♦ *The distance between the setting device - VAS 6430- and the centre point of the hub of the front wheels must be 2000 mm ± 25 mm.*
- ♦ *If available space is not adequate, move vehicle backwards onto alignment platform as required to achieve space required.*
- ♦ *The calibration board must be positioned centrally on the setting device.*
- ♦ *The event memory must be read before starting calibration. Any memory entries must be deleted.*

– Note testing preconditions for wheel alignment ⇒ page 163 .

– Drive car onto wheel alignment platform.

– Connect battery charger ⇒ Vehicle electrics; Rep. gr. 27 ; Battery; Charging battery .

– Connect ⇒ Vehicle diagnostic tester to vehicle, route diagnostic cable through open door window.

Note

During the calibration procedure, make sure that all the car doors remain closed and that the car's exterior lighting is switched off.

– Move front wheels to straight-ahead position.

– Select calibration run for front camera for driver assist systems -R242- on wheel alignment computer.

– Mount quick-release clamps on all four wheels.

– Mount wheel alignment sensors on wheels.

– Carry out wheel run-out compensation at rear wheels.

– Bounce springs.

Note

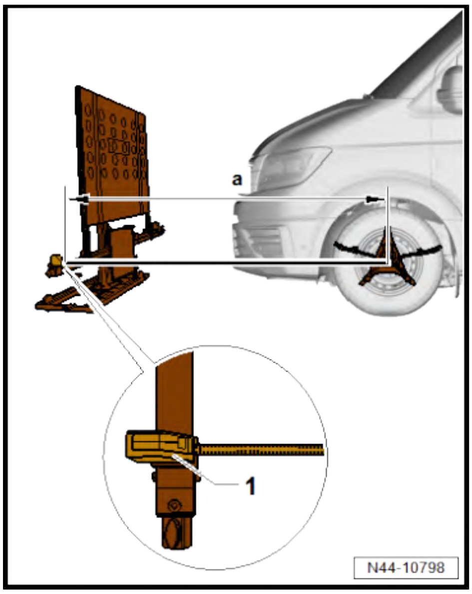
♦ *Setting device - VAS 6430- is not allowed to be moved on alignment beam.*

♦ *Wheel alignment platform must be in the lowermost levelled position for next work step.*

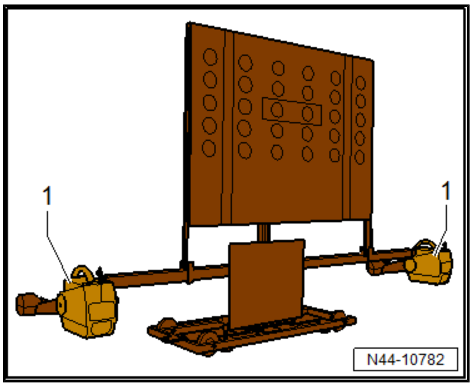
– Turn setting device - VAS 6430- upwards so that alignment beam is parallel to middle of measurement transducers on front wheels, in order to perform a correct measurement with distance measuring unit -1-.

1 - Distance measuring unit with tape measure and wheel centre mounting - VAS 6350/1-

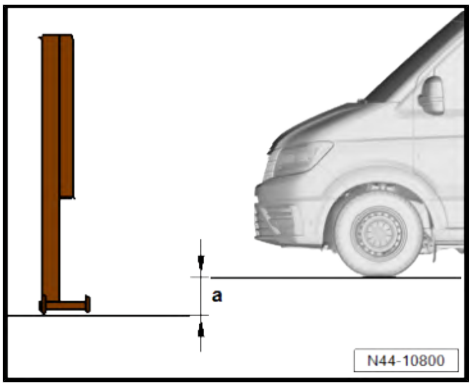
– Position setting device - VAS 6430- at distance -a- of 2000 mm ± 25 mm from centre point of hub of front wheels to beam of setting device - VAS 6430- .



– Mount measurement transducer -1- of front wheels onto setting device - VAS 6430- .



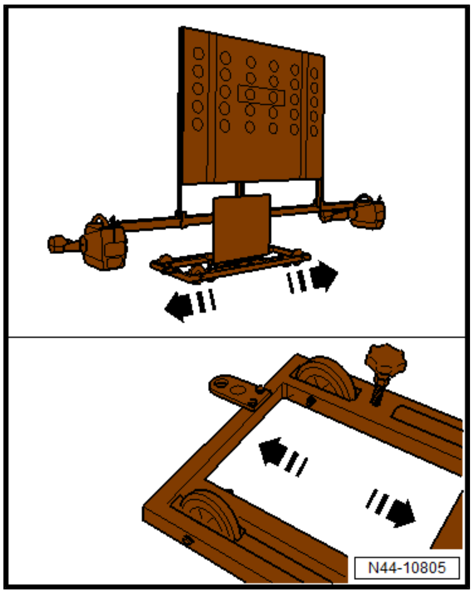
– Measure height value -a- between standing surface of setting device - VAS 6430- and wheel support surface on wheel alignment platform.
– Enter calculated value in wheel alignment computer.



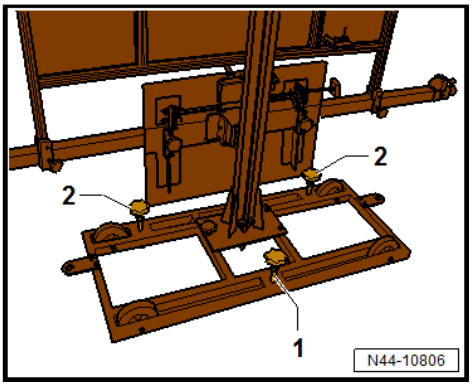
Note

If calibrating with a floor-mounted wheel alignment platform, ensure that adapter - VAS 6430/9- is used.

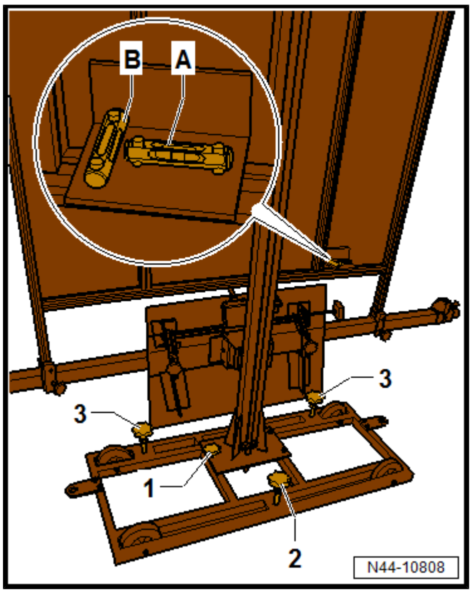
– Move setting device - VAS 6430- sideways in direction of -arrow- until display on wheel alignment computer is within tolerance.



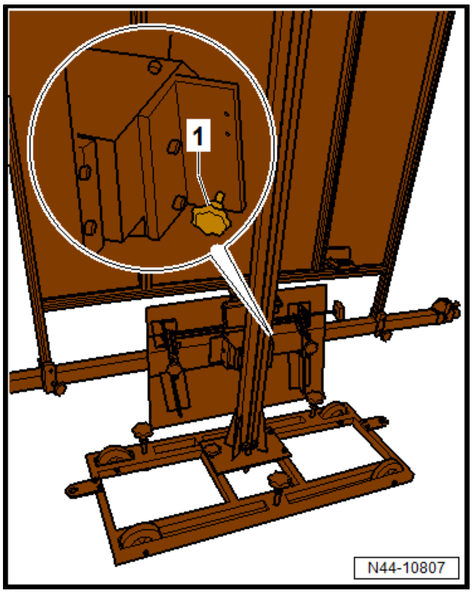
– Screw down adjuster screws -1- and -2- slightly to secure setting device - VAS 6430- against rolling away.



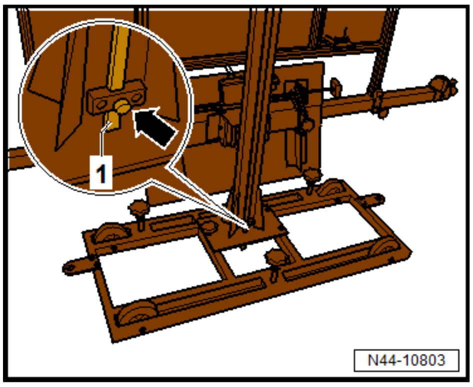
– Use adjuster screw -1- to level spirit level -A-.
– Use adjuster screws -2- and -3- to level spirit level -B-.



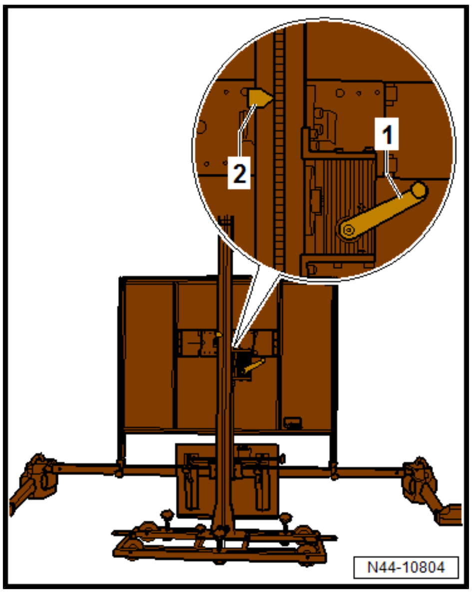
– Turn fine adjuster screw -1- until display on wheel alignment computer is within tolerance.



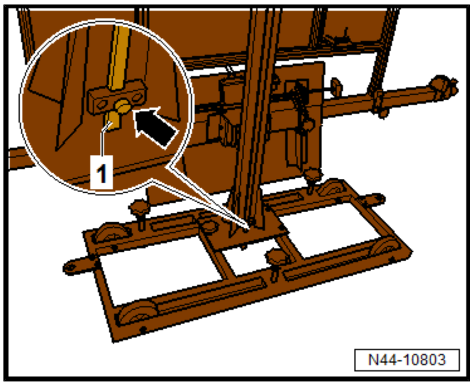
– Loosen clamping bolt -arrow-.
– Place measuring board -1- on the ground.



– Use crank -1- to set calibration panel - VAS 6430/4- to a nominal height -2- of 1850 mm.



When nominal height has been reached, measuring bar -1- must be moved upwards slightly and secured with clamping screw -arrow-.



– Continue calibration with ⇒ Vehicle diagnostic tester.

Height adjustment on vehicles with PR numbers K4A/K4B

– Determine height at front on left and right-hand side from edge of wheel housing to footprint.

– Determine height at rear on left and right-hand side from edge of wheel housing to footprint.

Height adjustment on vehicles with PR numbers K4F/K4G

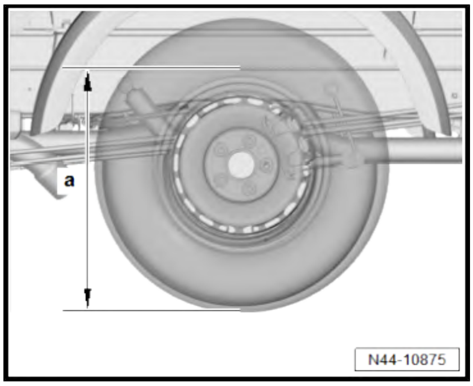
– Determine height at front on left and right-hand side from edge of wheel housing to footprint.

– Determine height at rear on left and right-hand side from lower edge of dropside frame to footprint.

Height adjustment on vehicles with PR numbers 5HJ/5HN

– Determine height at front on left and right-hand side from edge of wheel housing to footprint.

– Determine rear dimension -a- from height of body on left and right-hand side at lower edge of longitudinal member flange to footprint.



Note

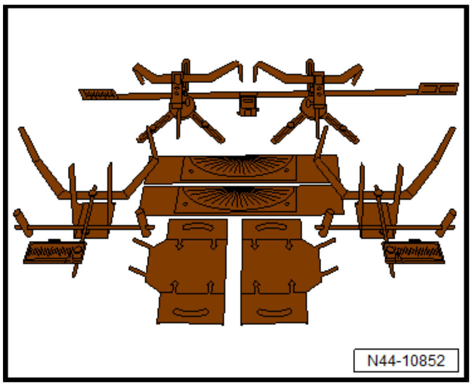
Dimension -a- must be determined in the direction of travel just behind the centres of the rear wheels.

Explanation of PR numbers ⇒ page 187 .

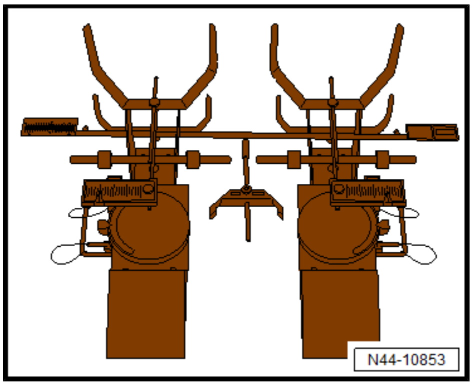
6.2 Calibrating front camera for assist systems, MAN TGE

Special tools and workshop equipment required

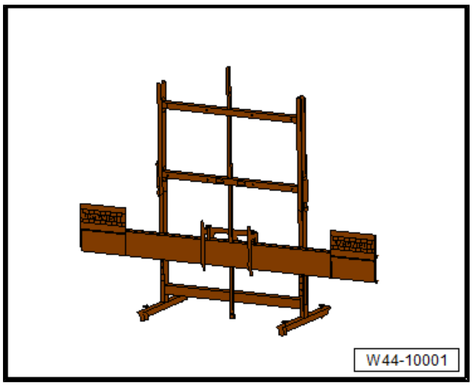
- Wheel alignment system HD-30 Easy Touch - 80.99607-6080-



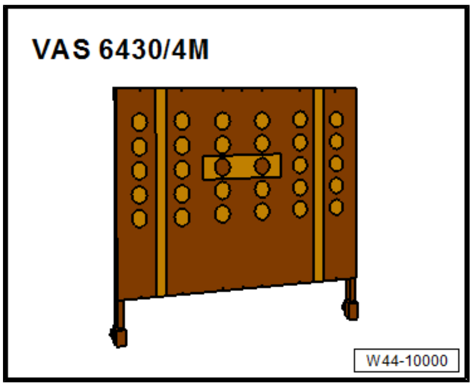
- Upgrade kit Transporter - 80.99607-6125-



- HD-10 light TGE Transporter - 80.99607-6126- can be used for different wheel alignment systems as an alternative to upgrade kit Transporter - 80.99607-6125- .
- Adjusting device CLS TGE - 80.99607-6120-



- Calibration board for lane departure warning VAS 6430/4M - 80.99607-0273-



- Vehicle diagnostic tester
- Battery charger

Note

The following situations can cause the camera function to be impaired by sustained poor visibility of the lane marker lines:

- The field of view of the camera is soiled or iced over. In this case, clean the affected area.
- The field of view of the camera is fogged over.

The camera’s viewing window must be cleaned manually if there is heavy soiling on the inside of the windscreen in the field of view of the camera. To do this, the control unit and lens hood must be removed and the windscreen cleaned using cleaning solution. Remove control unit and lens hood ⇒ Electrical system; Rep. gr. 96 ; Front camera for driver assist systems .

Correct basic setting is a precondition for trouble-free operation of front camera for driver assist systems - R242- .

The front camera for driver assist systems - R242- must be recalibrated if:

- Entry “no or incorrect basic setting / adaptation” is stored in event memory.
- Front camera for driver assist systems - R242- has been renewed.
- Windscreen has been renewed or removed.
- Rear axle has been loosened.
- Modifications having an effect on the body height have been made to the car's running gear.

Note

- *Before calibrating the front camera for driver assist systems, read out event memory and rectify faults as necessary.*
- *For calibration of the front camera for driver assist systems, it is only permissible to use setting device CLS TGE - 80.99607-6120- with calibration board for lane departure warning VAS 6430/4M - 80.99607-0273- .*

Note

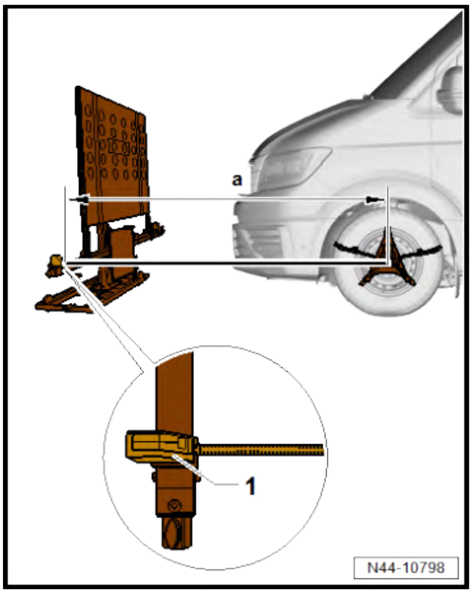
- *Park the vehicle on a flat hard standing. The distance between setting device CLS TGE - 80.99607-6120- and the wheel hub centre of the front axle must be 2000 mm ± 25 mm.*
- *Incorrect wheels or tyres and tyre inflation pressures can lead to false measuring and calibrating procedures.*
- *If the position of the calibration board for lane departure warning VAS 6430/4M - 80.99607-0273- on alignment beam is changed during the adjustment work, the setting of adjustment device CLS TGE - 80.99607-6120- must always be checked (e.g. spirit levels, individual toe values on adjusting beam, etc.).*
- *Observe the test prerequisites ⇒ page 163 .*

The following setting sequence is to be adhered to:

1 - Mount calibration board for lane departure warning VAS 6430/4M - 80.99607-0273- at a height of 1850 mm on setting device CLS TGE - 80.99607-6120- .

2 - Drive vehicle with distance of 2000 mm ± 25 mm to hub centre point centrally in front of calibration board for lane departure warning VAS 6430/4M - 80.99607-0273- .

3 - Position setting device CLS TGE - 80.99607-6120- at distance -a- in front of attached calibration board for lane departure warning VAS 6430/4M - 80.99607-0273- so that it is parallel.



-Dimension a- 2000 mm ± 25 mm

– -Dimension a- can be determined e.g. with tape measure -1-.

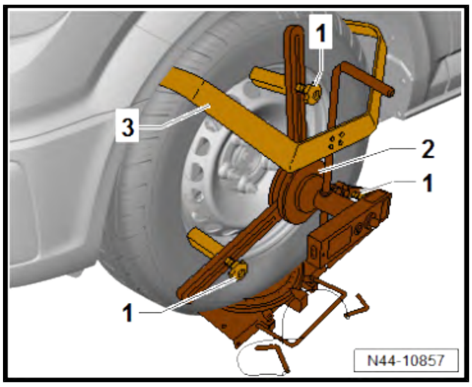
Mounting laser measuring head on rear axle

– Attach laser measuring head -2- with retaining bracket -3- loosely on rear wheel.

– Loosen measuring sensor -1-, relocate and mount on rim flange or tyre.

– Tighten threaded connection on measuring sensors.

– Adjust retaining brackets until all measuring sensors are in contact with rim. If necessary, choose contact surface of tyre if rims are aluminium.



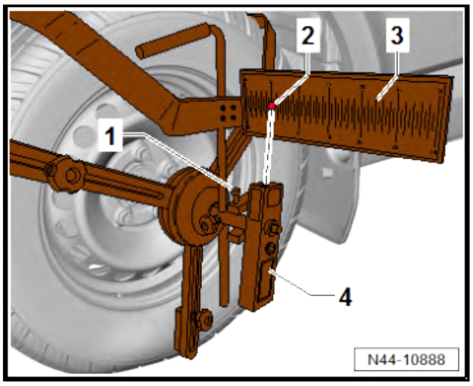
– Switch on laser measuring head -4- and turn upwards until point of laser -2- is on measuring scale -3-.

– Loosen locking screw -1- and move measuring scale -3- to value 5.

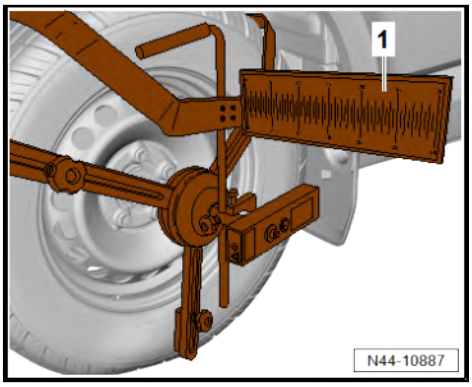
– Tighten locking screw -1-.

– Position laser measuring head in such a way that front mirror on setting device CLS TGE - 80.99607-6120- is hit and laser beam is reflected onto rear mirror.

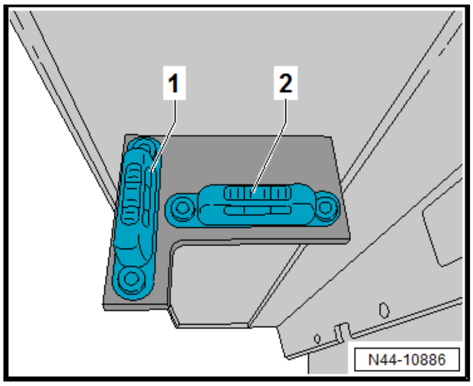
– Repeat procedure on other side of vehicle.



– To align setting device parallel to front of vehicle, move setting device CLS TGE - 80.99607-6120- so that same value is displayed on left and right rear axle mirror -1-.



– Use four setting screws on setting device CLS TGE - 80.99607-6120- with aid of upper levels -1- and -2- to set vertical position.



– Fine adjust via screws on calibration board for lane departure warning VAS 6430/4M - 80.99607-0273- to adjust board horizontally.

– Check height (1850 mm) of calibration board of lane departure warning VAS 6430/4M - 80.99607-0273- .

Note

During the setting procedure, make sure that all the car doors remain closed and that the car's exterior lighting is switched off.

– Connect battery charger ⇒ Electrical system; General information; Rep. gr. 27 ; Charging battery .

– Connect ⇒ Vehicle diagnostic tester to vehicle, route diagnostic cable through open door window.

– On ⇒ Vehicle diagnostic tester, press `GoTo` button and choose `Select function/component` function.

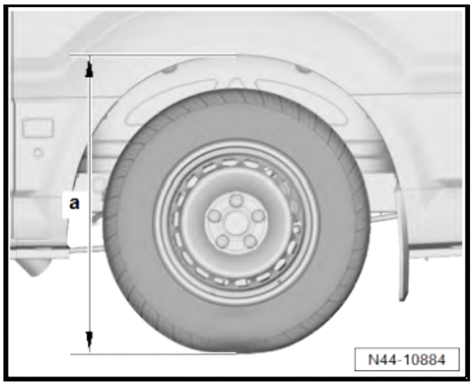
– Follow further instructions displayed on ⇒ Vehicle diagnostic tester.

Height measurement

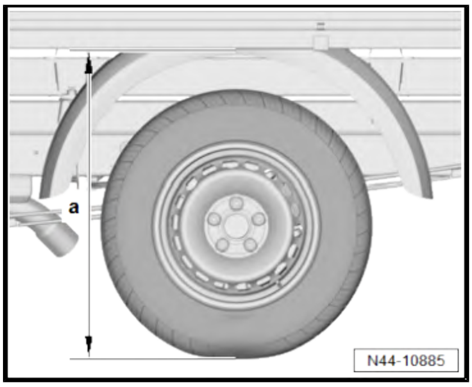
Vehicles with PR numbers K4A/K4B (box body)

– Determine height at front on left and right-hand side from edge of wheel housing to footprint.

– Determine rear dimension -a- on left and right-hand side from edge of wheel housing to footprint.

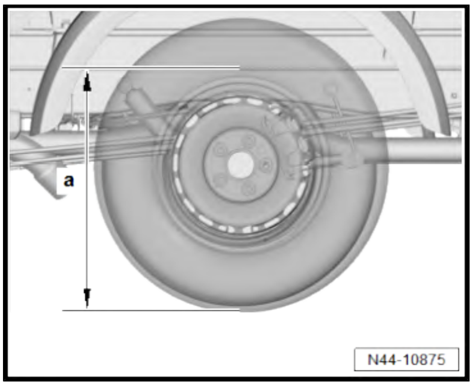


Vehicles with PR numbers K4F/K4G (dropside body)



– Determine height at front on left and right-hand side from edge of wheel housing to footprint.
– Determine rear dimension -a- on left and right-hand side from lower edge of dropside frame to footprint.

Vehicles with PR numbers 5HJ/5HN (chassis)



– Determine height at front on left and right-hand side from edge of wheel housing to footprint.
– Determine rear dimension -a- from height of body on left and right-hand side at lower edge of longitudinal member flange to footprint.

Note

Dimension -a- must be determined in the direction of travel just behind the centres of the rear wheels.

Explanation of PR numbers ⇒ page 187 .

48 – Steering

1 Steering wheel

⇒ “1.1 Assembly overview - steering wheel”, page 220

⇒ “1.2 Removing and installing steering wheel”, page 220

1.1 Assembly overview - steering wheel

1 - Steering column

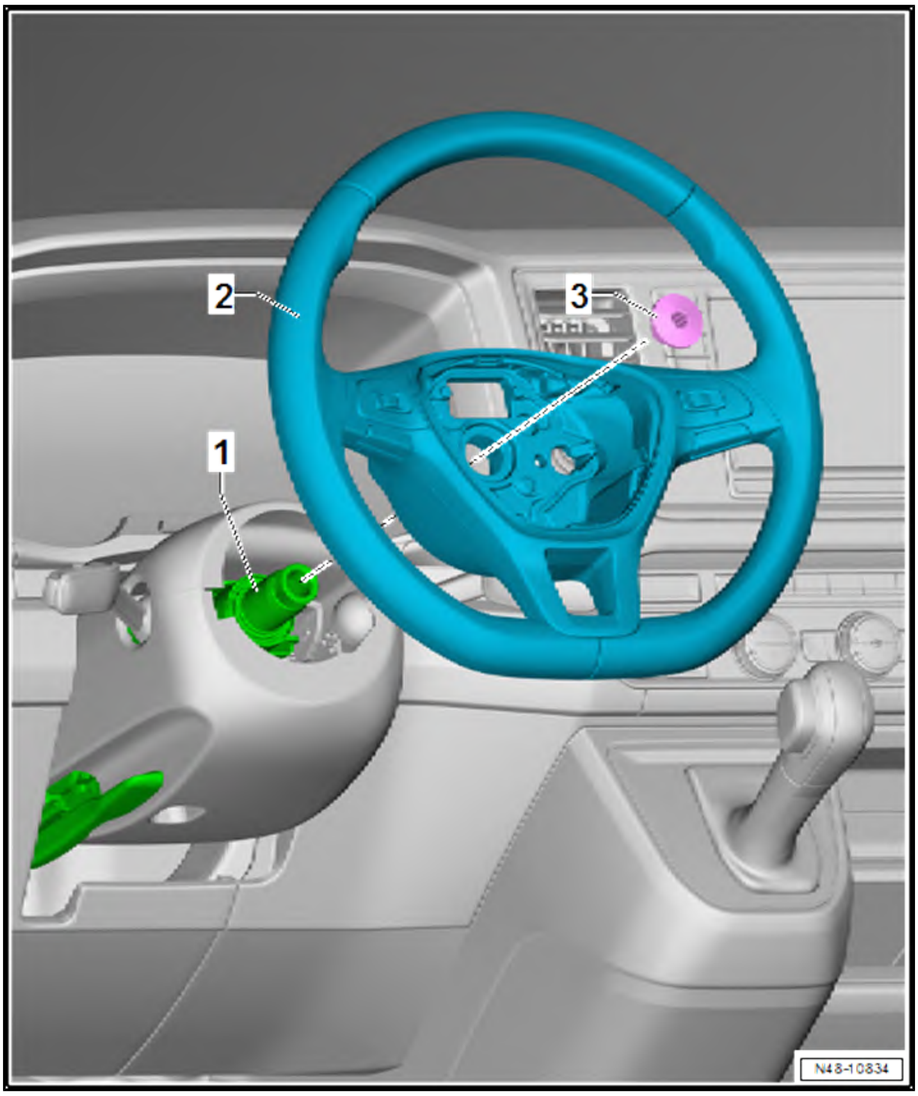
- ❑ Removing and installing ⇒ page 225

2 - Steering wheel

- ❑ Removing and installing ⇒ page 220

3 - Bolt

- ❑ Renew after removal
- ❑ 30 Nm +90°



1.2 Removing and installing steering wheel

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1331-

Removing

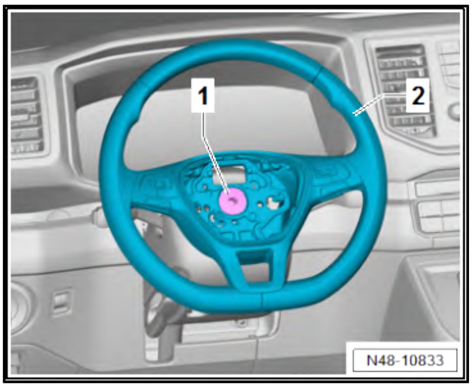
– Move steering column to middle height position.

– Remove airbag unit ⇒ General body repairs, interior; Rep. gr. 69 ; Driver side airbag; Removing and installing airbag unit with igniter .

Note

The steering wheel must be removed/installed in the middle position (wheels facing straight ahead).

– If fitted, disconnect electrical connector for steering wheel heating.



– Unscrew bolt -1-.
– Check that steering column has a centre punch mark that aligns with mark on steering wheel.
– If this is not the case, the position of the steering wheel to steering column must be marked on the steering column with a centre punch mark.
– Pull steering wheel -2- off steering column.

Installing

Install in the reverse order of removal, observing the following:

Make sure that the wheels are pointing straight ahead before fitting the steering wheel.

– When installing a steering wheel that has been removed, make sure that the markings on the steering column and steering wheel are in line.
– When installing a new steering wheel (without marking), the steering wheel must be mounted in middle position (steering wheel spoke must be horizontal and wheels in straight position).
– Install steering wheel.

Note

The bolt -1- must only be tightened when the actuating lever of the steering column is closed.

– Install airbag unit ⇒ General body repairs, interior; Rep. gr. 69 ; Driver side airbag; Removing and installing airbag unit with igniter .
– Carry out road test.
– If the steering wheel is skewed, the steering wheel must be removed again and fitted onto the splines of the steering column.

Specified torques

♦ ⇒ “1.1 Assembly overview - steering wheel”, page 220

2 Steering column

2.1 Assembly overview - steering column

1 - Steering column
- ❑ Removing and installing ⇒ page 225

2 - Bolts
- ❑ Qty. 4
- ❑ Observe specified torque and tightening sequence ⇒ page 224

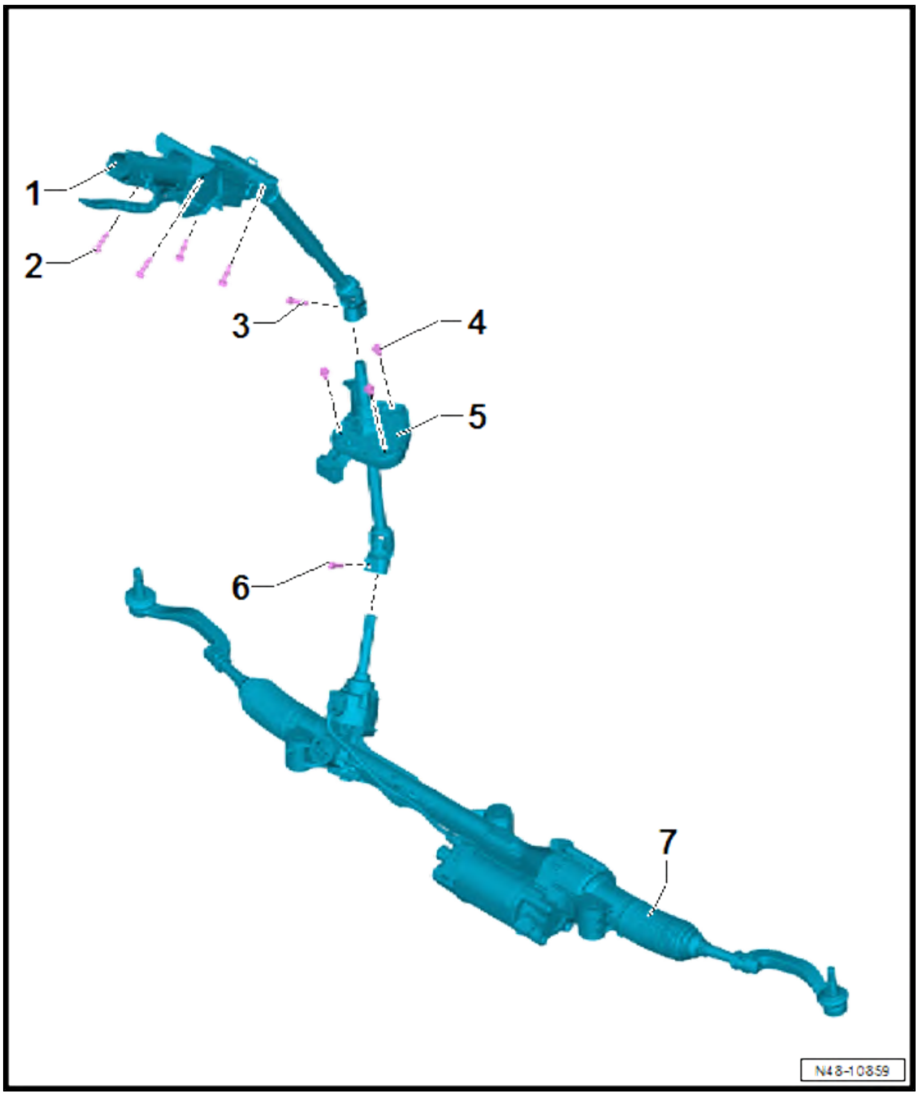
3 - Bolt
- ❑ Renew after removal
- ❑ 10 Nm +190°

4 - Bolts
- ❑ Qty. 4
- ❑ 25 Nm

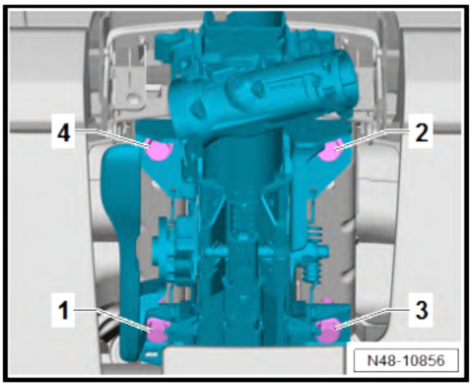
5 - Intermediate steering shaft
- ❑ Removing and installing ⇒ page 227

6 - Bolt
- ❑ Renew after removal
- ❑ 20 Nm +90°

7 - Steering rack
- ❑ Removing and installing ⇒ page 230



Specified torque and tightening sequence of steering column



Component	Specified torque
Bolts -1 … 4-	20 Nm

2.2 Checking steering column for damage

Visual check

– Check all steering column parts for damage.

Checking function

– Check that steering column turns smoothly and easily.

– Check whether steering column can be moved in the longitudinal direction and vertically.

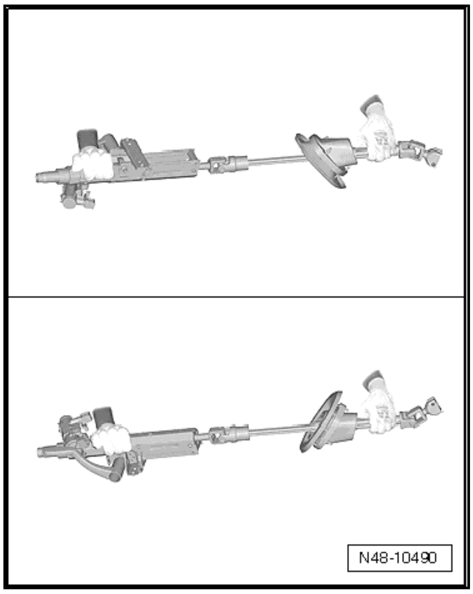
2.3 Handling and transporting steering column

Note

♦ *Adherence to proper steering column handling is essential.*

♦ *Improper handling of steering column may damage the steering column, leading to safety risks.*

Proper steering column handling and transport



♦ Use both hands to transport steering column.

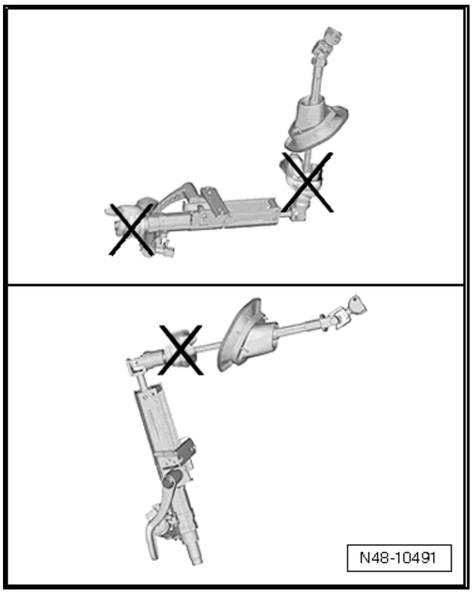
♦ Hold steering column at upper jacket tube and in area just above lower universal joint.

♦ Do not touch or contaminate plastic tube on intermediate steering shaft.

Improper handling of steering column with safety risks

Following methods of handling will damage universal joint bushes and lower steering column bearing:

- ♦ Transporting steering column with one hand on jointed shaft.
- ♦ Bending joints more than 90°.



2.4 Removing and installing steering column

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-

Removing

Note

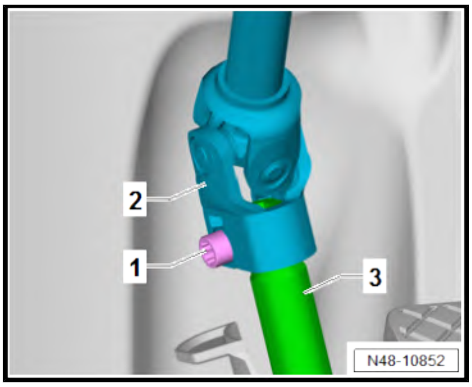
- ♦ *Observe instructions for use of steering column* *⇒ page 224* *.*
- ♦ *Only the complete steering column is supplied as a replacement part. Repair is not possible.*
- ♦ *Steering lock housing can be transferred ⇒ Electrical system; Rep. gr. 94 ; Steering column switch module; Removing and installing steering lock housing .*

– Turn wheels to straight-ahead position.

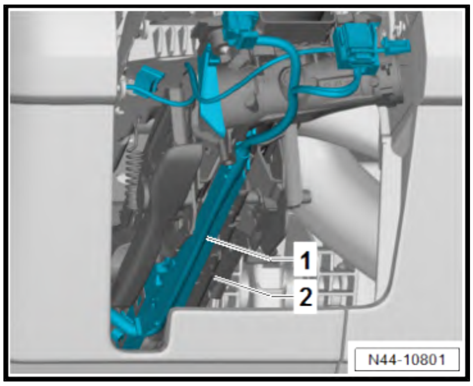
– Swing steering column down as far as possible and pull out.

– Remove steering column switch module ⇒ Electrical system; Rep. gr. 94 ; Steering column switch module; Removing and installing steering column combination switch .

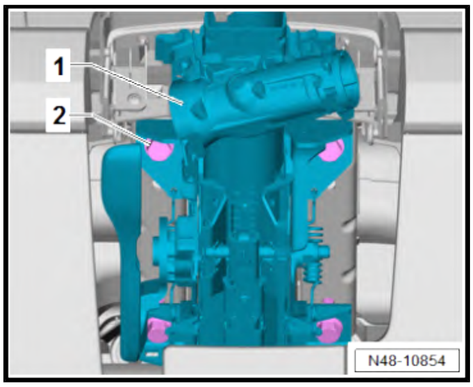
– Unscrew bolt -1-.
– Pull universal joint -2- off intermediate steering shaft -3-.



– Unclip cable duct -1- from steering column -2- and lay aside.



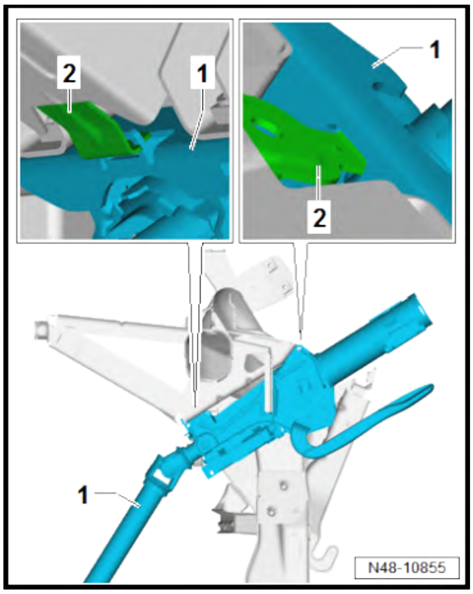
– Unscrew bolts -2-.
– Take out steering column -1-.



Installing

Install in the reverse order of removal, observing the following:

– Engage steering column -1- in brackets -2-.
– Check straight-ahead position of steering wheel during road test.
– Perform basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.



Specified torques

♦ ⇒ “2.1 Assembly overview - steering column”, page 223

2.5 Removing and installing intermediate steering shaft

Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1331-

Removing

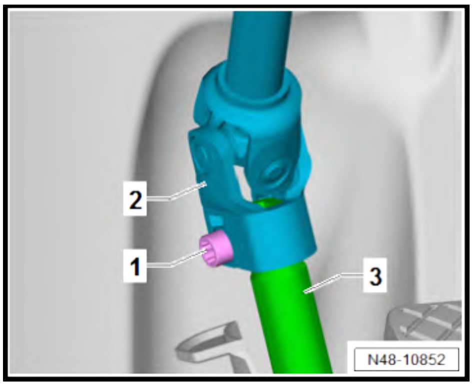
– Turn wheels to straight-ahead position.

– Remove floor covering ⇒ General body repairs, interior; Rep. gr. 70 ; Interior trim; Removing and installing floor covering .

– Clean connection of universal joint -2- and intermediate steering shaft -3-.

– Unscrew bolt -1-.

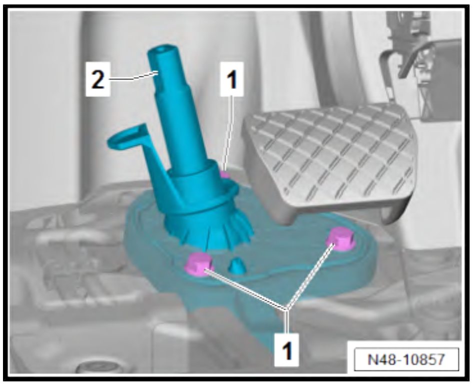
– Pull universal joint -2- off intermediate steering shaft -3-.

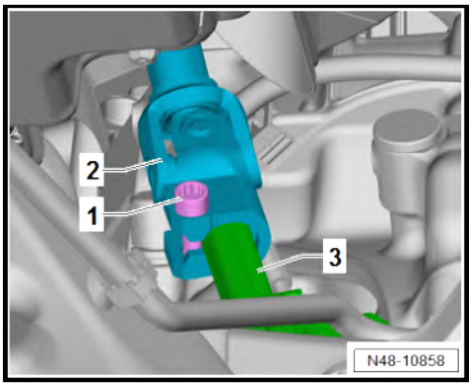


– Unscrew bolts -1-.

– Lower intermediate steering shaft -2- slightly.

– Clean connection of universal joint -2- and steering rack -3-.

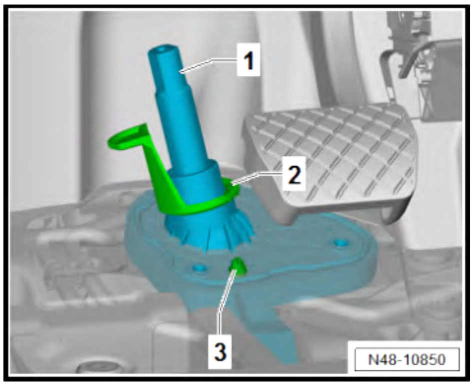




– Unscrew bolt -1-.
– Pull off intermediate steering shaft -2- from steering rack -3-.
– Remove intermediate steering shaft -2-.

Installing

Install in the reverse order of removal, observing the following:



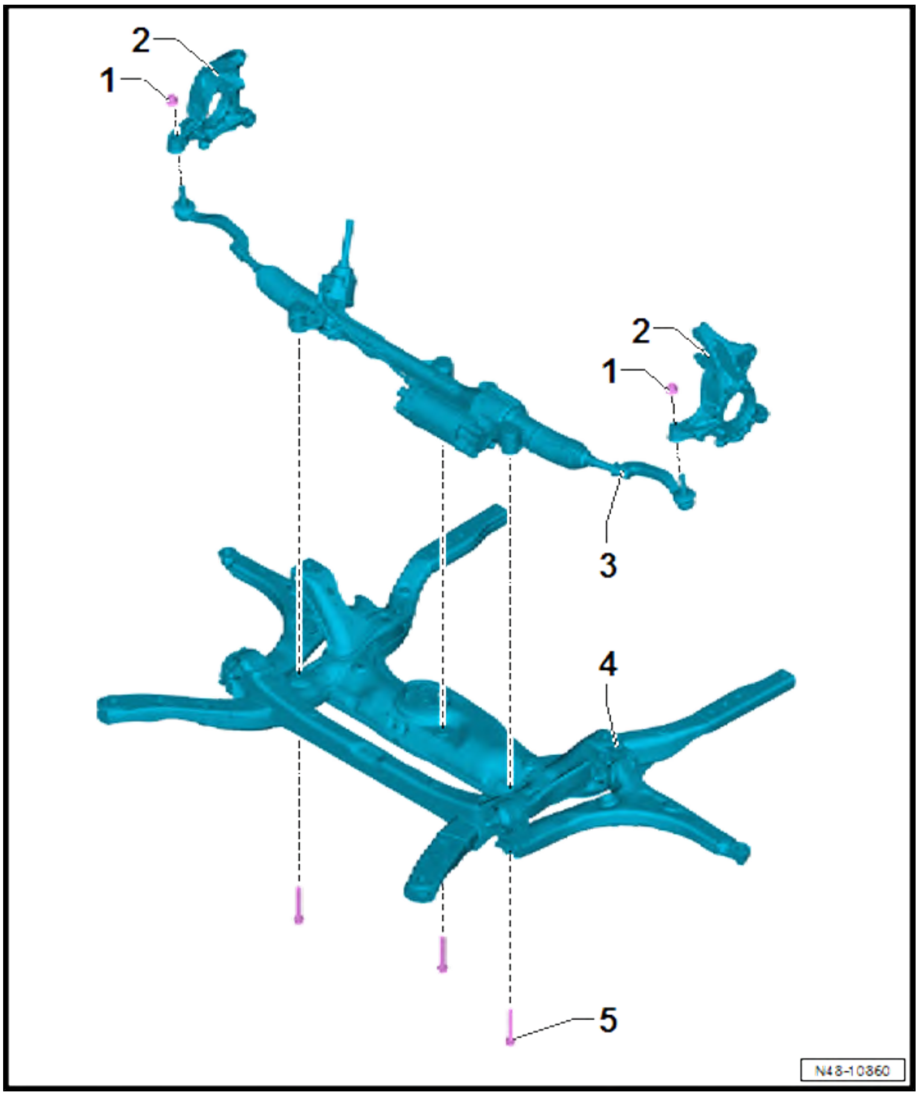
– Pull intermediate steering shaft -1- through opening by handle -2-.
– Align intermediate steering shaft -1- on centring pin -3-.
– Check straight-ahead position of steering wheel during road test.
– Perform basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.

Specified torques

♦ ⇒ “2.1 Assembly overview - steering column”, page 223

3 Steering rack

3.1 Assembly overview - steering rack



1 - Nuts
- ❑ Renew after removal
- ❑ Qty. 2
- ❑ 75 Nm +90°

2 - Wheel bearing housing
- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 75

3 - Steering rack
- ❑ Removing and installing ⇒ page 230

4 - Subframe
- ❑ Removing and installing ⇒ page 29

5 - Bolts
- ❑ Renew after removal
- ❑ Qty. 3
- ❑ 70 Nm +180°

3.2 Checking steering rack for damage

Visual check

– Check whether any parts, e.g. boots of steering rack, exhibit signs of damage.

– If the steering rack shows signs of corrosion, damage or wear, complete steering rack must be renewed ⇒ page 230 .

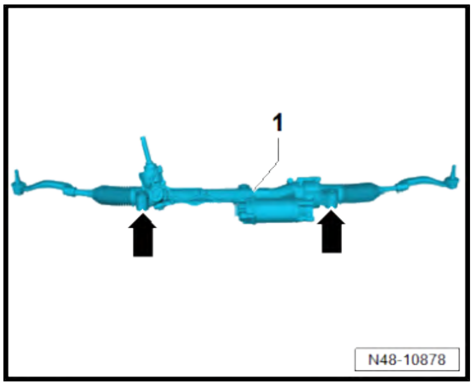
3.3 Handling and transporting steering rack

Note

- *Adherence to proper steering rack handling is essential.*
- *Before touching the steering rack, discharge any electrostatic energy.*
- *Improper handling of the steering rack may damage it, leading to safety risks.*

Proper steering rack handling and transport

- Use both hands to transport steering rack -1-.
- Only transport steering rack -1- in area marked by -arrows-.
- Boots must not be damaged.



3.4 Removing and installing steering rack

⇒ “3.4.1 Removing and installing steering rack, longitudinal engine”, page 230

⇒ “3.4.2 Removing and installing steering rack, transverse engine, without auxiliary coolant heater”, page 231

⇒ “3.4.3 Removing and installing steering rack, transverse engine, with auxiliary coolant heater”, page 233

⇒ “3.4.4 Removing and installing steering rack, transverse engine, right-hand drive vehicle”, page 234

⇒ “3.4.5 Removing and installing steering rack, e-Crafter or e-TGE”, page 236

3.4.1 Removing and installing steering rack, longitudinal engine

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-

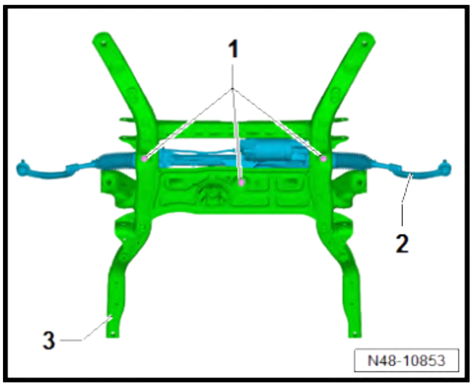
Removing

– Remove subframe with steering rack ⇒ page 29 .

– Unscrew bolts -1-.

– Observe steering rack handling and transporting information ⇒ page 230 .

– Remove steering rack -2- from subframe -3-.

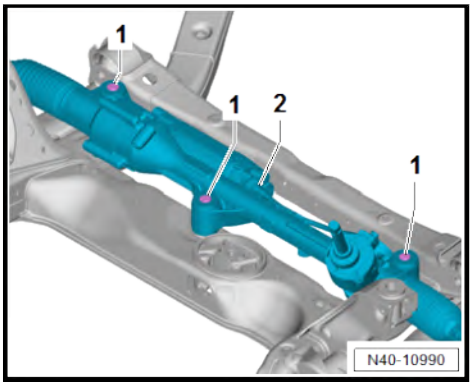


Installing

Install in the reverse order of removal, observing the following:

Note

Flats -1- of mountings must be parallel to axis of steering rack -2-.



– If necessary, move mountings to correct position if twisted.

– Counterhold steering rack threaded connection when tightening.

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

– Carry out basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.

– If new steering rack has been installed, adapt electromechanical power steering using ⇒ Vehicle diagnostic tester.

Specified torques

♦ ⇒ “3.1 Assembly overview - steering rack”, page 229

3.4.2 Removing and installing steering rack, transverse engine, without auxiliary coolant heater

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-

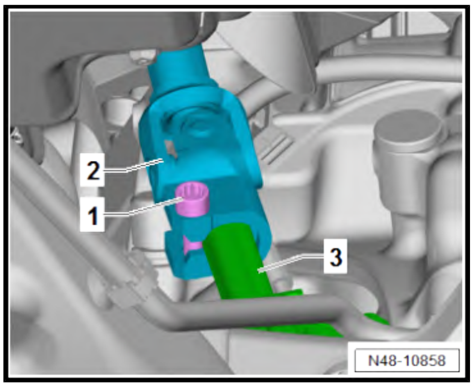
Removing

– Turn wheels to straight-ahead position.

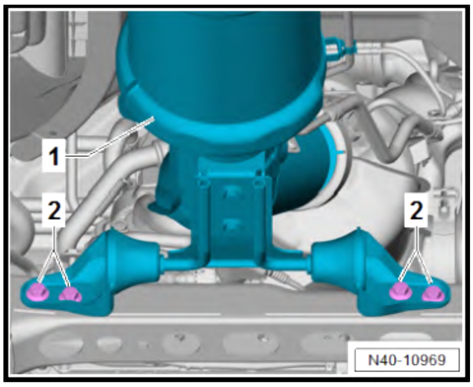
– Unscrew bolt -1-.

– Pull off intermediate steering shaft -2- from steering rack -3-.

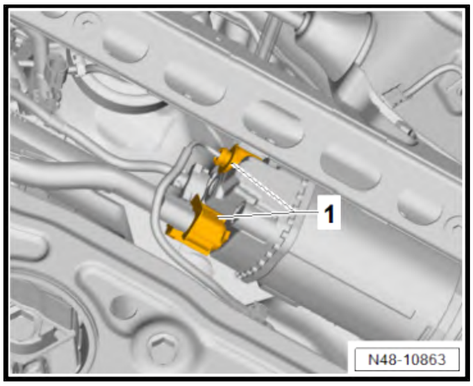
– Remove track rod ball joint ⇒ page 240 .



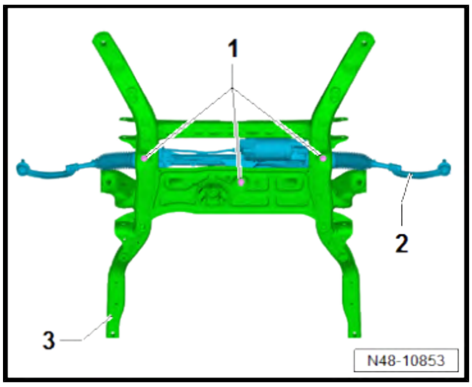
– Unscrew bolts -2- from exhaust system -1-.



– Separate electrical connectors -1-.



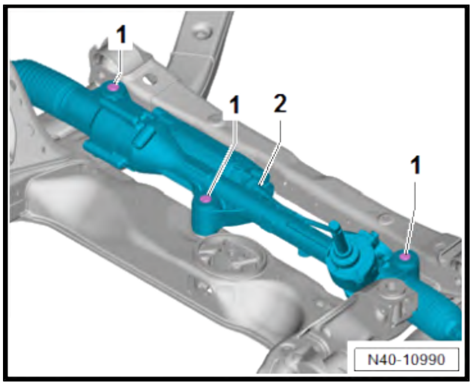
– Unscrew bolts -1-.



– Remove steering rack -2- from subframe -3- and swivel 180° backwards.
– Observe steering rack handling and transporting information ⇒ page 230 .
– Remove steering rack -2- to left.

Installing

Install in the reverse order of removal, observing the following:



Note

Flats -1- of mountings must be parallel to axis of steering rack -2-.

– If necessary, move mountings to correct position if twisted.
– Counterhold steering rack threaded connection when tightening.

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

– Perform basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.
– If new steering rack has been installed, adapt electromechanical power steering using ⇒ Vehicle diagnostic tester.

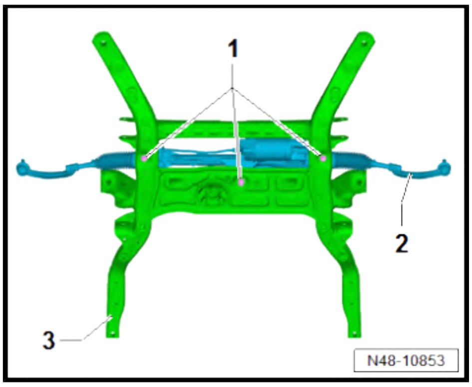
Specified torques

♦ Emission control; Assembly overview - emission control ⇒ Rep. gr. 26 ; Emission control; Assembly overview - emission control
♦ ⇒ “3.1 Assembly overview - steering rack”, page 229
♦ ⇒ “3.8 Repairing steering rack”, page 242
♦ ⇒ “2.1 Assembly overview - steering column”, page 223

3.4.3 Removing and installing steering rack, transverse engine, with auxiliary coolant heater

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-



Removing

– Remove subframe with steering rack ⇒ page 29 .

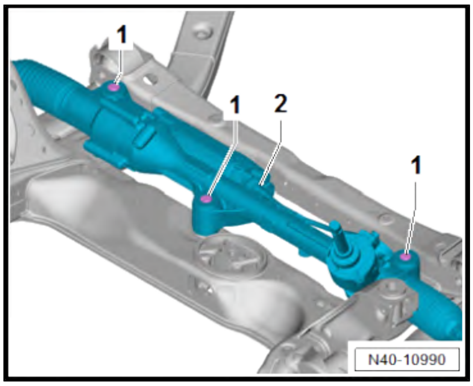
– Unscrew bolts -1-.

– Observe steering rack handling and transporting information ⇒ page 230 .

– Remove steering rack -2- from subframe -3-.

Installing

Install in the reverse order of removal, observing the following:



Note

Flats -1- of mountings must be parallel to axis of steering rack -2-.

– If necessary, move mountings to correct position if twisted.

– Counterhold steering rack threaded connection when tightening.

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

– Carry out basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.

– If new steering rack has been installed, adapt electromechanical power steering using ⇒ Vehicle diagnostic tester.

Specified torques

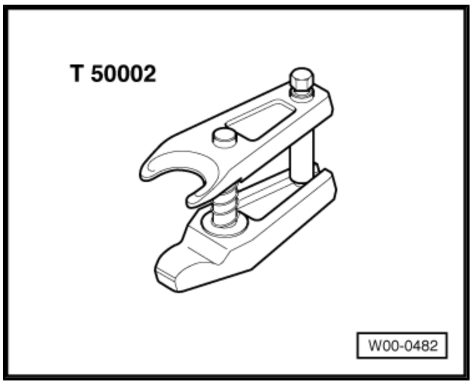
♦ ⇒ “3.1 Assembly overview - steering rack”, page 229

3.4.4 Removing and installing steering rack, transverse engine, right-hand drive vehicle

Special tools and workshop equipment required

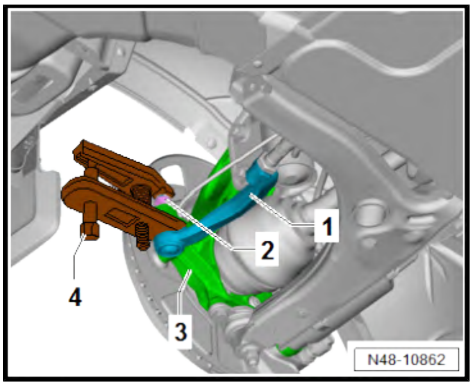
♦ Torque wrench - V.A.G 1332-

♦ Ball joint puller - T50002-

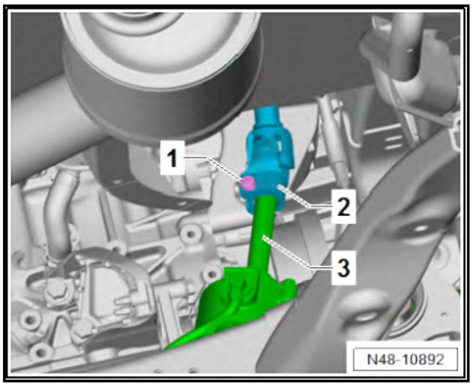


Removing

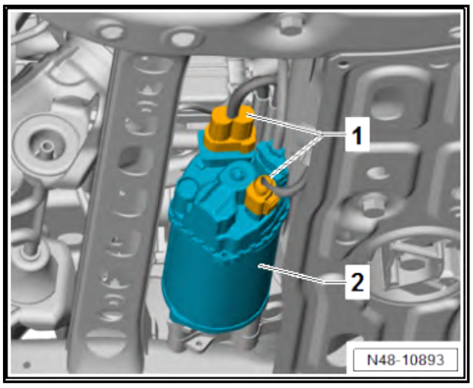
– Turn wheels to straight-ahead position.
– Remove front wheels ⇒ page 150 .
– Loosen nut -2-.
– Press off track rod ball joint -1- using ball joint puller - T50002- -4-.
– Unscrew nut -2-.
– Pull track rod ball joint -1- off swivel joint -3-.



– Unscrew bolt -1-.
– Pull off intermediate steering shaft -2- from steering rack -3-.



– Separate electrical connectors -1-.
– Unclip wiring harness from steering rack -2-.

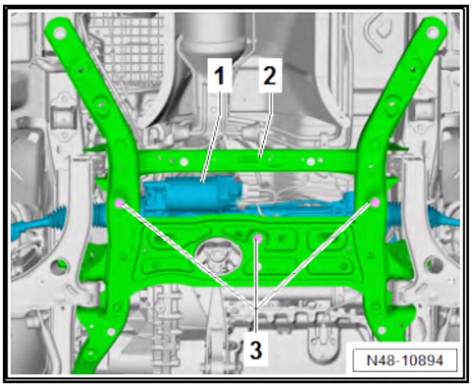


– Unscrew bolts -3-.

– Remove steering rack -1- from subframe -2- and swivel 180° backwards.

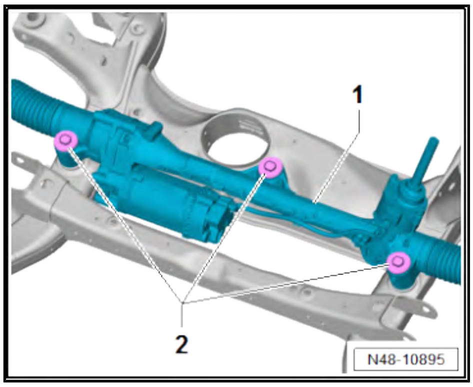
– Observe steering rack handling and transporting information ⇒ page 230 .

– Remove steering rack -1- to right.



Installing

Install in the reverse order of removal, observing the following:



Note

Flats -2- of mountings must be parallel to axis of steering rack -1-.

– If necessary, move mountings to correct position if twisted.

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

– Perform basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.

– If new steering rack has been installed, adapt electromechanical power steering using ⇒ Vehicle diagnostic tester.

Specified torques

♦ ⇒ “3.1 Assembly overview - steering rack”, page 229

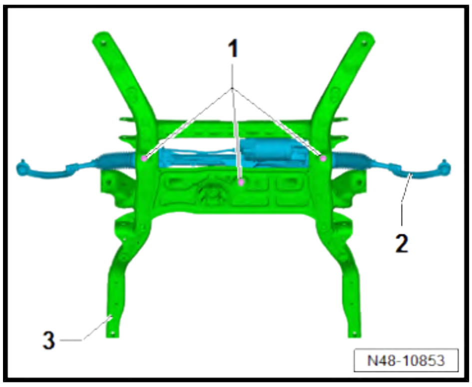
♦ ⇒ “3.8 Repairing steering rack”, page 242

♦ ⇒ “2.1 Assembly overview - steering column”, page 223

3.4.5 Removing and installing steering rack, e-Crafter or e-TGE

Special tools and workshop equipment required

♦ Torque wrench - V.A.G 1332-



Removing

– Remove subframe with steering rack ⇒ page 29 .

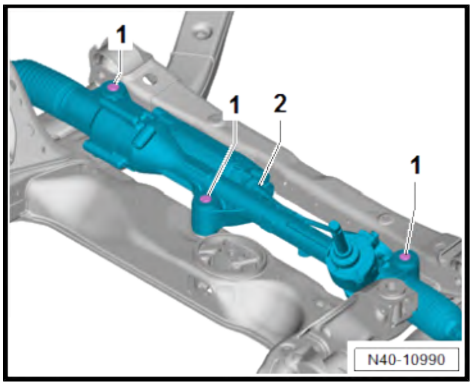
– Unscrew bolts -1-.

– Observe steering rack handling and transporting information ⇒ page 230 .

– Remove steering rack -2- from subframe -3-.

Installing

Install in the reverse order of removal, observing the following:



Note

Flats -1- of mountings must be parallel to axis of steering rack -2-.

– If necessary, move mountings to correct position if twisted.

– Counterhold steering rack threaded connection when tightening.

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

– Carry out basic setting for steering angle sender - G85- using ⇒ Vehicle diagnostic tester.

– If new steering rack has been installed, adapt electromechanical power steering using ⇒ Vehicle diagnostic tester.

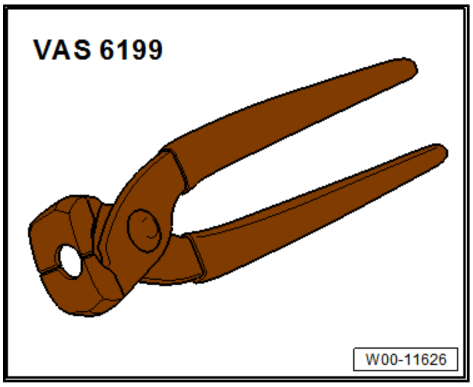
Specified torques

♦ ⇒ “3.1 Assembly overview - steering rack”, page 229

3.5 Removing and installing bellows

Special tools and workshop equipment required

♦ Locking pliers for steering rack - VAS 6199-



Removing

Vehicles with electric drive

DANGER

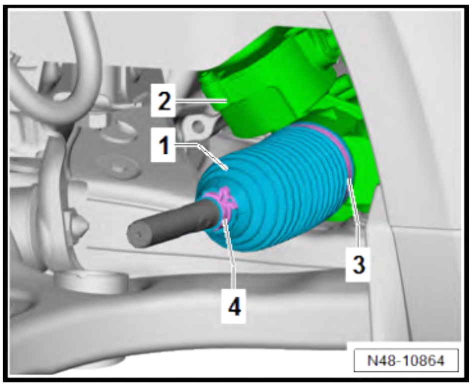
Danger to life from high voltage.

Severe or fatal injury from electric shock.

– The high-voltage system must be de-energised by a suitably qualified technician.

– De-energise high-voltage system ⇒ Rep. gr. 93 ; De-energising high-voltage system .

Continued for all vehicles



– Remove track rod ball joint ⇒ page 240 .

– Unscrew clamps -3- and -4-.

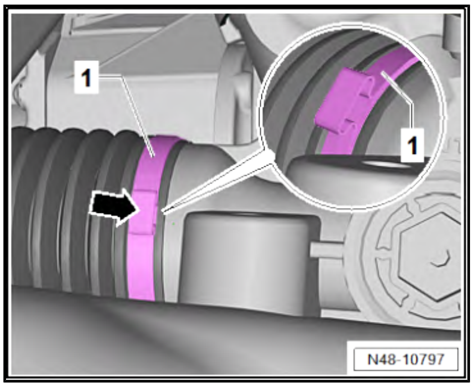
– Pull bellows -1- off steering rack -2-.

Installing

Install in the reverse order of removal, observing the following:

Note

If the steering rack shows signs of corrosion, damage or wear, renew the complete steering rack ⇒ page 230 .



– Secure new hose clip -1- using locking pliers for steering rack - VAS 6199- .

– Crimp together lug -arrow- of clamp -1- sufficiently as shown.

Vehicles with electric drive

WARNING

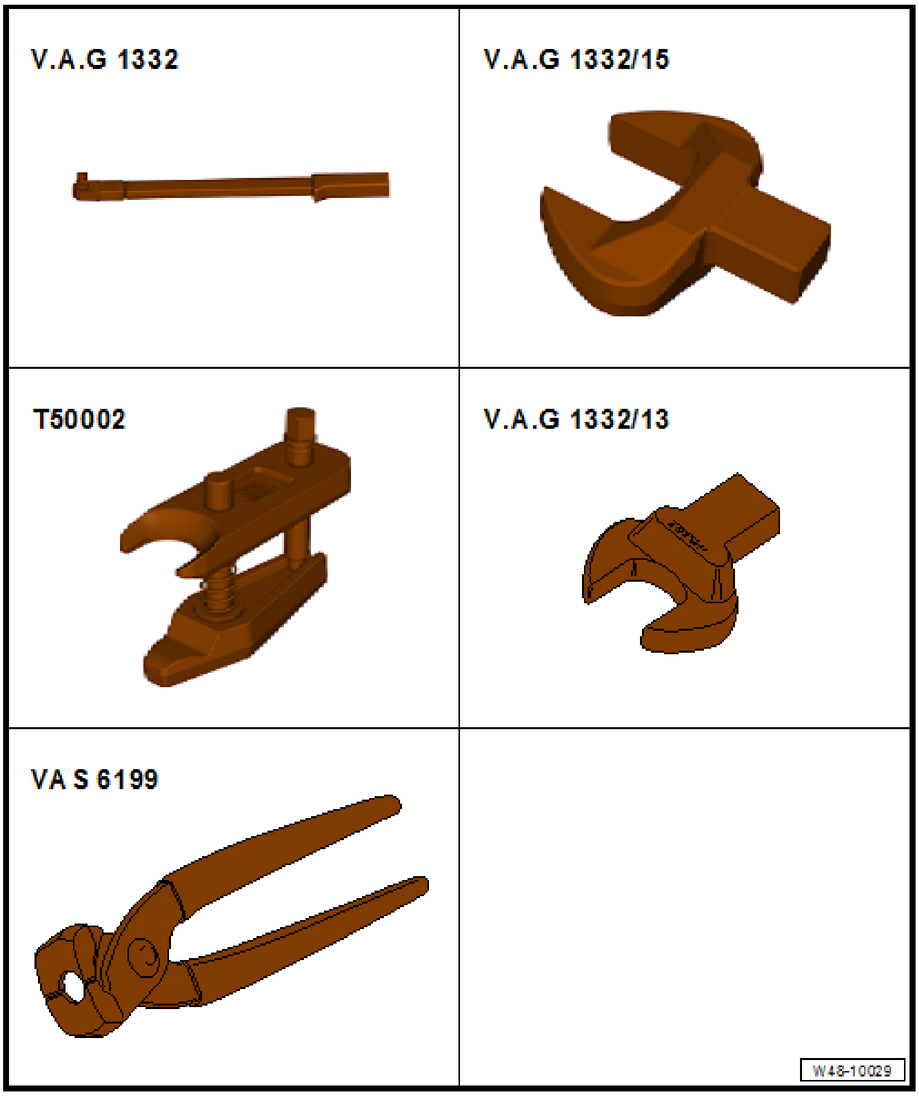
Danger to life from high voltage.

Risk of severe or fatal injury due to electric shock.

– Have a qualified technician re-energise the high-voltage system.

– Re-energise high-voltage system ⇒ Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

3.6 Removing and installing track rod



Special tools and workshop equipment required

- ♦ Torque wrench - V.A.G 1332-
- ♦ Open jaw socket insert AF 30 - V.A.G 1332/15-
- ♦ Ball joint puller - T50002-
- ♦ Open-end spanner insert AF 41 - V.A.G 1332/13-
- ♦ Locking pliers for steering rack - VAS 6199-

Removing

Vehicles with electric drive

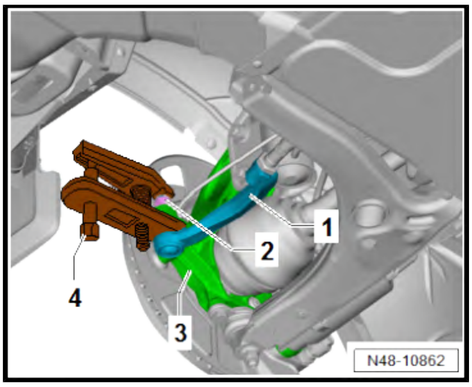
DANGER

Danger to life from high voltage.
Severe or fatal injury from electric shock.
– The high-voltage system must be de-energised by a suitably qualified technician.

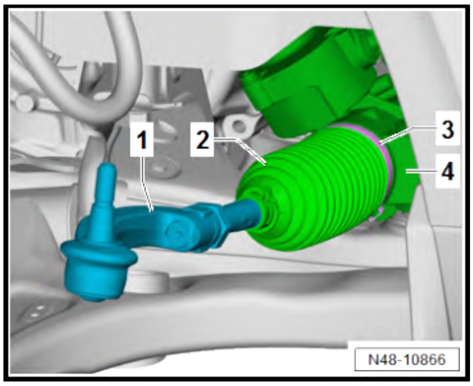
– De-energise high-voltage system ⇒ Rep. gr. 93 ; De-energising high-voltage system .

– Remove front wheel ⇒ page 150 .

– Unscrew nut -2- from track rod ball joint -1-.
– Press track rod ball joint -1- off swivel joint -3- using ball joint puller - T50002- -4-.
– Unscrew nut -2-.
– Pull track rod ball joint -1- off swivel joint -3-.



– Loosen clamp -3-.
– Pull bellows -2- off steering rack -4-.
– Unscrew track rod -1-.



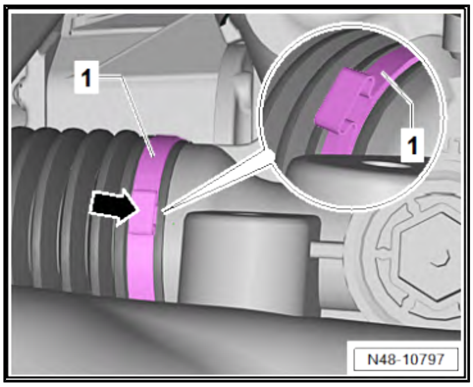
Installing

Install in the reverse order of removal, observing the following:

Note

If the steering rack shows signs of corrosion, damage or wear, renew the complete steering rack ⇒ page 230 .

– Secure new hose clip -1- using locking pliers for steering rack - VAS 6199- .
– Crimp lug -arrow- of clamp -1- sufficiently as shown.



To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

Vehicles with electric drive

WARNING

Danger to life from high voltage.
Risk of severe or fatal injury due to electric shock.
– **Have a qualified technician re-energise the high-voltage system.**

– Re-energise high-voltage system ⇒ Rep. gr. 93 ; Electric drive; Re-energising high-voltage system .

Specified torques

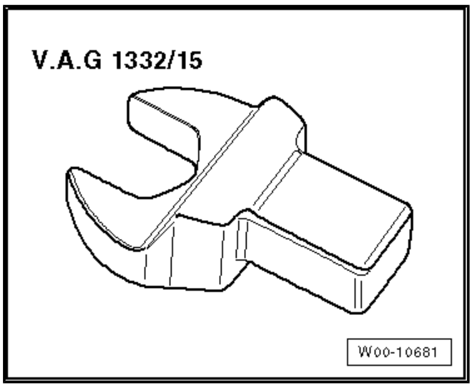
♦ ⇒ “3.8 Repairing steering rack”, page 242
♦ ⇒ “3.1 Assembly overview - steering rack”, page 229

3.7 Removing and installing track rod ball joint

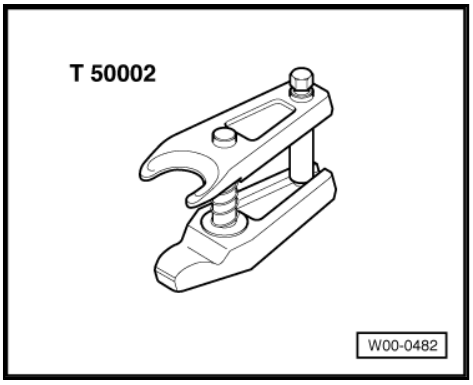
Special tools and workshop equipment required

- Torque wrench - V.A.G 1332-

- Open jaw socket insert AF 30 - V.A.G 1332/15-



- Ball joint puller - T50002-



Removing

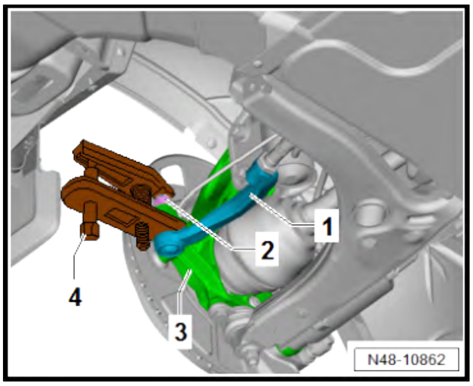
– Remove front wheel ⇒ page 150 .

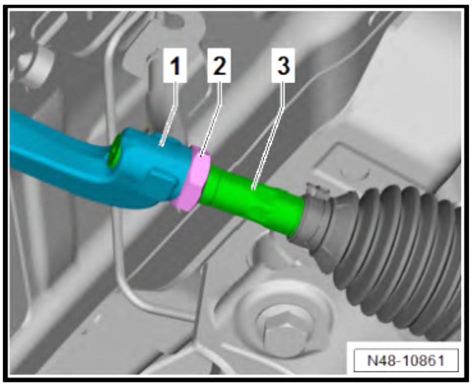
– Unscrew nut -2- from track rod ball joint -1-.

– Press track rod ball joint -1- off swivel joint -3- using ball joint puller - T50002- -4-.

– Unscrew nut -2-.

– Pull track rod ball joint -1- off swivel joint -3-.





– Loosen nut -2-.

– Unscrew track rod ball joint -1- from track rod -3-.

Installing

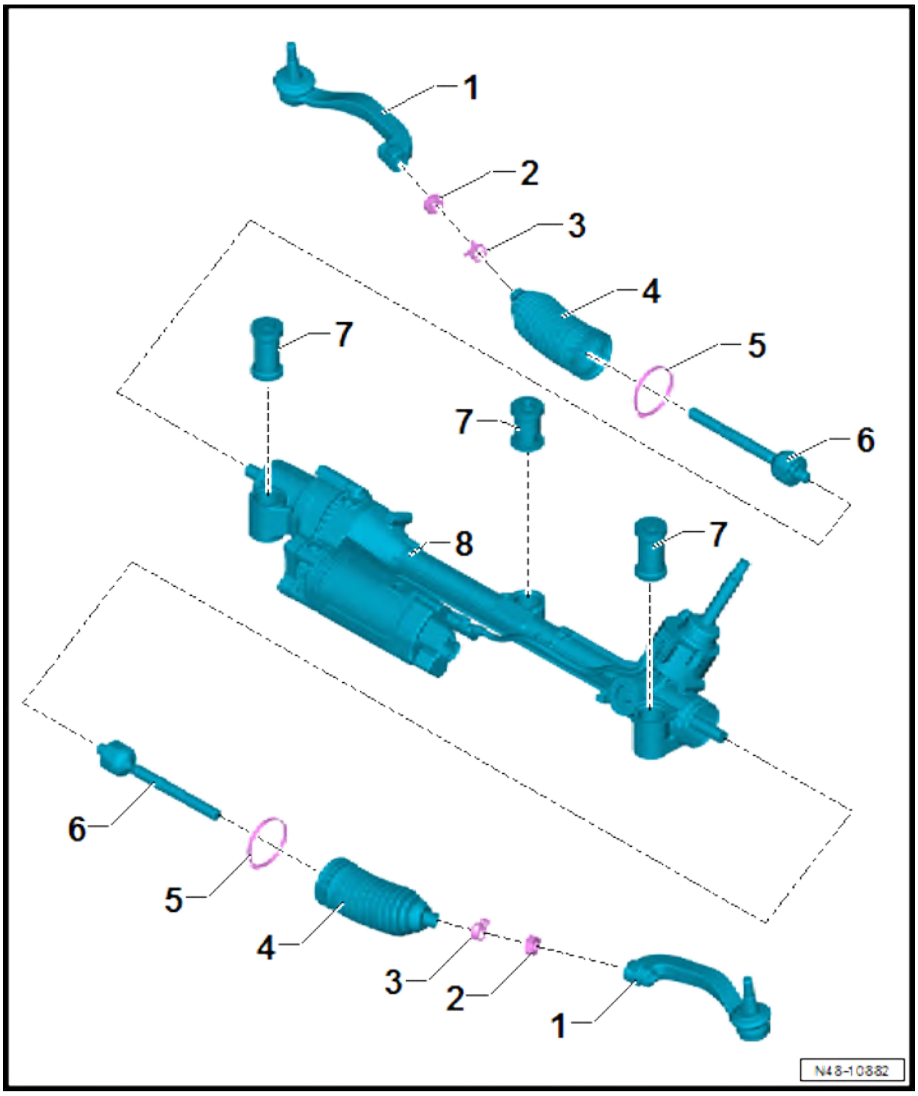
Install in the reverse order of removal, observing the following:

To determine in which cases wheel alignment is necessary, refer to table ⇒ page 186 .

Specified torques

- ⇒ “3.8 Repairing steering rack”, page 242
- ⇒ “3.1 Assembly overview - steering rack”, page 229

3.8 Repairing steering rack



1 - Track rod ball joints

- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 240

2 - Nuts

- ❑ Qty. 2
- ❑ 100 Nm

3 - Clips

- ❑ Qty. 2

4 - Bellows

- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 237

5 - Clips

- ❑ Renew after removal
- ❑ Qty. 2

6 - Track rods

- ❑ Qty. 2
- ❑ Removing and installing ⇒ page 238
- ❑ 115 Nm

7 - Mountings

- ❑ Qty. 3
- ❑ Removing and installing ⇒ page 242

8 - Steering rack

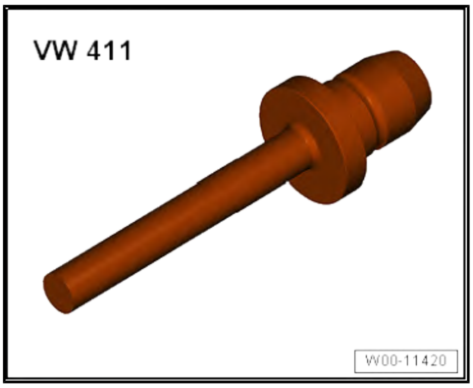
- ❑ Removing and installing ⇒ page 230

3.9 Removing and installing steering rack mountings

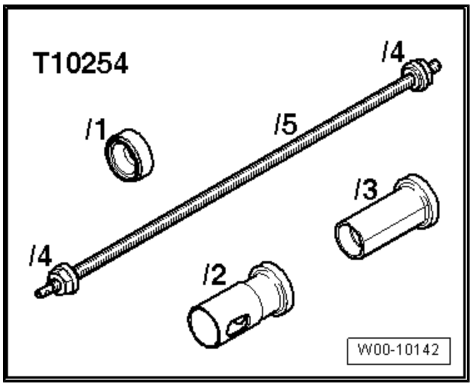
Special tools and workshop equipment required

◆ Thrust plate - VW401-

◆ Press tool - VW411-



◆ Assembly tool - T10254-



Removing

- Steering rack removed ⇒ page 230

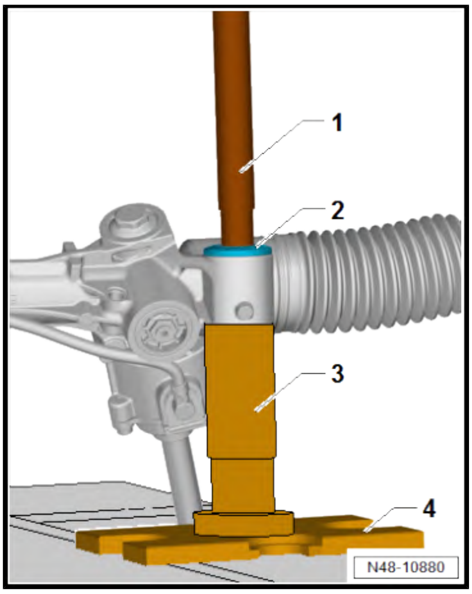
– Position tools as shown in illustration.

1 - Press tool - VW411-

3 - Assembly tool - T10254/2-

4 - Pressure plate - VW401-

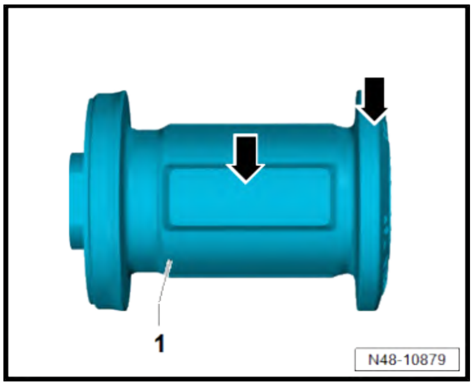
– Press out bonded rubber bush -2-.



Installing

Install in the reverse order of removal, observing the following:

– Coat new bonded rubber bush -1- with assembly lubricant ⇒ Electronic parts catalogue (ETKA) or ⇒ webMANTIS for MAN TGE in area marked by -arrows-.



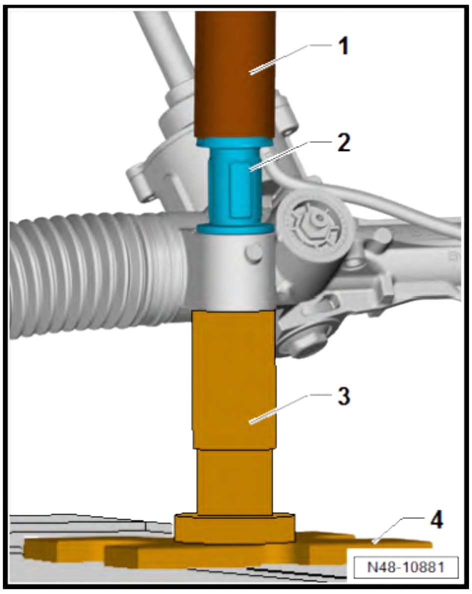
– Position tools as shown in illustration.

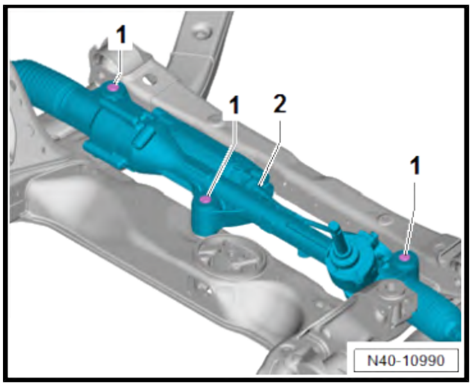
1 - Assembly tool - T10254/3-

3 - Assembly tool - T10254/2-

4 - Pressure plate - VW401-

– Pressing in bonded rubber bush -2-.





Note

Flats -1- of mountings must be parallel to axis of steering rack -2-.

– If necessary, move mountings to correct position if twisted.
– Counterhold steering rack threaded connection when tightening.

4 Sensors

⇒ “4.1 Removing and installing steering angle sender G85 ”, page 246

4.1 Removing and installing steering angle sender - G85-

Note

- *Steering angle sender - G85- is a component of the steering rack.*
- *It cannot be renewed as an individual part.*
- *If the steering angle sender - G85- is defective, the steering rack must be renewed ⇒ page 230 .*